



2020 K-Innovation ODA Program with Azerbaijan

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan (2nd Year): The Development of Strategic Plan for ANAS HTP

2020 K-Innovation ODA Program with Azerbaijan

Title	Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan (2 nd Year): The Development of Strategic Plan for ANAS HTP
Institute	Science and Technology Policy Institute (STEP I)
Partner Country	Republic of Azerbaijan
In Cooperation with	Azerbaijan National Academy of Sciences - High Technologies Park (ANAS HTP)
Program Investigators	Inkyoung Sun, Head, the Office of Development Cooperation Research, STEP I Wangdong Kim, Chief Director, Division of Global Innovation Strategy, STEP I
Project Manager	Wangdong Kim, Chief Director, Division of Global Innovation Strategy, STEP I
Senior Advisor	Myungdo Oh, Professor, University of Seoul
Researchers	Jong Hwa Choi, Head, Office of Future Industry Research, STEP I Yeojin Yoon, Researcher, , Division of Global Innovation Strategy, STEP I Jong-in Choi, Professor, Hanbat National University Sangho Choi, Head, Policy Planning Department, Daejeon Technopark Bunyamin Seyidov, Chief, Dept of Relations with Eastern Countries, ANAS HTP Ilaha Rzayeva, Head of Investment, Marketing and Commerce Department, ANAS HTP Vusal Suleymanli, Innovation Projects Coordinator, ANAS HTP
Project Coordinator	Yeojin Yoon, Researcher , Division of Global Innovation Strategy, STEP I

Survey Research 2020-05-04-01

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan (2nd Year): The Development of Strategic Plan for ANAS HTP

Published on 30 December 2020

Copyright © 2020 by Science and Technology Policy Institute (STEP I). All Rights Reserved.

5th-7th Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si,
30147. Republic of South Korea

Tel: +82-44-287-2000 Fax: +82-44-287-2068

www.stepi.re.kr

For more information, visit

iicc.stepi.re.kr

Printed by Mirae Media

Tel: +82-2-815-0407 Fax: +82-2-822-1269

ISBN 978-89-6112-736-3 93300



Executive Summary

The Science and Technology Policy Institute (STEPI) of South Korea and the Azerbaijan National Academy of Sciences – High Technologies Park (ANAS-HTP) have worked together to carry out the two-year long K-Innovation ODA Program (2019-2020) under the title of “Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan.” In the first year (2019), we analyzed Azerbaijan’s STI policy and the TT & TC ecosystem at the national level that affect ANAS-HTP’s TT & TC activities, and shared South Korea’s TT & TC experiences. For the second year (2020), the project team developed a strategic plan for the successful take-off of ANAS-HTP at the institutional level, based on the research outcomes of the first year.

The key contents of the proposed strategic plan are as follows. First, the project team has defined ANAS-HTP’s vision of “Becoming a Top High-Tech Park Company in Caspian Countries by 2035.” The project team has also presented structure, budget, workforce, and network as core elements of the internal system, which will help realize this vision. It has also identified key operational programs of i) TT & TC, ii) Start-up & Incubation, and iii) Test & Certification, and presented a roadmap to implement these programs.

In the second year, unlike the first year, all the project activities were inevitably carried out through video conference (e.g. Zoom) and social networks (e.g. What’s app) because of the COVID-19 pandemic. Following the preparatory meeting held in February, the Kick-off & Field Interview and the Capacity Development Workshop were held in May and in September, respectively. Afterwards, a joint team of the South Korean experts and the Azerbaijani counterparts worked together to develop a final report. The final outcomes of the project were then shared at the Dissemination Workshop held in December.

In this process, Dr. Ali Abbasov, former Director of ANAS-HTP, and Mr. Mammad Baghirzadeh, former Deputy Director on the Strategic Development, have provided their extensive support. At the Capacity Development Workshop held in September, some South Korean Technopark experts participated as speakers, kindly delivered their lectures, and offered their presentation materials to the project team. I would like to deliver my special thanks again to these experts, Ms. Yang-jin Kim (Entrepreneurship Consultant of Gyeongbuk TP), Ms. Kyoung Joo Nam (Team Leader of Gyeonggi Technopark), Mr. Jong-Wha Lee (Associate Researcher of Daejeon Technopark), and Mr. Seong Byung-Ho (Director General of Daegu Technopark). It is my ardent wish that this project would lead to a follow-up project between STEPI and ANAS-HTP, so that our continuous cooperation through the follow-up project would serve as a momentum for ANAS-HTP to emerge as a top high-tech park in the Caspian countries by the year 2035.

CONTENTS

Chapter 1. Project Overview	3
1. Introduction	3
2. Objectives.....	5
3. Project Framework	6
4. Project Team	10
5. Main Activities	12
6. Project Schedule	15
7. Project Outputs	16

Chapter 2. Current Status of ANAS HTP	19
1. Internal System.....	19
2. Technology Transfer & Commercialization	43
3. Start-up & Incubation	45
4. Test & Certification	52

Chapter 3. South Korean Experiences	63
1. Technopark (TPs) in South Korea: Internal System.....	63
2. Technology Transfer & Commercialization in South Korea	111
3. Start-up & Incubation in South Korea	130
4. Test & Certification in South Korea	145

Chapter 4. Capacity Development Process	159
1. Kick-off Meeting and Field Study.....	159
2. STI Capacity Development Workshop.....	191
3. The Dissemination Workshop.....	216

Chapter 5. Strategic Plan for ANAS HTP.....	227
1. Internal System.....	227
2. Technology Transfer & Commercialization	254
3. Start-up and Incubation	270
4. Test & Certification	284

Chapter 6. Conclusion	301
------------------------------------	------------

REFERENCES.....	305
------------------------	------------

List of Tables

[Table 1-1] STEPI Research Team	10
[Table 1-2] ANAS HTP Team	11
[Table 1-3] Project Schedule	15
[Table 2-1] Budget Distribution by Department at ANAS HTP	32
[Table 2-2] Amount of Budget by Source	33
[Table 2-3] Staff Schedule of ANAS HTP for 2020.....	35
[Table 2-4] Azerbaijan Team.....	53
[Table 2-5] List of SSATC Equipment.....	54
[Table 2-6] Description of Testing for Crude Oil, LPG, Gas and Other Products	57
[Table 3-1] Role of the Board of Directors in TP based on the Standard	69
[Table 3-2] Composition of the Board of Directors based on the Standard.....	71
[Table 3-3] Ex-officio and Elective Members in the Board of Directors based on the Standard	72
[Table 3-4] Composition of the CEO-president Recommendation Committee based on the Standard.....	73
[Table 3-5] Guidelines of Setting up an Organization Based on the Standard.....	74
[Table 3-6] CEOs/Presidents of South Korean TPs, June 2020	78
[Table 3-7] Budgets and Employees of South Korean TPs, Dec. 31, 2019.....	79
[Table 3-8] SME Support Programs and Grant Resources of GTP, 2020	83
[Table 3-9] Rates of Indirect Expenses in TPs	85
[Table 3-10] Annual Contribution by Local Governments for Sustainable Operation of TPs	87
[Table 3-11] 3-Year Rates of Employees' Continuous Working at GTP	89
[Table 3-12] HR in GTP, Dec. 2019	90
[Table 3-13] Benefits for Residents of Daejeon TP	102
[Table 3-14] System Operating Environment	107
[Table 3-15] Types of Technology Transfer	112
[Table 3-16] Types of Technology Commercialization	112
[Table 3-17] Types of South Korea's Technology Commercialization Support Policy	115
[Table 3-18] Current Status of TT & TC Laws	116

[Table 3-19] The 6 th TT & TC Promotion Plan	117
[Table 3-20] Projects for TC Support	120
[Table 3-21] Status of Technology Evaluation Laws	124
[Table 3-22] Status of Technology Evaluation and Transection Institutions	125
[Table 3-23] Current Status of Institutions Possessing and Transferring Technology	128
[Table 3-24] Number of TT Contracts	129
[Table 3-25] Number of Public Sector TT and TT Rate	129
[Table 3-26] Number of Technologies Transferred	129
[Table 3-27] Evolution of Business Incubation Center (BIC) in South Korea	131
[Table 3-28] Evolution of KISED	133
[Table 3-29] Incubation Performance Evaluation Index	137
[Table 3-30] KOLAS Accreditation Status	149
[Table 3-31] KAB Accreditation Status	149
[Table 3-32] Status of Conformity Assessment Laws	150
[Table 3-33] Status of Standards in South Korea	150
[Table 3-34] Conformity Assessment Requirement for ISO/IEC	151
[Table 3-35] Total Number of Technical Regulations	152
[Table 3-36] Certification Status in South Korea	153
[Table 3-37] Number of Certifications	153
[Table 3-38] Certification Systems and Laws in South Korea	154
[Table 6-1] An Example of Follow-up Project Activities	302

List of Figures

[Figure 1-1] Analytical Framework	7
[Figure 1-2] Strategic Roadmap	8
[Figure 1-3] Operational Framework	9
[Figure 2-1] Current Structure of ANAS HTP	27
[Figure 2-2] Budget Structure of ANAS HTP	32
[Figure 2-3] Azerbaijan Incubation Centers	45
[Figure 2-4] Performance Indicators	48
[Figure 2-5] Labs for Incubators	49
[Figure 2-6] ANAS Testing Center	56
[Figure 3-1] Concept of Technopark	64
[Figure 3-2] Changes in TP's Major Missions by Stage	65
[Figure 3-3] Roles & Functions of TP	67
[Figure 3-4] Major programs of TP	68
[Figure 3-5] Organization Chart of Gyeongnam (Prov.) TP	75
[Figure 3-6] Organization Chart of Chungnam (Prov.) TP	76
[Figure 3-7] Organization Chart of GTP	77
[Figure 3-8] Yearly Budget of GTP in 2020	81
[Figure 3-9] SME Supportive Programs of GTP, 2020	82
[Figure 3-10] Occupancy Rates and Ratio of Leasing Income in GTP (2016-2019)	85
[Figure 3-11] Bird's Eye View of the Smart Manufacturing Innovation Center of GTP	87
[Figure 3-12] Stakeholder Groups of Technopark	92
[Figure 3-13] Network Activation Program of TP	94
[Figure 3-14] Network Activation Program with Overseas Innovators	95
[Figure 3-15] Examples of representative international cooperative projects by Gyeongbuk TP	96
[Figure 3-16] DJTP's Internal System on Resident Support Projects	98
[Figure 3-17] Residence Selection and Management System	100
[Figure 3-18] Growth Management System by Growth Stages	103

[Figure 3-19] Residents Growth Management System Process.....	103
[Figure 3-20] Daegu TP Integrated Information System	106
[Figure 3-21] National TT & TC Policy Framework.....	114
[Figure 3-22] R&D Investment for SMEs.....	118
[Figure 3-23] Share of TC for SMEs	119
[Figure 3-24] Amount of Investment in TC	119
[Figure 3-25] TC Success Rate	121
[Figure 3-26] Technology Finance Structure	122
[Figure 3-27] South Korea's Technology Finance Scale	122
[Figure 3-28] Size of Venture Capital in Terms of GDP	122
[Figure 3-29] Technology Evaluation	123
[Figure 3-30] Technology Evaluation Infrastructure.....	124
[Figure 3-31] Technology Trading Platform.....	126
[Figure 3-32] Status of Technology Evaluation Platforms	126
[Figure 3-33] Status of National R&D and Patent Performance.....	127
[Figure 3-34] Patent Status in South Korea	128
[Figure 3-35] Startup Ecosystem.....	130
[Figure 3-36] Location and Performance of Business Incubation Centers.....	132
[Figure 3-37] Five Key Tasks of KISED	134
[Figure 3-38] TIPS Incubation Process	135
[Figure 3-39] Map of Daedeok Innopolis, South Korea	139
[Figure 3-40] 17 Business Incubation Centers in Daedeok Innopolis	140
[Figure 3-41] Research Institute Spin-off System: Incubation Method of Daedeok Innopolis.....	144
[Figure 3-42] Key Concepts of Test & Certification.....	146
[Figure 3-43] General Conformity Assessment Operating System.....	147
[Figure 3-44] National Conformity Assessment Operating System.....	147
[Figure 3-45] Status of National Accreditation Bodies	148
[Figure 3-46] Status of Laws Governing Quality Inspection System for Petroleum and LPG	155
[Figure 5-1] Current Status of ANAS HTP by Stage	229

[Figure 5-2] Vision Map of ANAS HTP (by 2035).....	233
[Figure 5-3] Roadmap of ANAS HTP by Phase	234
[Figure 5-4] Organizational Characteristics of ANAS HTP by Phase	235
[Figure 5-5] Structural Changes of ANAS HTP by Phase	238
[Figure 5-6] Inter-relationship between Current and New Structure	240
[Figure 5-7] The Sources and Uses of Budget (Year 2020)	242
[Figure 5-8] Roadmap for Budget	245
[Figure 5-9] Number of Staff by Department (as of October 2020)	246
[Figure 5-10] Roadmap for Workforce	249
[Figure 5-11] Stakeholders of ANAS HTP	251
[Figure 5-12] Roadmap for Network	253
[Figure 5-13] Roadmap for TT & TC	269
[Figure 5-14] The Structure of the New Global Entrepreneurship Index (GEI)	271
[Figure 5-15] INC-based Program Design	275
[Figure 5-16] TEC Algorithm.....	276
[Figure 5-17] Textbook and Translation	276
[Figure 5-18] Roadmap for Startup & Incubation.....	283
[Figure 5-19] Roadmap for Test & Certification	298



CHAPTER 1

Project Overview

Policy Consultation on the Promotion of Technology
Transfer and Commercialization in Azerbaijan

Chapter 1. Project Overview

1. Introduction

As of 2019, Azerbaijan is an upper middle-income country with a population of about 100 million people and a GDP per capita of USD 4,764 (IMF). After recording the highest GDP growth rate of 34.47% in 2006, it has continuously decreased to about 4% in 2019. In addition, Azerbaijan's oil and gas industry has been the basis of national economic development as a core industrial sector accounting for 45-50% of GDP and more than 90% of exports. However, in order to achieve massive social and economic development after the financial crisis in 2008 and 2011, the Azerbaijani government is pursuing policies focusing on escaping from excessive dependence on the oil industry, escaping the fiscal crisis due to low oil prices, diversification of industrial structure, fostering the non-oil sector, etc. In particular, the Azerbaijani government is striving to achieve development and industrial diversification policy implementation through industry-academia linkage in the areas of technological innovation, technology transfer and commercialization to strengthen national competitiveness.

In 2016, ANAS HTP (Azerbaijan National Academy of Science-High Technology Park) was established by Presidential Decree for sustainable economic development and national competitiveness. The goal of ANAS HTP is to support projects such as encouraging research and commercialization of technology innovation, building a science and technology ecosystem, business development activities and technology transfer. As of 2020, ANAS HTP, LLC, the management organization of ANAS HTP, is under operation with a total of 46 employees in 6 departments and a total annual budget of USD 300,000.

ANAS HTP, LLC requested STEPI to share South Korea's experiences in technology transfer and commercialization program development and operational system improvement in order to establish innovation system in Azerbaijan and strengthen the industry-university-research institutes cooperation system. STEPI and ANAS HTP have been working on the two-year long K-Innovation ODA Program (2019-2020) titled "Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan" with the approval of both countries.

2. Objectives

STEPI and ANAS agreed on the following directions in planning two-year long K-Innovation ODA Program (2019-2020): In 2019 (Year 1), two parties analyze the state of technology transfer and commercialization of Azerbaijan at the national level, and share related methodology. In 2020 (Year 2), two parties focus on developing strategic plan at an institutional level for the successful operation of the project.

In this regards, the objective of the first year (2019) was to enhance the understanding of the TT & TC ecosystem of Azerbaijan as well as the competence of relevant institutions including ANAS HTP at the national level. For this purpose, STEPI investigated the state of Azerbaijan's STI policy and TT & TC ecosystem and shared South Korea's TT & TC experiences, especially South Korean Technopark's methodology for TT & TC and Start-up & Incubation.

In the meantime, the objective of the second year (2020) was to develop a strategic plan for the successful operation of ANAS HTP at the institutional level. In other words, the project was aimed at analyzing vision, mission, principles, structure, budget, workforce, network, and operational programs of ANAS HTP and presenting a strategic plan. It also aimed to enhance capabilities for designing a strategic plan and operational programs through 'learning by participation'.

3. Project Framework

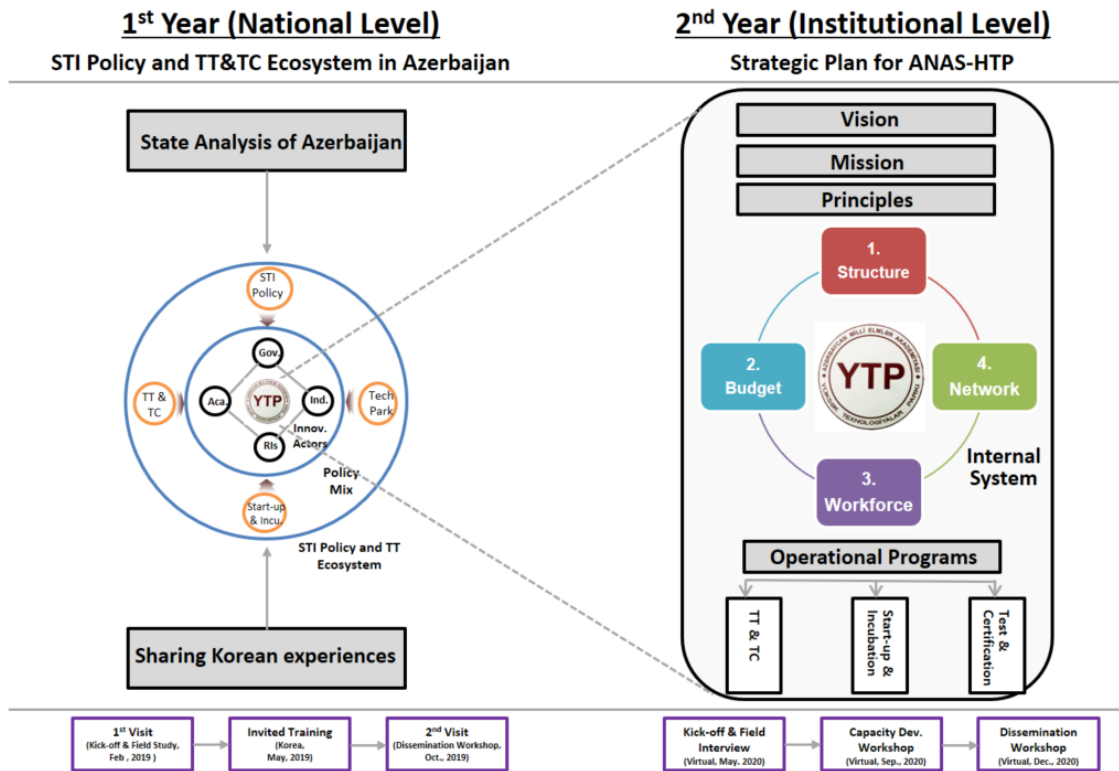
3.1 Analytical Framework

The analytical framework of the project is as follows.

In the first year (2019), the analysis of Azerbaijan's STI policy and TT & TC ecosystem affecting ANAS HTP's TT & TC activities was conducted at the national level. In addition, opportunities to learn relevant theories and methodologies were provided through sharing of South Korean experiences. In particular, the project analyzed the capabilities of government, academia, industry, and research institutes in Azerbaijan and focused on strengthening Azerbaijan's capabilities from the perspective of a policy mix such as STI policy, TT & TC, Start-up & Incubation, Technopark, and etc. To this end, the Kick-off & Field Study Visit was held in Baku in February 2019, and the Capacity Development Workshop was held in South Korea in May 2019 through the invitation of Azerbaijani participants to South Korea. In October 2019, the Dissemination Workshop was held in Baku to disseminate the research outcomes.

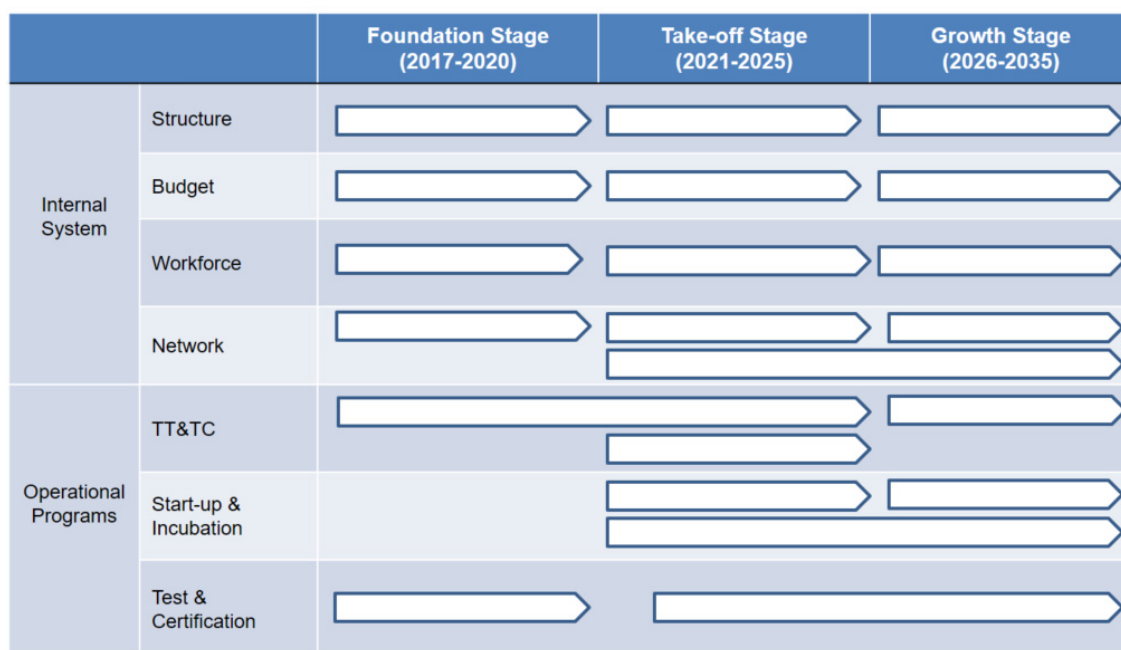
Meanwhile, in the second year of the project, the research team focused on the development of a strategic plan at the institutional level for the successful operation of ANAS HTP based on the research outcomes from the first year. ANAS HTP's vision, mission, principles and an internal system covering organizational structure, budget, workforce, and network were analyzed and future plans were presented accordingly. As a result of the research in the first year, three topics, TT & TC, Start-up & Incubation, and Test & Certification, have been identified as critical operational programs for ANAS HTP to achieve its vision, mission and principles. Therefore, in the second year, we analyzed the current status of ANAS HTP based on these three operational programs and suggested future plans accordingly. In the second year, unlike the first year, the project was inevitably carried out through video conference because of the COVID-19 pandemic. Specifically, the Kick-off & Field Interview in April, the Capacity Development Workshop in September, and the Dissemination Workshop in December were all held via video conference.

[Figure 1-1] Analytical Framework



The strategic plan of ANAS HTP, which is the goal of this project, is largely composed of an internal system and operational programs. The roadmap for this project is divided into three stages of; i) Foundation, ii) Take-off, and iii) Growth (see [Figure 1-2]). The first stage (2017-2020) is from the establishment of ANAS HTP to the present time, while the second stage (2021-2025) is a stage for an acceleration and a take-off. Finally, during the third stage (2026-2035), ANAS HTP's system for sustainable growth is further elaborate and expanded. In this project, strategic directions for each of these three stages have been presented.

[Figure 1-2] Strategic Roadmap



3.2 Operational Framework

The conceptual framework of the operational programs that ANAS HTP will concentrate in the future is as follows (see [Figure 1-3]).

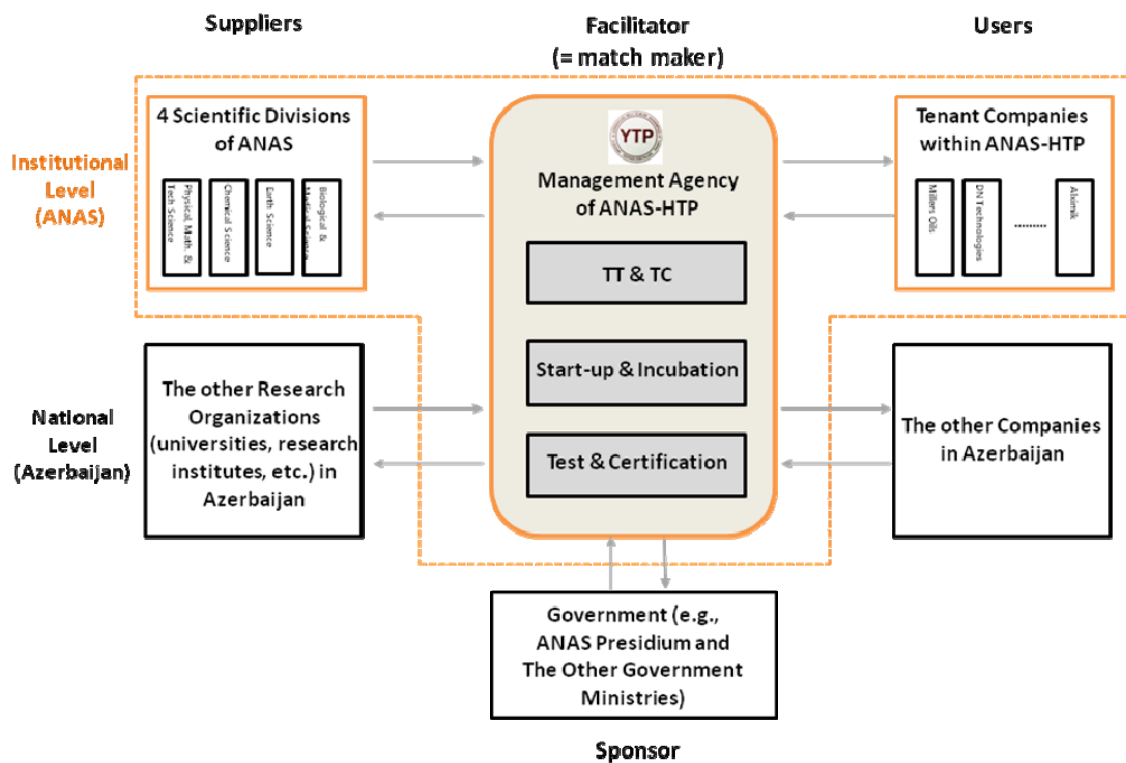
ANAS HTP, LLC¹, the Management Agency of ANAS HTP, facilitates interactions among diverse suppliers, users, and sponsors of Azerbaijan's research outcomes. When it comes to TT & TC, the Agency transfers research achievements of four science-related divisions of ANAS (of 6 divisions) to resident companies of ANAS HTP or other companies in Azerbaijan. It also helps facilitate the commercialization of research outcomes. With regard to Start-up & Incubation, the Agency supports start-up and incubation efforts of students or researchers from the four science-related divisions of ANAS and other universities and research institutes in Azerbaijan. Finally, in the area of Test & Certification, the Agency supports ANAS HTP's resident companies and other companies in Azerbaijan by taking

¹ In this report, ANAS HTP means ANAS HTP, LLC hereafter.

advantage of ANAS HTP's certification service, using its own equipment or ones from four science-related divisions of ANAS and other universities and research institutes in Azerbaijan.

As explained above, the Management Agency of ANAS HTP is expected to play a pivotal role in Azerbaijan's three service sectors, and make a remarkable contribution to the STI development in Azerbaijan. Therefore, in this project, we will focus on the aforementioned three service areas as the key operating programs of ANAS HTP.

[Figure 1-3] Operational Framework



4. Project Team

4.1 Research Team

This project was carried out through collaboration between STEPI on the South Korean side and ANAS HTP on the Azerbaijani side. The South Korean Team was led by STEPI and external experts also joined for specific topics. Dr. Wangdong Kim (Chief Director, Division of Global Innovation Strategy, STEPI) led the project as a project manager, and Professor Myungdo Oh from University of Seoul served as a senior advisor.

For specific areas of expertise, Dr. Jong Hwa Choi from STEPI worked on public procurement, Mr. Jong Hwa Choi, the head of Policy Planning Department from Daejeon Technopark, worked on TT & TC as well as Certification. In addition, Professor Jongin Choi worked on Start-up and Incubation. Finally, Yeojin Yoon, researcher from STEPI, assumed a coordinator role for this project.

[Table 1-1] STEPI Research Team

Name	Organization	Position	Area of Expertise	Contact Information
Dr. Myungdo Oh	University of Seoul	Professor	Senior Advisor	mdoh@uos.ac.kr
Dr. Wangdong Kim	STEPI	Chief Director of Global Innovation Strategy Division	Project Manager, Internal System	wangdkim@stepi.re.kr
Dr. Jong Hwa Choi	STEPI	Research Fellow	Public Procurement	jhchoi@stepi.re.kr
Dr. Sangho Choi	Daejeon TP	Head of Policy Planning Department	TT & TC, Test & Certification	shchoi@ditp.or.kr
Dr. Jongin Choi	Hanbat University	Professor	Start-up & Incubation	jongchoi@hanbat.ac.kr
Ms. Yeojin Yoon	STEPI	Researcher	Coordinator,	yjyoon@stepi.re.kr

As this year's project focuses on establishment of a strategic plan of ANAS HTP at the institutional level, the Azerbaijani team has been primarily consisted of ANAS HTP personnel. First of all, Mr. Ali Abbasov, Director of ANAS HTP, supervised the overall project. Mr. Buyanmin Seydov from ANAS was in charge of the internal system of ANAS HTP, and Dr. Ilana Rzayeva of ANAS HTP was in charge of Test & Certification. Finally, Mr. Vusal Suleymanli, Innovation Projects Coordinator of ANAS HTP, was in charge of TT & TC along with Start-up & Incubation.

[Table 1-2] ANAS HTP Team

Name	Organization	Position	Area of Expertise	Contact Information
Mr. Vasif Abbasov	ANAS HTP	Director	Engineering	office@ameaytp.az
Mr. Bunyamin Seyidov	ANAS	Head of Division of International Scientific Cooperation, Foreign Relations Department of ANAS	Internal System of ANAS HTP	bunyamin.seyidov@science.az
Ms. Ilaha Rzayeva	ANAS HTP	Independent expert	Test & Certification of ANAS HTP	ilaha.rza@gmail.com
Mr. Vusal Suleymanli	ANAS HTP	Independent expert	Technology Transfer & Commercialization, Start-up & Incubation	vusalsuleyman@hotmail.com

5. Main Activities

5.1 Activity 1: Preparation of the Project

In January, the STEPI team selected Project Manager (PM) and organized a project team. The direction of the project for the second year was planned based on the review of the project report of the first year and the meetings between the two organizations. In February, STEPI and ANAS HTP held a video conference to verify the feasibility of the direction and contents of the project for the second year. At the conference, both parties confirmed the project direction and process for the second year and the “ANAS HTP Strategic Plan Development” was derived as the topic for the second year.

Based on the meeting outcomes, the STEPI team prepared a Project Proposal for the second year and sent it to ANAS HTP in March and soon received feedback. Finally, the two organizations agreed to hold a kick-off meeting in Baku in April, a Capacity Development Workshop in South Korea in July, and a Dissemination Workshop in Baku around in October.

After that, the STEPI team conducted a literature review on the related subject before their first business trip to Azerbaijan to conduct the kickoff and field interview. In addition, for efficient analysis of the current status of ANAS HTP, the STEPI team requested that three local Azerbaijani consultants be recommended. Then, a final contract for the project was signed afterwards.

5.2 Activity 2: Kick-off and Field Interview

Initially, it was planned to hold a Kick-off and Field Interview in Baku in April. However, it was inevitably conducted via video conference in early May due to the COVID-19 pandemic. For the interview, the STEPI team prepared and sent an interview questionnaire in advance for each topic, and then the ANAS HTP team presented their answers at the conference.

5.3 Activity 3: Capacity Development Workshop

STEPI planned a Capacity Development Workshop to reinforce the strategic planning capacity of ANAS HTP, a partner organization of this project. Initially, this workshop was planned to be held in South Korea around in August, but it was inevitably carried out via online conference in early September due to the COVID-19 situation.

The two-day workshop was held in two sessions. Out of a belief that South Korea's Technopark-related experiences would greatly help ANAS HTP solidify its position as a leading high-tech park in Azerbaijan, STEPI invited four South Korean Technopark experts to the workshop. They gave lectures on such topics as business model, fundraising, resident hunting, and digitalization of the Technopark, which are considered as the most important issues in the operation of ANAS HTP in the future.

On the second day, an interview & discussion session was held to obtain additional information because the field interview was not held in Baku in May due to the COVID-19 pandemic. For each topic, the STEPI team prepared and sent an interview questionnaire in advance, of which the ANAS HTP team prepared their answers. For additional requests that could not be answered immediately at the workshop, answers have been supplemented afterwards through frequent interaction.

5.4 Activity 4: Writing the Final Report

The final report of this project is composed of three parts; the analysis of ANAS HTP, the South Korean experiences, and a development of a strategic plan for ANAS HTP. Among these three parts, the analysis of the current status of ANAS HTP was written by three local consultants from ANAS HTP. After reviewing the draft from ANAS HTP, experts of the STEPI team finalized the report, incorporating input from the Azerbaijani local consultants upon the request of STEPI.

A strategic plan for ANAS HTP was created in reference to the South Korean experiences along with analyses by local consultants in Azerbaijan. A draft of the final report was first

sent back to ANAS HTP for the preliminary review and the final report was prepared based on the comments from ANAS HTP.

5.5 Activity 5: Dissemination Workshop

The STEPI team held a video workshop in December to disseminate the outcomes of this policy consultation project. The workshop was mainly targeted at ANAS HTP and other related organizations. At the workshop, the project results of the first year were presented by topic. ANAS HTP concentrated on the final results of the project by topic along with South Korean experiences and shared their additional questions on the presentations.

5.6 Activity 6: Final Reporting

STEPI published the final report incorporating the comments made at the Dissemination Workshop and delivered it to ANAS HTP in both formats, an electronic file and a hard copy version, at the end of December 2020.

6. Project Schedule

The project has been carried out in the following schedule. Project activities were carried out online because of the COVID-19 pandemic.

[Table 1-3] Project Schedule

Activities	Programs	Jan.-Apr. '20	May '20	Jun. '20	Jul. '20	Aug. '20	Sep. '20	Oct. '20	Nov. '20	Dec. '20
Preparation of the Project	Needs Confirmation									
Kick-off & Field Interview (Virtual)	Kick-off Seminar									
	Field Study									
Research on South Korean Cases	South Korean Team									
Capacity Development Workshop (Virtual)	Lecture									
	Interview & Discussion									
Writhing the Final Report	Azerbaijan & South Korea team									
Final Report (Virtual)	Dissemination Workshop									
	Final Reporting									

7. Project Outputs

STEPI provided the following deliverables:

- **Deliverable 1:** Project Proposal (February 2020)
- **Deliverable 2:** Guidelines for the Kick-off and Field Interview (April 2020)
- **Deliverable 3:** Presentation Materials and Interview Questions for the Kick-off and Field Interview (May 2020)
- **Deliverable 4:** Guidelines for the Capacity Development Workshop and the Interview Questions (October 2020)
- **Deliverable 5:** Training(Lecture) Materials for the Capacity Development Workshop
- **Deliverable 6:** Storylines for the Final Report (September 2020)
- **Deliverable 7:** Draft of the Final Report (November 2020)
- **Deliverable 8:** Guidelines and Presentation for the Dissemination Workshop (November 2020)
- **Deliverable 9:** Submission of the Final Report (December 2020)



CHAPTER 2

Current Status of ANAS HTP

Policy Consultation on the Promotion of
Technology Transfer and Commercialization in
Azerbaijan

Chapter 2. Current Status of ANAS HTP

1. Internal System²

1.1 Vision & Functions

The High Technology Park of Azerbaijan National Academy of Sciences (ANAS HTP) was established on the basis of the presidential decree of the Republic of Azerbaijan dated on November 8, 2016 (No. 2425).³

The main goals of ANAS HTP are; the development of the national economy, the enhancement of national competitiveness, the expansion of innovation and high technologies based on S&T achievements, the creation of modern R&D complexes of developing new technologies, the establishment of new innovative enterprises, the development of competitive science-based production on the basis of innovative high technologies, and the support for entrepreneurship in this field.

In accordance with its goals, ANAS HTP strives to promote innovation, research, experimental work, inventions and discoveries, transfer high technologies, and increase the proportion of S&T-based products in the economy. The development of innovative entrepreneurship, the creation of favorable conditions for new activities and products, the implementation of measures for the transfer and development of advanced technologies, the creation of innovative zones, incubation services, technology transfer centers for the development and the application of high technology products, as well as conducting scientific research and development work and their commercialization and ensuring their

² This chapter was written by Mr. Bunyamin Seyidov of ANAS

³ President of the Republic of Azerbaijan, <https://president.az/articles/21637>

application in production are important elements of ANAS HTP's key activities.

In addition, ANAS HTP's key principles include; strengthening the ties between science and industry in order to create an innovative economy, the formation of necessary mechanisms for conducting applied research in accordance with market needs, the stimulation of the production of competitive products by increasing innovative activity, and implementing measures to ensure the effective utilization and development of innovation potential, the establishment of an innovative system that ensures the use, and the development and dissemination of new knowledge and technologies in Azerbaijan in accordance with the requirements of the information society by referring to international experiences.

The Republic of Azerbaijan is swiftly moving to implement a national innovation strategy. The government, research institutes, industry and entrepreneurs have all embarked on their path to innovative development. ANAS has also directed the nation's scientific research into innovation. With the support of the President of the Republic of Azerbaijan, a unique ecosystem has been created in ANAS. Many platforms have already been created on the territory of Azerbaijan for the implementation of innovations. ANAS HTP is one of the key actors in this platform. The organizational structuring and the creation of legislative mechanisms for the implementation of innovations have already been completed. However, additional resources were in need for further landscaping since this ecosystem was created on the ruins of a Soviet-era plant and there were still a lot of old buildings of little use. So, it was necessary to renovate the entire complex. For this reason, an innovation hub of ANAS was created in parallel. ANAS was designated as an innovation hub because of some advantages like its location and stable infrastructure and its possession of scientific, technical, administrative and investment knowledge.

It is clear that ANAS HTP is a key stakeholder of the country's ecosystem, which is a key driving force for the related issues. However, due to the lack of commitment and dedicated mindset of the top management and shortage of human resources (though most of the employees are engineers and technical clerks) at the lower managerial level, ANAS HTP's role remains as a mediocre player in the ecosystem. Though ANAS HTP has well-organized infrastructure (the plant and other supplementary buildings in the

technopark area in Ahmedli district), it is difficult to claim that it has the full capacity to represent the country's overall innovation system.

Though the research potential of ANAS (especially in petrochemicals, chemistry, geology, etc.) is quite enormous, it is still important to consider other players within the "ANAS HTP+" format (e.g., ADA, INNOLAND, BEU Technopark, etc.).

ANAS HTP forms a strong basis for the implementation of key tasks by linking science, industry, and government. Key functions of a Technopark are to support the implementation of science and innovation projects and the commercialization of science. As ANAS HTP meets the requirements of that national economy through its daily operation, it is capable of implementing research projects based on the entrepreneurs' proposals, developing innovative projects, and transferring them to production. The projects being implemented in the Technopark are often distinguished by their competitiveness, export potential and import substitution functions. ANAS HTP can surely create a favorable environment for the development of innovative high-tech products in Azerbaijan's science community.

The key findings of this project imply that there are no major challenges threatening the very existence of ANAS HTP. In fact, ANAS HTP owns a large-scale infrastructure, whose operation is now expanding every year. The only threat may be a decrease in the number of residents, which may weaken ANAS HTP's innovation activities and reduce its budget revenues.

The project team believes that, in order to develop its future activities over the next 10 years, ANAS HTP needs to attract extraterritorial residents to widen the scope of its residents in the Technopark, create innovation funds, promote innovation, research and technology commercialization in Azerbaijan, expand its technology incubation service using the country's science and innovation potential that has been enhanced significantly as result of its reduced dependency on oil and gas. In addition, for the development of the ecosystem in Azerbaijan, the country needs to focus on the following activities over the next 10 years; promoting an exchange of knowledge and experience in S&T innovation

with other countries and ensuring the involvement of developers or creators in increasing the production of high-tech products with export potential in Azerbaijan.

As a result of the above-mentioned activities, ANAS will be able to strengthen its role in diversifying the economy by reducing the cost burden on its state budget, preventing the inflow of foreign currency from the country, developing import substitute products, supporting SMEs, and increasing the share of science in the national economy. The main goal is to turn ANAS HTP into a center of science and high technology as quickly as possible.

Besides, the future activities of ANAS HTP include the cooperation with higher education institutions, the establishment of ANAS HTP's Investment Fund, the expansion of digital services, the formation of a register of intellectual products, the development of an appropriate legislative framework to strengthen the innovation ecosystem, the restoration of incubation activities, and the establishment of new enterprises with the joint participation of other institutes under ANAS.

1.2 Structure

ANAS is a supervising body responsible for handling the issues of ANAS HTP. ANAS is the highest scientific organization implementing the state policy in the field of science in the Republic of Azerbaijan. ANAS assists the Managing Organization in organizing, managing and developing activities related to HTP. It also prepares an annual report on HTP's activities and submits this report to the President of the Republic of Azerbaijan. ANAS conducts R&D in accordance with the ANAS HTP profile and decides priorities in developing and improving innovative products and high technology.⁴

The supreme governing body of ANAS HTP is the General Assembly of its members. The authority of the General Assembly related to ANAS is exercised by the Presidium of ANAS. ANAS participates in the Azerbaijani government's policymaking and has mandate to develop and implement the nation's STI policies. Regarding to this mission, ANAS carries out the following functions:

⁴ ANAS HTP, <https://www.ameaytp.az/index.php/qanun/nizam>

- To strengthen the links between science and industry in order to create an innovative economy in the country, to create the necessary mechanisms for conducting applied scientific research in accordance with market needs;
- To take measures to stimulate the production of competitive products by increasing innovation activity, to establish the necessary mechanisms for the implementation of innovation activities, to ensure the effective usage and development of innovation potential;
- To take measures for the transfer and mastering of advanced technologies, along with the creation of a favorable environment for the development of innovative entrepreneurship and related activities, to make proposals on the organization of technoparks, innovation zones, incubation services, technology transfer centers for the development and application of science-based products and technologies, etc.⁵

The authorities of the General Assembly on the issues solved by ANAS include the following:

- Determination of the development directions of ANAS HTP;
- Election of the Supervisory Board of ANAS HTP;
- Determination of the structure, number of employees, salary system and approval of cost estimates of ANAS HTP;
- Establishment of executive bodies of ANAS HTP, determination and early termination of their authorities;
- Appointment and dismissal of the head of the executive body of ANAS HTP, its deputies, heads of representative offices and branches, as well as subsidiaries;
- Approval of HTP's annual reports, balance sheets, profit and loss accounts, and distribution of profit and loss.⁶

The exclusive authorities of the General Assembly include:

⁵ ANAS HTP, <https://www.ameaytp.az/index.php/qanun/nizam>

⁶ ANAS HTP, <https://www.ameaytp.az/index.php/qanun/nizam>

- To certify the charter, structure, size of the authorized capital and amendments;
- To consent for the creation or liquidation of ANAS HTP's branches, representative offices and to agree for the creation, reorganization or liquidation of subsidiaries;
- To agree for the pledge of the state property (immovable and movable) owned by ANAS HTP and register transactions related to this process.⁷

The Supervisory Board controls the activities of the executive body of ANAS HTP in the period between the General Assemblies within the limits of its authorities. The Board consists of five members. Members of the Board are appointed by ANAS for the 5 years period. They may be re-appointed. The head of the executive body cannot be a member of the Board. The Chairman of the Board is elected by the members of the Board from among the members of the Board. The Chairman of the Board presides over the Board. Board meetings may be regular or extraordinary. The next meetings of the Board shall be convened by the Chairman of the Board at least once every three months. Extraordinary (emergency) meetings of the Board are convened by the Chairman with the request of the members of the Board, and new members are appointed within 30 (thirty) calendar days if the number of members is reduced to half. Decisions of the Board shall be taken by a simple majority of votes, with each member having one vote. Members are not allowed to abstain from voting. In the case of a tie, the vote of the Chairman of the Board shall be decisive.

The organization, management and development of the activities of ANAS HTP is carried out by the Managing Organization. The Managing Organization of ANAS HTP is the state commercial legal entity "High Technology Park" (HTP) which organizes, manages and develops its activities. To ensure more efficient management of ANAS HTP, the operator can be hired on a competitive basis by decision of the competent authority. The Managing Organization and operator are responsible for the operation of ANAS HTP. The main activity of the operator is to perform the functions according to the agreement with ANAS HTP.

The functions and rights of Managing Organization are quite extensive as followings:

⁷ ANAS HTP, <https://www.ameaytp.az/index.php/qanun/nizam>

- Preparation and submission of proposals to the Authority Body on the development and regulation of the activities of the ANAS HTP;
- Negotiation and involvement of legal entities and individuals in the implementation of scientific research and development in order to improve innovative products and high technology in ANAS HTP;
- Preparation of HTP development concept in coordination with ANAS;
- Obtaining statistical reports and forecast information on the activities of HTP's residents and monitoring their activities;
- Participation in inspections carried out by state bodies on HTP residents;
- Preparation of the proposals for the improvement of legislation in the field of high technology and innovations, and making recommendations on the improvement of the legislation on protection of intellectual property rights in the field of high technology;
- Participation in the process of applying the results of scientific research in the field of high technology and innovations in different areas of entrepreneurship;
- Establishment of institutions related to the activities of HTP, including funds, as well as innovation funds, business incubators, research and technology transfer centers and other specialized service centers;
- Investing in the infrastructure of the ANAS HTP, obtaining income from those investments and the management of the ANAS HTP;
- Implementation of a free tariff policy based on the proposals of the Authority Body for the services provided by the Managing Organization;
- Negotiating with foreign organizations and technoparks of foreign countries, as well as with foreign credit organizations and donors, signing cooperation agreements, as well as joining their associations within the scope of its authorities and in accordance with the legislation;
- Issuance of decisions and orders related to the activities of the HTP, as well as the adoption of other documents and monitoring their implementation;
- Involvement of independent experts and specialists in their activities;

- Participating in international events, including forums, conferences, exhibitions and other events, and organizing such events in the country and abroad;
- Supporting for high-tech innovative initiatives (start-ups), competitions, advocacy events, seminars and trainings;
- Submission of proposals on improvement of working and social-living conditions, material security and health protection of the employees;
- Ensuring of rights on state property in use.⁸

The Managing Organization realizes the management and development of ANAS HTP, as well as maintenance of the park's infrastructure, implementation of current and capital repairs, the promotion of the Park's activities, development and improvement of innovative products and high technology, provision of the technopark with relevant infrastructure, including transport, communication lines, utilities and other necessary services, the establishment and maintenance of relevant structural units, office, consulting, training, finance, marketing, law, logistics and etc. The Managing Organization also carries out the provision of communication, security, technical support, advocacy services and other services for the effective operation of ANAS HTP.

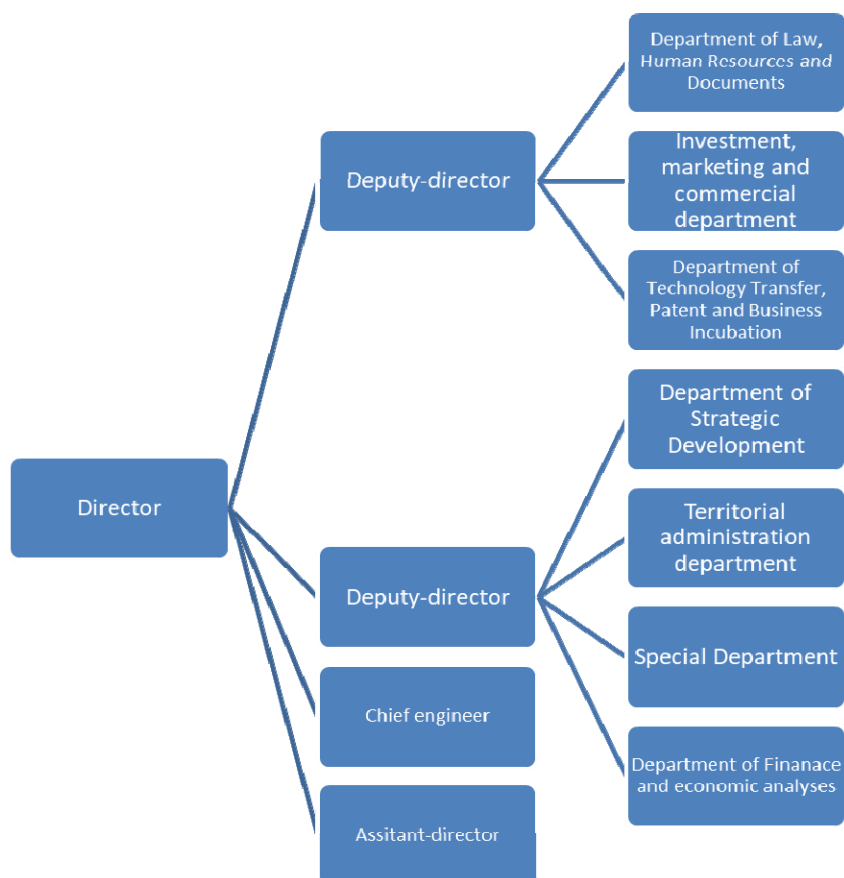
In addition, the Managing Organization assists its residents in complying with the legislation, coordinates their interaction with state and local self-government bodies, develops public-private partnership in the relevant field, maintains a register and certifies them in accordance with the Cabinet of Ministers, organizes the interaction of science and industry in the field of high technology and innovations, maintains and develops the scientific and technical potential of HTP, supports the role of innovative technologies in the economy, as well as increases the state's information resources in this area.

The Managing Organization organizes the analysis of new scientific achievements on high technology and innovations at higher education institutions, as well as other research

⁸ ANAS HTP, <https://www.ameaytp.az/index.php/qanun/esasn>

institutions and the application of these results in ANAS HTP.⁹

[Figure 2-1] Current Structure of ANAS HTP



The activity of the technopark is managed by the Director. The Director is fully responsible for the results of all activities of HTP. It acts on behalf of the enterprise without a power of attorney, carries out operational management, represents HTP in relation to third parties including in court, prepares documents for approval of the general meeting, signs official documents, appoints, reassigns and dismisses employees.

The Deputies-Director oversees the attraction of investments to ANAS HTP, realizes commercial activities of the park and organizes effective work with the production process

⁹ ANAS HTP, <https://www.ameaytp.az/index.php/qanun/esasn>

carried out by residents. They are also responsible for the organization and maintenance of the HTP's infrastructure, as well as the implementation of strategic goals and oversees the implementation of international projects including the establishment of cooperation with foreign partners and realization of international events.

The Chief Engineer carries out the technical management of HTP, development prospects and technical policy of specialization of production, high rate of increase of labor productivity, constant increase of level of engineering training of production, reduction of expenses, ensuring efficient use of production funds. The Chief Engineer also supervises all work on occupational safety and health. They execute control over observance of technological discipline, observance of construction norms and rules, labor protection and safety. They also provide engineering training, modernization of production, full application of the principles of full economic accounting and self-financing.

The Assistant-Director provides opinions for the implementation of work to ensure the activities of the Director, participates in the review, study, collection and analysis of materials on certain issues of the activities of ANAS HTP. Assistant-Director also obtains the necessary information for the Director from the departments of HTP, executors of his orders and instructions, if necessary, summarizes systematizes and analyzes the obtained information.

The Department of Law, Human Resources and Documents organizes and implements the preparation of proposals to improve the legal framework of HTP, reviews and comments on draft normative legal acts and other normative documents and controls the systematization of the HTP's human resources activities.

The Special Department provides the necessary official correspondence with law enforcement agencies and ensures the registration and constant control of confidential documents.

The Department of Finance and Economic Analyses carries out the registration of HTP's property, liabilities and all economic operations, organizes accounting policy, ensures the implementation of budget forecasts, ensures the implementation of inventory in the

manner prescribed by law, monitors the financial and economic activities of HTP and operates in other areas defined by law.

The Department of Technology Transfer, Patent and Business Incubation systematizes patents obtained on the basis of inventions of ANAS scientists, creates a patent bank, provides written confirmation of compliance of products produced by ANAS HTP residents with official requirements, supports new business of entrepreneurs wishing to cooperate with ANAS HTP. The department also provides identification of consumer needs for ANAS HTP exchange activities, preparation of products in accordance with these needs, marketing, warehousing, transportation, evaluation of products, organization of pre-sales and after-sales service, as well as advertising activities

The main purpose of the Department of Strategic Development is to develop draft strategies and implementation mechanisms related to the activities of HTP, to develop the proposals for planning the existing work process, increase efficiency and ensure sustainable development, to study and apply international experience and innovations, relevant foreign agencies and relevant international organizations to ensure the implementation of cooperation with these organizations. The department prepares and submits to the management proposals on strategic goals and development directions of HTP activity, strategies to achieve the set goals, mechanisms to be applied, and future complex measures.

The Territorial Administration Department controls the procurement, supply and use of goods (works, services) required for HTP, as well as ensuring labor protection. It also takes measures to ensure their legal use and documentation of property. This department provides orders for HTP, organizes the delivery of goods, devices, equipment and other inventory in the prescribed manner. In addition, it provides the organization of security of HTP administrative building and other constructions, current and capital repairs, organization of contracts with relevant departments, enterprises and organizations on economic and supply works, placing orders with relevant bodies and distribution of purchased goods in accordance with requirements of HTP structural units to perform their work, to give an opinion in accordance with the economic activities of the Technopark, to

conduct analyses and generalizations, to prepare analytical materials, and to make proposals, prepare correspondence with other organizations related to the economic issues of HTP and organize their sending.¹⁰

The Department of Investment, Marketing and Commerce was established to attract investment and organize, regulate and ensure the sustainability of sales and commercial issues that are part of the production process. The presentation of innovative products in accordance with the aggressive marketing policy of ANAS HTP, development of plans for the implementation of marketing strategy and policy, organization of control over cooperation between government agencies and private companies are the main activity directions of HTP. Besides above-mentioned, the department:

- Investigates the market situation, to constantly study the demand and supply of products and services of interest to it;
- Prepares of annual marketing plan;
- Expands the customer base;
- Studies the market situation;
- Develops and implements an advertising strategy

The Strategic Scientific Analytical Testing Center of ANAS operates as a unique laboratory in the territory of ANAS HTP, which has no analogues in the Caspian littoral states. The innovative laboratory, created with a multifaceted scientific approach, is internationally accredited and provided with advanced equipment. According to the existing resource balance in the laboratory, especially gas, oil and oil products are accurately tested.

In addition to the structure of the HTP, the Internal Auditor operates under the supervision of ANAS for the organization of audit work and carries out continuous control over the effectiveness of the internal control and risk management systems in cooperation with the leadership of the HTP. Standards and rules of the internal auditor's activity are based on the Law of the Azerbaijan Republic "On Internal Audit", normative documents of

¹⁰ ANAS HTP, <https://ameaytp.az>

ANAS, relevant normative acts of the Central Bank and international internal audit standards.

The formation of the structure of ANAS HTP can be considered optimal. Because the activities of the organization are largely reflected in the structure. But some shortcomings occur along the work process. So, if the activity on international relations is in the format of a separate department, the strategy of international relations of ANAS HTP will be effective. A team consisted of 4-5 persons can ensure the implementation of international projects in groups, as well as the attraction of foreign residents, the exchange of experience and other activities with high quality.

It would be better to combine the legal and human resources issues under the name of the Legal Support and HR Department. Because personnel issues are mostly regulated by legal acts, and it is expedient to unify these two areas.

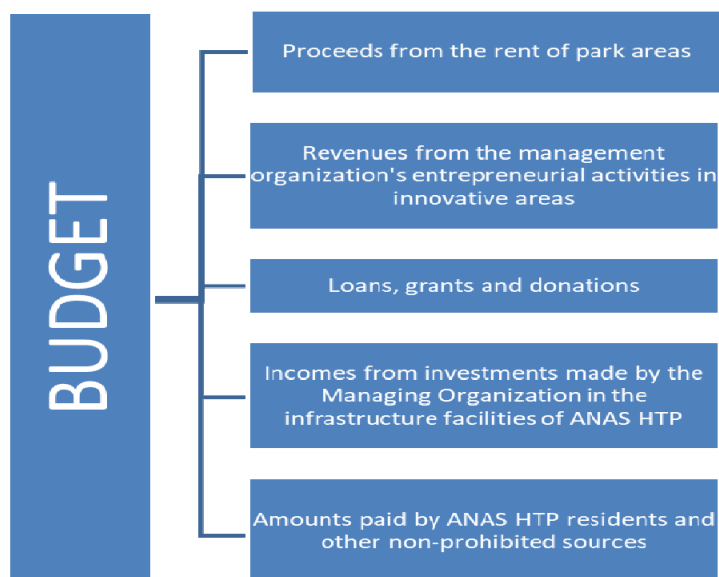
The Department of Patents, Certification, Marketing and Business Incubation combines four areas. Since each of these areas is a separate broad activity, it is necessary to systematize this department as two separate departments.

1.3 Budget

The total budget of ANS HTP is about 500,000 AZN (Approximately USD 300,000).

The organization of ANAS HTP's activity is financed from the revenues of the Managing Organization and other non-prohibited sources. The Managing Organization creates conditions for funds, government agencies, foreign investors and international financial institutions to attract extra-budgetary funds. According to the legislation of the Republic of Azerbaijan, on the basis of an agreement signed with the residents of ANAS HTP, a certain amount is paid monthly to ANAS HTP from the total income from innovative projects implemented by them. The formation of HTP's budget is as follows:

[Figure 2-2] Budget Structure of ANAS HTP



[Table 2-1] Budget Distribution by Department at ANAS HTP

Name of Department	Amount of Budget (USD) /Year 2020	Proportion (%)
Leadership	35,294	17.1
Department of Law, Human Resources and Documents	22,588	10.9
Investment, Marketing and Commercial Department	22,588	10.9
Department of Finance and Economic Analyses	18,352	8.9
Department of Technology Transfer, Patent and Business Incubation	22,588	10.9
Department of Strategic Development	18,352	8.9
Territorial Administration Department	55,058	26.8
Special Department	10,588	5.6
Total	205,411	100

[Table 2-2] Amount of Budget by Source

Name of sources	Amount of Budget (USD) /Year 2020	Number of Cases
Proceeds from the rent of park areas	20,000	1
Revenues from the management organization's entrepreneurial activities in innovative areas	0	0
Loans, grants and donations	5,411	1
Incomes from investment made by the Managing Organization in the infrastructure facilities of ANAS HTP	0	0
Amounts paid by ANAS HTP residents and other non-prohibited sources	180,000	12
Total	205,411	14

The budget formed from these sources is used to maintain the infrastructure of ANAS HTP, staff salaries, household expenses and other expenses. It can be considered that the current budget, formed from the expenses of the above-mentioned sources, is partially sufficient for the maintenance of the infrastructure of ANAS HTP, staff salaries, household expenses and other expenses. However, it should be noted that one of the main tasks is to provide additional income by providing ANAS with extra-budgetary funds. From this point of view, it is impossible to say that the budget is sufficient in general, as the predicted results have not been achieved yet.¹¹

In order to optimize the budget of ANAS HTP during the next 10 years, the number of residents should be increased, potential donors should be attracted, and the activities of ANAS HTP's own production facilities should be strengthened. It should be noted that high-tech products create high prices. This, in turn, means earnings, employment, high incomes for scientists, increasing the living standards of scientists and the population, and thus enriching the budget. In addition, special attention should be paid to international grants and projects, through which the ANAS HTP budget should be enriched with

¹¹ ANAS HTP, <https://ameaytp.az>

additional revenues. In a globalizing world, there are sample opportunities to join and benefit from projects. The above-mentioned issues can be direct proposals that ensure the sustainability of the budget. Additionally, it is possible to note the following factors that indirectly affect the formation of the budget:

- Use of state guarantees as a means of stimulating innovation, when obtaining loans from commercial banks or indirect financing of the innovation process by the government;
- Application of budget subsidies and soft loans to state corporations and centers, private firms based on the principle of public-private partnership, as well as state industrial research institutes that create innovations;
- Providing direct funding for long-term research in priority research areas characterized by high uncertainty.

The process of budget distribution in ANAS HTP is an internal matter of the organization and is carried out by the Department of Finance and Economic Analysis. The distribution of the budget is carried out in two simultaneous and closely interrelated stages: Creation of budget revenues and Use of budget funds.

It is envisaged that all expenditures provided for in the use of budget revenues will be covered from the budget.

In order to mobilize the ANAS HTP's budget, the results of budget preparation and realization should be evaluated, financial and budget strategy should be substantiated, factors which hinder the implementation of the plan should be identified, full, timely and stable collection of budget revenues should be ensured, and inefficient use of budget fund's factors should be eliminated.

1.4 Workforce

[Table 2-3] Staff Schedule of ANAS HTP for 2020

No.	Position	Staff unit	Proportion (%)
Leadership			
1	Director	1	
2	Deputy-Director	1	
3	Deputy-Director	1	
4	Chief Engineer	1	
5	Assistant Director	1	
	Total	5	10.86
Department of Law, Human resources and for Work with Documents			
6	Chief Lawyer-Head of Department	1	
7	Lawyer-Senior Specialist	1	
Division of Human Resources and for Work with Documents			
8	Head of Division	1	
9	Senior Specialist on the Work for the Documents	1	
10	Specialist	1	
	Total	5	10.86
Department of Finance and Economic Analyses			
11	Chief Accountant	1	
12	Accountant-Treasurer	1	
13	Economist	1	
	Total	3	6.5
Department of Investment, Marketing and Commercial			
14	Head of Department	1	
15	Senior Specialist	1	
16	Marketing Manager	1	
17	Senior Specialist (on commercial issues)	1	
18	Advertising and PR Manager	1	
	Total	5	10.86
Department of Technology Transfer, Patent and Business Incubation			
19	Head of Department	1	
20	Technology Transfer and Business Incubation Manager	1	
21	Senior Specialist on Patent	1	
22	Leading Specialist on Business with Entrepreneurs	1	
23	Specialist	1	
	Total	5	10.86
Department of Strategic Development			
24	Head of Department	1	

No.	Position	Staff unit	Proportion (%)
25	Senior Specialist-Manager on International Relations	1	
26	Senior Specialist- Manager on Innovative Projects	1	
27	Leading Specialist- Designer	1	
	Total	4	8.69
Territorial Administration Department			
28	Head of Department	1	
29	Network Administrator	1	
30	Leading Specialist on Occupational Safety, Security, Fire Protection and Civil Defense	1	
31	Power Engineering Specialist	1	
32	Administrator	1	
33	Specialist on Buildings and Devices	1	
34	Electrician	1	
35	Plumber	1	
36	Driver-Mechanic	1	
37	Driver	1	
38	Driver	1	
39	Housekeeper	1	
40	Housekeeper	1	
41	Housekeeper	1	
42	Worker	1	
43	Worker	1	
44	Gardener	1	
	Total	17	36.95
Special Department			
45	Head of Department	1	
46	Specialist	1	
	Total	2	4.42
	FINAL	46	100

Currently, MoT programs are implemented in some universities in the form of separate courses, but this is not sustainable. Only ADA University has a Bachelor of Science in Information Technology (BSIT) program .This program provides students with a solid foundation in information technology (IT) theory and practice with emphasis on developing IT solutions. Combined with the math and science foundations, the program covers a wide range of areas including computer architecture, web and mobile technologies, systems analysis and design, IT project management, and cyber-security

fundamentals. Equipped with theoretical knowledge and hands-on experience, upon graduation our high-impact graduates will be ready to deliver data driven solutions which will challenge traditional ways of organizational behavior. In the future, the organization of such programs with STEPI and ANAS HTP would be very effective.¹²

Looking at the current structure of ANAS HTP, we can observe that the number of existing employees is sufficient. The number of employees can be increased if additional departments are created. It is estimated that the number of employees could increase from 40 to 60 during the next 10 years.

The Director appoints, reassigns, dismisses, and orders incentives and penalties for the ANAS HTP employees. The Director, his Deputies, Chief Accountant and the Internal Auditor of HTP are appointed and dismissed by ANAS. Heads of departments of ANAS HTP, as well as other employees are appointed and dismissed by the Director. If there is a need for vacancy for any position in HTP, a competition is announced through the website of ANAS and is carried out transparently. The appropriate candidates are appointed for different positions.

1.5 Network

The science and technology focus of ANAS HTP is directed to biotechnology, nanotechnology, petrochemicals, robotics, agriculture, ICT-Software development, metallurgy, mechanics and health sciences. This focus should be modified, either reduced or expanded depending on certain factors. These factors are demand, availability of resources, profitability and capacity. Based on the above-mentioned factors, ANAS HTP closely cooperates with the Scientific Departments of ANAS. In order to regularly commercialize scientific ideas based on the proposals of the institutes, some projects are submitted and their application opportunities are evaluated. Information on patents obtained by scientists representing various scientific departments are submitted to ANAS HTP and are systematized on this basis of these documents. ANAS HTP interacts with four main departments of ANAS (Department of Physics, Mathematics and Technical Sciences,

¹² ADA University, <https://www.ada.edu.az/en/schools/programs/site/170-information-technology>

Department of Chemical Sciences, Department of Earth Sciences and Department of Biological and Medical Sciences).¹³

In order to operate in HTP, legal entities and individuals must register and obtain the HTP registration certificate. The registration of residents as well as extraterritorial residents of HTP and the issuance of the relevant registration certificate to them are carried out by the Managing Organization of ANAS HTP. Currently, there are 11 residents:

- **“ANAS Experimental-Industrial Plant”** is the first and only experimental test facility in Azerbaijan. The main function of the plant is to establish direct links between science and industry, to adopt new technological processes in practice and experience. This enterprise produces a number of petrochemical products (industrial, motor, hydraulic, transmission, compressor and lubricating oils, additives and engine oils). These heat-resistant products and other quality parameters are very interesting for consumers. Production capacity is up to 25,000 tons per year in starting phase. The amount of the profit is about USD 30 million. It is possible to produce up to 75,000 tons of products during a year in three shifts. Considering that about 125,000 tons of these products are imported to the country, this means the opening of new opportunities in the field of protection of foreign exchange reserves, employment, etc.
- **“Millers Oils Azerbaijan”** LLC is a UK based internationally known premium class engine oil company that established jointly by Millers Oils UK and ANAS. At Millers Oils Azerbaijan, the process of mixing oils during production is carried out in five directions at the same time, as a result of which you can get a wide range of chemical products for industry, automobile oils, and motor oils. There are also diesel and gasoline additives that not only increase the octane and titanium content, but also help clean the entire fuel system and injectors in parallel. Such technologies have already begun to spread in Europe and are still represented in Azerbaijan owing to Millers Oils Azerbaijan LLC. Firstly, the production capacities of the already launched phase allow producing up to 15,000 tons of products per year.

¹³ ANAS HTP, <https://www.ameaytp.az/index.php/component/content/article/88-son-x-b-r/569-milli-innovasiya-siyas-ti-h-yata-kecirilir-kaspi-d-d-rc?Itemid=539>

Most importantly, all this was done with the participation of Azerbaijani scientists, and, owing to Millers Oils Azerbaijan it became possible to significantly improve the quality of base oils, which are raw materials for other industries¹⁴.

- **“Science-Technology Park”** LLC carries out researches, experimental-design works for the preparation, development and improvement of innovative products and high technologies.
- **“EPC Group”** LLC is one of the leading Venture Capital company in the region. It develops heavy industry technologies, especially in alunite/ aluminum extraction.
- **“DN Technologies”** LLC operates a 3D modeling & simulator and robot technology company, which produces and sells software products.
- **“Azelttech”** LLC realizes the production of Radiation Detector Devices.
- **“Algoritmika”** LLC is an international programming platform for children, operating in eight countries. Users learn programming on the online platform. They also create their own games and cartoons with the classroom teacher. Thus, users develop logical and analytical thinking.
- **“Alximik”** LLC produces cheaper and better quality detergents and chemistry products with the use of high technology.
- **“BUTA ARMOR”** LLC is manufacturing armored garments, helmets and bulletproof vests for military purpose. The bulletproof vests produced by BUTA ARMOR are on the top three in the world and the first in Europe. The models are appropriate NATO standards. They are ergonomic, light and bulletproof. The individual defense of each fighter is a priority in the Azerbaijani army. At the end of 2018, all permits were obtained. This project attracts an innovative investment of 8 million AZN to the country's economy.
- **“AZMONBAT”** LLC produces batteries for military and civilian purposes. AZMONBAT will build a plant for the production and disposal of lead acid batteries. These include reproduction, ecology, employment, protection of foreign exchange

¹⁴ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/9834>

reserves and other macroeconomic issues. The battery project attracts an innovative investment of 30 million AZN to the economy of the country.

- **“INCOR”** Limited Liability Company produces high-quality anti-corrosion agents in the fields of oil and gas production, oil and gas processing, in the chemical industry and agriculture, also the production of salts, paraffin inhibitors, inhibitory bactericides, demulsifiers, alkaline solvents, blowing agents, organic solvents, anti-corrosion dyes, plant growth stimulants, complex fertilizers, anti-harm drugs, tele, micronutrients, as well as the production of cosmetics, drugs, adhesion additives to the bitumen, and emulsifiers.

Various privileges have been applied to the Managing Organization and residents of ANAS HTP and these privileges create conditions for increasing the efficiency of interaction, ensuring interest in attracting investors and production of competitive products.

The Managing Organization takes advantages as listed below:

- Income Tax – part of the park's infrastructure to be built and maintained;
- Value-added tax – import of equipment, technological equipment and facilities for creation and construction of infrastructure and production areas of parks, as well as with the purpose of carrying out scientific-research and experimental-design works (based on the supporting document of the relevant executive authority);
- Property tax – according to the property of the parks;
- Land tax – according to the lands used in the territory of parks;
- Customs fee – import of equipment, technological equipment and facilities for creation and construction of infrastructure and production areas of parks, as well as with the purpose of carrying out scientific-research and experimental-design works (based on the supporting document of the relevant executive authority)¹⁵.

¹⁵ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/10374>

Resident takes advantages listed below for 7 years from the date of registration:

- Tax on profits – profit from the activity of the physical person engaged in entrepreneurial activities without the creation of a legal entity (excluding income tax);
- Income tax – tax coming from the activity in the park;
- Value-added tax – import of machinery, technological equipment and tools for the construction of industrial sites in the park, the implementation of researches and development and production (based on the supporting document of the relevant executive authority);
- Property tax – according to the property of the parks;
- Land tax – according to the lands used in the territory of parks;
- Customs fee – import of equipment, technological equipment and facilities for creation and construction of infrastructure and production areas of parks, as well as with the purpose of carrying out scientific-research and experimental-design works (based on the supporting document of the relevant executive authority).¹⁶

ANAS HTP closely cooperates with universities, research centers and companies operating in Azerbaijan. For joint implementation of innovative projects, granting the status of "resident of ANAS," implementation of aggressive marketing policy, organization of procurement of scientific, competitive and innovative products, industrial mineral and lubricants produced by HTP residents in state enterprises, application of new technologies in agriculture and for implementation of other purposes, the cooperation network was established with SOCAR Baku Oil Refinery Plant, Institute of Petrochemical Processes of ANAS, "Smart Home" LLC, Western Caspian University, "Azerishig" OJSC, AzScienceNet science computer network of ANAS Institute of Information Technologies, Ministry of Ecology and Natural Resources, Ministry of Agriculture, Agroleasing Open Joint-Stock Company, Baku Stock Exchange, KUTVAN LLC and Baku Engineering University. Additionally, in order to support the development and procurement of innovative products in ANAS HTP, the Agreements have been signed with Azerbaijan Technical

¹⁶ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/6173>

University, Economic Reforms Research Institute of the Ministry of Economy of the Republic of Azerbaijan, Azerbaijan State University of Oil and Industry, and EPC Group LLC¹⁷.

Currently, ANAS HTP cooperates with the Science and Technology Policy Institute (STEPI) of South Korea, the International Association of Science Parks and Areas of Innovation (IASP) Center for Analysis of Economic Reforms and Communication, South Korea Innopolis Foundation, European Union TAIEX Program, Islamic Corporation for the Development of the Private Sector, Moscow Center for Structural Transformation of Industry, Saint Petersburg Technopark, EPC Group of Companies, China-Russian Innovation Technopark, Anzali-Astara Technopark of Iran, Azerbaijan Office of the South Korea International Cooperation Agency (KOICA), SGS (Switzerland), BP, ACTIVA of Spain, General Robotics of Israel”, Egyptian Company CHIVY, British Company Micro Focus & HPE Software, Czech Institute of Experimental and Applied Physics and other organizations. The scale of these relationships is expanding every year. The Memorandums of Understanding were also signed with China's Ningbo State Technology and Industry Company Hua Pei and with the Islamic Development Bank's ICD.

¹⁷ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/11860>

2. Technology Transfer & Commercialization¹⁸

ANAS HTP conducts TT & TC operations under the partnership with coordinators of the Presidium of ANAS and the Institutes of ANAS. In the previous years, ANAS had the Patent Department. After restructuring, this department has been canceled and its functions have been assigned to ANAS HTP. During the first year after the restructuring, ANAS HTP centralized the patents information of all ANAS Institutes and coordinated the functions of the previous department of ANAS HTP; Patent, Certification, Marketing and Business Incubation. Today, as result of the restructuring, a new department dedicated to TT & TC is planned to be set up with 3-5 people.

Our planning in terms of core business with the new department for TT & TC is as follows:

- Management of all patent/ licensing operations of ANAS (40 Institutes and organizations);
- Realizing commercialization of science via a well-developed investor/ tech broker network.

Currently, the operating budget of TT & TC is about 20,000 AZN per year, which is only sufficient to cover the labor costs. The team is now seeking for extra budget. Hopefully, the budget will be doubled next year. Through a centralized coordination, ANAS Institutes get new patents each year and the number is expected to increase. In addition, with technology transfer, ANAS itself can take a role as tech broker as well.¹⁹

ANAS has projects with major government ministries and organizations like the Ministry of Agriculture, the Ministry of Economy, and State Oil Company (SOCAR). Main tech transfer partners from the government side are mentioned above. At this time, there are no government sponsored projects underway. Generally, ANAS itself conducts research projects with the support of the Science Fund of Azerbaijan. But ANAS HTP is not a part of

¹⁸ This chapter was written by Mr. Vusal Suleymanli

¹⁹ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/13089>

these operations in terms of research. ANAS HTP plays a key role in the stage of implementation of research projects to industry. ANAS HTP has a plan to increase the performance of the TT & TC Center via government support. The areas of planned projects include ICT and agro innovations.

TT & TC is a very important topic for ANAS HTP and for the entire Azerbaijani innovation ecosystem in general. Of course, the professional education and training program in this field is crucial to improve TT & TC operations.

ANAS HTP is maintaining strong networks with government and academy. Now, it's important to engage private sector in this operations, so ANAS HTP plans to active engagement programs like B2G meetings, conferences, match-making sessions.

What is needed for ANAS HTP's TT & TC team from an organizational point of view based on the results of this project include expert network, system, platform, and organizational capability.

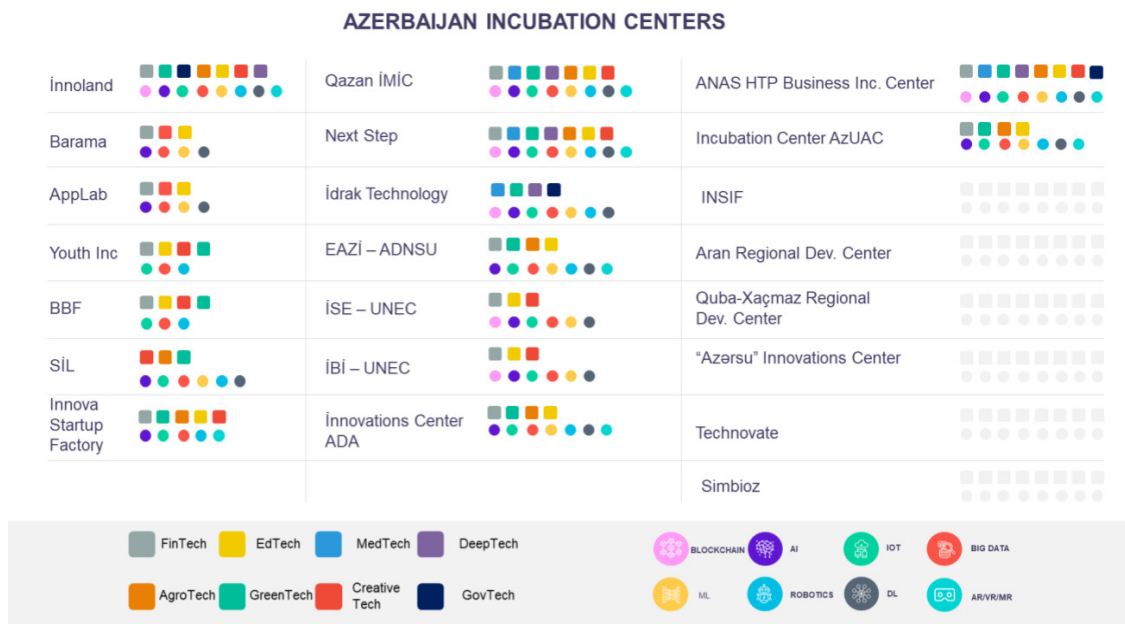
The key requirements for the activation of technology transfer and technology commercialization in Azerbaijan are 1) law and policy base, 2) government R&D investment, 3) government TT & TC project, 4) public and private technology finance base, 5) technology evaluation/technology transaction system base, etc. All of these requirements are crucial for the local TT & TC system. Because, it is necessary to build a modern system to engage all ecosystem stakeholders in an open innovation ecosystem and promote commercialization opportunities. In this regard, following major challenges have been identified:

- Legal system related with TT & TC;
- Non-vibrant science / tech commercialization system;
- Small market for this operations;
- Low interest for modernization and innovations

3. Start-up & Incubation²⁰

Azerbaijan is a country with 10 million population and over USD 40 billion GDP. It is one of the lead developing nations in the Caucasus region. The country's economy is boosted by natural resources and a significant part of its GDP comes from outputs of resource based economy. The transformation from resource-based economy to knowledge-based one was initiated in the early 2010s. In 2012, the first High-Tech Park of Azerbaijan was established and special support for startups started. Today, the country's startup ecosystem is developing rapidly. The current status of this ecosystem is summarized in the list given below²¹.

[Figure 2-3] Azerbaijan Incubation Centers



Generally, the history of incubation policy in Azerbaijan and ANAS HTP dates back to 2012. As of today, there are nearly 20 business incubation centers in Azerbaijan and 5 of them are located in the local provinces of the country. Half of these incubators are supported by

²⁰ This chapter was written by Mr. Vusal Suleymanli.

²¹ Ministry of Transport, Communication, High-Technologies (MTCHT) in Azerbaijan

the government. The rest of them are being operated mainly as part of the corporate social responsibility projects of big corporations. In recent years, hundreds of startups have received support from these incubators and they have produced some successful outputs. Generally, these incubators made fruitful outcomes in terms of increased awareness of tech entrepreneurship. In addition, many qualified experts have been developed through these entrepreneurial support centers and their programs.

ANAS HTP's Business Incubation Center (BIC) was established in 2017. The following is general information of ANAS HTP BIC.

- Building: ANAS HTP Business Incubation Center (BIC) is located in the city center, on the 4th floor of ANAS Central Library.
- Budget: Current budget is over 2,500 AZN per month (800 AZN for coordinator, 600 AZN for Operations Manager, 600 AZN for Technical/Digital Manager)
- Workforce: 3 personnel (Their educational background is mainly economics, business, marketing and ICT)
- System: Local management system, basically match-making between idea owners and resource owners through coordination
- Network: strong networks with most of the players in the local startup ecosystem. For international networking, BIC's networks are limited to South Korea, China and Turkey.

The target customers of Start-up & Incubation service of ANAS HTP are basically all the players participating in the Azerbaijani startup ecosystem including ANAS as well as students and colleagues of ANAS Institutes and other universities.

ANAS HTP BIC provides the following services:

- Training (for students)
- Consulting: public / civil (for organizations)
- Tech business services (for tech brokers)
- Lab services

ANAS HTP BIC offers a convenient space for startups. At the current time, there is no new building plan, but based on the demand of the ecosystem, additional buildings can be constructed in the future. Actually, there is a huge space in the current building under the Central Library. ANAS HTP BIC plans to expand its office area in the upcoming year.

Currently, there are no active startup companies under operation at BIC due to the current pandemic situation. But ANAS HTP has a total of 12 resident companies, of whom BIC resident companies have not be identified yet. Generally, priorities are placed on ICT (Cloud, SaaS) companies. Also, the current startup ecosystem also show such tendency of focusing on ICT industry. This explains why a majority of startups prefers to join such ecosystem.

Regarding investment, a key player of the investment ecosystem is the Innovation Agency of Azerbaijan. Current, ANAS HTP has no special fund for investment. But the Presidium of ANAS is working on to build internal Innovation Investment Fund using government budget.

Generally, when it comes to ANAS HTP Incubation Managers' careers, their background are science, engineering or business and they continue their career as an innovation manager, business development manager, marketing manager, or growth manager.

The following are key performance indicators defined for startups and incubation centers.

[Figure 2-4] Performance Indicators

Ecosystem element	KPIs
Startup	-Team Potential -Growth Potential -Technology Potential -IP-Intellectual Property
Incubation Center	-Number of Supported Projects -Number of Monetized Projects -Tech Resources Capacity -Active Startup Rate (graduated) -Raised Investment Amount -ROI – Return on Investment -Rate of Exits (graduated) -Network Connection intensity of alumni -Number of new jobs creation -Social / Economic Impact of Alumni

ANAS HTP BIC is an internal part of ANAS and it has strong relations with Institutes of ANAS. It has access to over 360 labs. But its existing capacity is relatively weak to accelerate the technology movement and gain a leading position in the market. That's why, from the technology sector perspective, ANAS-HT BIC is still in the beginning stage in terms of meeting the demand and supply of the technology market. In terms of the operational volume, ANAS HTP has planned to incubate 10 startups for the current year.

ANAS HTP BIC conducts self-evaluation based on the following five criteria; ① the number of startups or projects, ② revenues and expenditures balance, ③ the quality of business network, ④ relations with financial stakeholders, and ⑤ timely response to the market demand.

The Azerbaijani incubation model is characterized as being based on an integrated public-private partnership (PPP). In this perspective, state or public companies should play a strong role in the market and the market should support open market principles. Innoland (<http://innoland.az/en>) is a good model for this approach. Of course, the current incubation model can be further improved. Information on the incubation model was already provided in the previous report. More information is available at <https://economy.gov.az/en/bi-ink/>

Another unique aspect of ANAS HTP BIC is its strong linkage with academia and science community.

The performance of ANAS HTP includes sales, employment, products, history , etc. Resident companies who used ANAS HTP BIC service in the past were mainly ICT companies and their average staff was about 10 people. Their main products were hardware and software solutions. Many of them were newly established companies and their average sales were in the range of 10,000 AZN – 50,000 AZN per month.

In Azerbaijan, incubation centers are active in training and development programs. They transform university students to entrepreneurs with their intensive boot camps, workshops and face-to-face networking events. The financial support is provided through grant programs like the contest, startup competitions and Youth Foundation, and Innovation Agency. Also, there are some labs, from whom entrepreneurs receive technical support to build their products and solutions. Some of these labs are listed below (Info - MTCHT):

[Figure 2-5] Labs for Incubators

LABS FOR INCUBATORS

Labs		Standard equipments	Straight. machien	PCB Design	3D Printer	CNC	Ranking
Sumaks	IoT	✓	✓	✓	✗	✗	★ ★ ★ ☆
Helind	Alternative Energy	✓	✗	✓	✓	✓	★ ★ ★ ★
ELSMART	Smart home	✓	✓	✗	✗	✓	★ ★ ☆ ☆
AMEA	General, Defense	✓	✗	✓	✗	✓	★ ★ ★ ☆
AMEA YTP	General, Dron	✓	✗	✓	✓	✓	★ ★ ★ ★
Dr.robot	Equipments	✓	✗	✗	✗	✗	★ ☆ ☆ ☆
Schwatech	Electronics	✓	✗	✓	✗	✓	★ ★ ★ ☆

Mechanic Lab

ANAS HTP	Molding equipments, CNC
STP	
IDRAK TT	
Printo	3D Printer/ Silikon molding

Standard equipments:

- * Ossilograf
- * Multimeter
- * Soldering devices
- * Electronic Device Parts
- * And etc.

In Azerbaijan, a majority of business incubation centers has amazing workspaces and co-working centers. Kitchen, phone center, entertainment center, meeting rooms, high – speed Internet are general services provided to their resident companies. For incubators, government has no support package yet, but discussions are underway regarding how to increase the interest of other civil and private organizations in the incubation centers through government support for the incubation ecosystem with their services and solutions. So, government’s financial or other technical support packages are expected to be prepared in the upcoming years. Also, business incubation centers have a wide range of mentor pools from technology to business experts and from academicians to government officers. Currently, there is no centralized online tools for mentor matchmaking. By the end of this year, however, the government plans to build a centralized information database of the incubation ecosystem in Azerbaijan.

When it comes to the mater of a certificate for incubation centers, there is specific resident certificate issued to residents of incubation centers of ANAS HTP, which is a general phenomenon for other incubation centers in Azerbaijan. But government agencies are considering to provide one-stop service for startups at incubation centers by issuing a startup passport and startup certificate for them. At the same time, by the end of this year, a centralized database for innovation projects will be established in Azerbaijan. Once registered with this database, startups can enter their company profile including projects into the database and they can get their startup passports. Using this startup passport, companies can receive government support.

An incubation program to enhance the competitiveness of the incubation center residents will be based on a practical approach. General curriculum can be formalized into three modules: (1)Technical skills development, (2)Business skills development, (3)Communication skills development. The planned tools for workshops are Business Model Canvas (BMC), Value Proposition Canvas (VPC), and Lean Canvas Incubation Program (LC). The period of the program is for 3 months. One training session based on each module of the curriculum will be organized per week. ANAS HTP BIC focuses on ICT, tourism, agriculture, and energy.

By taking advantage of ANAS HTP BIC's extensive mentor pool of experts from Institutes of ANAS (2000+ young researchers) and business sectors, a good guidance can be provided for startups. In every two weeks, a demo presentation will be made on the current status of development. At the end of each month, there will be a rehearsal of the demo day. At the end of the incubation program, there will be a demo day with pitch deck on the startup ecosystem in Azerbaijan, where startups are introduced. In addition, networking will be provided for the stakeholders of the ecosystem for venture matching. Each batch of incubation program will be open for 8 startup projects. Key requirements for applying for this program is that the applicant should be a group of at least 2 people with ideas in MVP (Minimal Viable Product) stage. In a year, there will be 4 batches with a total of 32 startup projects. An annual target of ANAS HTP BIC is to convert at least 8 startups to technopark residents.

ANAS HTP BIC is active in promoting the awareness of entrepreneurship in the Azerbaijani startup ecosystem. For this purpose, it has established partnership with given projects:

- Climate LaunchPad
- World's Largest Green Tech Competition supported by EU - Institute of Innovation & Technology
- Social Innovation Lab - SIL
- "Yeni Fikir – New idea" Startup Competition Program for universities of Azerbaijan
- Republic Innovation Competition – Ministry of Economy's Initiative
- Hult Prize – Social Innovation Program
- Science Week Azerbaijan Program
- United Nations SPECA Program
- European Commission TAIEX / TWINNING Program

4. Test & Certification²²

According to the needs and demand of the information society of the Republic of Azerbaijan and its policy of “converting the black gold into the human gold,” and the decision of the Presidium of ANAS No. 9/22 of July 5, 2017, the Centralized Analytical Laboratory was established under ANAS HTP as its structural unit equipped with modern and cutting-edge technologies for conducting R&D and testing and certification operations. The Centralized Analytical Laboratory was renamed as the Strategic Scientific Analytical Testing Center of ANAS (SSATC).

SSATC operates in accordance with the Laws of the Republic of Azerbaijan, other normative legal acts, and normative documents of the State Agency of the Republic of Azerbaijan on Antimonopoly and Consumer Market Control and ANAS HTP approved by the Azerbaijan Accreditation Center on January 21, 2019. SSATC is funded by ANAS and provides its services for the purpose of improving the quality of scientific research and fundamental research conducted in the scientific institutions and organizations of ANAS²³. The center is conducting its activities independently from producers and consumers.

The purpose of SSATC is to concentrate scientific equipment, devices and other means on the balance of scientific institutions and organizations of ANAS, meeting the modernization requirements. It mobilizes these resources in a more effective manner in conducting scientific research and development using new and existing local and interstate technologies. Its key mission is to conduct high-level testing of gas, oil and petrochemical products in accordance with the requirements of the Control Quality and to provide customers with real test results.

Equipped with modern analytical devices and equipment from leading manufacturers in accordance with the project, the Center for Quality Control of Gas, Oil and Petrochemical Products received relevant certificates.

²² This chapter was written by Dr. Ilaha Rzayeva

²³ Above and further information is verified and taken from internal source of ANAS HTP

An accreditation document was prepared in accordance with the purpose of accreditation of the Center in accordance with the standard AZS ISO / IEC 17025: 2009 "General requirements for the competence of testing and calibration laboratories." So, a metrological inspection certificate of equipment and measuring instruments and AZ 01.0382 for 21/01/2019-21/01/2022 Accreditation Certificate No. 01.19 was obtained. The Center has the right to carry out conformity assessment activities on the basis of the accreditation document attached to the Certificate.

ANAS SSATC, ANAS HTP's testing organization, currently consists of 6 people, and its annual budget is estimated at around 30,000 to 50,000 AZN, only sufficient to cover the salaries of its employees. However, it now plans to outsource more professionals from ANAS or universities with whom it cooperates. ANAS SSATC's main challenge is securing government funding and support for stabilizing its operation including R&D and S&T activities.

Though it is a core unit of ANAS HTP, funding from the government is not likely in the near future. Instead, it is likely to get some support from ANAS including equipment support. Though it is well equipped with modern technologies, it still needs to expand its equipment for future R&D, testing and analysis activities.

[Table 2-4] Azerbaijan Team

Period	20201-2023	2024-2026	2027-2029
Goals	• Become a centralized certification center; • Receive several ISO standards; • Equipping the Center with cutting-edge technologies	• Slowly but planned to diversify the fields of testing; • Implement and develop new strategies and visions; • Invite global experts and scientists.	• Meanwhile, SSATC is supporting small term development-but long term development is under consideration.

ANAS SSATC started its operations nearly a year ago, but it has completed few successful projects. Now, the center plans to expand its customer base through public-private partnership. ANAS SSATC strive to generate profits to cover its operations. Its annual labor cost is between 30,000 AZN – 50,000 AZN. Currently, the testing capacity of ANAS SSATC can handle the testing of a total of 13 test standards in gas and LPG test fields and a total of 14 test standards in crude oil and other by-products test fields.

ANAS SSATC is planning to be certified in additional standards like ISO 14001/2015, ISO 9001/2015 and OHSAS 1800//2015. SSATC currently holds a total of 25 kinds of equipment conforming to 27 test standards as presented in [Table 2-5] and it is utilizing them for its effective services.

[Table 2-5] List of SSATC Equipment

Laboratory's equipment	
No	Name of equipment
1	Determination of sulfur compounds in natural gas, liquefied gas and light petroleum fuels. Gas chromatograph (GC-SCD) equipped with a sulfur chemical-luminescence detector. Test method ASTM D5504-12 and ASTM D5623-14. Manufacturer USA, Agilent Technologies Ltd.
2	Determination of arsenic and phosphine in natural gas and liquefied gases. Determination of arsine and phosphine gas chromatograph. Test method ASTM D6159-17 and ASTM D7833-14. Manufacturer USA, Agilent Technologies Ltd.
3	Determination of individual hydrocarbon content (PONA) of fuels used in internal combustion engines. Agilent GC Analyzer. Test methods ASTM D6730. Manufacturer USA, Agilent Technologies Ltd.
4	Determination of hydrocarbon content in liquid petroleum gases and propane / propylene mixture. RGA Gas chromatograph. Test method ASTM D2163-14e1, ASTM D1945, UOP 539, EN 15984. Manufacturer USA, Agilent Technologies Ltd.
5	Determination of low and high heat value. RGA Gas chromatograph. Test method ASTM D2163-14e1. Manufacturer USA, Agilent Technologies Ltd.
6	Determination of demetanized liquid hydrocarbon mixtures containing nitrogen and carbon dioxide. RGA Gas chromatograph. Test method ASTM D2597-10. Manufacturer USA, Agilent Technologies Ltd.
7	Determination of dew point in gaseous fuels. DP-172000 / M device. Test method ASTM D1142-12. Manufacturer Switzerland, Linetronic.
8	Determination of the density or relative density of compressed gas petroleum hydrocarbons under pressure. Pressure Hydrometer. Test method ASTM D1657-17. Manufacturer Switzerland, Linetronic.
9	Determination of saturated vapor pressure in liquid petroleum gases. Vapor Pressure determination system, LP-Gas Method. Test method ASTM D1267-18. Manufacturer USA, Koehler Instruments Company Inc.
10	Test on copper plate in liquefied petroleum gases. Copper Strip Corrosion Test for LPG. Test method ASTM D1838-16. Manufacturer USA, Koehler Instruments Company Inc.

Laboratory's equipment	
No	Name of equipment
11	Determination of the mass fraction of water in liquefied petroleum gases. AMETEK 3050-RM Moisture Analyzer. Test method ISO 10101-R2014. Manufacturer USA, AMETEK Progress Instruments.
12	Determination of volatility in liquefied petroleum gases. Determination of volatility system. Test method GOST R56868-16. Manufacturer USA, Koehler Instruments Company Inc.
13	Determination of the amount of residue after evaporation in liquid petroleum gases. Determination of volatility system. Test method ASTM D2158-16ae1. Manufacturer USA, Koehler Instruments Company Inc.
14	Determination of oxygenates in hydrocarbon matrices C2, C3, C4 and C5. Agilent 7890B Series GC Custom. Test method ASTM D7423-17. Manufacturer USA, Agilent Technologies Ltd.
15	Determination of oxygenates in gasoline by gas chromatograph. Agilent 7890B Series GC Custom. Test method ASTM D4815-15b. Manufacturer USA, Agilent Technologies Ltd.
16	Determination of the fractional composition of crude oil in a fully automated PILODIST distillation unit (rectification column equipped with 20 theoretical plates). PETRODIST PD 100CC 20l. Test method ASTM D2892-18a. Manufacturer Germany, PILODIST Laboratory & process technology.
17	Fractional determination device for fully automated distillation of heavy hydrocarbon mixtures (vacuum potstill method). PETRODIST PD 200CC 10l. Test method ASTM D5236-18a. Manufacturer Germany, PILODIST Laboratory & process technology.
18	Determination of aromatic hydrocarbons in medium petroleum distillates by the method of determination of the refractive index on a high-efficiency liquid chromatograph (HPLC). Agilent 1260 Infinity Binary HPLC System with RID Detection. Test method ASTM D6591, IP391-2000. Manufacturer USA, Agilent Technologies Ltd.
19	Determination of density and relative density in oil and oil products in the digital densitometer DMA. Density / Specific Gravity Meter DA-650. Test method ASTM D4052-18 and ASTM D5002-16. Manufacturer Japan, KEM Kyoto Electronics Manufacturing Co., Ltd.
20	Determination of electrical conductivity in aviation and distillate fuels. DBSO118 EMCEE Model 1152. Test method ASTM D2624-15. Manufacturer USA, EMCEE Electronics, inc.
21	Determination of oxidation resistance in aviation fuel at 260 degrees. JFTOT 230 Mark IV, PAC Alcor. Test method ASTM D3241, IP 323, ISO 6249. Manufacturer USA, PAC.
22	Determination of the degree of purity of isopropyl alcohol, acetone and di-isopropyl ether. Gas chromatograph equipped with Flame Ionization Detector (FID). Test method GOST R 51698-01. Manufacturer USA, Agilent Technologies Ltd.
23	Determination of the composition of pesticides, drugs, petroleum distillates and petrochemical products. 7890B GC system-5977B MSD. Manufacturer USA, Agilent Technologies Ltd.
24	Determination of toxic elements in consumer goods (children's toys, textiles, plastic materials, leather and wood products, paint products) on HDXRF spectrometer. XOS HD Prime Analyzer. Test method ASTM F963-17, ASTM F2853-15, ASTM F2617-15, ASTM F2923-14. Manufacturer USA, XOS.
25	Water cleaning system and Bi-distillation plant. Manufacturer Germany, Thermo Scientific.

Though its equipment is supporting 27 test standards, ANAS SSATC is planning to purchase more equipment including serum test equipment –H₂SO₃, related to chemical products, for providing better testing service. Despite its relatively small size, the center possesses cutting-edge equipment, which is quite extraordinary in Azerbaijan.

ANAS SSATC has been recognized as a testing house and has been operated by an accreditation body in the ISO/IEC 17025 field. Certainly, the test field is known as gas, LPG, crude oil and by-products. Some pictures of the center are depicted as below:

[Figure 2-6] ANAS Testing Center



The center provides testing services using its modernized equipment conforming to 27 test standards as below:

[Table 2-6] Description of Testing for Crude Oil, LPG, Gas and Other Products

No.	METHODS		TEST DESCRIPTION
1	ASTM UOP	2163 or 1945 539	Analysis of NG, LPG and Propane concentration by Gas Chromatography (GC)
2	ASTM D	2597	Analysis of Demethanized hydrocarbon liquid mixtures containing nitrogen and carbon dioxide by Gas Chromatography (GC)
3	ASTM D	5504	Standard Test Method for Determination of Sulfur Compounds in NG and Gaseous Fuels by Gas CG and Chemiluminescence. Detection range 0,1 ppb
4	ASTM D	1142	Water Vapor Content of Gaseous Fuels by Measurement of Dew-Point Temperature method
5	ASTM D	1657	Standard Test Method for Density or Relative Density of Light Hydrocarbons by Pressure Hydrometer
6	ASTM D	1267	Method for Gauge Vapor Pressure of Liquid -field Petroleum (LP) Gases
7	EPA	6020A	Determination of trace elements (from lithium to uranium) by the ICP-MS method in crude oil and petroleum products
8	ASTM D	4052	Method for Density and relative density of liquids by digital density meter
9	ASTM D	5002	Method for Density and Relative Density of Crude oils by Digital Density Analyzer
10	ASTM D	3241	Standard Test Method for Thermal Oxidation Stability at 260 dec.C of Aviation Turbine Fuels (JFTOT Procedure)
11	ASTM D	2624	Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels
12	-	-	Determination of the mass fraction of water (ppm) in LPG propylene (AMATEK 3050-RM)
13	GC/FID		Analysis of Methanol and Isopropyl alcohols, Acetone, di-isopropyl either by GC/FID
14	GC-MS		Analysis of pesticides by GC-MS
15	ASTM F	963 2853 2617 2923	Analysis toxic elements in toys, shoes, polymeric materials, furniture, paint layers

Currently, ANAS SSATC provides limited certification services in some areas. In the near future, it is highly likely that ANAS HTP not only promotes its test house, but also expand its service into certification based on ISO/IEC 17065. For the time being, the certification service will be limited to gas and LPG fields. In the long-term, ANAS SSATC plans to roll out its certification service into other areas like ICT, chemicals, and pharmaceuticals.

ANAS SSATC maintains close cooperation with universities, research institutes and companies in various industries. However, testing demand analysis has not been conducted yet with universities, research institutes, or companies. So it is now planning to conduct this demand analysis. In fact, ANAS SSATC is a quite young organization with a history of only three years. So, the results of such demand analysis can serve as useful information for securing test facilities, increasing test profits, and expanding test scope. It is recommended that ANAS SSATC conduct the test demand survey annually with various service users like universities, research institutes, and companies. These can be taken into consideration to take a systematic and professional approach while delivering these services to the resident companies and local market.

One of the needs that ANAS HTP wants to address through this project is to build well designed systematic and professional testing & certification service. Still, ANAS SSATC is quite a young organization with a short history, so South Korean experience in this field would be a great help for the center to upgrade its system and develop business opportunities.

ANAS SSATC is being operated based on AZS ISO/IEC 17025:2009, for 21/01/2019-21/01/2022 period (Accreditation No: AZ 01.0382.01.19). The center can be a key partner of the local ecosystem. ANAS SSATC is in compliance with various standards and is an operating body of US (ASTM), Britain (IP), Germany (DIN), Russia (QOST), ISO, EN and AZS. The center is also a key partner of various private and public organizations and it has various opportunities to collaborate with its stakeholders. Currently, however, its testing service is focused on petrochemicals and toxicology.

Though ANAS SSATC has the right to conduct conformity assessment, it is not the key

organization in charge of that. There is an accreditation body in Azerbaijan, officially responsible for conformity assessment: <http://www.accreditation.gov.az/>. A conformity assessment determines whether goods and methods of their manufacturing meet relevant technical standards and fulfill relevant requirements set out by technical laws. The assessment is procured by a manufacturer prior to releasing its goods into the market. The assessment shall be carried out by the relevant accredited entities through: (i) certification and (ii) declaration of conformity. The high-risk products require mandatory certification and obtaining of conformity certificate while the goods categorized as medium- and low-risk require declaration of conformity approved by the conformity assessment bodies. Rules of issuance of certificates and declarations of conformity will be determined by the Cabinet of Ministers and information about the issued conformity documents will be entered into a special register accessible online. Foreign conformity documents will be recognized subject to their validity and correspondence to the terms of international agreements that Azerbaijan is a party to.

Producers located abroad will have a right to pass the conformity assessment of their goods through their representatives in Azerbaijan. Products, subject to a technical regulation control, unassessed or having failed to pass the conformity assessment, shall be restricted from being manufactured, imported, sold, or used in Azerbaijan. The Law refers to the criminal, administrative and civil liability of the persons who fail to comply with it. It depends on situations of course. Operation based on law, can be acceptable.

From the organizational point of view of ANAS HTP, the top priority to enable testing and certification is ① securing professional workforce and training for workforce, ② efficient operation of quality system, ③ increasing test revenue by creating networks to expand test demand, and ④ expanding test areas into ICT, chemical and medical areas other than gas and LPG. ANAS HTP's certification body has a long term plan and initiative in terms of international cooperation and it needs technical assistance from OECD, WIPO and ITU.



CHAPTER 3

South Korean Experiences



Policy Consultation on the Promotion of Technology
Transfer and Commercialization in Azerbaijan

Chapter 3.

South Korean Experiences

1. Technopark (TPs) in South Korea: Internal System

1.1 Mission & Functions of Technoparks²⁴

1.1.1 Legal and Institutional Basis²⁵

The first Technopark in South Korea was established in 1998 based on a special act. According to Article 32 of the “Civil Act” and Article 4 of the “Special Cases Act on Support for Industrial Technology Complexes,” Technopark is designated and supported by the Minister of SMEs as a project implementer that creates and operates industrial technology complexes.

In order to enhance the competitiveness of local companies and innovate local industrial technologies, industrial technology complexes started to be established from 1998 and the Technopark foundation was established as the project operator.

1.1.2 Mission of Technopark

1.1.2.1 Background

Technopark (TP) is defined as a South Korean version of science and technology park (STP). Technopark builds an organic network of cooperation with regional innovation institutions including local industries, academia, research institutes, and government. TP is a base organization for promoting local industries, who discovers and fosters knowledge-based

²⁴ This chapter was written by Dr. Yang-jin Kim of the Gyeongbuk Technopark.

²⁵ KTPA (2016), Techno Park, The growth engine of Korea

small and medium-sized technology companies by establishing industrial development strategies and policies that fit the local situations²⁶.

South Korean TPs have distinct operational features different from those of existing overseas STPs such as agreement and cooperation between each execution entities, integrated budget operation and management by region, including competition between regions and central management.

[Figure 3-1] Concept of Technopark



1.1.2.2 Mission and Goals

Purpose

The purpose of the establishment of TP is to serve as a base institution to discover and foster knowledge-based small and medium-sized technology companies through an organic cooperation network with university-industry-government associations.

Mission²⁷

Initially, it focused on corporate incubating and technical support, but gradually expanded its functions to local policies, business planning, and corporate support services. Phase 1 (1998-2002) focused on space creation (hardware), Phase 2 (2003-2012) focused on software projects, and Phase 3 (2013-) promoted its basic functions.

²⁶ KTPA, <http://www.technopark.kr>

²⁷ KTPA (2015), Establishment of Korean Techno Park (TP) model and establishment of KVIP operation know-how transfer plan

[Figure 3-2] Changes in TP's Major Missions by Stage



1.1.2.3 Strategy

Five key success strategies of South Korean TPs have been identified as follows:

- **Cooperation between central and local governments:** The Board of Directors of a Technopark is composed of central and local government personnel.
- **Guaranteed business continuity and stability under the special laws:** Maintaining stability and sustainability of the TP operation is secured via the enactment of special laws.
- **Integrated budget execution and management:** The budget for regional strategic industries and plans to foster regional industries are integrated and operated centering on TP.
- **Consistent competition-based efficiency:** The central government annually evaluates management performance and regional industrial promotion plans for TPs nationwide.
- **Cooperation with CCEI, TFC & HEIs, etc.:** Cooperation with Creative Economy Innovation Center (CCEI), Tech Financing Center (TFC), and regional universities (HEIs) can have a good synergy effect in fostering local businesses.

1.1.3 Roles & Functions of Technopark²⁸

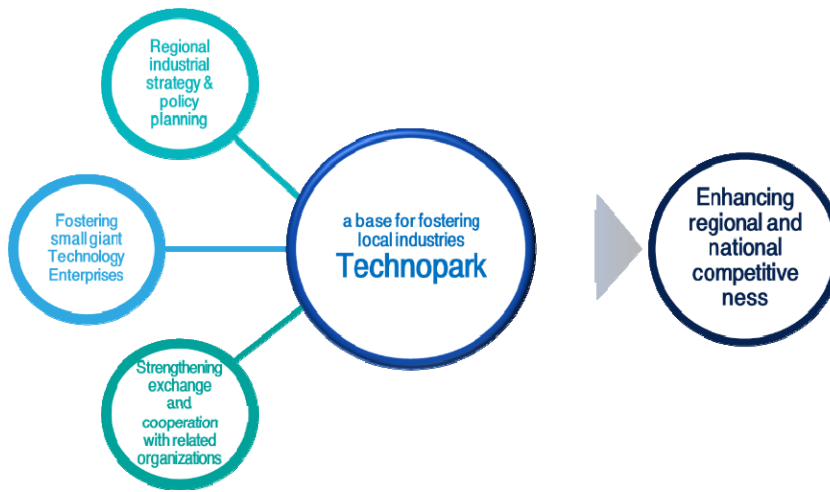
The key roles and functions of TP are to plan strategies and policies, foster small-scale technological enterprises, and strengthen cooperation networking.

TP carries out the function of strategy planning and policy development for the purpose of fostering knowledge-based local industries. TP plans mid- and long-term development strategies and policies for local industries. It also performs the function of discovering and planning new regional growth engine industries.

TP also has the function of fostering small-scale technology companies through the discovery and support of technology innovation companies. Firstly, corporate support includes securing of advanced equipment and support for joint use, support for product trial production, the provision of space for corporate occupancy, the provision of comprehensive technical/ management consulting, support for domestic and international marketing and overseas expansion, training and supply of excellent human resources, etc. Secondly, business creation includes the discovery of promising technology companies and support for start-ups, the commercialization of technology development and transfer technologies, attracting high-tech companies, etc. Finally, technology innovation includes the promotion of innovative technology development through industry-academia joint research and development, the revitalization of technology trade and transfer, new technology education and training, etc. Another function of TP is to strengthen exchange and cooperation among related organizations through the establishment of regional networks.

²⁸ KTPA(2016), Techno Park, The growth engine of Korea

[Figure 3-3] Roles & Functions of TP



1.1.4 Major Programs²⁹

Depending on the growth stage of a company, TP implements customized business support programs for the company to improve its efficiency.

The first is the Startup stage. In this stage, the most representative programs is startup consulting. The main support programs performed during this stage are (1) startup consulting (mentoring), and start-up training, (2) startup space support, (3) market research support, and (4) linking investment and policy funds

The second is the Growth stage. The representative programs of this stage are big data-based smart support and R&D. The main support programs performed at this stage are (1) big data-based industry status analysis/provision, (2) big data-based customized company analysis, (3) support project matching, (4) management consulting and growth strategy establishment, and (5) R&D and technology strategy establishment.

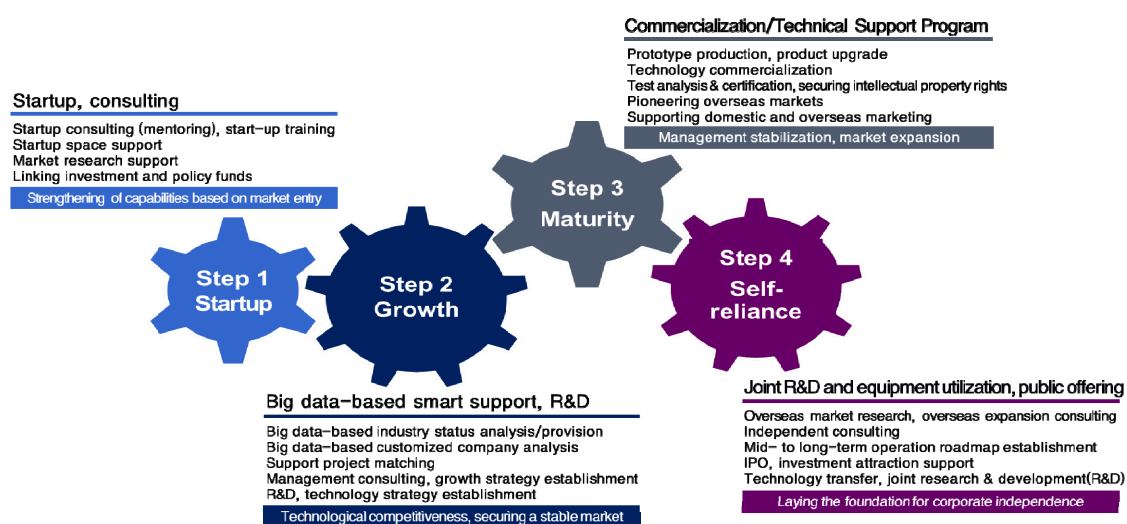
The third is the Maturity stage. The representative program is the commercialization and technical support program. The main support programs performed at this stage are (1) prototype production and product upgrade, (2) technology commercialization, (3) test

²⁹ Ministry of SMEs and Startups(2018), Technopark (TP) function enhancement plan

analysis & certification and securing intellectual property rights, (4) pioneering overseas markets, and (5) supporting domestic and overseas marketing.

The fourth is the Self-reliance stage. The representative programs are joint R&D, equipment utilization and public offering. The main support programs performed during this stage are (1) overseas market research and overseas expansion consulting, (2) independent consulting, (3) development of a mid- and long-term operational roadmap, (4) IPO and investment attraction support, (5) technology transfer and joint research & development (R&D)

[Figure 3-4] Major programs of TP



1.2 Structure of Technopark³⁰

The South Korean government laid down a set of laws, implementing ordinances, and standards for operating TPs after the establishment of TPs. For more detailed guidelines, the South Korean government announced a standard called the “Standard on Implementer’s Operation of South Korean Technopark”. The standard covers the board of directors, structure of organization as implementer, promotion and hiring of employees,

³⁰ This chapter was written by Kyoung Joo Nam of the Gyeonggi Technopark.

pay rule, budgets and accounting, settlement of accounts, supervision of implementer, evaluation of implementer, and management disclosure to the public. All 18 South Korean TPs follow the standard which guides the entirety of their organizational operations.

1.2.1 Governance of Technopark

According to the standard, every TP should include the core matters in its bylaws, and the board of directors makes decisions on important matters as follows:

[Table 3-1] Role of the Board of Directors in TP based on the Standard

Standard Article	Contents
Matters to be entered in the bylaws	- The project implementer shall describe the following matters in the bylaws of TP. 1. Name 2. Purpose 3. The location of the main office 4. Project scope, contents, and execution 5. Matters concerning executives (board members) and employees 6. Operation of the board of directors 7. Direct departments and affiliated organizations 8. Committee composition and operation 9. About the organization 10. Matters concerning property and accounting 11. Changes in the bylaws 12. Matters concerning the announcement and method 13. Dissolution of corporation and disposal of property 14. Audit and accounting inspection 15. Other basic matters related to corporate operation
The role of the Board of Directors	- The project operator shall establish a board of directors to decide on important matters in each of the following subparagraphs. The board of directors opens with the attendance of a majority of the current directors, and decisions are made with the approval of a majority of the attending directors. However, in the case of the following items 1 and 6, the decision is made with the approval of two-thirds or more of the attending directors. 1. Matters concerning the dissolution and liquidation of a corporation

Standard Article	Contents
	2. Matters concerning deliberation of the business plan and budget 3. Matters concerning the approval of business performance and settlement 4. Matters concerning the disposal and management of borrowings and basic assets 5. Matters concerning the enactment and amendment of the bylaws and important regulations 6. Matters concerning the appointment, dismissal, and dismissal of executives 7. Dismissal or dismissal of the head of a direct department or affiliated institution 8. Matters concerning profits into basic assets and its use 9. Matters on management of employee quota 10. Matters concerning the CEO/president's income 11. Matters requiring resolution by the board of directors according to the bylaws

1.2.2 Composition of the Board

The board members have the power to make decisions on TP's core matters. Therefore, the composition of the board (e.g. the number of directors and their affiliated organizations) is very important. The ratio of the numbers from each institution could highly affect the decision making on implementer's major practices of businesses and the long term directions.

Certain members of the board are positioned as ex-officio, meaning a position someone automatically gains because of another job or position that he/she already holds. Also the contribution money provided by certain institutions could reserve their seats on the board by their ratio.

[Table 3-2] Composition of the Board of Directors based on the Standard

Standard Article	Contents
Number of chairperson, director, president and auditors	- The project implementer shall have the following directors. - No more than 2 chairpersons - One president (CEO/president in implementer organization) 15 directors or less (including the chairperson and president) - Within 2 auditors
Ex-officio directors and elected directors by contribution ratio in board members	- The board of directors appoints directors. However, a person falling under any of the following subparagraph will naturally become a director (hereinafter referred to as "ex-officio director") without the appointment by the board of directors. 1. Chairperson 2. President (CEO in implementer organization) 3. Director of the Ministry of Small and Medium Venture Business (South Korean Gov) 4. Regional Small and Medium Venture Business Administration (South Korean Gov) 5. Director of South Korea Industrial Technology Promotion Agency (South Korean Gov. agency) 6. If the institution that has contributed the most to the implementer is a city government or a private institution, the head of the relevant institution (Local Gov) - In principle, directors who are not "ex-officio director" are to be appointed after the organizations that have contributed to the implementer may recommend candidates for directors following the contribution ratio. However, if a member of an institution that has contributed to the implementer of TP is an ex officio director, the number of candidates for non ex-officio directors may be recommended excluding the number of the ex-officio directors. - Non ex-officio directors are elected by the board of directors after recommendation from among experts in various fields such as business, academia, science and technology, law, and media.

As you see, the number of members from the government is more than 1/3 (3.-5. in the above table) of the entire numbers in the board. Also the head of metropolitan or provincial government has a right to be chairperson of the board of directors by the standard and the very nature of TP's seed money.

In case of Gyeonggi Technopark (GTP), seven members are ex-officios from the governments, and even the elected director, the president of GTP, is also a former government official. Also, seven members are elected by either city or provincial government though these elected members are from industry or academia. In fact, South Korean TPs' operation is deeply affected by the public sector, especially the government sector's direction and its policies.

[Table 3-3] Ex-officio and Elective Members in the Board of Directors based on the Standard

Standard Article	Contents
Board of Directors	• Ex-officio and persons by contribution ratio (7): ① Governor of Gyeonggi Province (ex-officio), ② Regional Corporate Policy Officer of the Ministry of SMEs and Startups (ex-officio) ③ Director of the Gyeonggi Regional Small and Medium Business Administration (ex-officio) ④ Director of the Gyeonggi Province Economic Office (ex-officio), ⑤ Director of the South Korea Institute of Industrial Technology (ex-officio), ⑥ Mayor of Ansan (ex-officio), ⑦ Head of Industrial Support Headquarters in Ansan City (ex-officio) → 3 (South Korean Gov.), 2 (Prov. Gov), 2 (City Gov.) • Ex-officio elected(1) : President of Gyeonggi TechnoPark (elected), • Elected (7): CEO of local SME, Vice-President of Seoul Arts University, President of Hanyang University, President of Smart Hub CEO Association, CEO of local SME, Chairman of Ansan CCI, CEO of local SME → 6 recommended by City, 1 recommended by Prov.

1.2.3 Recommendation Committee for Appointing CEO-President

As you see, there are at least seven ex-officio board members from the governments. Another seven elected directors are also appointed by the governments. The election of CEO-president of TP is practiced likewise. A TP forms a recommendation committee following the standard article. As you see the Table below, the South Korean central and local (provincial) governments have power to pick a CEO-president of the TP implementer.

[Table 3-4] Composition of the CEO-president Recommendation Committee based on the Standard

Standard Article	Contents
CEO-president Recommendation Committee	• The CEO-President Recommendation Committee consists of 2 recommended by the chairperson, 2 recommended by the Minister of SMEs, and a head person in economy group designated by the provincial governor. It consists of eight people, including the head of the Startup Growth Support Division of the Small and Medium Venture Business Administration and the head of the local government (or director). However, if the institution that has contributed the most to a TP is a city government or a private institution, it may be composed of nine members including one recommended by the relevant institution.

In case of GTP, 2 members are recommended by the chairperson (Provincial Governor), 2 by the central government, and 1 head person by the chairperson (Provincial Governor) again. The two ex-officios are from the central and provincial governments. GTP has one “additional” member from Ansan City since the city is the most contributing entity to fund GTP. Therefore, all the recommendation committee members to pick a new CEO/President are government officials. They are either 100% government officials or they are all recommended by committee members.

In the GTP’s history of electing its president, the first president appointed in 1992 was a professor from Hanyang University. The second president was appointed in 2011 based upon Ansan City’s recommendation and he was a former high-ranking government official of the Ministry of Science and Technology. The third president appointed in 2014 was a former high-ranking government official of the Gyeonggi Provincial Government. The fourth and fifth presidents, both former high-ranking government officials from the Gyeonggi Provincial Government, were appointed in 2017 and 2019, respectively. The presidential term was three years in the past and it has now reduced to two years and presidents can serve two consecutive terms.

During an official interview, some resident companies of GTP expressed their wish that one of the residents be entitled to be a member of the committee. Though their request makes a full sense, it would take quite a long time for their request to be realized.

1.2.4 Organizational Structure in Implementer

Most South Korean TPs follow the standard to set up their organization since the standard requires each TP to get an approval from the board of directors, when president wants to change the organization. There are a few exceptions to the standard. Some TP such as Pohang, Seoul, Gyeonggi, Daejeon and Busan were initiated by private sector and/or local governments, which means no seed money from the central government at the foundation stage. In case of Gyeonggi TP (GTP), Ansan City's contribution is over 52%. These TPs who have less influence from the central government qualify for exceptions in the arrangement in their organizations.

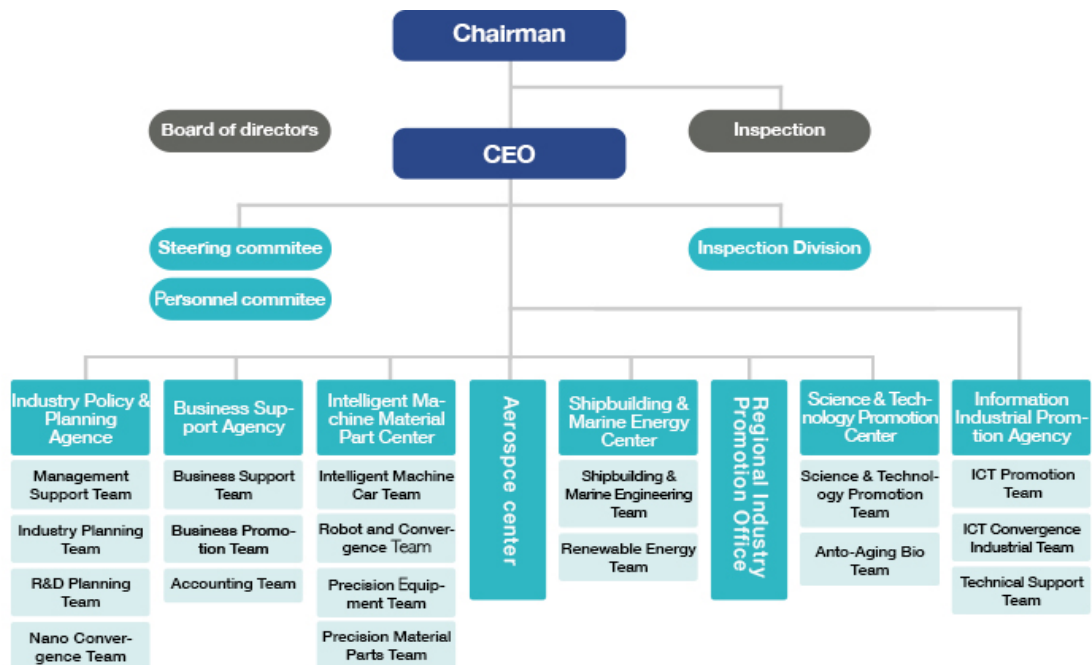
[Table 3-5] Guidelines of Setting up an Organization Based on the Standard

Article in Standard	Contents
Organization	• The president of each TP has a policy-planning group, an SME support group, and a specialized technology center as direct departments, and may establish an administrative support office, audit office, and regional industry promotion office as a department under direct control. • If a separate organizational operation is required, the president may establish an affiliated institution through the decision by the board of directors. • The president may establish an affiliated center (direct department) for fostering and supporting specific projects after the approval of the board of directors when South Korean government does not contribute to the infrastructure such as buildings and equipment for the affiliated center. If universities contribute, the center can be set up at the university site. • The policy-planning group performs tasks such as regional industrial policy planning, regional R&D planning, TP management strategy, mid-to long-term development plan, etc. • The SME support group performs duties such as surveying job information, establishing a regional innovation system, operating the industry-academia-research cooperation council, and comprehensive support for local companies, and operating a contact center to support local companies, supporting resident companies within the TP complex, technology transfer, technology investment promotion, and policy-designated projects. • Specialized [Technology] centers perform joint R&D, R&D support, technical support, industrialization support through equipment and pilot production support, and work to cultivate professional technical

Article in Standard	Contents
	workforce. • The president may operate the administrative support office to support administrative support work and facility management. • The president may establish a regional industry promotion office to carry out consignment projects, etc. • The president may establish an audit office to improve the internal organization's control system, promote the appropriateness of operation, fairness, and efficient use of the budget, and achieve management improvement.

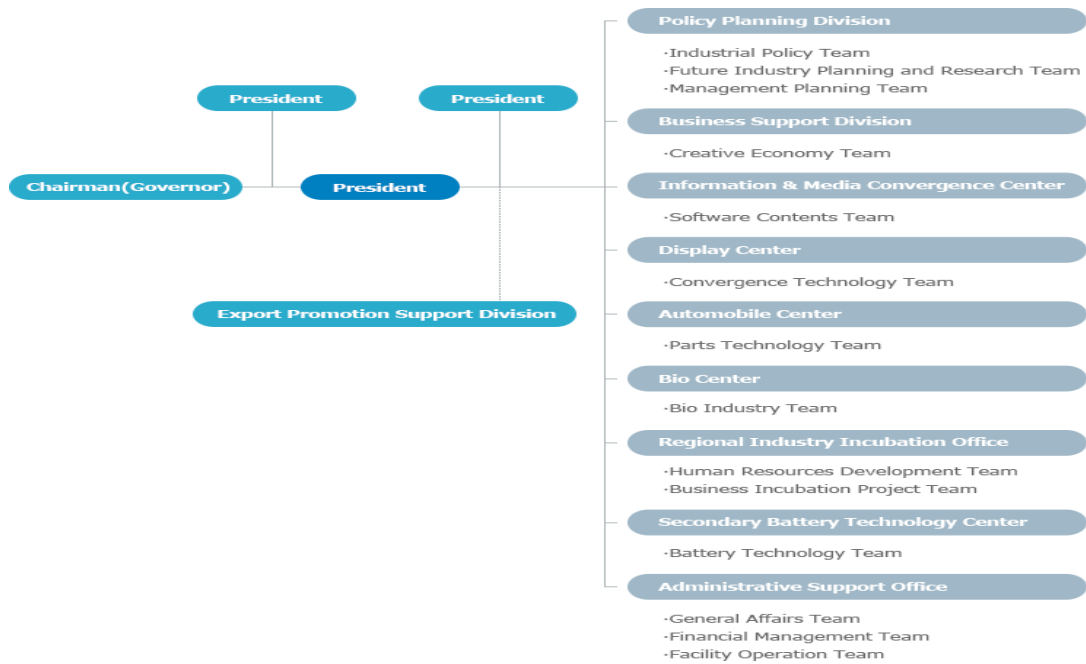
The organization charts of Gyeongnam Technopark and Chungnam Technopark below are examples of a standard organizational chart.

[Figure 3-5] Organization Chart of Gyeongnam (Prov.) TP



Source: GNTTP webpage

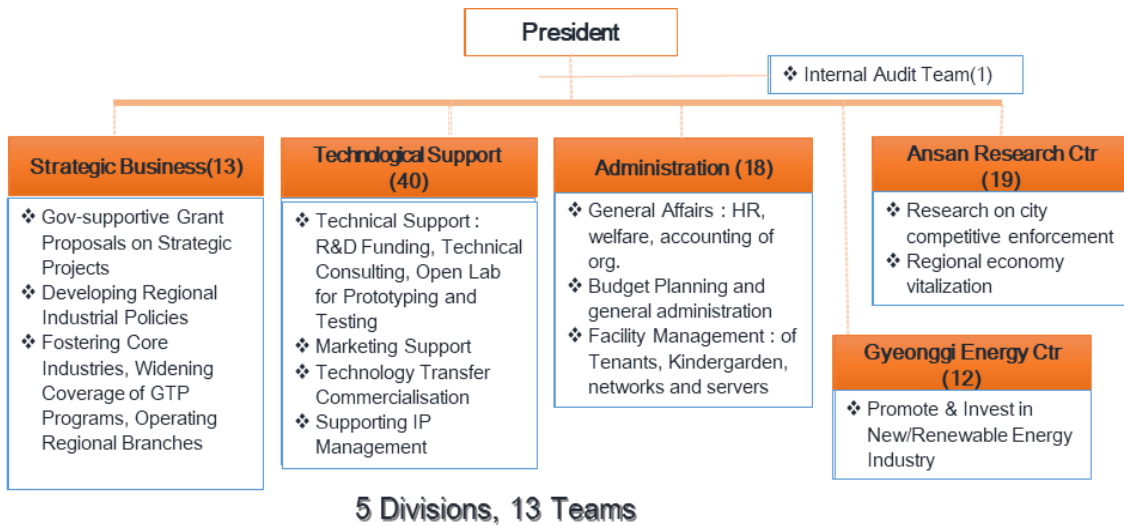
[Figure 3-6] Organization Chart of Chungnam (Prov.) TP



Source: GNTP webpage

GTP's organizational structure is different from the typical ones found in other TPs. The South Korean government invested more to regional TPs, not to ones in the Seoul-Metropolitan area. In fact, the country failed to develop its land in a well-balanced manner during the period of its rapid economic development from the 1960s to the 1990s driven by the national economic development plan. Since 2006, however, investments have been channeled into regional TPs in non-Seoul-Metropolitan area. A huge difference between the TPs in the Seoul-Metropolitan area and the regional TPs outside the Seoul-Metropolitan area is the existence of specialized technology development centers. Regional TPs in non-Seoul-Metropolitan area have 3-4 regionally specialized technology-based development centers. Some of them have evolved into accreditation centers while others have become test-bed centers. GTP does not have these technology development centers and its size is smaller than that of other regional TPs.

[Figure 3-7] Organization Chart of GTP



Source: GTPs internal doc.

1.2.5 CEO-President of Technopark

In principle, the presidents of South Korean TPs are appointed through the public recruitment. However, the key persons in the President/CEO Recommendation Committee consisting of government officials or government affiliated persons since the standard regulates the committee membership (e.g. members of specific entities qualified for the committee members). As expected, most CEO-Presidents of South Korean TPs are former government officials and/or ones from the organizations making high contributions to TPs.

Here is a full list of presidents in TPs. Eight of them are from academia, eight from the central government or government agencies, two from the provincial or city governments, and one from another TP. Many applicants for this position are from private sector such as big enterprises, but it is not easy for applicants with business career to be appointed. The CEO/President Recommendation Committee considers the harmonious working relationship with the public offices and the research sector as the most important qualification of the candidates.

[Table 3-6] CEOs/Presidents of South Korean TPs, June 2020

TP	Name (birth yr)	Former Career	Relation
Seoul	Kim (58)	- Research professor, Hanyang University - Research Fellow/Executive Director, Pohang Institute of Industrial Science and Technology, Managing Director, POSCO	Academia
Busan	Choi (55)	Professor, Department of Business Administration, Pusan National University	Academia
Daegu	Kwon (54)	Director of Planning, Keimyung University	Academia
Incheon	Seo	- President, South Korea Information Society Promotion Agency - Director of Operational Support, National Information Strategy Committee	South Korean government
Gwangju	Kim (63)	- Head, Trade and Investment Office, Ministry of Trade, Industry and Energy - Director of Industrial Science and Mid-term Policy Bureau, Gov.	South Korean government
Daejeon	Choi (61)	- Vice President, OB Beer Corporation - President, South Korea Radio Promotion Agency - Policy advisor to the Minister of Information and Communication, Gov.	South Korean government
Ulsan	Cha (62)	Industrial Technology Policy Officer, Ministry of Trade, Industry and Energy	South Korean government
Sejong	Kim (61)	Vice-Chairman, Small and Medium Venture Business Corporation	Government agency
Gyeonggi	Bae (60)	- Vice Mayor, Anyang City - Deputy Mayor, Goyang City	Prov. government
Gangwon	Kim (59)	- President, Cheorwon Plasma Industrial Technology Research Institute - Professor, Kwangwoon University	Academia
Chungbuk	Song (57)	- Head, Chungbuk Regional Small and Medium Business Administration - Director, Technology Management Innovation Division, SMBA - CEO, South Korea Institute of Construction Living Environment Test	South Korean government
Chungnam	Lee (69)	- Professor, Mechanical and Automotive Engineering, Gongju University - Assistant professor, mechanical engineering,	Academia

TP	Name (birth yr)	Former Career	Relation
		Suncheon National University	
Jeonbuk	Yang (52)	• President, Camptic Institute of Technology • Professor, Department of Mechanical Engineering, Chonbuk National University	Academia
Jeonnam	Yoo (63)	• Jeonnam Ombudsman, South Korea Industrial Technology Agency • President, South South Korea Regional Economic Industry Research Institute	Gwangju Technopark
Gyeongbuk	Lee	• Professor, Business Administration, Yongnam Univ	Academia
Gyeongnam	Ahn (64)	• Vice president/president, South Korea Gas Corporation • American attorney at Kim & Chang Law Office	South Korean government
Jeju	Tae	• Director, Jeju Technopark Corporate Support • Co-representative, South Korea Della Glass Co., Ltd.	Government agency
Gyeong D.	Yim (63)	• Professor, Gangneung-wonju Nat'l University	Academia
Pohang	Lee (59)	• Director of Nam-gu, Pohang City • Regional Vice Director, Pohang City	City government

Source: Internal doc of South Korea Technopark Association (KTPA)

1.3 Budget of Technopark³¹

1.3.1 Budgets and Employees of South Korean TPs

[Table 3-7] Budgets and Employees of South Korean TPs, Dec. 31, 2019

TP	Total Budget (K USD)	Budget for SME Support* (K USD) (% out of total)	Employees** (person)
Incheon	217,056	172,975 (80)	270
Gyeonggi	144,597	126,265 (87)	150
Gyeonggi D.	94,808	92,586 (98)	71

³¹ This chapter was written by Ms. Kyoung Joo Nam of the Gyeonggi Technopark.

TP	Total Budget (K USD)	Budget for SME Support* (K USD) (% out of total)	Employees** (person)
Pohang	40,295	27,605 (69)	63
Seoul	27,538	21,225 (77)	52
Gyeongnam	252,510	239,804 (95)	202
Daegu	188,966	151,824 (80)	216
Busan	156,848	132,136 (84)	201
Daejeon	141,471	126,265 (89)	141
Chungnam	139,536	124,461 (89)	132
Gyeongbuk	138,431	111,954 (81)	160
Chungbuk	130,794	101,704 (78)	150
Jeonbuk	92,291	86,622 (94)	140
Jeonnam	108,311	83,342 (77)	149
Ulsan	109,127	80,918 (74)	171
Jeju	94,405	80,473 (85)	142
Gangwon	78,531	68,541 (87)	92
Gwangju	71,147	58,686 (82)	95

* Budgets to run SME supportive programs only

** All employees (including contract based employees)

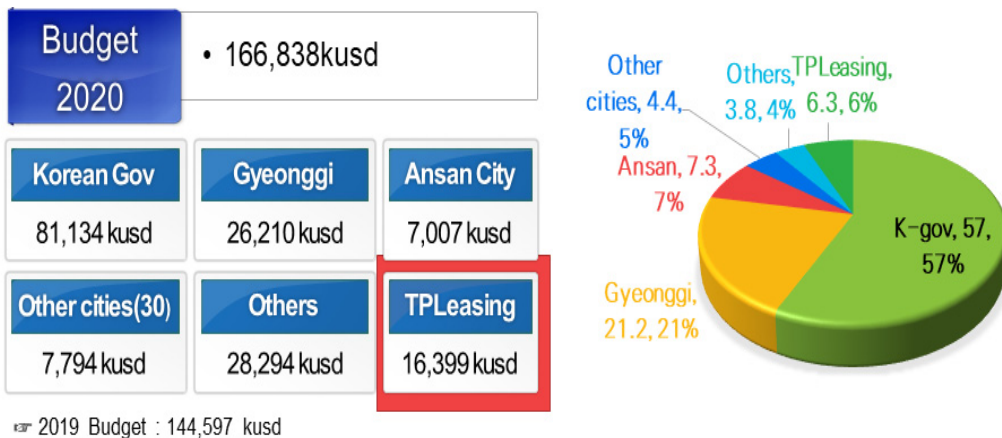
Source: Internal doc of South Korea Technopark Association(KTPA)

The budgets of South Korean TPs vary widely. Their funding resources are different though their way of using the budget is unilaterally regulated by the standard. In case of Incheon TP that has the most budget though Incheon is located in the Seoul-Metropolitan area, 3 large institutions, a TP, an economy promotion agency, and an IT promotion agency were merged together to form one mega-size institution called Incheon TP in 2016. Their budgets were also merged to Incheon TP as well. In case of Gyeongnam TP and Gyeonggi TP, industry 4.0 (smart factory to local SMEs or digital transformation) programs have been a major driving force to increase the entire budget since these two regions have

more manufacturing sites than others. In most TP programs, the South Korean government always considers a balanced development of the nation, which implies more budgets to regional TPs located in non-Seoul-Metropolitan area. However, in relation to the smart factory, the South Korean government has allocated budget in proportionate to the number of factory sites.

1.3.2 Yearly Budget of GTP in 2020

[Figure 3-8] Yearly Budget of GTP in 2020



Source: Internal document of GTP

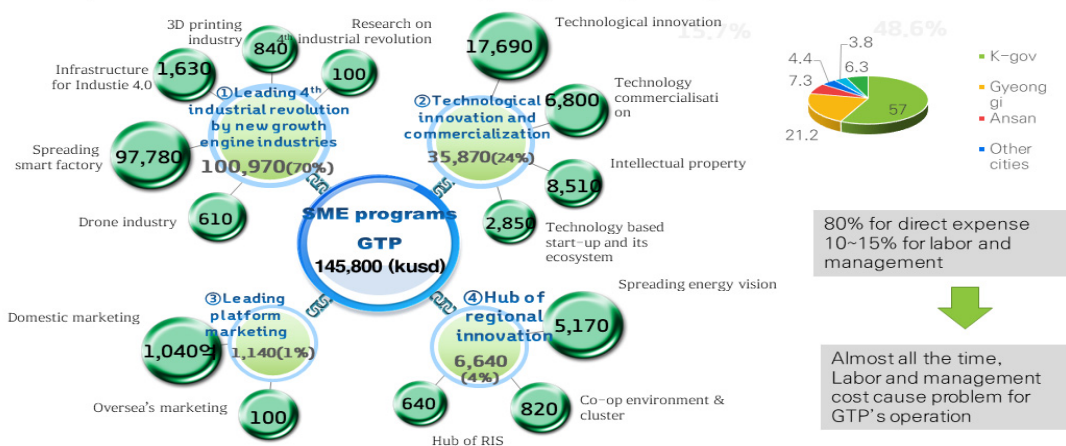
GTP's annual budget in 2020 is USD 166,838 K. Nearly 87% of its yearly budget is being spent for SME supportive programs. The leasing income from residents account for only 6% of its total budget. The government's grant takes up the biggest, occupying 57% because the smart factor project grant for local SMEs accounts for more than 50% of the entire yearly budget. For several years before the start of the smart factory project in 2019, the budget from the Gyeonggi Provincial Government had exceeded 50%.

1.3.3 Budget for SME Support Program

The budget for SME supportive programs are divided into four categories such as (1) Leading 4th industrial revolution by new growth engine industries, (2) Technological innovation, (3) Leading platform marketing and (4) GTP as a hub of regional innovation. All four areas are equally important but the size of budget varies across categories. (1) and (2) are the key areas that GTP has been focusing on with spending a major share of the grant money.

[Figure 3-9] SME Supportive Programs of GTP, 2020

4 major fields · 14 divisions · 50 programs(2020)



Source: Internal document of GTP

All funds come from the central and local governments. GTP offers proposals to relevant government bodies every year (usually in October) in order to secure the next year's budget. Here is the problem of budget expenditure structure. 87% of the entire budget comes to SMEs, of which more than 80% goes to SMEs directly while the remaining 20% or less is spent for GTP's labor and management cost to operate the SME supportive programs. Because of this non-proportionate expenditure structure, labor and management shortage threatens GTP's stable operation all the time.

Here is the full list of the SME supportive programs in 2020. Members of city or provincial parliament often complain about the overlaps between projects. However all 50 programs

have their own domains and they don't share grant resources, so they cannot be merged into a single project even though some of the project contents are overlapped.

[Table 3-8] SME Support Programs and Grant Resources of GTP, 2020

	Resources(K USD)							
	total	K.Gov	Prov.	Ansan	Other City	Others	Carry-over	Return
Total	145,883	80,894	22,335	6,807	7,793	328	22,488	5,238
① Leading 4th industrial revolution	100,986	76,353	2,585	1,400	1,400	300	16,763	2,185
Operation of demo-factory test bed	1,630		300	1,000		300		30
Smart factory to local SMEs (2020)	93,700	76,233					15,467	2,000
Smart factory to local SMEs (Prov.)	1,968		1,000				868	100
Smart factory to local SMEs (Ansan)	717			400			317	
Smart factory to local SMEs (Gimpo)	500				500			
Smart factory to local SMEs (Hwaseong)	900				900			
Training on 3D modeling for 3D printer	724		700					24
Operation of the Maker Space	122	120						2
Drone industry support	610		585					25
Training of marine leisure technician	8						5	3
Fostering marine leisure industry	7						6	1
Study with working group on the Fourth Industrial Revolution	100						100	
② Tech innovation and commercialization	35,872	4,312	15,012	3,957	5,481	8	5,377	1,725
Young adult start-up incubation	1,633			1,590				43
Supporting Start-up based tech transfer	610		600					10
Consulting for start-up	276						226	50
Agile speedy support for commercialization	574	572						2
Boost up of start-ups in incubation Ctr.	21			20				1
Operation of Ansan IT industry promotion ctr	315			300				15
Techno doctor	8,432		4,713	538	2,781			400
Localization of parts & materials	2,538		2,038				400	100
Ansan Smart Hub R&D support	548			500				48
Operation of local branches of GTP	2,445		1,595		790			60
Customized support on environmental issue of local SMEs in Gimpo	320				300			20
Support of printing workshops	756	400			300		26	30
Plating & mold industry support with facility enhancement	1,550		800	240	410			100
Improving on handling harmful chemical substances in local SMEs	1,103						843	260
IP support for export enterprises	540		500					40
Manufacturing-service convergence support	310		300					10
Converging IT·SW to local SMEs	303			300				3
Servitization of SW in local SMEs' manufacturing process	667	302		364				1
Gyeonggi R&D	4,291		606				3,547	138
Supporting small hidden champion SMEs	69			55		8		6

2020 K-Innovation ODA Program with Azerbaijan

	Resources(K USD)							
	total	K.Gov	Prov.	Ansan	Other City	Others	Carry-over	Return
Supporting regulation-free demonstration	53						43	10
Accelerating IP acquisition and start-up	6,448	3,038	2,300	50	860			200
Fostering IP experts and job creation	390		300					90
Prov. owned IP management	125		110					15
Hanam city IP doctor	40				40			
IP protection	1,515		1,150				292	73
③ Leading platform marketing	1,546	0	630	390	166	0	348	12
Support on SMEs joining in Prov. owned online shopping mall	555		550					5
Support of social enterprises' marketing on On-line platform	82		80					2
Supporting technology transfer on offline market	314						314	
Customized support for local SMEs (Ansan)	394			390				4
Support on online overseas marketing (Yongin)	34						34	
Export support for local SMEs (Yongin)	67				66			1
Supporting online store start-up (Yangpyeong)	100				100			
④ Hub of regional innovation	6,646	229	4,108	1,060	746	0	0	503
GTP as local innovation hub	642	229	258	150				5
Ansan Science Valley as innovative cluster	610			450				160
ASV Science Festival (Cancelled-COVID 19)	215			200				15
Renewable energy to community	1,099			260	746			93
Operation of Gyeonggi Energy Center	4,080		3,850					230
Returnable grant after completion of project	833	0	0	0	0	20	0	813
Returnable grant after completion of project	833	0	0	0	0	20	0	813

Source: Internal document of GTP

Securing budget to cover indirect expenses and labor cost is a critical issue when the project is kicked-off. Indirect expenses refers to management and operational cost. It may or may not include labor cost. Fund providers often try to give TPs lesser amounts of the operational expenses while TPs always try to get more for its operational cost. South Korean TPs are no exception though there are some minute differences. There is no “the fixed” rate for indirect expenses, thus causing frequent disputes over this matter.

[Table 3-9] Rates of Indirect Expenses in TPs

No	TP	Base rule/law/ordinance /internal rule	Rate	Remarks
1	Busan	none		Different programs with different charging rates
2	Jeonnam	Internal rule (Financial Management)	>8% of direct expense	-
3	Incheon	Internal rule (Financial Management)	>10-17% of DE	-
4	Gwangju	Internal rule (Financial Management)	<17% of DE	In case of programs with additional regulation/rule, exceptional
5	Pohang	Internal rule (Financial Management)	10-17% of DE	With additional regulation of programs, highest rate can be charged
6	Ulsan	Internal rule (Financial Management)	17% of DE	In case of lower than 17% in programs, employees should talk with management team in advance
7	Gyeonggi	Internal rule (Financial Management)	10% of DE	Province programs: 2% (Labour+DE can be 10-15% of total amount)

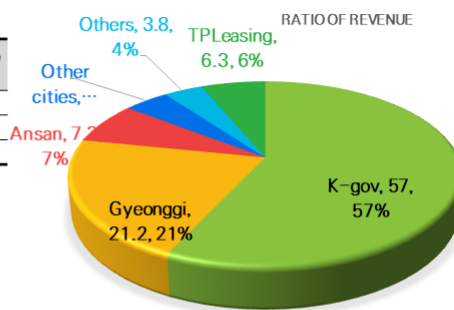
Source: Internal document of KTPA

1.3.4 Leasing Space for Tenants: The Earliest and the Oldest Business Model

[Figure 3-10] Occupancy Rates and Ratio of Leasing Income in GTP (2016-2019)

Usage ratio (㎡, %)

Year	2016	2017	2018	2019	Goal	Increase Rate
Occupancy(A)	36,791.0	36,915.7	37,173.6	39,199	-	-
Availability(B)	38,891.5	38,891.5	38,716.6	41,657	-	-
Occupancy rate(A/B)	94.6	94.9	96.0	94.1	100%	-5.9%



Growth of Tenants (kUSD, %)

Year	2016	2017	2018	2019	Goal	Increase Rate
No. of Tenants	105	107	106	107	-	-
Sales by Tenants	2,143,785	1,962,380	1,588,940	1,386,506	-	-
Average Sales	20,417	18,340	14,990	12,958	21,499	-39.7%

Source: Internal document of GTP

This is GTP's budget breakdown for this year. GTP is located in the Seoul-Metropolitan area. Most companies do not want to leave the Seoul-Metropolitan area. That means it is relatively easier for GTP to find resident companies than other regional TPs. Therefore, the occupancy rate of GTP always remains over 94%, which may be the highest possible rate that can be achieved since 100% is realistically impossible. The profit from residents allows TPs to maintain the complex. Since the entire leasing profit is only 6% of GTP's entire budget, it is a very meager portion. In case where GTP makes losses in other projects, the income from leasing would not be of much help for GTP. GTP has to hire quite a lot of personnel for administrative and operational work, but leasing profits are way too small to cover their labor cost.

GTP spends leasing income from residents for maintenance (70%), repairs, and the labor cost of the facility team employees (4). As the facilities get older, more cost is required for maintenance and repair. Currently, GTP's earnings is not sufficient to cover 100% of the repair cost.

GTP made a decision to construct a new building with its term deposit money along with some government funding in 2016, proposing a plan to build a test bed as part of a scheme related to the Fourth Industrial Revolution. And a proposal was set forth with funding from local governments. Still it took 5 years for all the stakeholders to finally decide their contributions to the construction of the building. When the construction of the building is completed, more leasing income can be secured, which then can be used to support maintenance expenses.

[Figure 3-11] Bird's Eye View of the Smart Manufacturing Innovation Center of GTP



Planning for expanding complex with fundraising
 Construction : Oct. 2020~2023
 Smart Manufacturing Innovation Center

Source: Internal document of GTP

1.3.5 Sustainable Funds from Local Governments

[Table 3-10] Annual Contribution by Local Governments for Sustainable Operation of TPs

TP	Annual contribution (k usd)	Source	Usage	Re
Gwangju	500	Gwangju City	Maintenance & repair of old facilities ('16 400, '17 500, '18 500, '19 700)	4yrs
Gyeonggi	1,100	Gyeonggi-do Ansan city	Labor (Team of finance, Team of facility management), Repair of old facilities	
Gyeonggi D	922	Gyeonggi-do Daejin Univ.	Labor (Team of finance, Team of facility management) Free leasing of land from Daejin Univ.	
Gyeongbuk	802	Gyeongbuk-do	Managing cost for the Green car part research institute	
Daejeon	4,800	Daejeon City	Labor & operation (no charging for additional labor and indirect expense in SME supportive programs)	
Busan	1,461	Busan City	Labor	
Jeonnam	1,309	Jeonnam-do	Labor & operation Operation cost of 4 specialized ctrs.	
Jeju	4,125	Jeju-do	Labor & operation	

TP	Annual contribution (k usd)	Source	Usage	Re
Chungnam	200	Chungnam-do	Labor Shortage of labor by parenting encouragement	
Chungbuk	1,045	Cheongju City etc	Exploring new programs and upgrading equipment	
Incheon	3,500	Incheon City	Labor & operation	
Sejong	960	Sejong City	Labor & operation	5yrs
Others			None	

Source: Internal document of KTPA

SME supportive programs cannot cover enough of the cost of labor and management of technopark. Therefore, some South Korean technoparks ask local governments to give annual operation-supportive contribution money. From some local governments like Daejeon and Incheon, all labor and management is covered by the governments' annual contribution. In case of nothing given to technoparks, their indirect expense and labor for SME supportive programs are sufficient, as found in Ulsan's case.

1.4 Workforce of Technopark³²

1.4.1 Rate of Continuous Working at Technopark

Since TP's initiation and operation is handled by the national law and ordinances, the job stability of its employees is high. When any new entry level person in his/her late 20s or early 30s, the job is secured until the retirement age of 60. Due to its stability, more young people want to be hired by TPs. The rate of continuous working is as high as 98.58% and the rate has been reduced a bit by some other issues like the delayed promotion. The average income of employees is similar to that of employees working at mid-sized enterprises.

For three years since 2016, the average rate of employees' continuous working stood at 96.81%, 1.77%p less than the average. The rate is still higher than that of the government

³² This chapter was written by Ms. Kyoung Joo Nam of the Gyeonggi Technopark.

officials, which remains at 95.9%(The Year Book of Statistics in HR Innovation, 2019).

[Table 3-11] 3-Year Rates of Employees' Continuous Working at GTP

Year	2016	2017	2018	2019
No. of quits (A) out of (B)	-	1	2	3
No. of full-time employee (B)	54	71	70	94
Rate of continuous working $[(1-(A/B)) \times 100]$	100%	98.59%	97.14%	96.81%

Source: Internal doc. of GTP

1.4.2 Arrangement of Workforce in Organization

Some employees responsible for facility management and intellectual property management are required to have expertise with licenses, while other employees are from various educational and career backgrounds. Employees are encouraged to study higher degree diploma courses and also are required to take annual training courses according to the organizational rules. Including the contract based employees, the entire employees' status is as follows:

[Table 3-12] HR in GTP, Dec. 2019

	Team	Major	HR										
			Position/Rank							Degree		BS/BA /HS	Total
			1	2	3	4	5	6	7	Ph.D.	Master		
Team of Audit		Eng./Sci.	-	-	-	-	-	-	-	-	-	-	0
		Others	-	-	1		-	-	-	-	1	-	1
	Sub total	Eng./Sci.	0	0	0	0	0	0	0	0	0	0	0
		Others	0	0	1	0	0	0	0	0	1	0	1
Div. of Strategic Business		Eng./Sci.	-	-	-	-	-	-	-	-	-	-	0
		Others	1	-	-	-	-	-	-	-	-	1	1
	Strategic	Eng./Sci.	-	-	-	-	-	-	-	-	-	-	0
	Planning	Others	-	1	-	2	1	-	-	1	1	2	4
	Future	Eng./Sci.	-	-	-	1	-	-	-	-	1	-	1
	Business	Others	-	-	1	-	1	2	-	-	2	2	4
	Smart	Eng./Sci.				1			1	-	1	1	2
	Factory to local SMEs	Others			1		1	1	10	-	4	9	13
	Sub total	Eng./Sci.	0	0	0	2	0	0	1	0	2	1	3
		Others	1	1	2	2	3	3	10	1	7	14	22
Div. of Technology Support		Eng./Sci.	1	-	-	-	-	-	-	1	-	-	1
		Others	-	-	-	-	-	-	-	-	-	-	0
	Technology	Eng./Sci.	-	-	-	1	1	1	1	-	-	4	4
	Support	Others	-	1	1	3	2	1	5	1	3	9	13
	Technology	Eng./Sci.	-	-	-	2	-	1	-	-	1	2	3
	Commercialization	Others	-	-	1	1	1	1	4	-	2	6	8
	Intellectual	Eng./Sci.	-	-	1	5	4	-	2	3	1	8	12
	Property Ctr	Others	-	-	-	2	2	1	-	1	4	-	5
	Sub total	Eng./Sci.	1	0	1	8	5	2	3	4	2	14	20
		Others	0	1	2	6	5	3	9	2	9	15	26
Div. of Administration		Eng./Sci.	-	-	-	-	-	-	-	-	-	-	0
		Others	1	-	-	-	-	-	-	-	-	1	1
	Internal	Eng./Sci.	-	1	1	1	-	1	-	-	1	3	4

	Team	Major	HR										
			Position/Rank							Degree			Total
			1	2	3	4	5	6	7	Ph.D.	Master	BS/BA	
												/HS	
	Administration	Others	-	-	-	-	1	-	1	-	1	2	3
	Accounting	Eng./Sci.	-	-	-	-	-	-	-	-	-	-	0
		Others	-	-	1	1	1	-	2	-	3	2	5
	Facility	Eng./Sci.	-	-	-	-	-	1	-	-	-	1	1
	Management	Others	-	-	1	-	1	1	-	-	3	35*	38
	Sub total	Eng./Sci.	0	1	1	1	0	2	0	0	1	4	5
		Others	1	0	2	1	3	1	3	0	7	40	47
Gyeonggi Energy Center		Eng./Sci.	-	-	-	-	-	-	-	-	-	-	0
		Others	-	-	-	-	-	-	-	-	-	-	0
	Energy Vision	Eng./Sci.	-	-	-	-	-	1				1	1
		Others	-	1		1			1		1	2	3
	Investment for	Eng./Sci.	-	-	-	1	1	-	1	-	2	1	3
	Renewable Energy	Others	-	-	-	1	-	-	1	-	1	1	2
	Sub total	Eng./Sci.	0	0	0	1	1	1	1	0	2	2	4
		Others	0	1	0	2	0	0	2	0	2	3	5
		Eng/Scie	-	-	-	-	-	-	-	-	-	-	0
Others		-	-	-	-	-	-	-	-	-	-	0	
Ansan Innovation Center	Policy Research	Eng./Sci.				3				2	1		3
		Others				2	1	1	1	2	2	1	5
	Ansan IT	Eng./Sci.			1	1	1					3	3
	Promotion Ctr	Others		1				1	1		1	4	5
	Sub total	Eng./Sci.	0	0	1	4	1	0	0	2	1	3	6
		Others	0	1	0	2	1	2	2	2	3	5	10
	Total		Eng./Sci.	1	1	3	16	7	5	5	6	8	24
		Others	2	4	7	13	12	9	26	5	29	77	111

* Facility maintenance in gas, electricity, security and cleaning job included

※ 25.5% out of all employees including maintenance job are engineering and science majors. Except maintenance job, 34.2% of engineering science majors and 41.4% in BA/BS/Master/Ph.D. educational background.

Source: Internal document of GTP

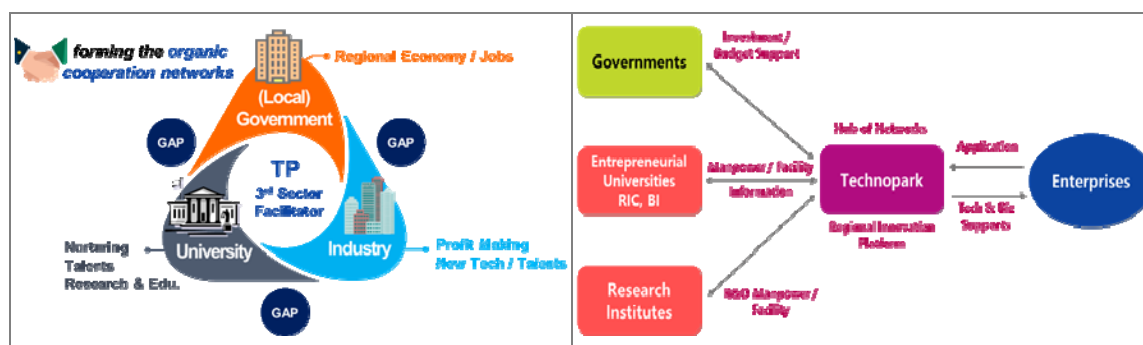
In the area of intellectual property, employees are required to take courses run by a higher institution called KIPA, the South Korea Invention Promotion Association, and even to take exams in order to get a certificate of IP consultation. In the area of technology transfer, many private training institutions offer annual programs that anyone can get a certificate of technology transactions. At least 15 employees have certificates of technology valuation. However, state licenses such as a patent attorney tend to be more recognized than other license issued by private institutions, so GTP hired 2 employees with the state license.

1.5 Network of Technopark³³

1.5.1 Stakeholder Groups (network activation plan between groups)³⁴

In fact, a TP is a facilitator of university-industry-government (UIG) cooperation. TP was introduced to further foster the institutional systems of UIG relations that technologically intensive industry park has not been able to fully meet. Centering around TP, research and education function of universities, new technologies of industries and economy, and jobs of local governments form organic cooperation networks. Through TP, a hub of these networks, the government provides budget, universities cultivate professionals, research institutes strengthen their R&D capabilities, and companies receive technology and commercialization support for their products. In short, TP serves as a platform.

[Figure 3-12] Stakeholder Groups of Technopark



³³ This chapter was written by Dr. Yang-jin Kim of the Gyeongbuk Technopark.

³⁴ KTPA(2015), Establishment of Korean Techno Park (TP) model and establishment of KVIP operation know-how transfer plan

1.5.2 Network Activation Program between Resident Companies and External Innovators (Universities, Companies, etc.)

TP is an innovation networking hub. It serves as a platform based on enabling value-creating interactions between startups and SMEs, between SMEs and large firms, and between service providers and users. TP works for regional industries, conducts policy planning, and fosters small-but-strong enterprises, forming an organic cooperation network with universities, industries, and academia, and collaborating with government ministries and agencies. The final goal of TP is to strengthen industrial competitiveness both at the regional and national levels.

TP also functions as the Industry-Academia Collaboration One-Stop Solution Center. TP is also the Fusion-Collaboration Promotor among industries, universities, and research institutes. Another important mission of TP is to overcome the differences of awareness between industries and academy as well as recover the broken collaboration system.

Some key networking promotion programs of TP includes i) a database construction and connection, ii) R&D support for on-site problem solving, iii) technology commercialization customized to companies, and iv) matching new technologies and people with potential³⁵.

³⁵ KTPA(2017), The role and function of TP for fostering local businesses

[Figure 3-13] Network Activation Program of TP



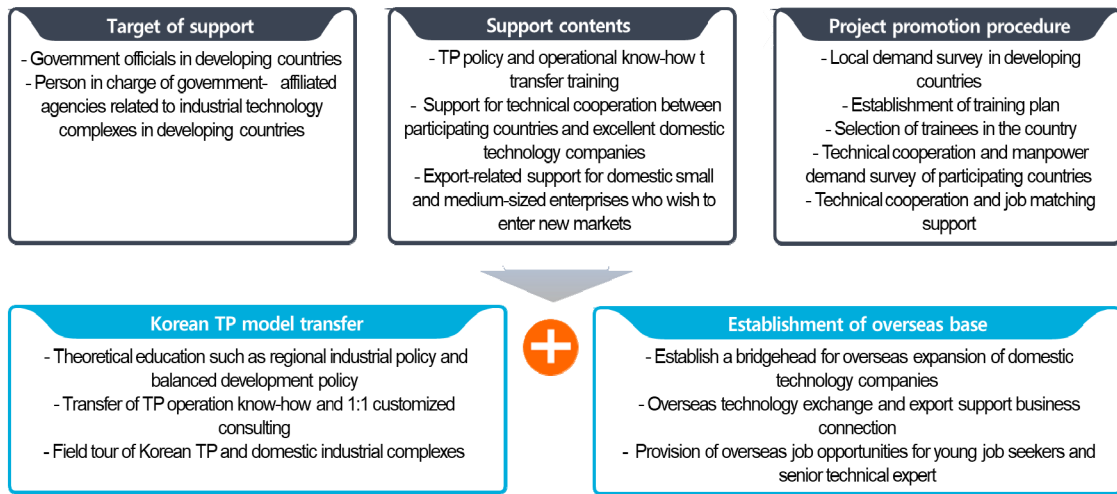
1.5.3 Network Activation Program with Overseas Innovators

South Korean TPs transfer the local TP model to developing countries and establish overseas cooperation networks to support local SMEs' overseas advancement and creation of overseas employment opportunities for the South Korean youths. In this efforts, key targets are government officials as well as personnel in charge of government-affiliated agencies related to industrial technology complexes in developing countries.

The supportive programs are i) TP policy and operational know-how transfer training program, ii) support for technical cooperation between participating countries and excellent local technology companies, and iii) export-related support for local SMEs who wish to enter new markets.

The process of the project promotion is ① conducting a local demand survey in developing countries, ② establishment of a training plan, ③ a selection of trainees in the participating countries, ④ conducting technical cooperation and workforce demand surveys of participating countries, ⑤ providing support for technical cooperation and job matching.

[Figure 3-14] Network Activation Program with Overseas Innovators



The benefits of the South Korean TP model is the transfer of knowledge and knowhow through; i) theoretical education on regional industrial policies and balanced development policies, ii) transfer of TP operation know-how, iii) one-to-one customized consulting, and iv) field trips to South Korean TPs as well as local industrial complexes in South Korea. Another benefit is the establishment of an overseas expansion base. This includes i) the establishment of a strong foothold for overseas expansion of domestic technology companies, ii) provision of opportunities for global cooperation and export support business connection, and iii) provision of overseas job opportunities for young job seekers and senior technical experts.

The followings are examples of representative international cooperative projects being carried out by Gyeongbuk TP.

The first is the Ethiopia Textile Technopark. It is currently being built in an Bole-Lemi Industrial complex in Addis Ababa in Ethiopia. The South Korean government is providing various supports including its support for the establishment of the Test & Evaluation Center for certification documents, training of specialist through equipment support and technical education, construction of the pilot plant and the Business Incubation center, and the transfer of TP operational know-how.

The second project is the Uzbekistan Textile Technopark. Through the connection between universities of South Korea and Uzbekistan, it is operated mainly to support industry-academic joint R&D and establish a support platform for companies between the two countries. In particular, it is focused on strengthening the organizational capacity of Uzbekistan to foster companies through the transfer of the TP operating system.

[Figure 3-15] Examples of representative international cooperative projects by Gyeongbuk TP



The third and fourth projects are under way in Uzbekistan and Guatemala, respectively. These projects also include equipment support, technical workforce training, Business Incubation center support, Start-up & Scale-up support, investment attraction from South Korean companies, resident company management, and transfer of TP operational manuals.

1.6 Resident Hunting of Technopark³⁶

It is important for TPs in South Korea to establish effective resident hunting policies with a relevant scope of strategies. By designing and practicing strong resident hunting operations, we can effectively find promising SMEs, support their growth, and lead them to become the hidden champions who can ultimately contribute to the development of regional economy.

In this section, we discuss the actual practice of TP in the City of Daejeon, South Korea, in the field of resident hunting and its implementation schemes. Daejeon Technopark (DJTP) is one of the nineteen TPs in South Korea that attempt to embrace regional startups and enterprises.

At DJTP, resident hunting projects are designed and implemented within a frame of a continuous process - startup promotion, business incubation (BI), and post-BI support projects - so that an enterprise along the course of a well-designed program may be nurtured properly and subsequently achieve a successful business status in the regional innovation ecosystem. In turn, even after graduation, the enterprises are expected to continue business in the region and contribute to the regional economic development.

The discussions on the resident hunting in this section will extend over the institutional structures and actual practice cases by DJTP for resident move in and out process and the related support policies in general format.

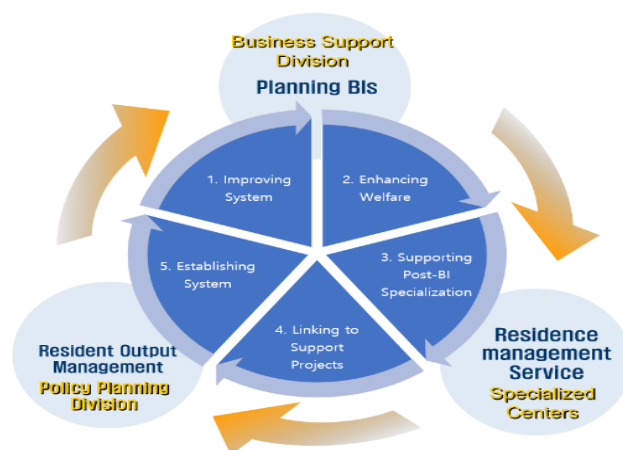
1.6.1 Internal System on Resident Support Projects

The resident enterprises at DJTP are systematically supported and managed under the TP policy and regulations. Four areas of strategies are set forth for promoting enterprises: networking, providing services for invigoration, enterprise monitoring, and strengthening enterprise capacities. Under the basis of these strategy frames, DJTP has established a comprehensive management system for resident enterprises in order to promote their growth.

³⁶ This chapter was written by Dr. Jong-Wha Lee of the Daejeon Technopark.

In general, DJTP's management of residents is structured through the systematic efforts of creating a virtuous cycle of planning and implementation by two Divisions and four industry specific Specialized Centers as shown in [Figure 3-16]. The Policy Planning Division and the Enterprise Support Division at DJTP have their own policies towards fulfilling their internal roles that involve planning the management of the resident enterprises of TP.

[Figure 3-16] DJTP's Internal System on Resident Support Projects



Based on their policies, the two Divisions cooperate to produce harmonized policies and strategies for establishing an enterprise management system and improving the institutional system as illustrated in [Figure 3-16]. The resulting outcomes, which follow the guidelines of the central and local governments, affect the way support projects are carried out in the Specialized Centers of DJTP. Accordingly, a series of progressive approaches are made towards establishing a system of managing resident enterprises, and, if necessary, involve cooperation with other BI institutions and professionals in the region for certain promotional projects.

Overall, the two Divisions, along with the other Centers, closely cooperate by carrying out their individual roles to ultimately organize the structural system of information management related to DJTP's support projects and the relevant information of enterprises that take on these projects. The collaboration also establishes a comprehensive support platform that can be utilized for collecting and analyzing data.

Hereupon DJTP tries to establish the growth management system (discussed below) on the premise of having a data-based Project Information Management System (PIMS) that allows for an enterprises-centered management on the project and their performance data.

By improving the institutional system for resident enterprises in this way, a plan is generated and utilized to support the current residents and further the enterprises even after their graduation. It should be noted that by successfully establishing a structure for the resident hunting system, DJTP establishes a connected cooperative system between the Centers that mostly carry out their separate individual support projects for their residents.

Each Center, in order to comprehend the needs of the residents and to support them considering their growth stages, tries to eliminate the walls of the silos between each support project being carried out separately in each Center, and designs performance goals by growth stages of enterprises; and finally they promote the connective follow-up milestone projects. For this purpose, DJTP strengthens the system for monitoring the performance goals; proceeding with subsequent diagnosing and consulting for enterprises to achieve their goals; and preparing the supporting system of, the so called, 'growth ladder model.'

The resulting policy operations affect the practices in two ways: first, improving the welfare of the resident enterprises in TP; and, second, supporting them effectively based on their customized needs. Thus, concrete promotional projects are designed to meet the support needs by category: ① Establishing a comprehensive online support platform, ② Providing a customized commercialization support system, ③ Diagnosing technology commercialization, ④ Promoting a star-enterprise promotion program, ⑤ Supporting certification acquisition by growth stages, and ⑥ Supporting financing and funding connections.

Considering the fundamental schemes for the resident support projects introduced here, it is necessary to explain the actual administrative process of the resident hunting in detail. In the following discussions, we will introduce the details of the administrative process on

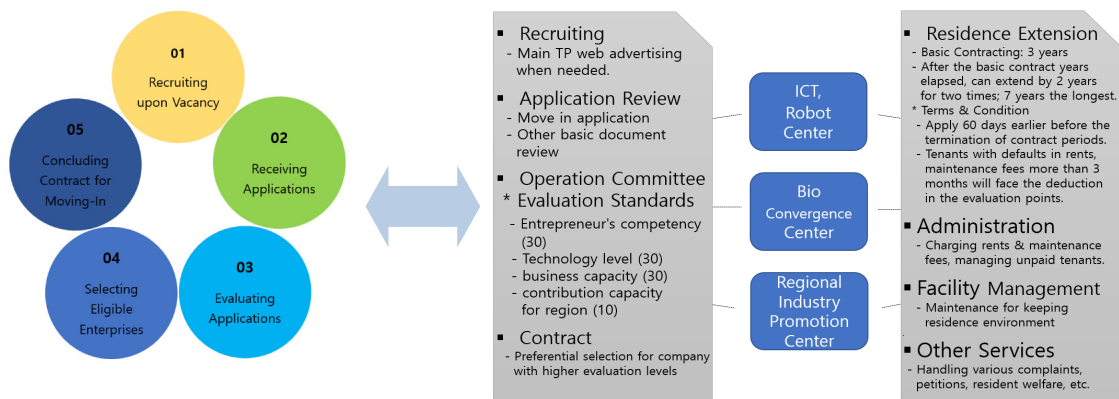
the selection of the resident and corresponding institutional systems and further the explanations on the growth management systems for resident enterprises.

1.6.2 Residents Selection and Management System

All the administrative processes of the resident selection are based on the DJTP's "Rules for the Management and Operation of the Industrial Technology Complex," which is also based on the "Act of Special Cases Concerning the Support for the Industrial Technology Complex" in South Korea.

The resident selection process at DJTP is carried out at all times upon any vacancy created due to the voluntary move-out or graduation of a current resident enterprise. Thus, as vacancies occur in residential spaces, the TP initiates the recruitment process for new resident enterprises under an open competition basis for selection. [Figure 3-17] below outlines the selection process and the detailed summary of the administration process of moving in and out.

[Figure 3-17] Residence Selection and Management System



First, the recruiting process begins with a notice on the DJTP's website, which serves as an advertisement. Potential residents can submit their application documents including a business project plan, by mail or by dropping off the application directly to the Technopark Center office.

Eligible enterprises include industry affiliated companies, institutions, and foreign investment companies. Currently, the related industries of Specialized Centers at DJTP are high frequency, wireless communications, defense ICT, IT/SW, robot, and bio-technologies.

Next, the evaluation process begins based on the initial application documents. The TP holds an Operation Committee meeting for selecting potential residents. Screening involves an application review and a presentation including the interview process. Evaluation standards prepared based on the DJTP Rules are illustrated in the four categorical rubrics (total points of 100): Entrepreneur's competency (30), Technology level (30), Business capacity (30), and Contribution potential for the regional economy (10). Each category has several sub-categories for evaluation.

The TP can execute interim evaluations of residents in multiple times per year and can also ask the relevant residents to submit related documents. The residents would be required to comply with any requests. The evaluation results would be categorized as 'excellent', 'satisfactory', 'average', or 'poor'. Residents who receive 'poor' evaluation results more than three times can be notified to vacate the residence.

The TP also carries out site inspections on the residents two times per year for the following areas: (1) Residential space utilization, (2) Implementation status of business projects, (3) Difficulties or bottlenecks by enterprises, (4) Employment status (including the youth employment requirement if necessary), and (5) Any other areas that require a checkup by the TP's Chief Operating Officer.

In addition, existing residents who complete their initial three years of residency have the option to apply for a lease extension. Prior to the lease termination, the TP office sends residents a notice asking for their intention regarding residency extension. Residents can stay for up to seven years in the TP facilities - three years for the first basic contract, two additional years upon successful contract renewal, and two more years with any special reason for their business promotion. Residents should apply for an extension at least 60 days before the conclusion of their contracts. The criteria for approving extensions is based on the same evaluation processes discussed previously.

The move-out enforcement for cases other than lease termination will be based on

resident state of affairs. Residents with a voluntary intention to vacate facilities within a contract period should submit a document describing their move-out plan. On the other hand, residents could face eviction due to various reasons: defaults in rent or maintenance & public utilities more than three months, contract violations, or incorrectly implemented business plans deviating from the initial proposals.

In general, benefits for the TP residents are categorized in [Table 3-13].

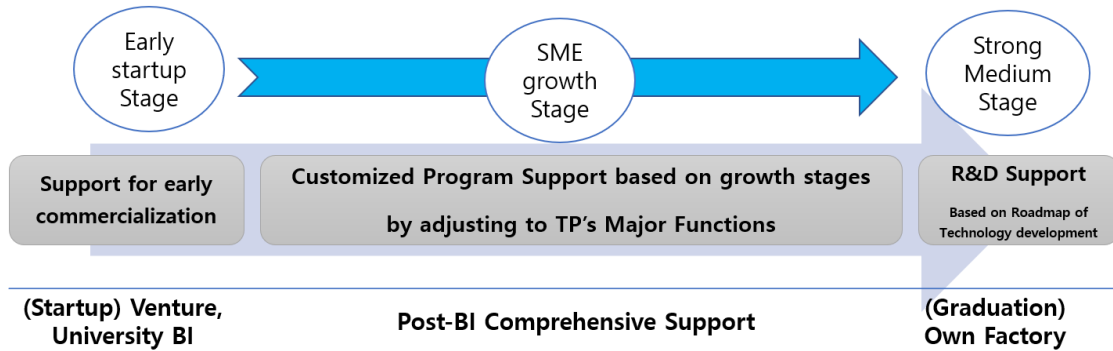
[Table 3-13] Benefits for Residents of Daejeon TP

Benefits	Support Details
Marketing	Participation in Exhibits, Licensing, Products Advertising
Counselling	Technical Assistance, Patent, Marketing
Technology Transfer	Commercialization Support utilizing Technologies by Universities, Research Institutes and Companies.
Networking Support	Industry-University-Institute Networking
TP Facility Utilization	Conference Hall, Seminar Room, Co-working Offices, Fitness Center, etc.
Rental Fee Support	50% deduction of rental fee (upon needs of applicable conditions)

1.6.3 Establishing Growth Management System for Residents

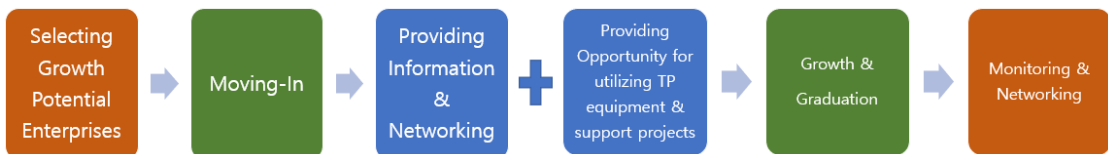
The purpose of having a well-improved internal process for comprehensively supporting residents is to apply various systems for intensive specialized support structures that depend on move-in stages and enterprise types. This effectively attracts promising enterprises into the system and helps them to grow and graduate successfully towards every successive stage of growth and eventually settle down in the region's industrial complex.

[Figure 3-18] Growth Management System by Growth Stages



In order to maintain a productive resident hunting structure, projects should continuously be linked from the early start-up BI stage to the post-graduation stage. For that to be successful, a well-designed growth management system must be established and applied. Then, it is expected that the system will attract enterprises outside the region as well as promising companies in the city who could finally settle their businesses in the local industry complex and contribute to the regional economy.

[Figure 3-19] Residents Growth Management System Process



DJTP, as discussed above, tries to select the promising small enterprises with growth potential, and this is reflected in the residence selection and management system by placing technology capacity and growth as the key indicators. These days, with upgraded systems of growth management, DJTP is making its endeavor with concentrated support for resident enterprises with growth potential capacity, and with further support through projects and networking even after their graduation, as illustrated in [Figures 3-18 & 3-19].

In this way, the BI promotion projects are provided through various opportunities and support formats. Current project operations in DJTP focus on newly designed follow up projects of the Post-BI, so DJTP hereby develops resident enterprises and lets them move into a new advanced residence system. Based on the needs of the regional industry

policies, the post-BI projects are designed, and promising enterprises inside and even outside the regions are attracted and hunted to become residents and nurtured to be small hidden champions. Post graduate enterprises, with the designed program of a strengthened BI, will have incentives to settle down in the region as regional star enterprises and put their businesses in a good position to contribute to regional economic development.

1.6.4 Concluding Remarks

The discussions introduced in this section are somewhat particular to Daejeon TP, but the general features are applicable to TPs of other regions in South Korea. However, most details presented in this section are roughly sketched to give readers the general insights regarding the larger picture on the “resident hunting” processes with the relevant administrative procedures.

DJTP continuously attempts to make its efforts to be fruitful for the success of the resident enterprises and for policy methodologies to evolve toward finding appropriate policies and strategies. Making these efforts effective can make the resident hunting projects successful in the end.

Finally, the discussions here on resident hunting are expected to be understood in connection with the BI policies and practices. Based on the visions for BI, the directions and methodologies for the resident hunting can be defined and designed effectively. Thus, in setting related regional industrial policies, the resident hunting scheme needs to be set based on the connective operation of the “startup - BI - Post BI” promotion. Strategic support in the continuous process will shape the prospective enterprises to be more promising, and therefore, will make them contribute to the regional economic development.

1.7 Digitalization of Technopark³⁷

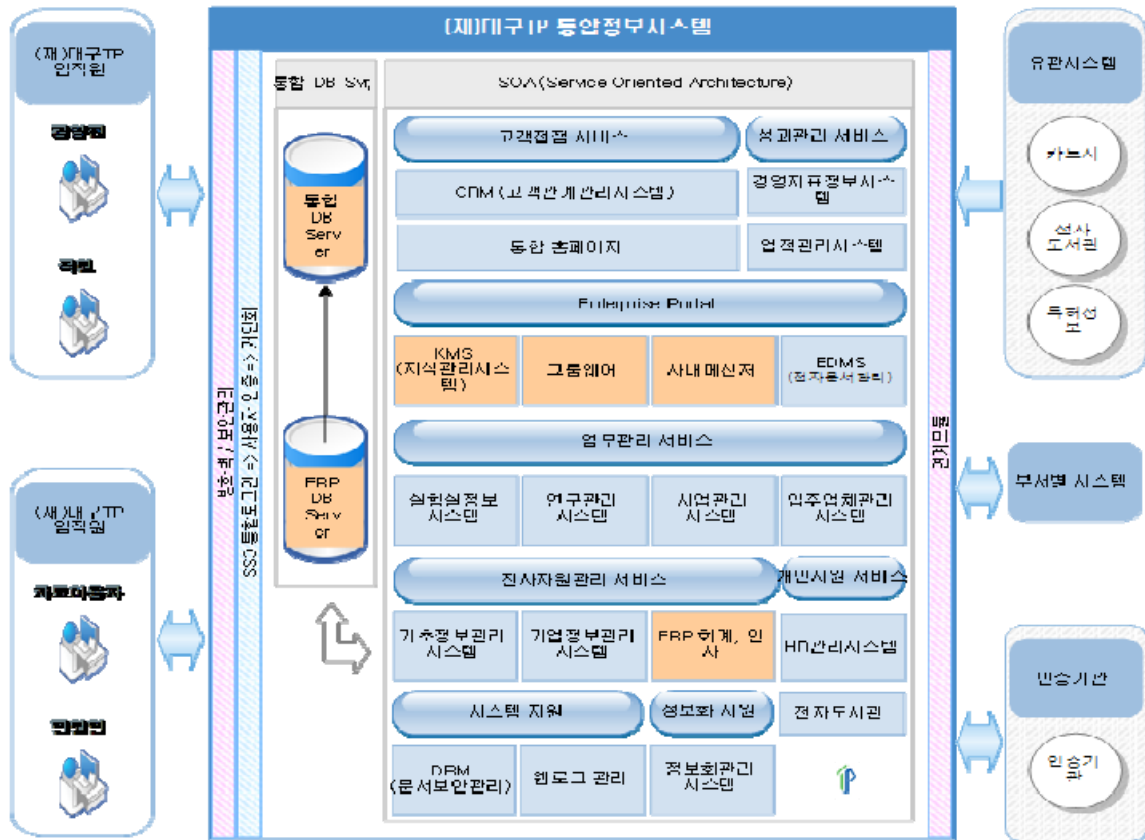
1.7.1 Intranet: KMS

The information vision of Daegu Technopark is becoming ‘a representative model agency for support in the information sector’ and providing ‘support for achieving vision through informatization’. To this end, all departments must be integrated and a system must be established that can provide comprehensive and customized services to companies. In addition, it is necessary to establish a system that can respond appropriately by making it possible to grasp the status of the foundation, department, and individual performance indicators in real time.

After the informatization is completed, (1) all information in the Daegu TP is integrated, (2) important tasks are computerized, (3) all information systems are linked and (4) the status of the foundation, department, and individual performance indicators are monitored and identified in real time, and (5) comprehensive customized services can be provided to companies. The existing information systems such as the Hanaro Support Center are operated in connection and linkage with the foundation's integrated information system.

³⁷ This chapter was written by Dr. Seong Byung-Ho of the Daegu Technopark.

[Figure 3-20] Daegu TP Integrated Information System



In order to achieve the informatization vision of Daegu TP, it is necessary to proceed with informatization in the following four promotional steps. The first stage of informatization (“knowledge sharing and communication activation”) and the second stage (“establishment of an integrated management system”) are the basis of the foundation's informatization, and in particular, it is necessary to organize an integrated management system by integrating data. The third stage is to establish the performance management base for work efficiency and the fourth is to establish the system to maximize the business performance.

1.7.2 Enterprise Resource Planning (ERP)

ERP-IU is to use the latest information technology to support enterprise-wide resource planning and management using a series of work flows from order receipt to shipment, production management, sales and purchase management, management and financial accounting, and personnel and salary management. It is an integrated management information system. Basically, ERP features are service-oriented business functions, its data-oriented processes can be implemented, business processes of various customers ranging from small and medium-sized enterprises to mid-sized companies are effectively implemented, corporate productivity is improved by introducing standard business processes, and unit work is modularized and integrated. It is a pursuit of the integration of management with the field.

ERP-IU provides its users with various useful functions such as convenient search, Excel conversion, column hiding, and column order change among the searched data by clicking the right mouse button in all individual menus (except for some specific menus). To run the program, registering the basic registration items is needed in the following order: registration of organization information, authorization and environment setting, account setting, registration of basic information in your account, and slip input.

[Table 3-14] System Operating Environment

Software Configuration	• Server: IIS (Internet Information Server), MTS(Microsoft Transaction Server), APP svr(Application server) • Client: web Browser
Hardware Configuration	• Server: one more computer for Server, minimum requirement –CPU Pentium dual core, RAM over 4G, HDD over 100GB • Client: PC for over Window XP
Data Base	• O/S (Operating System) • Server: Microsoft Windows 2003 Server, Advanced Server, NET Server • Client: More than Microsoft Windows XP • Develop tool: Visual Studio .NET (C#, ASP.NET) • Data Base: > SQL Server 2005 • Server environment: Microsoft .NET Framework, COM+, .NET Remoting, IIS • Client environment: Microsoft .NET Framework, COM+, .NET Remoting

1.7.3 R&D Supporting

The R&D Supporting Programs of Daegu TP consists of “Daegu Science & Technology Information Service” and “Daegu R&D Equipment Information Service.”

1.7.3.1 Daegu Science & Technology Information Service

The Daegu Science & Technology Information Service (www.dtis.re.kr) was developed by the Science and Technology Promotion Center. This is the nation's first regional science and technology information portal that provides information on investments, achievements, and support companies for science and technology (R&D & non-R&D) projects supported by Daegu City's budget. The purpose of this system is to improve the investment efficiency of the city government's budget through the provision of information service for the entire period of R&D projects.

The system is composed of databases of company information and project data including project outcomes, project support history, national R&D project, the status of project under Daegu city's budget support, and researcher workforce. Various forms are used for running the system, for example, dashboard, infographic (table, graph, map), real-time customized statistics, and etc. Using these forms, the system provides diverse statistics on the status and performance of national and Daegu city R&D projects, publicizes the outcomes of national and Daegu city R&D projects, provides support history and researcher information, and provides an overview of institutions supporting Daegu City R&D projects.

1.7.3.2 Daegu R&D Equipment Information

The Daegu Research Equipment Information System (www.dris.or.kr) was developed by the Daegu TP Nano Convergence Practical Application Center. This system is an integrated research equipment management system that provides information on the status, inquiry, reservation, performance, and corporate support of R&D equipment built in the region. The purpose of this system is to improve the convenience and accessibility of regional companies and research institutes by establishing a research equipment information

system and, thus, to promote the effective management and utilization of R&D equipment in the region.

The system contains data on regional institutions possessing R&D equipment as well as their R&D equipment. The system offers an infographic (table) data. The data can be used for analyzing the Daegu R&D equipment construction status and providing the Daegu TP Nano Convergence Practical Application Center's equipment reservation service and other information services such as project support and education.

1.7.4 Company Support Service

Daegu TP's information support services for companies consist of "Daegu Industrial Economic Trend Service" and "Daegu Star Company Platform Service."

1.7.4.1 Daegu Industrial Economic Trend

The Daegu Industrial Economic Trends Online System (distat.daegu.go.kr) was developed by the Daegu City and the regional innovation agency of Daegu TP. This system is called the Daegu Industrial Economic Statistics Portal for providing regional industrial economic statistics and trend information. In addition, the purpose of this system is to provide a systematic analysis of data-based economic flows in Daegu area, various statistics and trends related to the real economy and business and industry status, and customized information to consumers in real time.

This system is composed of 34 major indicators (10 major industries, business economy, employment, etc.), 3 major categories (economy, industry, income), and 229 indicator statistics database for providing such data as Daegu industrial trend, Daegu economic trend, and the results of panel database investigation of Daegu enterprises. The implementation form of this system is a dashboard and infographic (table, graph) and real-time customized statistics inquiry service is also available.

Data are provided in such formats as key indicators, statistics database dashboard, infographic, and real-time statistical reports. In addition, this information system provides

industrial economic news, industrial economic calendar, industrial economic issue report, and survey result reports using the Daegu Enterprise Panel database.

1.7.4.2 Daegu Star Company Platform

The Daegu Star Enterprise Platform (star.daegu.kr) is being developed by the business promotion agency of Daegu TP. This system manages the basic information and support history of target companies, and also provides information on corporate trends and support policies. The reason for constructing this platform is to manage the Daegu City Growth Ladder Corporate Promotion System well. As mentioned earlier, the purpose of this platform is to establish an open platform for multilateral linkage and cooperation to systematically foster and manage the target group of companies for joint fostering in Daegu City, and to establish an organic linkage and cooperation system between participating organizations.

The data in the platform covers the basic status of the target company (sales, employment, etc.), the target company support history and performance, the target company's difficulties and support needs, and the status of the participating and cooperating organizations (PM). The data are available in infographics (tables, graphs). Real-time information on target companies and statistics on the nurturing and participating organizations are also available.

Data can be used for analyzing the basic status of target companies and their support history, direct support projects to the secretariat and hotline consultations, support policies (projects) for participating organizations, and demand surveys and statistical data such as target companies and organizations participating in the promotional projects.

2. Technology Transfer & Commercialization in South Korea³⁸

2.1 General Overview of TT & TC

Technology commercialization is defined as the process of converting knowledge created from research into new or improved products, processes and services, and selling them in the market to create economic benefits from them. Technology commercialization is an innovation process that adds value and can be classified into technology commercialization from the public sector or from the private sector according to the technology source.

Based on the “Act on Promotion of Technology Transfer and Commercialization,” technology transfer is defined as the transfer of technology from the technology holder (including those who have the authority to dispose of the technology) to others by means of transfer, licensing, technology guidance, joint research, joint venture, or merger and acquisition. Technology commercialization is defined as the use of technology to develop, produce, or sell products, or to improve technology in the process.

In the era of the Fourth Industrial Revolution, as technology competition between countries and companies intensifies, new technologies emerge, and the speed of technology development is also rapidly advancing exponentially. As efforts to cope with the rapidly changing technological environment are required, the amount of investment in R&D is constantly increasing at home and abroad, and patent applications are also on the rise at the same time. In contrast to the increase in R&D investment and patents outcomes, the life cycle of products and markets is being gradually shortened. To proactively respond to the changing market, it is essential to acquire and use technology in a more efficient way, and global companies are actively using ‘Buy R&D’ strategy to preempt the initiative

³⁸ This chapter was written by Dr. Sangho Choi of the Daejeon Technopark

in the new convergence market.

The types of TT are as follows³⁹.

[Table 3-15] Types of Technology Transfer

Types	Contents
Tech Assignment	It is important for technology sellers to set royalties in consideration of technology development costs, and technology adopters must consider risks and profits related to technology commercialization and economic feasibility.
Joint Research	In general, joint research is the development of new products or new technologies by providing technologies that a company possesses, complementing mutual technologies with other companies or research institutes.
Licensing	Licensing is relatively less risky for cost and success in terms of investment costs and small resources than in the form of a joint venture.
Spin-off	In most countries, various support systems are being prepared by placing importance on spin-off technology transfer.
Joint Venture	In the case of a joint venture, it is completely separated from a company or research institute and operated as an independent corporation
M&A	In the case of only the purpose of acquiring technology, the effectiveness may be inferior except for M&A of small venture companies.

The types of TC can be largely divided into two categories for the public sector and the private sector, respectively.⁴⁰

[Table 3-16] Types of Technology Commercialization

Sectors	Contents
Public Sector	(Commercialization of public technology transfer) A type to increase the efficiency and economic effect of government R&D projects by transferring and commercializing the technology developed through the investment of government R&D funds to private companies.
	(Public technology developer startup) A method that allows professors,

³⁹ Ministry of Science and ICT(2018), A study on ways to revitalize technology commercialization through cooperation between the public and the private sector

⁴⁰ Ministry of Science and ICT(2018), A study on ways to revitalize technology commercialization through cooperation between the public and the private sector

Sectors	Contents
	researchers, etc. who participated in technology development, to start up and commercialize, led by universities and public research institutes
Private Sector	• (Commercialization of own technology) A plan to directly commercialize and sell technologies developed by private companies or jointly developed • (Commercialization of transfer technology) A plan to commercialize the technology by connecting the person who wishes to sell the technology and the person who wishes to purchase the technology.

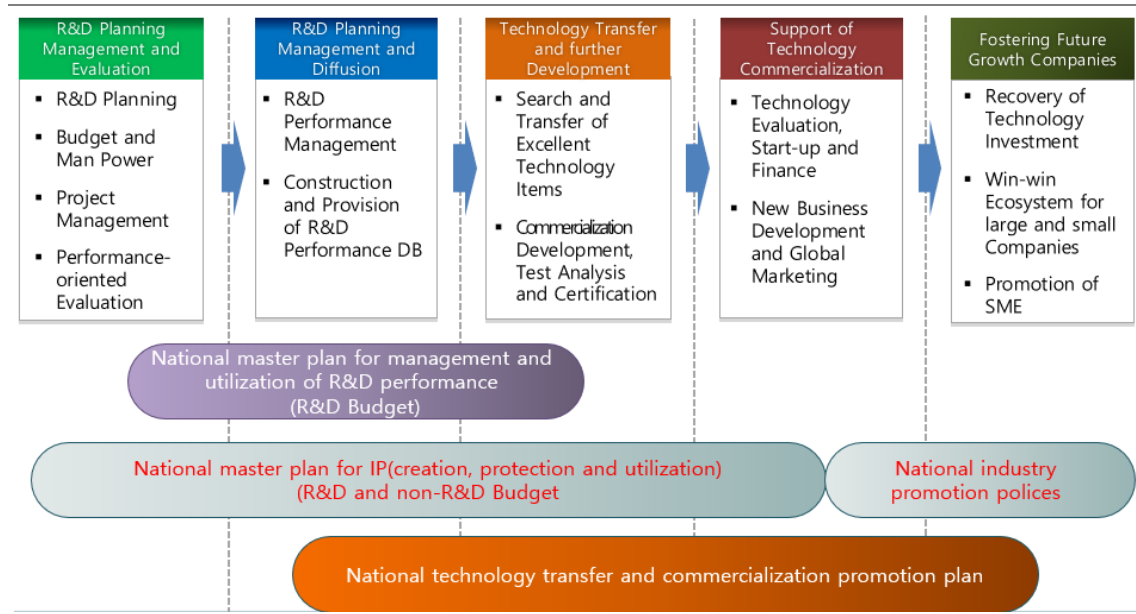
2.2 South Korea's TT & TC Policy Status

The support for TT & TC in South Korea is promoted by focusing on the “Technology Transfer and Commercialization Promotion Plan,” the “Research Performance Management and Utilization Basic Plan,” and the “National Intellectual Property Basic Plan” according to the stages of R&D, commercialization and enterprise development⁴¹.

As a start-up and technology financing support policy, South Korea is promoting the “2nd Venture Boom Expansion Strategy,” the “Innovative Start-up Boom Creation Plan,” and the “Innovative Start-up Ecosystem Creation Plan,” and is providing the Technology Bank Service that supports matching between investors and companies, government industrial fund for technology and services (GIFTs), and technology finance education, etc.

⁴¹ KISTEP(2019), Technology Commercialization Policy Trend, KISTEP Technology Trend Brief, 2019-8

[Figure 3-21] National TT & TC Policy Framework



The types of South Korea's TC support policy are as follows.⁴²

⁴² KIAT(2019), Public technology transfer and commercialization status and tasks

[Table 3-17] Types of South Korea's Technology Commercialization Support Policy

Supporting Policies	Contents
R&BD plan	• Support product development and the development of technology commercialization model
Technology consulting	• Provide professional support for technology valuation
Tech information DB support	• Comprehensively build and provide information on technologies
Technical certification and standardization support	• Certify the performance and system of products, processes, and services, and support the standardization of technologies
Technology transfer support	• Transmit and support know-how for the application of technologies
Tech trade support	• Support intermediary service and trading through technology trading institutions
Support for strengthening the competency of a dedicated organization for TC	• Provide support for fostering commercialization workforce • Support for a dedicated organization for commercialization (Establishment and operation of technology holding companies, etc.)
On-site support for technical personnel	• Match and dispatch R&D experts to the field for support
Marketing and sales consulting support	• (Marketing promotion) Support strategy development, etc. to activate trading • (Pioneering domestic market) Support the market entry to increase domestic market acceptance • (Entering the global market) Support the development of export channels
Legal and institutional improvement support	• Provide support for improving infrastructure to promote commercialization and eliminate factors that hinder market entry
Financial support and fund creation	• Implement the diversification of investments and loans and creation of funds

The current status of laws related to TT & TC are as follows.

[Table 3-18] Current Status of TT & TC Laws

Laws	Master Plan	Contents related to TT & TC
Technology Transfer and Commercialization Promotion Act	Technology transfer and commercialization promotion plan	Regulations on matters related to TT and creation of the foundation for TC
Basic Science and Technology Law	National Science and Technology Master Plan	Provision of matters related to TT and promotion of TC and improvement of success rate
National R&D Project Performance Evaluation and Performance Management Act	Master plan for management and utilization of research outcomes	Regulations on the matters of the utilization plan, comprehensive management of performance DB, outcomes management and legal system improvement, etc., regarding the use and spread of research outcomes.
Intellectual Property Basic Law	National Intellectual Property Master Plan	Stipulating basic policies for the creation, protection, and utilization of intellectual properties
Industrial Technology Innovation Promotion Act	Industrial technology innovation plan	Regulations on the strategies for innovating TC support system (Spreading and protecting industrial technology innovation achievements, promoting TT & TC, etc.)
Act on Fostering Special R&D Zones	-	Regulation of commercialization of R&D results and support for startups

The “Technology Transfer and Commercialization Promotion Plan” is a mid-term promotional plan established every three years in accordance with Article 5 of the “Technology Transfer and Commercialization Promotion Act.”

The goal is to contribute to linking the development of mid-sized companies and creating a win-win ecosystem between companies by intensively supporting research planning, management, diffusion, TT, and additional development, and TC in the stages of R&D,

commercialization and corporate development. The details of the “6th Technology Transfer and Commercialization Promotion Plan (2017-2019)” are as follows.⁴³

[Table 3-19] The 6th TT & TC Promotion Plan

Promotion Strategies	Main Contents of TT & TC
Creation of a demand infra for 'Buy R&D'	• Introduce open innovation B&D (Buy and Develop) system • Expand technology demand search (Support for coaching technology buyers for SMEs, and technology export revitalization project to expand global demand) • Improve technology transactions promotion system (Introduce R&D tax benefit system and reinforce restrictions on abnormal theft of technology)
Supply of technology desired by demanding companies (Buyable R&D)	• Expand and introduce the legal entity support model for TC projects • Strengthen the marketability of public R&D (the introduction of technology management from universities and research institutes and fast track to commercialize knowledge assets) • Promote high value-added technology startups (Reinforce public technology startups and discovery and fostering of founders)
Bridging the gap between consumers and suppliers	• Construct on/off-line convergence technology market platform (Information network for the TT and M&A) • Create private technology trading market • Help companies overcome death-valley through technology finance • Buy R&D awareness through education and promotion
Establishment of a collaboration system for technology commercialization across ministries	• Build cross-ministerial cooperative governance • Prepare a regulation-friendly system related to TC

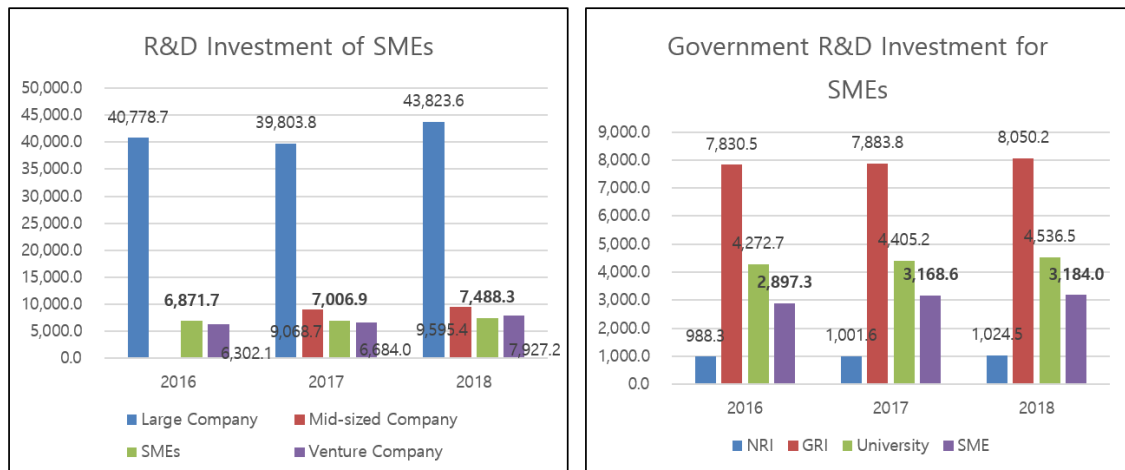
⁴³ Ministries(2017), The 6th technology transfer and commercialization promotion plan

2.3 Major Activities of TT & TC in South Korea

2.3.1 TT & TC Support Activities for SMEs

The government has doubled R&D for SMEs to promote technological innovation and commercialization of SMEs and continued to provide R&D investments support. The government's R&D investment in SMEs increased from KRW 2.4 trillion in 2014 to KRW 3.18 trillion in 2018⁴⁴.

[Figure 3-22] R&D Investment for SMEs



Source : KISTEP (2019b)

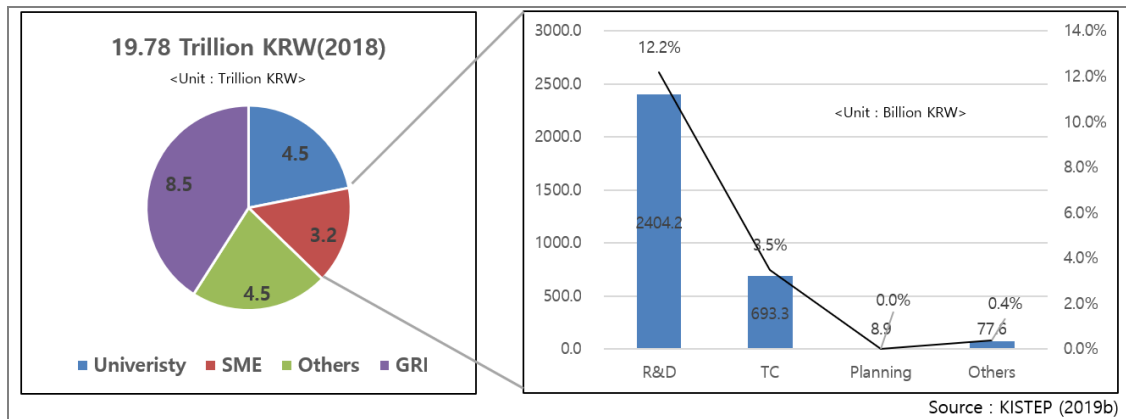
The total government R&D investment stands at KRW 19.8 trillion as of 2018. In the meantime, the total amount of government R&D investment to support SMEs is KRW 3.18 trillion, with R&D (KRW 2,44.2 trillion), TC (KRW 693 billion), planning (KRW 8.9 billion), workforce training (KRW 1 billion), and others (KRW 76.5 billion).^{45,46}

⁴⁴ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

⁴⁵ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

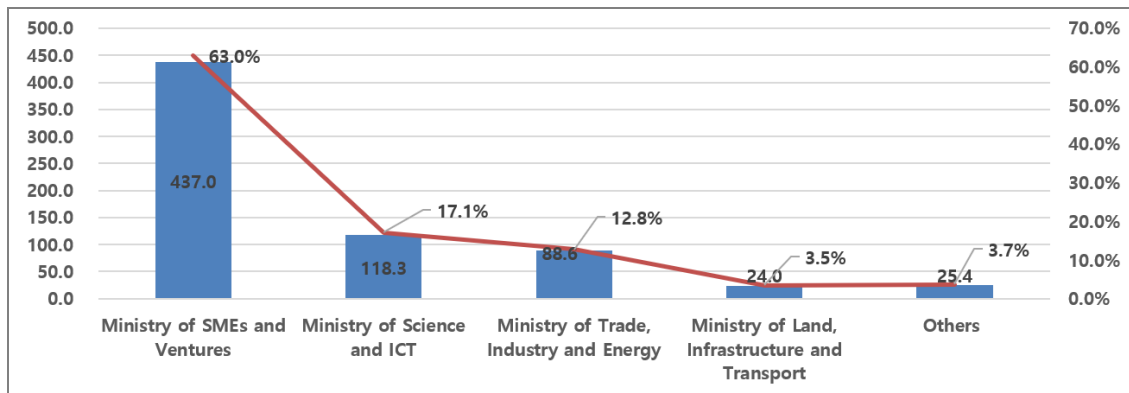
⁴⁶ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

[Figure 3-23] Share of TC for SMEs



The amount of investment in TC to support SMEs was KRW 693.3 billion, of which the Ministry of SMEs and Ventures, Ministry of Science and ICT, and Ministry of Trade, Industry and Energy accounted for 92.9% (643.9 billion KRW).⁴⁷

[Figure 3-24] Amount of Investment in TC



Of the detailed types of TC, technology transfer and additional development (KRW 386.9 billion) accounted for the largest share, followed by marketing sales consulting (KRW 186.9 billion), investment linkage (KRW 81.9 billion), research planning and diffusion (KRW 48.8 billion), and technology certification and standardization support (KRW 6.2 billion).

TC support for SMEs in 2018 promoted 18 projects in a total of 7 ministries, and the total budget was KRW 597 billion. The project distribution by department is as follows.

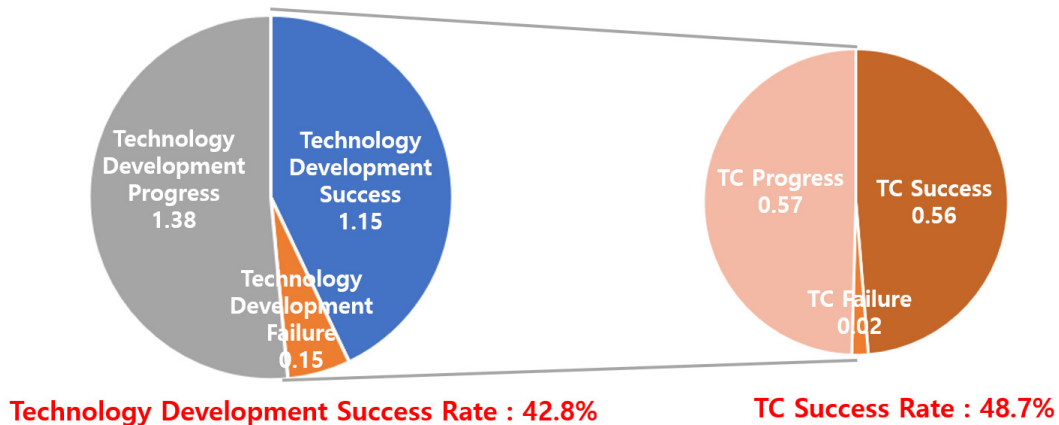
⁴⁷ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

[Table 3-20] Projects for TC Support

Ministries	Projects	Budget Amount	Budget for SME
Ministry of Small and Medium Venture Business	SME TC capability enhancement	4.6	3.6
	SME R&D capability enhancement	10.8	4.5
	SME commercialization technology development	170.2	162.9
	Startup growth technology development	280.0	266.0
Ministry of Science and ICT	TC support for public research achievements	26.5	3.4
	Support for vitalization of industry-academia-research cooperation	19.9	8.1
	Support for corporate growth in commercialization of public technology in connection with investment	3.8	3.6
	Support for R&D special zones	76.3	46.2
Ministry of Trade, Industry and Energy	Promotion of utilization of technological results	23.4	12.0
	Development of technology linked to commercialization	42.0	40.1
Ministry of Land and Transport	Support for commercialization of national land transport technology	24.8	23.5
	Support for utilization of national land transport research results	1.3	0.5
Ministry of Culture, Sports and Tourism	Sports service commercialization support	2.1	1.0
	Convergence tourism service commercialization support	2.0	0.9
Ministry of Agriculture and Food	TC Support	8.9	5.1
	Agricultural food research results follow-up support	3.0	2.1
	Agricultural commercialization technology R&D support	11.9	10.7
Ministry of Environment	Future promising green environment TC support	3.0	2.8
Total		714.5	597

SMEs promoted an average of 2.68 technology developments per company in 2018, and completed 1.30 cases, of which 0.56 were successfully commercialized. The technology development success rate averaged 42.8%, and the commercialization success rate averaged 48.7% compared to the number of technology development successes.⁴⁸

[Figure 3-25] TC Success Rate



Source: KISTEP Issue Paper, 2020-07

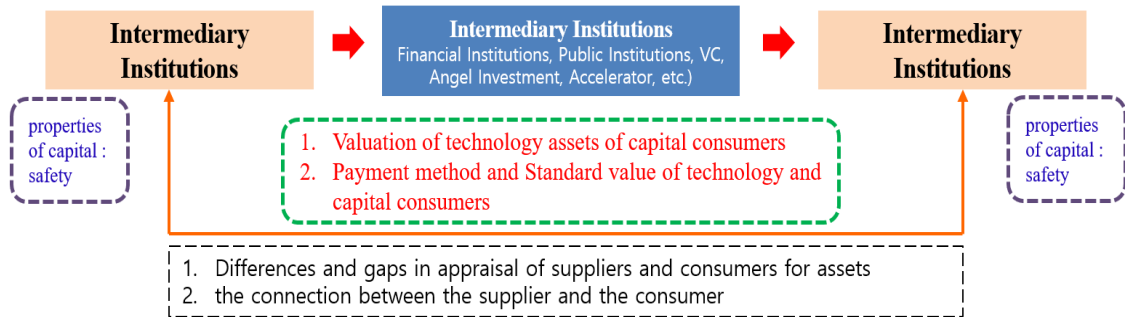
2.3.2 South Korea's Technology Finance Infrastructure

Technology finance is a transaction of funds centered on technology assets, and is an element that is injected to complement technology and marketability in TC. Technology finance can be said to be a financing of capital based on qualitative and quantitative evaluation of intangible assets of technology. Capital supplied through technology evaluation is representative of government subsidies, guaranteed loans from the Technology Guarantee Fund, indirect investment through fund of funds, and direct investment from venture capitals. Technology finance has a structure in the form of a platform, that is, an intermediary organization that mediates not only the person who supplies the capital and the demander of the capital.⁴⁹

⁴⁸ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

⁴⁹ STEPI(2019), Technology financing proposal for efficient technology commercialization, STEPI Insight

[Figure 3-26] Technology Finance Structure



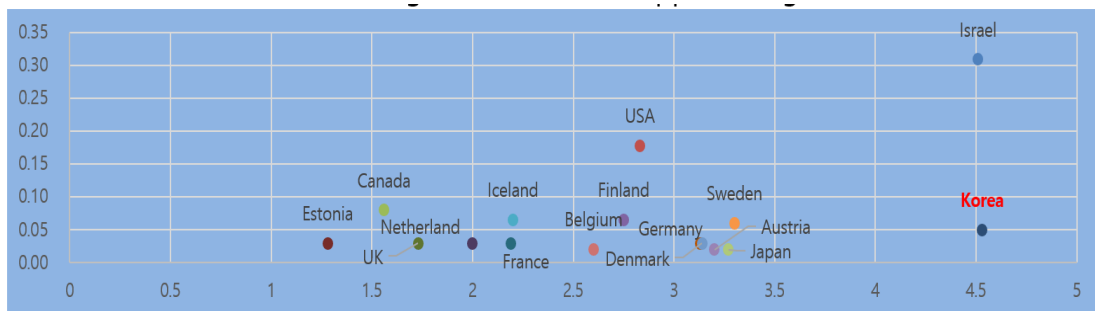
Domestic technology finance has grown from KRW 100 trillion in 2017 to KRW 200 trillion in 2020, leading the domestic technology commercialization.

[Figure 3-27] South Korea's Technology Finance Scale

Technology Finance					
Capital for Tech Advancement	Capital for fostering ventures <unit : Trillion KRW>				
R&D investment	Loan (Tech credit loan)	Investment			Guarantee fund
		Angel Investment	Start-up Investment	New technology financial company	
1	164	0.2	9.5	5.0	22.4

South Korea's R&D investment is among the world's highest in terms of GDP, while the size of VC is within the 10th largest in the world, approaching 0.05% of GDP.

[Figure 3-28] Size of Venture Capital in Terms of GDP

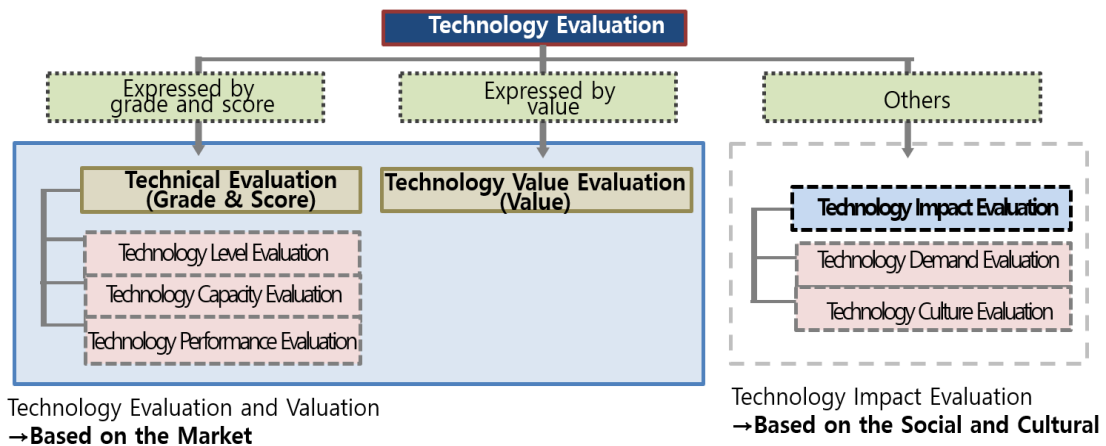


Source : STEPI Insight 2019

2.3.3 South Korea's Technology Evaluation Systems

Technology evaluation is defined as ‘expressing the economic value of technology that can occur through commercialization in terms of value, grade, or score’ based on Article 2 of the “Technology Transfer and Technology Commercialization Promotion Act.” Although it may include technology impact assessment, technology evaluation means only ‘technical (level, capacity, performance) evaluation’ and ‘technology value evaluation’ that can convert technology into an asset based on a market perspective.⁵⁰

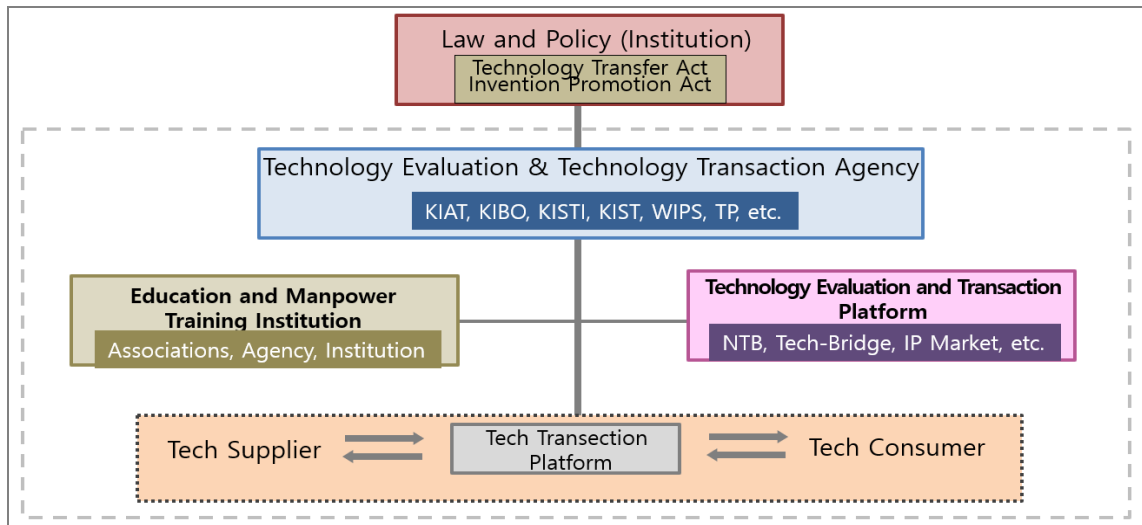
[Figure 3-29] Technology Evaluation



The technology evaluation infrastructure includes not only laws and policies required for technology evaluation activities, technology evaluation institutions/technology trading institutions, but also technology evaluation/transaction systems, education and personnel training institutions, etc.

⁵⁰ National Assembly Budget Office(2019), Analysis of national research and development projects

[Figure 3-30] Technology Evaluation Infrastructure



A number of laws such as the Technology Transfer Act, the Venture Business Act, the Technology Guarantee Fund Act, and the Invention Promotion Act specify the basis and organization of technology evaluation.

[Table 3-21] Status of Technology Evaluation Laws

Status of laws	Technology Evaluation Designated Organization	Main contents of the law
TT and TC Promotion Act	Ministry of Trade, Industry and Energy designated organization	Technology evaluation for technology transfer and commercialization
Industrial Education Promotion and Industry-Academic Cooperation Promotion Act		
Act on Special Measures for Fostering Venture Businesses		
Foreign Investment Promotion Act	KIAT, KIBO, KEIT, South Korea Environment Corporation, KATS, KIST, KECO, KISTI, NIPA	Valuation of industrial property rights, etc.
Technology Guarantee Fund Act	KIBO	Technology guarantee and technology evaluation for the purpose of financing new technology projects
Invention Promotion Act	NRI, GRI, Private Research Institutes	Evaluation of technology and business feasibility for invention
Geospatial Information Industry Promotion Act	KIST, KIAT, KEIT, KIBO	Technology evaluation when investing in kind related to

		geospatial information business
--	--	---------------------------------

The status of technology evaluation institutions: Article 35 of the Technology Transfer Act allows the Minister of Trade, Industry and Energy (MOTIE) to designate a technology evaluation agency. As of October 2019, there are a total of 26 institutions (18 public and 8 private).

The national technology evaluation: Based on the performance of 25 institutions, the technology evaluation results decreased from 13,335 cases in 2016 to 7,594 cases in 2018.

[Table 3-22] Status of Technology Evaluation and Transection Institutions

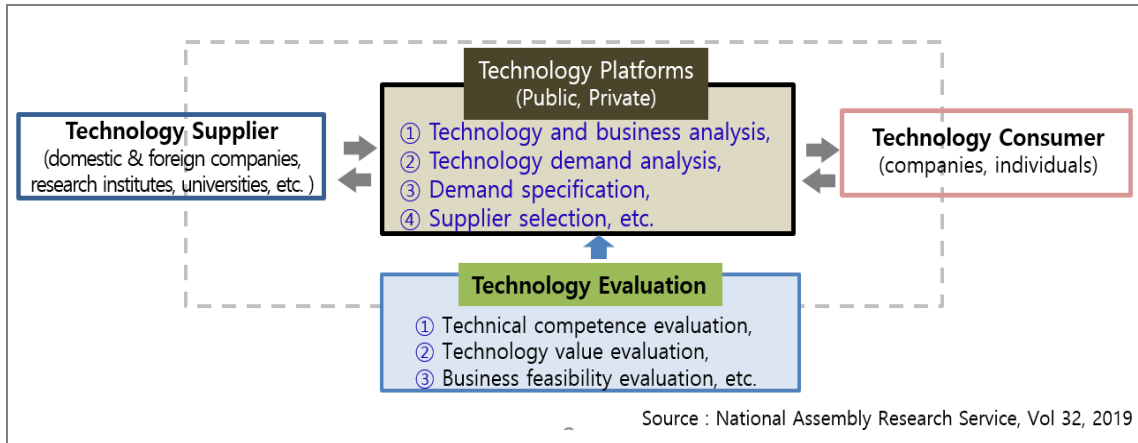
Category		2016	2017	2018
Public Sector	17 Technology Evaluation Institutions	13,335	13,614	6,830
Private Sector	8 Technology Evaluation Institutions	0	368	764
Total		13,335	13,982	7,594

The status of technology trading institutions: Based on Article 10 of the Technology Transfer Act, technology trading institutions can be designated by MOTIE, and as of October 2019, there are a total of 133 institutions (36 public, 97 private).

The Importance of the technology trading platform: The technology trading platform, a place where matching and trading activities between technology providers and technology consumers are conducted through technology as a medium, plays a role as an engine in the operation of the technology market.

Technology transaction is a transfer from the technology holder (or supplier) to others through the transfer of technology or license grant while accompanying financial transaction from the market point of view. When a transaction is made between a technology supplier and a technology consumer through a technology trading platform, the results of technology evaluation are used according to the purpose of the transaction.

[Figure 3-31] Technology Trading Platform



Technology evaluation and technology trading platform status: There are a number of platforms opened for the purpose of smooth technology transfer and transaction relaying, and in the past, the permanent South Korea Technology Exchange is currently operating as a web-based platform.

[Figure 3-32] Status of Technology Evaluation Platforms

Name of the platform	Relevant government departments	Web site
NTB Tech Bank	Ministry of Trade, Industry and Energy	https://www.ntb.kr
KDB Tech Trade Mart		https://ipmart.kdb.co.kr
Tech-Bridge	Ministry of Small and Medium Venture Business	https://tb.kibo.or.kr/
National IP Trade Platform	Korean Intellectual Property Office	https://www.ipmarket.or.kr
Future Tech Place	Ministry of Science and ICT	http://rnd.compa.re.kr
Star Value		http://www.starvalue.or.kr
Military Tech Trade Market	Defense Acquisition Program Administration	http://techmart.dtaq.re.kr/

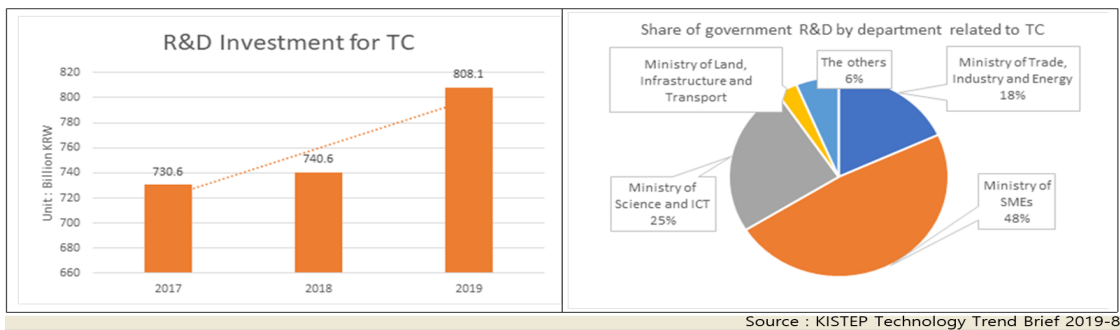


Source : National Assembly Research Service, Vol 32, 2019

2.4 Major Statistics in TT & TC

The amount of government R&D investment related to TC is expanding with an annual average of 5.2% increase (2017-2019) to KRW 730.6 billion in 2017, KRW 747.6 billion in 2018, and KRW 8,081 billion in 2019. The Ministry of SMEs (48.0%), the Ministry of Science and ICT (24.5%), and the Ministry of Trade, Industry and Energy (18.1%) accounted for 90.6% of government R&D investment related to TC.⁵¹⁵²

[Figure 3-33] Status of National R&D and Patent Performance



Government R&D Budget: In 2017, the government R&D budget executed by universities and public research institutes was about 13.3 trillion won, 68.5% of the total government R&D (about KRW 19.4 trillion).⁵³

The government-funded research institutes (GRI): KRW 7.9 trillion (40.7%), universities: KRW 4.4 trillion (22.7%), national research institutes (NRI): KRW 1.2 (5.2%).

Patent Performance: As results of the government R&D investment, the number of patent applications by universities and public research institutions in 2017 was 19,907 (61.3% of the total), and the number of patent registrations was 13,468 (68.6% of the total),⁵⁴ 15.9% (application) and 16.3% (registration) of the South Korean patents (32,501 applications, 19,641 registrations).

⁵¹ KISTEP(2019), Technology Commercialization Policy Trend, KISTEP Technology Trend Brief, 2019-8

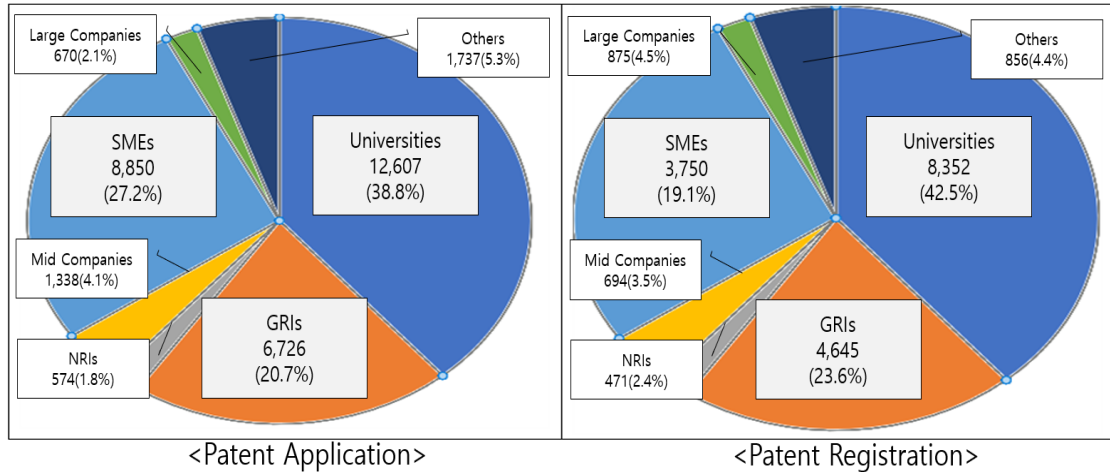
⁵² KIIP(2019), Government R&D Patent Management Status and Implications

⁵³ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

⁵⁴ KIIP(2019), Government R&D Patent Management Status and Implications

- University's patent performance: 38.8% of applications, 42.5% of registrations, GRI (20.7%, 23.6%) and NRI (1.8%, 2.4%).

[Figure 3-34] Patent Status in South Korea



Source : Korea Institute of Intellectual Property, 2019

In the public sector in 2018, there were a total of 269 institutions with technology, and 209 institutions with TT. The current status of institutions possessing and transferring technologies in the public sector is follows.⁵⁵

[Table 3-23] Current Status of Institutions Possessing and Transferring Technology

Institutes	Number of Institutes with Technology	Number of Institutes with Technology Transfer
Public Research Institutes	129	94
Universities	140	115
Total	269	209

The number of newly signed technology transfer contracts in 2018 totaled 8,105, an increase of 8.4% from the previous year (7,477). By contract type, paid technology implementation (license) was the largest with 5,158 (63.6%), and technology assignment (trade) was 1,796 (22.2%), etc.⁵⁶

⁵⁵ National Assembly Budget Office(2019), Analysis of national research and development projects

⁵⁶ Ministry of Trade, Industry and Energy(2018), Public technology transfer commercialization status survey, Statistical Information Report

[Table 3-24] Number of TT Contracts

Years	2014	2015	2016	2017	2018
Technology assignment (trade)	718	1,107	1,587	1,466	1,796
Free transfer (technology sharing)	419	550	511	282	159
Paid technology implementation (license)	4,098	5,113	5,385	4,905	5,158
Free technology implementation (license)	639	322	327	278	340
Option contracts	0	0	1	0	0
Others	107	207	226	546	652
Total	5,981	7,299	8,037	7,477	8,105

Source : 2019 Technology Transfer Commercialization Report

In 2018, the number of TT by public research institutes was 11,002, a decrease of about 12.0% compared to the previous year (12,503). The TT rate in 2018 was 34.3%, down 3.6%p from the previous year (37.9%).

[Table 3-25] Number of Public Sector TT and TT Rate

Institutes	No. of New acquisition technology	No. of TT	TT Rate
Public Research Institutes	13,459	5,331	39.6
Universities	18,587	5,671	30.5
Total	32,046	11,002	34.3

* Tech Transfer Rate = No. of TT / Number of new acquisition (development) technologies

Among the TT contracts, TT from public research institutes to large companies was 2.5%, mid-sized companies 2.0%, and SMEs 89.1%, indicating that SMEs are major technology consumers of universities and public research institutes.

[Table 3-26] Number of Technologies Transferred

Company type	Number of TT contracts signed in 2018	Number of technologies transferred in 2018
large companies	206 (2.5%)	299 (2.7%)
mid-sized companies	162 (2.0%)	232 (2.1%)
SMEs	7,224 (89.1%)	9,474 (86.1%)
overseas companies	66 (0.8%)	534 (4.9%)
others	447 (5.5%)	463 (4.2%)

Source : 2019 Technology Transfer Commercialization Report

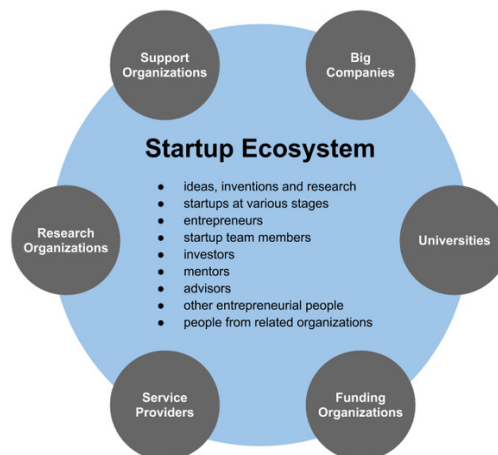
3. Start-up & Incubation in South Korea⁵⁷

3.1 Incubation for the Start-up in South Korea

Business incubation is defined as an organization designed to accelerate the growth and success of entrepreneurial companies through an array of business support resources and services that could include physical space, capital, coaching, common services, and networking connections.⁵⁸ Since 1998, after the IMF bailout crisis, South Korea has continuously tried to make an incubation for the new startup and job creation throughout the 23 years of incubation history.

A startup ecosystem is formed by people, startups in their various stages and various types of organizations in a location (physical or virtual), interacting as a system to create and scale up new startup companies. These organizations can be divided into different categories such as universities, funding organizations, supporting organizations (e.g. incubators, accelerators, co-working spaces, etc.), research organizations, service providers (e.g. legal and financial services) and large corporations.

[Figure 3-35] Startup Ecosystem



Source: https://en.wikipedia.org/wiki/Startup_ecosystem

⁵⁷ This chapter was written by Dr. Jongin Choi of Hanbat University

⁵⁸ <https://www.entrepreneur.com/encyclopedia/business-incubator>

3.1.1 Evolution of Business Incubation

Among the incubators, university has a large portion of incubation, about 74%, while research institutes like ETRI and KRIBB and local government account for 5.4 % and 4.7%, respectively.

[Table 3-27] Evolution of Business Incubation Center (BIC) in South Korea

Opening Status of University BIC (1998-2017)

Year	98	99	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17
Total	6	77	82	45	23	10	0	4	0	1	4	1	4	1	6	2	6	1	1	3

Source : Korea Business Incubation Association, Korean Business Incubation White Paper (2018).

Status of BICs Operation Status by Year (1998-2018)

Year	98	99	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18
Total	18	105	201	253	277	290	290	278	264	269	278	270	280	269	284	271	280	271	265	258	262

Source : Korea Business Incubation Association, Korean Business Incubation White Paper, Bi-net (2019).

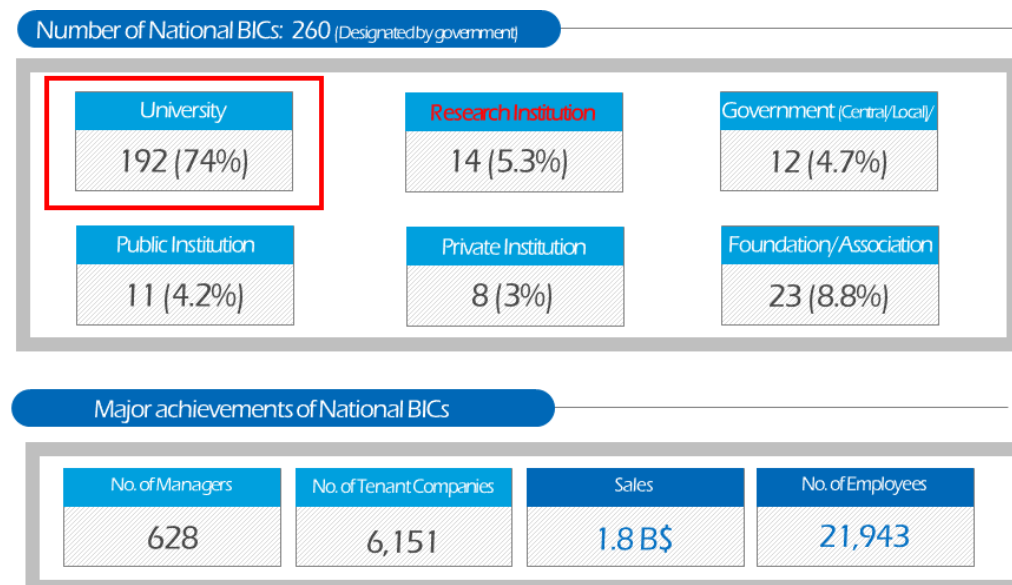
Status of BICs by Region (2019)

Region	Center	Ratio	Region	Center	Ratio	Region	Center	Ratio	Region	Center	Ratio
Seoul	32	12.3	Chungnam	14	5.3	Busan/Ulsan	17/2	7.3	Gwangju	12	4.6
Gyeonggi	49	18.8	Daejeon	13	5	Daegu	12	4.6	Jeonbuk	15	5.8
Incheon	5	1.9	Chungbuk	15	5.8	Gyungbuk	22	8.5	Jeonnam	13	5
Gangwon	15	5.7	Sejong	3	1.1	Gyungnam	17	6.5	Jeju	4	1.5

Source : Bi-net (2019).

Major achievement of the incubation centers are 6,151 resident companies, whose value is estimated at around USD 1.8 billion with 21,943 job creation and economic development for the future.

[Figure 3-36] Location and Performance of Business Incubation Centers



3.1.2 South Korean Institute of Startup and Entrepreneurship Development (KISED)

Where can get information on startups in South Korea ? There are some useful sites such as Bi-net and KISED (South Korea Institute of Startup and Entrepreneurship Development). The BI-network System (Bi-net) provides general information on incubation including BI centers, BI infrastructure support, manager capacity building, certificate of manager, and so on.⁵⁹ KISED, a public institute supporting startup activities, was launched in 2008 based on Article 39 of the Act for Supporting the Establishment of Small and Medium Businesses.⁶⁰ KISED aims to contribute to the development of the national economy by promoting the growth of startup businesses, job opportunities, and technology based on startups of future entrepreneurs through the cultivation of entrepreneurial spirit.

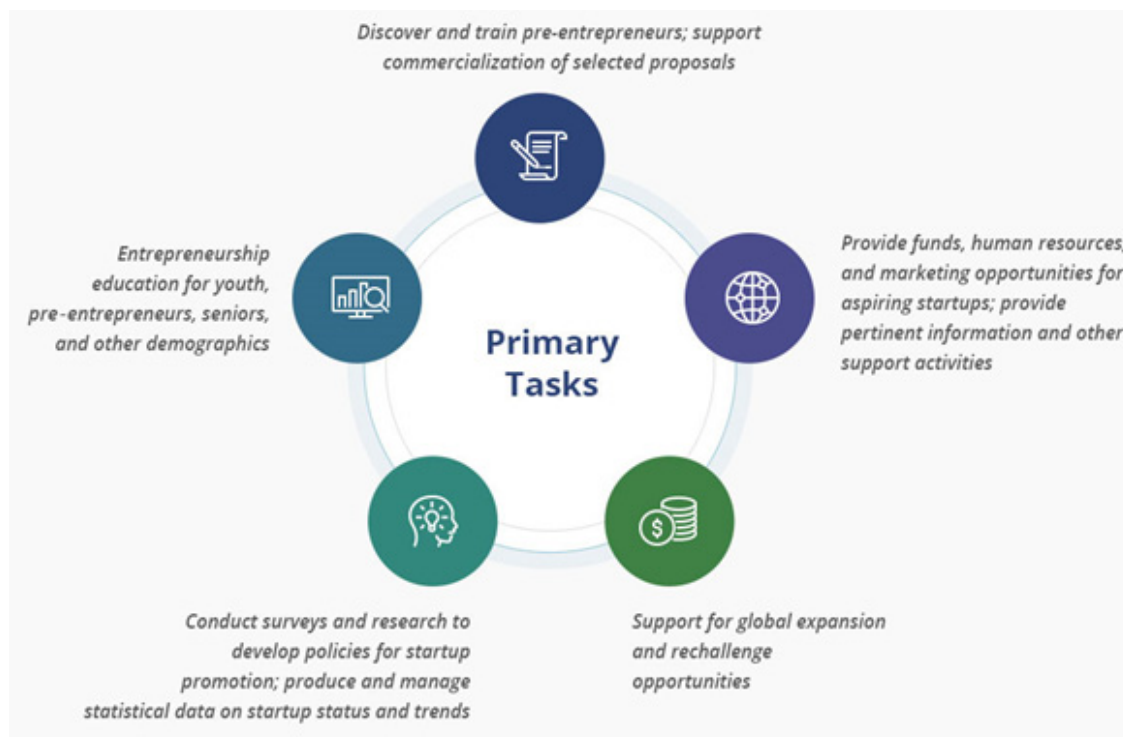
⁵⁹ <https://www.bi.go.kr/main/main.do>

⁶⁰ https://www.kised.or.kr/_eng/

[Table 3-28] Evolution of KISED

Year	Contents	Workforce / Budget(\$)
2008.12	The Institution of South Korean Entrepreneurship Development (IKED) established in the beginning	13 / 4.2 million
2010.10	President/CEO is executive position	32 / 91 million
2011.02	Name is changed (IKED → KISED) *KISED (South Korea Institute of Startup and Entrepreneurship Development)	41 / 114 million
2017.01	Public Accelerator	124 / 377 million
2019.10	Article 39, Act for Supporting the Establishment of Small and Medium Businesses	198 / 577 million
2020.12	Construction of a new building and moving to Sejong City from Daejeon	209 / 988 million

[Figure 3-37] Five Key Tasks of KISED



3.1.3 TIPS (Accelerator Investment-Driven Tech Incubator Program for Startup)

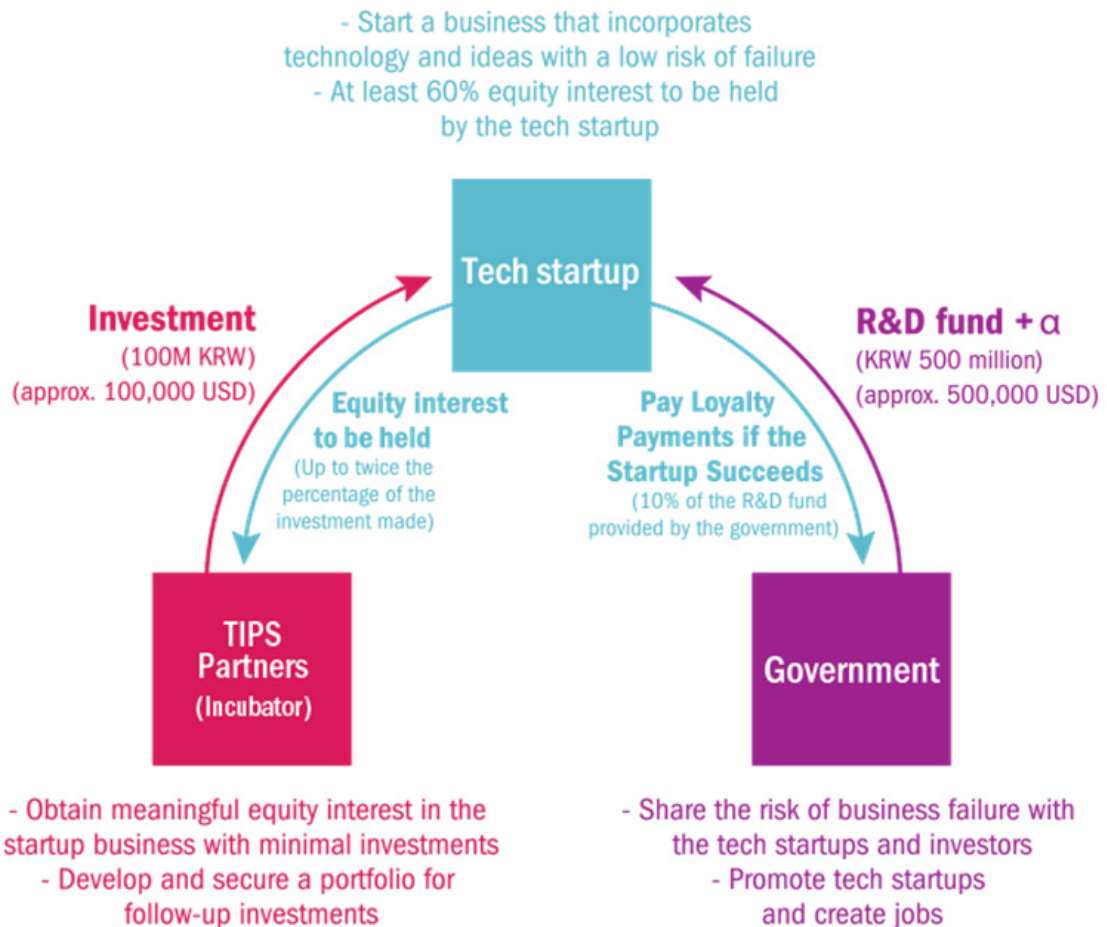
TIPS is designed to identify and nurture the most promising startups with innovative ideas and groundbreaking technologies. In order to support them when entering the global marketplace, it appoints and designates successful venture founders – who are now angel investors and leaders of technological enterprises – as their incubators and accelerators. It then offers seamless service encompassing angel investor networking, incubating, mentoring and professional support and matching R&D funds.

TIPS creates the best odds for success through financial investment, in addition to tested and proven mentoring and business incubation by some of the most successful angel investors. TIPS aims to promote and support talents related to technological innovation where start-ups cannot grow their business due to a lack of funding or a fear of failure.

TIPS also provides start-ups with business incubation spaces through consortiums with

universities and research institutes. The angel matching fund scheme offers KRW 100 million (approx. USD 0.1million) in an initial investment, KRW 500 million (approx. USD 0.5 million) in R&D, and up to KRW 400 million (approx. USD 0.4 million) in additional investments per start-up up for up to three years (KRW 100 million (approx. USD 0.1 million)) for startup funding. It also provides KRW 200 million (approx. USD 0.2 million) in angel matching funds and KRW 100 million (approx. USD 0.1 million) in overseas marketing assistance.

[Figure 3-38] TIPS Incubation Process



Source : <http://www.jointips.or.kr/global/>

3.1.4 The K-Startup Grand Challenge

The K-Startup Grand Challenge is a systematic startup competition program in South Korea.

Of more than 1,700 competing startups from 108 countries, 73 startups are selected to participate in a three and a half month acceleration program and receive initial funding and office space.

Then, top 40 teams are selected at the Demo Day and are given additional funds for 6 months. The startups are judged on their level of creativity, business concept, technologies and their commitment to base their business in South Korea. The aim of this competition program is to promote collaboration and exchange of ideas among startups not just from South Korea but from all around the world. At the end of the acceleration program, the South Korean government will host a demo day to select top 20 startups. Then, these startups will get additional financial incentives, and if they choose to establish their businesses in South Korea, they will get additional support from the South Korean government.

The extensive benefits of this competition program include the following: 4-month accelerating program, free office space at Pangyo Startup Campus, an access to production space filled with 3D printers, hardware testing platforms and more, USD 840 reward per team for top 50 teams, USD 27,000 for top 25 startups if they set up a legal entity in South Korea, grants between USD 6,000 to USD 100,000 for top 4 teams, potential for equity investment opportunities from accelerators and local VCs, seminars, workshops, and one-on-one mentoring from South Korea's largest tech companies.

3.1.5 Incubation Evaluation Index

Performance evaluation is defined as a formal and productive procedure to measure an incubation's outcomes based on job responsibilities. It is used to gauge the amount of value added by an incubator in terms of increased business revenue, job creation, capacity building and so on. The Ministry of SMEs and Startups of the South Korean government has developed the following evaluation criteria.

[Table 3-29] Incubation Performance Evaluation Index

Item	Fator	Method	Quan.	Qual.	Contents	TH Perspective
Institutional expertise (30)	Agency's will and interest	Qual.		7	Evaluating whether or not efforts are being made to secure the organization's expertise and reinforce expertise, such as the agency's will and interest, and the percentage of regular managers	The level of the university's role and system implementation as a subject of innovation activities
	Expertise of dedicated personnel (15)	Percentage of full-time managers	Quan.	5		
		Manager work experience	Quan.	4		
	Efforts to strengthen the expertise of dedicated personnel (8)	Number of managers	Quan.	6		
		Training performance	Quan.	5		
		Efforts to strengthen professionalism	Qual.	3		
Operating infrastructure (15)	Performance of securing operating funds	Quan.	5		Evaluating the performance and status of operating infrastructure, such as the performance of securing the operating fund of the incubation center, the rate of turnover of the nursery, and the satisfaction of the tenant company.	The level on building an environment for sustainable growth
	Space turnover	Quan.	5			
	Measurement of satisfaction	Quan.	5			
Effectiveness of the business plan (30)	Degree of achievement of the previous year's business plan	Qual.		10	Evaluating of the achievement of the previous year's business plan, the appropriateness of the current year's business plan, and the composition of the tenant company support program	The level on the promotion of new strategies and idea creation projects
	Appropriateness of the business plan for the year	Qual.		10		
	Composition of the tenant company support program	Qual.		10		
Linkage with related organizations (open collaboration) (20)	The degree of connection with external organizations such as innovation center, TP, VC, and internal resource utilization	Qual.		20	Evaluating the degree of cooperation with related organizations inside and outside the region and the utilization of internal resources to support tenant companies	The level at which innovation activities are carried out through interaction with innovation actors
Operation performance management (5)	BI-Net database management fidelity	Quan.	5		Evaluating the data management fidelity	The level of rotation and management of innovation activities
Total (100)			40	60		

Source : Ministry of SMEs and Startups, 2019 Business Incubation Center Management Evaluation Plan (2019).

There are 5 categories of incubation activities such as institutional expertise, operation infrastructure, effectiveness of business plan, linkage with related organization, and operation performance management. Each category has a specific measurement criteria with proportional values attached. They are measured by both the quantity and quality factors.

3.2 Case: Daedeok Innopolis

Daedeok Innopolis, formerly known as Daedeok Science Town (DST), is the research and development district in the Yuseong-gu district in Daejeon. Located 150 kilometers South of Seoul, Daedeok Innopolis is an hour distance from Seoul by high-speed train. Daedeok Innopolis evolved from a research cluster established by President Park, Chunghee in 1973, which started with the establishment of KAIST. Over 20 major research institutes and 40 corporate research centers participated in establishing this science and technology cluster.

Over the last few years, a number of IT venture companies have sprung up in this region, which has a high concentration of Ph.Ds. in the field of applied sciences. Currently, there are 232 research and educational institutions found in Daejeon, especially in the Daedeok area. The Electronics and Telecommunications Research Institute (ETRI) and the South Korea Aerospace Research Institute(KARI) are key players among them. The Daedeok area took a core role for the International Science and Business Belt. The Daedeok Innopolis logo was created by the industrial design company INNO Design in Palo Alto, USA.

Daedeok Innopolis is located at 32 legal administrative districts of dong of Yuseong-gu and Daedeok-gu in Daejeon Metropolitan City and the area is about 70.4 km² based on the legal administrative districts, regulated in the enforcement ordinance. There are 17 business incubation centers at research institutes and universities. Besides, industry-academia-research institute exchange centers and techno business centers are located at the park operation facilities. Daedeok Innopolis Foundation is also located in the center of Daedeok. The park is divided into general residential area, semi-residential area, commercial zone, green zone, education/research zone, business facility zone and industrial zone. Restrictions are set on construction work, business activities and development activities within the park (Seo, 2013).

Followings are the entities that form Daedeok Innopolis: 29 government-sponsored institutions (among them 18 GRIs), 14 government and public agencies, 7 public institutions, 5 education institutions, 17 venture incubation centers and about 1,100 firms (about 580 venture firms). The number of employees at Daedeok Innopolis was 45,526 in 2010. Daedeok Innopolis is the area of the top research capacity in South Korea, whose focus is on the government research institutes. There are 29 government-funded research institutes, including 18 GRIs under the South Korea Research Council for Industrial Science & Technology (KRCI) and the South South Research Council of Fundamental Science & Technology (KRCF). R&D expenses of government-sponsored institutions at Daedeok Innopolis accounted for 31.9 % of total government R&D budget for 2017 and 10.4 % of gross expenditure in R&D (GERD). The R&D expense of Daedeok Innopolis was about USD 7 billion in 2017.

[Figure 3-39] Map of Daedeok Innopolis, South Korea

Characteristics of each sector at INNOPOLIS Daedeok**Sector I (27.8km²)
Daedeok Research Complex**

- A cluster of research entities, including government-sponsored research and private sector institutions
- An R&D cluster of research-focused universities, including KAIST, and venture company collaboration zones

**Sector II (4.3km²)
Daedeok Techno Valley**

- The research and production base of support for advanced businesses in the Daedeok Research Complex

**Sector III (3.2km²)
Daedeok Industry Complex**

- The industrial base of INNOPOLIS Daedeok

**Sector IV (28.6km²)
Northern Green Belt Area**

- The northern area near the Daedeok Research Complex which includes the core science belt areas of Shindong and Dungok

**Sector V (3.9km²)
The Agency for Defense Development area**

- Area for military defense which includes ADD and Hanwha (defense contractors)

Source: Daedeok Innopolis (2020)

3.2.1 Incubation Supporting Agency: The Innopolis Foundation

Since 2005, the Innopolis Foundation has promoted Daedeok Innopolis to become the home to over 1,100 SMEs through its major programs as below after Daedeok science park was transformed into an innovation cluster with newly developed industrial area, Zone 2. With these programs, the Innopolis Foundation directs substantial effort to transform research outcomes at Daedeok Innopolis into companies and products generating revenue. The supportive programs consist of four main sub-programs such as venture incubation project, three-up program, special clustering project and total design project. And these programs are mobilized for activities promoting research institutes' spin-off mechanism (Seo, 2013).

[Figure 3-40] 17 Business Incubation Centers in Daedeok Innopolis



3.2.1.1 Venture Incubation Project

It has solved difficulties and cultivated the business incubation and star business through systematic business support of small and middle venture enterprises in Daedeok Innopolis. This technology business creation and management services provide customized consulting service and solutions to address difficulties encountering enterprises in the process of the commercialization of research outcomes.

3.2.1.2 Three-Up Program : Start-up, High-up and Lift-up

As a life-time business supporting program for entrepreneurs, the Innopolis Foundation provides three-up program consisting of 3 sub-programs like start-up, high-up and lift-up. The start-up stage is to educate prep entrepreneurs starting high-tech based businesses to grow the entrepreneurship and help develop a basic business plan. The high-up stage

supports the early production through continuous monitoring of the business plan. The last stage of lift-up supports the implementation of the business expansion strategy with active commercialization of excellent business items identified through special technology joint development projects.

3.2.1.3 Special Clustering Project

Special clustering project has been facilitated in connection with the enterprises with big demands for technologies held by public research institutes, facilitating the early commercialization of research outcomes and market entry. This project is positioning Daedeok Innopolis as the center of new business creation through hub-spoke type business promotion in connection with other strategic business areas.

3.2.1.4 Total Design Project

This project is to maximize the successful commercialization of advanced technologies and to facilitate the success of enterprises through the operation of comprehensive support system from the planning of a new business model to design development and marketing support. Through this process, designs of a new business model are developed by merging excellent technologies of research institutes or enterprises located in the Special District.

Marketing support is provided in connection with prominent enterprises at home and abroad and the production of prototypes including technical design of the developed model are implemented. The project has facilitated the commercialization and transfer of public technologies by supporting TT and marketing of technologies jointly developed by private commercialization enterprises and public TLOs (Technology Licensing Office) by identifying excellent technologies with high TC potential.

This project has an element of a technology marketing project based on patent packaging. So, it has contributed to the activation of TT and the reduction of TT costs, which allows a bulk purchase of technologies in huge demand. Also as a strategic business commercialization project, the project has supported solutions required for the

development of a new business model or an advancement of businesses, utilizing public technologies.

3.2.1.5 Research Institute Spin-off Companies

Spin-offs from either research institutes or universities are not a new idea for TT. The survival rate of the spin-offs in 3 years after the start of their business stands at around 75 % (Chong 2012). This is far higher than that of the average 3-year start-up survival rate of 46 %. Spin-offs are a means of TT from a parent organization, representing a mechanism for creating jobs and new wealth (Steffensen, et al 1999). And Steffensen (1999) suggested that there are two types of spin-offs: (1) planned, when the new venture results from an organized effort by the parent organization, and (2) spontaneously occurring, when the company is established by an entrepreneur who identifies a market opportunity and who founds the spin-off with little encouragement (Seo, 2013).

However, performances of spin-offs after their establishment vary significantly across organizations. The cross-organization variation in start-up activity is explained for four reasons such as i) understanding source of knowledge spillovers, ii) proximity to economic development and agglomeration economies, iii) supports from TLO, inventor and CEO of the organization, and iv) norms and policies of the organization (Gregorio and Shane 2003). Based on the “Special Act on Promoting Daedeok Innopolis,” Daedeok Innopolis is granted to have a special policy to create spin-offs from government research institutes in Daedeok Innopolis. This special policy allows spin-offs from GRIs to have tax benefits after meeting several conditions like ownership and company location. The legal definition of “Research Institute Spin-off” is presented as below.

The “research institute spin-off ” refers to a company established in Daedeok Innopolis with investment in kind of more than 20 % by appropriate entities set forth in the law, on its own, or jointly, for the purpose of starting a business with the technology government research institutes developed and retained (Article 9 & 3 of the Enforcement Act, Article 13 of the Special Act on Daedeok Innopolis). This move of creating policies for promoting research institute spin-offs first started when there arose a criticism that the

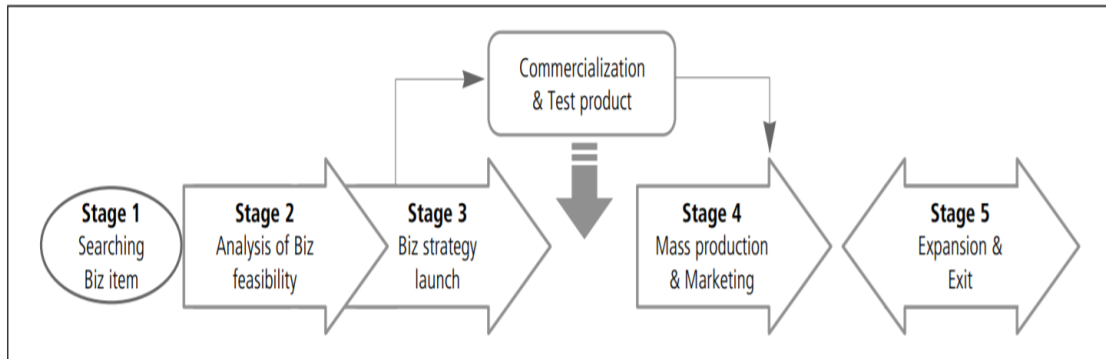
commercialization of government research institutes' R&D outcomes into real products or businesses had been sluggish despite a consistent increase of government R&D investment since the 1990s. The underlying view is indicating that the primary reason of low rate of successful technology commercialization of public R&D results can be found in the inefficient business mechanism of the R&D institutes (e.g., lack of diversified commercialization approaches simply focusing on a license). As an attempt to improve such shortcomings, it is being emphasized that appropriate strategic support is required to optimize the success rate by implementing a wide range of technology commercialization methods (e.g., by technology capital investment, joint research company, or professor's or researcher's start-up), in consideration of the characteristics of developed technologies and qualifications of business entities collectively.

In addition, the project is expected to encourage innovation networks between private companies as the users and government research institutes as technology providers by combining excellent government R&D outcomes, private capital, and management know-how. The entity which is entitled to establish a research institute spin-off in accordance with the law of Daedeok Innopolis is "a government research institute, an educational & industrial cooperation tech-holding company, or a technology start-up supporting company." Research institute spin-offs can receive two major benefits, of which the bigger and the more important is tax benefits as presented below (Seo, 2013).

- National taxes: income tax and corporate tax exemptions (100 % exemption for three years and 50 % reduction for the next two years).
- Local taxes: acquisition and property tax exemptions (100 % exemption for seven years and 50 % reduction for the next three years).

The second benefit is the life-cycle business process support program, which helps spin-offs to prepare their business action plan and commercialize the technology from a research institute with supporting matching fund. This program supposedly provides life time support to spin-offs for the whole business process from stage 1 to stage 5 as below.

[Figure 3-41] Research Institute Spin-off System: Incubation Method of Daedeok Innopolis



Source: Seo (2013)

The Funding Ecosystem in Daedeok Innopolis has evolved by the efforts of the government, industries, and universities. The major achievement of Daedeok Innopolis for the past 40 years is its contribution to the nation's economic development through the development of diverse technologies.

4. Test & Certification in South Korea⁶¹

4.1 General Overview of Test & Certification

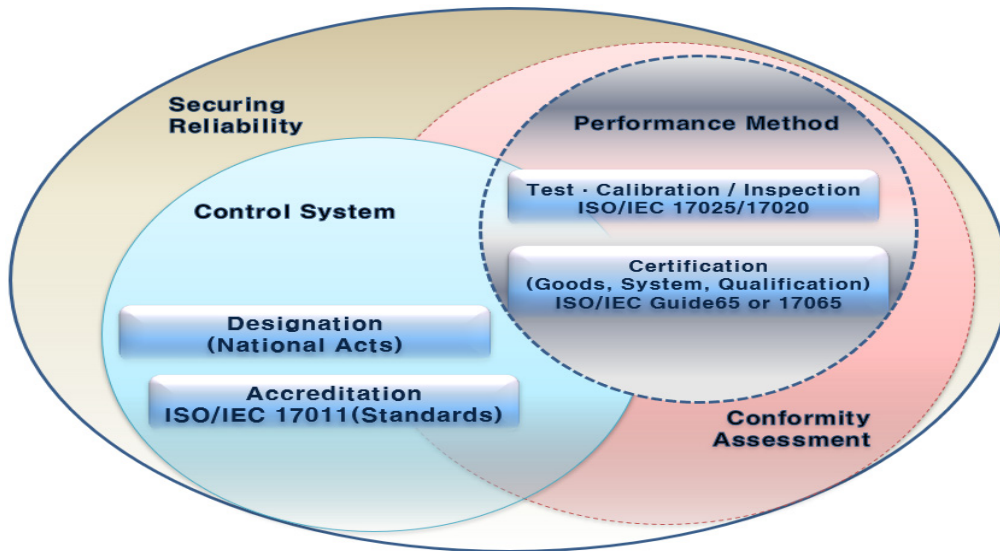
Conformity assessment based on the “Framework Act on National Standards” refers to evaluating whether a product, service, process, system, etc. meets the requirements stipulated in standards, product standards, and technical regulations. Certification refers to the process of certifying that a specific product has passed the performance quality verification test, and conforms to contracts, regulations or quality standards stipulated in the regulation.

Designation refers to the administrative action of the designated authority giving specific qualifications to the conformity assessment body (certification body) that performs the conformity assessment procedure. Accreditation refers to (administrative) activities in which an accreditation body with official authority, based on the “Framework Act on National Standards,” evaluates inspection institutions with the ability to conduct specific tests and inspections and guarantees their related capabilities.

The purpose of conformity assessment is to secure the reliability of products through regulations on matters related to relevant business and technology development promotion, and to improve national competitiveness and the safety and welfare of the people.

⁶¹ This chapter was written by Dr. Sangho Choi of the Daejeon Technopark

[Figure 3-42] Key Concepts of Test & Certification



4.2 Status of Laws on Conformity Assessment in South Korea

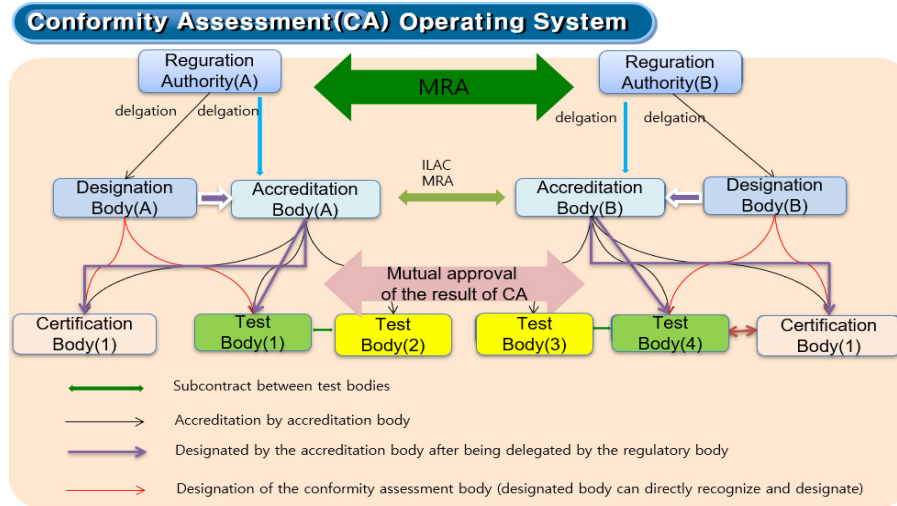
In South Korea, tests, certifications, and inspections are operated based on separate laws and regulations for each industry area. Representative laws related to conformity assessment in South Korea include the “Framework Act on National Standards” and the “Act on Conformity Assessment Management.” In particular, the “Act on Conformity Assessment Management” was enacted in April 2020 for the purpose of securing the reliability of conformity assessment and improving national competitiveness and public safety and welfare.

The enactment of this law aims at: (1) Establishment of policies for conformity assessment work, (2) Establishment of regulations for conformity assessment work, (3) Establishment of accreditation bodies and their accreditation, and the regulation of the expiration date of accredited organizations, use of accreditation marks, and periodic inspection, etc., (4) Regulations on penalties and fines for accredited institutions, and (5) Supportive policies for enhancing competency in conformity assessment activities such as fostering professional workforce, international cooperation, and technology development.

4.3 Conformity Assessment (CA) Operating System in South Korea

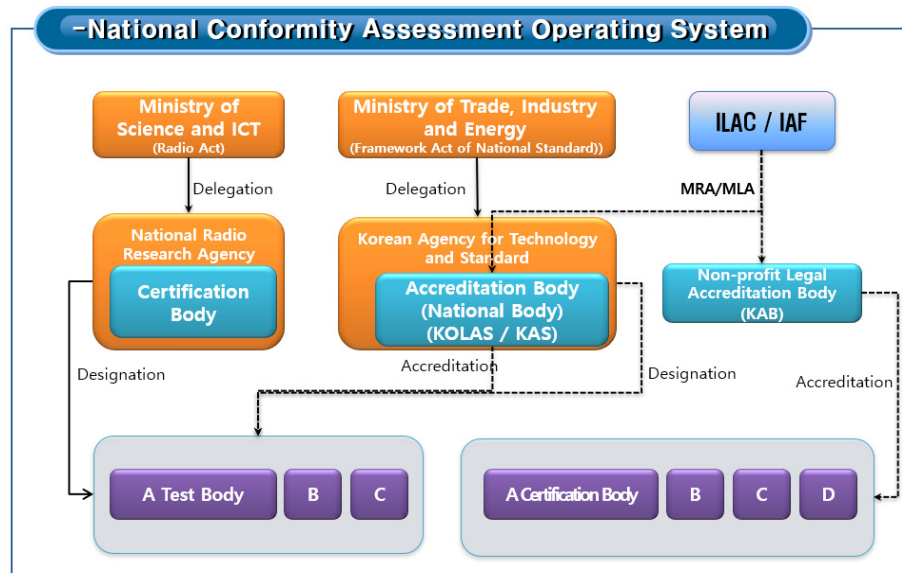
The following is the general operating system of conformity assessment.

[Figure 3-43] General Conformity Assessment Operating System



In the meantime, the national conformity assessment operating system is depicted as follows.

[Figure 3-44] National Conformity Assessment Operating System



As for national accreditation bodies, there are; i) KOLAS which accredits test bodies, calibration bodies, inspection bodies, standard reference material production bodies and medical inspection bodies; ii) KAS which accredits product certification bodies; and iii) KAB which accredits 17 fields of accreditation including quality management and environmental management. The following is the operational status of these accreditation bodies.

[Figure 3-45] Status of National Accreditation Bodies

National Accreditation Body			
Body	Korea Laboratory Accreditation Scheme KOLAS	Korea Accreditation System KAS	Korea Accreditation Board KAB
Accreditation Mark			
Acts	Articles 23 of the Framework Act on National Standards	Articles 21 of the Framework Act on National Standards	Articles 24 of the Framework Act on National Standards and Article 32 of the Civil Law
Office	Korean Agency for Technology and Standard	Korean Agency for Technology and Standard	(Foundation)Korea Accreditation Board
Accreditation Scope	· test bodies, calibration bodies, · inspection bodies, · standard reference material production bodies, · medical inspection bodies	· product certification bodies	· 17 fields such as quality management/environmental management
MRA Status	ILAC MRA APLAC MRA	IAF MLA PAC MLA	IAF MLA PAC MLA
Accreditation Results	889 bodies	19 bodies	99 bodies

The accreditation status of KOLAS is as follows⁶².

[Table 3-30] KOLAS Accreditation Status

(2018.12.31)

Body	2011	2012	2013	2014	2015	2016	2017	2018
Test Body	403	416	460	468	487	496	532	564
Inspection Body	46	49	52	50	55	58	61	62
Calibration Body	193	189	192	201	209	212	226	231
Standard Reference Material Body	9	11	11	13	13	12	13	13
Medical Body	-	-	-	2	5	6	8	8
Proficiency Bodies	-	-	7	7	8	9	9	11
계	651	665	722	741	777	793	849	889

The accreditation status of KAB is as follows.⁶³

[Table 3-31] KAB Accreditation Status

(2018.12.31)

Body	2011	2012	2013	2014	2015	2016	2017	2018
Product Certification Body	15	16	17	20	20	20	19	19

Regarding the conformity assessment in South Korea, there are two ways to designate a testing or a certification body. The first method is to directly designate the testing bodies by the relevant ministry, and the second method is to substitute them with KOLAS/KAB, South Korea's representative accreditation bodies. South Korea has the following laws and regulations on the use of KOLAS accreditation bodies.

⁶² Korean Standards and Certifications(<https://standard.go.kr>)

⁶³ Korean Standards and Certifications(<https://standard.go.kr>)

[Table 3-32] Status of Conformity Assessment Laws

No.	Ministry	Laws
1	Ministry of Employment and Labor	Laws related to occupational safety and health (2)
2	Fair Trade Commission	Basic Consumer Act (1)
3	Ministry of Land, Infrastructure and Transport	Construction Technology Promotion Act (1)
4	Ministry of Land, Transport and Maritime Affairs	Act on the Construction and Management of Geospatial Information (1)
5	Meteorological Administration	Meteorological Observation Standardization Act, etc. (3)
6	Ministry of Strategy and Finance	Tobacco Business Act, etc. (2)
7	Ministry of Health and Welfare	Public Health Management Act, etc. (2)
8	Ministry of Trade, Industry and Energy	Quantification Act, etc. (17)
9	Fire Service Administration	Laws and ordinances related to quality control of firefighting supplies (3)
10	Ministry of Food and Drug Safety	Act on Test and Inspection in the Food and Drug Field (1)
11	Public Procurement Service	Administration Procurement Business Act (2)
12	Ministry of Oceans and Fisheries	Ship Safety Law, etc. (7)
13	Ministry of Public Administration and Security	Elevator Safety Management Act, etc. (6)
14	Ministry of Environment	Construction Waste Recycling Promotion Act (15)
Total	13 Ministries	Total 62 Laws

In particular, the conformity assessment for ICT products is based on the Radio Wave Act, and the Ministry of Science, ICT and ICT directly designates the testing agency.

4.4 Status of Certification in South Korea

As of 2018, there are 20,507 South Korean standards in all industries. The status of enactment and revision of standards over the last three years is as follows.⁶⁴

[Table 3-33] Status of Standards in South Korea

Year	Notification Status by Year				No. of Standards Held (as of year-end)
	Enactment	Revision	Confirmation	Abolition	
2016	311	1,482	1,963	554	20,149
2017	341	1,407	3,505	208	20,282
2018	441	1,392	2,733	216	20,507

⁶⁴ Korean Standards and Certifications(<https://standard.go.kr>)

The international standard used for the accreditation of testing and calibration bodies is ISO/IEC 17025, while the international standard used for the accreditation of certification bodies is ISO/IEC 17065. These are requirements established and recommended by the Conformity Assessment Committee and the CASCO (Committee on Conformity Assessment) of the International Organization for Standardization (ISO), in relation to conformity assessment activities.

[Table 3-34] Conformity Assessment Requirement for ISO/IEC

Conformity Assessment requirements for ISO/IEC		
Field	Published	Projects
Testing & Calibration	17025:2019	General requirements for the competence of testing and calibration laboratories
Inspection	17020:2015	Conformity assessment — Requirements for the operation of various types of bodies performing inspection
Certification	17065:2012	Conformity assessment — Requirements for bodies certifying products, processes and services
Accreditation	17011:2014	Conformity assessment — General requirements for accreditation bodies accrediting conformity assessment bodies

Technical standards refer to “standards that are legally binding adopted by governments or organizations or adopted by contracts” (KS A 0014), or “the characteristics of products or related processes and production methods including applicable administrative regulations” as stipulated in the related regulations. It defines that compliance is mandatory and may include or address terms, symbols, packaging, labeling, or product attachment requirements that apply to products, processes and production methods (WTO/TBT). In South Korea, each ministry manages and operates standards individually according to its jurisdiction. The total number of technical regulations (standards) in South Korea is 2,473 as of 2020.⁶⁵

⁶⁵ Korean Standards and Certifications(<https://standard.go.kr>)

[Table 3-35] Total Number of Technical Regulations

No.	Ministry	No. of Technical Regulation	No.	Ministry	No. of Technical Regulation
1	Ministry of Employment and Labor	41	15	Ministry of Health and Welfare	16
2	Fair Trade Commission	2	16	Forest Service	24
3	Ministry of Science and ICT	62	17	Ministry of Trade, Industry and Energy	1338
4	Customs Service	2	18	Fire Department	152
5	Ministry of Education	2	19	Ministry of Food and Drug Safety	113
6	Ministry of Land, Infrastructure and Transport	218	20	Ministry of Gender Equality and Family	2
7	Meteorological Administration	19	21	Nuclear Safety Committee	57
8	Ministry of Strategy and Finance	1	22	Public Procurement Service	2
9	Ministry of Agriculture, Food and Rural Affairs	88	23	Ministry of Small and Medium Venture Business	3
10	ural Development Administration	22	24	Intellectual Property Office	1
11	Cultural Heritage Administration	4	25	Maritime Police Agency	5
12	Ministry of Culture, Sports and Tourism	7	26	Ministry of Oceans and Fisheries	112
13	Korea Communications Commission	2	27	Ministry of Public Administration and Security	26
14	Defense Program Administration	3	28	Ministry of Environment	149
Total		2,473			

In addition, South Korea has introduced and is operating a technology regulation impact assessment system as a measure to continuously ease regulations, relieve burden on manufacturing companies, and promote development.

The certification system is divided into the legal certification system and the private certification system according to the existence of legal basis. The legal certification system is then further divided into a mandatory certification and a voluntary certification depending on whether it has a compulsory nature. In addition, the certification system implemented by each ministry is operated under various names depending on the characteristics of the subject of certification, such as certification, type approval, verification, type verification, and type registration. The status of legal compulsory and voluntary certification systems is as follows.⁶⁶

⁶⁶ Korean Standards and Certifications(<https://standard.go.kr>)

[Table 3-36] Certification Status in South Korea

Year	2012	2013	2014	2015	2016	2017	2018	2019
Compulsory Certification	68	69	73	73	69	71	70	81
Voluntary Certification	115	132	136	130	97	104	104	106

The certifications operated by 24 South Korean government ministries are as follows.⁶⁷

[Table 3-37] Number of Certifications

No.	Ministry	No. of Certification	Compulsory	Voluntary
1	Ministry of Land, Infrastructure and Transport	32	17	15
2	Ministry of Trade, Industry and Energy	26	15	11
3	Ministry of Oceans and Fisheries	21	10	11
4	Ministry of Agriculture, Forestry and Livestock	18	2	16
5	Ministry of Environment	15	11	4
6	Ministry of Science and Technology, Information and Communication	9	2	7
7	Forest Service Administration	11	2	9
8	Ministry of Public Administration and Security	6	3	3
9	Ministry of Culture, Sports and Tourism	6	3	3
10	Ministry of Cultural Heritage	1	0	1
11	Ministry of Health and Welfare	9	2	7
12	Ministry of Education	2	0	2
13	Ministry of Food and Drug Safety	5	4	1
14	Ministry of Employment and Labor	4	2	2
15	Meteorological Administration	3	2	1
16	Defense Project Administration	4	2	2
17	Ministry of SMEs and Startups	1	0	1
18	Fair Trade Commission	2	0	2
19	Ministry of Gender Equality and Family	2	0	2
20	Fire Department	5	3	2
21	Customs Service	1	0	1
22	Intellectual Property Office	1	0	1
23	South South Korea Communications Commission	1	0	1
24	Maritime Police Agency	2	1	1

⁶⁷ Korean Standards and Certifications(<https://standard.go.kr>)

Currently, there are 81 compulsory legal certification systems in South Korea. Even with the same purpose of ‘product safety’, there was an inconvenience of having to obtain duplicate certification because different ministries require different certification marks. As a result, time and money were wasted, and the international credibility of South Korean products has been degraded and the issue of the wasted national wealth was raised, such as the need for recertification because mutual recognition was not allowed in cross-border transactions. Accordingly, 13 legally obligated certification marks were integrated into one national integrated certification mark in November, 2011.

The National Integrated Certification Mark (KC Mark, ) was launched in July 2009 by integrating 10 certification marks from the Ministry of Knowledge Economy and the Ministry of Labor, and as of January 2011, 13 certification marks from 5 ministries were integrated into the KC Mark. Since then, government ministries all have adopted the KC mark when creating or changing their legal compulsory certification system. As of September 2017, 23 statutory compulsory certification systems in 8 departments are using the KC mark.

[Table 3-38] Certification Systems and Laws in South Korea

No.	Ministry	Certification System	Laws
1	Ministry of Trade, Industry and Energy	Type approval and verification of meters	Metering Act
2		Safety certification of industrial products	Electrical Appliance Safety Management Act
3		Safety management of children's products	Children's Product Safety Special Act
4		Safety certification of electrical appliances	Electrical Appliance Safety Management Act
5		High-pressure gas safety management	High Pressure Gas Safety Control Act
6		Gas supply inspection	Liquefied Petroleum Gas Safety Management and Business Act
7		Energy consumption efficiency grade indication	Energy Use Rationalization Act
8	Ministry of Land, Infrastructure and Transport	Self-certification of automobiles and parts	Automobile Management Law
9		Mounting inspection of pressure-resistant containers	Automobile Management Law
10		Inspection of hydrological survey equipment	River Law
11		Certification of fireproof structure	Fireproof Structure Certification and Management Standard
12	Ministry of Oceans and Fisheries	Certification of wall sound insulation structure	Wall Sound Insulation Structure Certification and Management Standard
13		Type approval of marine environment measuring equipment	Marine Environment Management Law
14	Ministry of Environment	Water purifier quality inspection	Drinking Water Management Law
15		Hygiene safety standard certification	Waterworks Law
16	Ministry of Science and ICT	ICT equipment conformity certification, etc.	Radio Wave Act
17	Ministry of Public Administration and Security	Fire fighting equipment type approval	Firefighting Facility Installation Maintenance and Safety Management Act
18		Flame retardant performance inspection	Firefighting Facility Installation Maintenance and Safety Management Act
19	Ministry of Oceans and Fisheries	Marine pollution control performance certification	Marine Environment Management Act
20	Ministry of Employment and Labor	Safety certification of dangerous machinery and equipment,	Industrial Safety and Health Act,
21		Safety of protective devices and protective equipment Certification	
22	Ministry of National Defense	Application of KC mark for munitions of textiles and clothing	Defense Business Act
23	Meteorological Administration	Approval of meteorological test	Meteorological Observation Standardization Act

There is a total of 564 accredited testing institutes in South Korea, and are distributed in all industries to provide test services. Most of the testing bodies are operated by income from test fees, and in some cases, government-sponsored public institutions are being operated. The size of the local test market is estimated to be approximately KRW 1 trillion.

4.5 Current Status of Quality Inspection System for Petroleum and LPG in South Korea

The quality inspection system for petroleum and petroleum alternative fuels is based on the “Petroleum and Petroleum Alternative Fuel Business Act.” In particular, this Act stipulates Article 24 on Quality Standards for Petroleum Products and Article 25 on Quality Inspection and Designation of Inspection Agency. The LPG quality inspection system is based on Article 27 of the “Liquefied Petroleum Gas Safety Management and Business Act.” In particular, the South Korea Petroleum Institute (Kpetro, <https://www.kpetro.or.kr/eng/index.do>) is the representative inspection agency.

[Figure 3-46] Status of Laws Governing Quality Inspection System for Petroleum and LPG

» Petroleum and Alternative Fuel Business Act

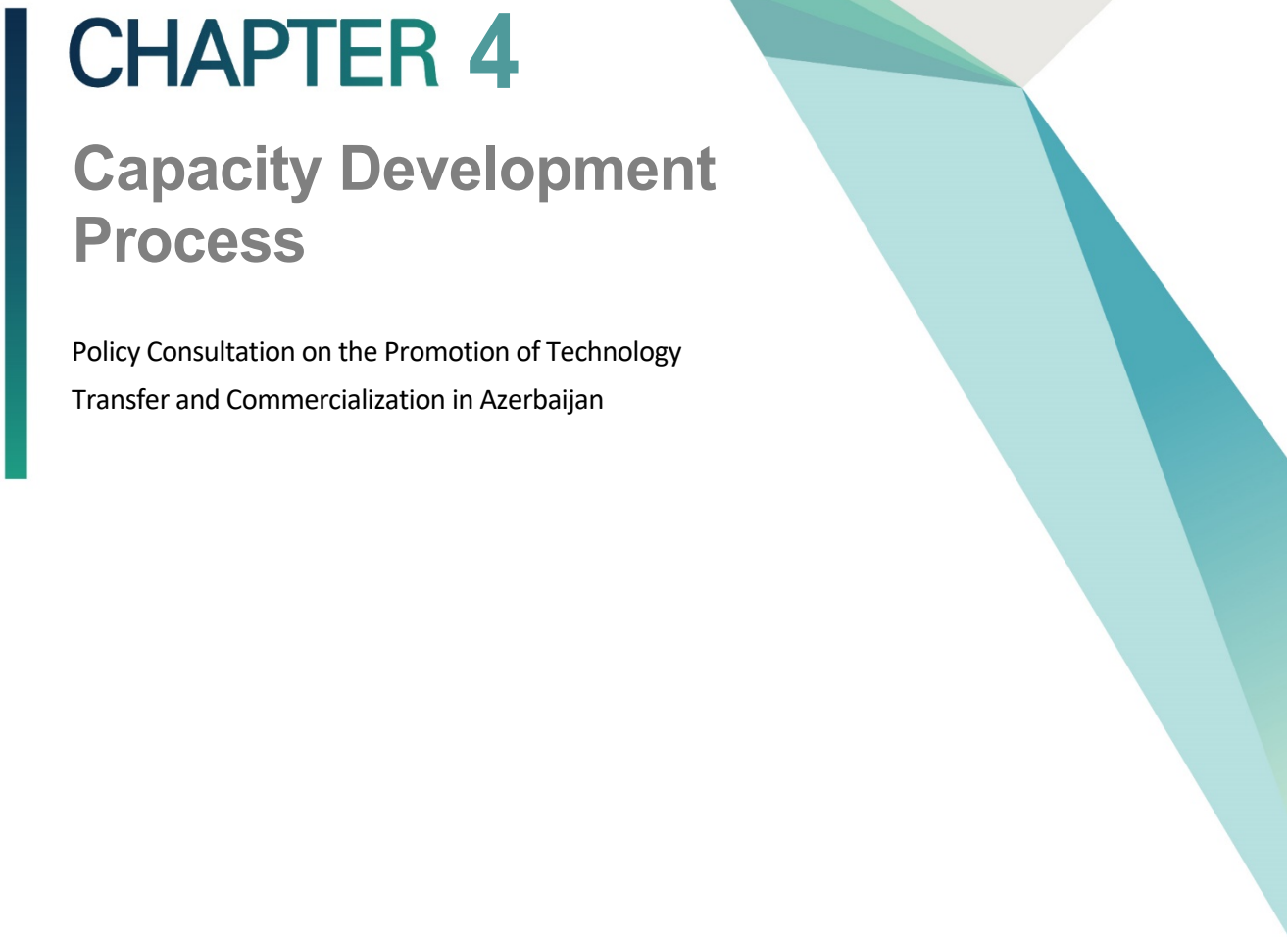
○ Quality Inspection system of oil product

Quality Inspection organization for legal mandatory examination from the court (Article 25, Paragraph 1, in the law)	Consigned by the Minister of Trade, Industry, and Energy for frequent inspection (K-Petro) (Article 45, paragraph 4-9, in the Enforcement Decree)
--	---

» Petroleum and Alternative Fuel Business Act

○ Oil alternative quality inspection system

Quality Inspection organization for legal mandatory examination from the court (Article 31, paragraph 3, in the law)	Consigned by the Minister of Trade, Industry, and Energy for frequent inspection (K-Petro) (Article 45, paragraph 4-11, in the Enforcement Decree)
--	--

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and green, also pointing downwards.

CHAPTER 4

Capacity Development Process

Policy Consultation on the Promotion of Technology
Transfer and Commercialization in Azerbaijan

Chapter 4. Capacity Development Process

1. Kick-off Meeting and Field Study

1.1 Background

In January 2020, the STEPI team selected PM of the project and organized a project team consisting of various experts both inside and outside of STEPI. In February, STEPI and ANAS HTP had an online conference to reaffirm the feasibility of the project and determine specific directions and procedures of the project for the second year (2020). At the meeting, both parties shared in-depth opinions and suggestions and finally confirmed “ANAS HTP Strategic Plan Development” as a topic for the second year.

According to the meeting outcomes, the STEPI team prepared a project proposal for the second year and sent it to ANAS HTP in March. After ANAS HTP presented its feedback on the project proposal offered by STEPI, the two organizations finally agreed to hold the Kick-off Meeting in Baku in April, and the Capacity Development Workshop in South Korea in July, and the Dissemination Workshop in Baku around in October.

After several times of discussion among experts within the STEPI team, four sub-topics were drawn as strategic areas to deal with for the development of ANAS HTP. The selected sub-topics were (1) TT & TC, (2) Start-up & Incubation, (3) Public Procurement, and (4) Test & Certification. In order to provide in-depth consulting and develop strategic plans for these four areas, STEPI and ANAS HTP agreed to have a kick-off and field interview. The interview was to grasp the current status of ANAS HTP along with actual and critical demands from it.

1.2 Overview of the Kick-off Workshop and Field Interview

- **Purpose**
 - To establish a specific direction for the project through interviews with local consultants from ANAS HTP
 - To review of the framework conditions of Azerbaijan's technology transfer ecosystem and its current status
 - To identify the specific demand for the 2020 consultation project
- **Date: May 8, 2020 (FRI) 15:00 - 18:00**
- **Venue: Video Conference (in consideration of the COVID-19 situation)**

1.3 Attendees

STEPI Research Team (total of 6)

Name	Position/Organization	Role/Area of Expertise
Wangdon Kim	Chief Director, Division of Global Innovation Strategy, STEPI	Project management and planning Consultation and strategy development for the internal system of ANAS HTP
Jong Hwa Choi	Head, Office of Future Industry Research, STEPI	Consultation and strategy development for public procurement
Yejin Yoon	Researcher, Division of Global Innovation Strategy, STEPI	Research and project coordinating
Myungdo Oh	Professor, Department of Mechanical and Information Engineering, University of Seoul	Reviewing and advising overall project strategies
Jongin Choi	Professor, Department of Business Accounting, Hanbat National University	Consultation and strategy development for Start-up & Incubation
Sangho Choi	Head, Policy Planning Department, Daejeon Technopark	Consultation and strategy development for TT & TC and Test & Certification

Azerbaijan Research Team (total of 4)

Name	Position/Organization	Role/Area of Expertise
Ali Abbasov	Director, ANAS HTP	Project management
Bunyamin Seyidov,	Chief, Department of Relations with Eastern Countries, ANAS HTP	Consulting on internal system of ANAS HTP
Mammad Baghir-zadeh	Deputy Director on the Strategic Development, ANAS HTP	Consulting on Test & Certification and Public Procurement of ANAS HTP
Vusal Suleymanli	Innovation Projects Coordinator, ANAS HTP	Consulting on TT & TC

1.4 Main Activities and Agendas

▪ Opening / Closing Ceremony

- Opening Ceremony: Prof. Myungdo Oh (University of Seoul), Director. Ali Abbaov (ANAS HTP)
- Closing Ceremony: Dir. Wangdong Kim (STEPI)

▪ Kick-off

- Introduction to participating organizations, project details, and project procedure

▪ Local Consultant Interview

- (Procedure) Interview between one South Korean and one Azerbaijani experts per session
- (Session composition) Internal system, TT & TC, Test & Certification, startup Promotion, and public procurement

The initial plan was to hold the interviews in Baku in April, but due to the COVID-19 situation that aggravated in February, it was decided to have interviews via video conferencing in early May.

1.5 Schedule

Kick-off Session			
15:00-15:05	Opening Remarks	Myungdo Oh	Professor, Department of Mechanical and Information Engineering, University of Seoul
15:05-15:10		Ali Abbasov	Director, ANAS HTP
15:10-15:20	Program Overview	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
Field Interview Session			
15:20-15:50	Internal System	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
		Bunyamin Seyidov	Chief, Department of Relations with Eastern Countries, ANAS HTP
15:50-16:20	Technology Transfer & Commercialization	Sangho Choi	Head, Policy Planning Department, Daejeon Technopark
		Vusal Suleymanli	Innovation Projects Coordinator, ANAS HTP
16:20-16:50	Test & Certification	Sangho Choi	Head, Policy Planning Department, Daejeon Technopark
		Mammad Baghir-zadeh	Deputy Director on the Strategic Development, ANAS HTP
16:50-17:20	Start-up & Incubation	Jongin Choi	Professor, Department of Business Accounting, Hanbat National University
		Vusal Suleymanli	Innovation Projects Coordinator, ANAS HTP
17:20-17:50	Public Procurement	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
		Mammad Baghir-zadeh	Deputy Director on the Strategic Development, ANAS HTP
Closing Session			
17:50-18:00	Closing Remarks	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI

1.6 Interview Results

1.6.1 Internal System of ANAS HTP

- **Have the vision, mission, operating principles, mandates, roles & functions, and strategies of ANAS HTP been granted by ANAS?**

ANAS HTP was established on the basis of Decree of the President of the Republic of Azerbaijan dated November 08, 2016 No. 2425. The main goals of ANAS HTP are; the development of the national economy, the enhancement of national competitiveness, the expansion of innovation and high technologies based on S&T achievements, the creation of modern R&D complexes of developing new technologies, the establishment of new innovative enterprises, the development of competitive science-based production on the basis of innovative high technologies, and the support for entrepreneurship in this field.

In addition, ANAS HTP's key principles include; strengthening the ties between science and industry in order to create an innovative economy, the formation of necessary mechanisms for conducting applied research in accordance with market needs, the stimulation of the production of competitive products by increasing innovative activity, and implementing measures to ensure the effective utilization and development of innovation potential, the establishment of an innovative system that ensures the use, and the development and dissemination of new knowledge and technologies in Azerbaijan in accordance with the requirements of the information society by referring to international experiences.

- **Given the changing environments, what do you think should be a vision, mission, and principles of ANAS HTP in 10 years (Year 2030)?**

In my opinion, in order to develop its future activities in the next 10 years, ANAS HTP needs to attract extraterritorial residents to widen the scope of its residents in the Technopark, create innovation funds, promote innovation, research and technology commercialization in Azerbaijan, expand its technology incubator using the country's science and innovation potential that has been enhanced significantly as result of its reduced dependency on oil and gas.

In addition, for the development of the ecosystem in Azerbaijan, the country needs to focus on the following activities over the next 10 years; promoting an exchange of knowledge and experience in S&T innovation with other countries and ensuring the involvement of developers or creators in increasing the production of high-tech products with export potential in Azerbaijan.

As a result of the above-mentioned activities, ANAS will be able to strengthen its role in diversifying the economy by reducing the cost burden on its state budget, preventing the inflow of foreign currency from the country, developing import substitute products, supporting SMEs, and increasing the share of science in the national economy. The main goal is to turn ANAS HTP into a center of science and high technology as quickly as possible.

Besides, the future activities of ANAS HTP include the cooperation with higher education institutions, the establishment of ANAS HTP's Investment Fund, the expansion of digital services, the formation of a register of intellectual products, the development of an appropriate legislative framework to strengthen the innovation ecosystem, the restoration of incubation activities, and the establishment of new enterprises with the joint participation of other institutes under ANAS.

- **Current status of governance (including various committees)**

(AUTHORIZED BODY) ANAS is a supervising body responsible for handling the issues of ANAS HTP. ANAS is the highest scientific organization implementing the state policy in the field of science in the Republic of Azerbaijan. ANAS assists the Managing Organization in organizing, managing and developing activities related to HTP. It also prepares an annual report on HTP's activities and submits this report to the President of the Republic of Azerbaijan. ANAS conducts R&D in accordance with the ANAS HTP profile and decides priorities in developing and improving innovative products and high technology.

(THE SUPREME GOVERNING BODY) The supreme governing body of ANAS HTP is the General Assembly of its members. The authority of the General Assembly related to ANAS is exercised by the Presidium of ANAS. ANAS participates in the Azerbaijani government's policymaking and has mandate to develop and implement the nation's STI policies.

Regarding to this mission, ANAS carries out the following functions:

(SUPERVISORY BOARD) The Supervisory Board controls the activities of the executive body of ANAS HTP in the period between the General Assemblies within the limits of its authorities. The Board consists of five members. Members of the Board are appointed by ANAS for the 5 years period. They may be re-appointed. The head of the executive body cannot be a member of the Board. The Chairman of the Board is elected by the members of the Board from among the members of the Board. The Chairman of the Board presides over the Board. Board meetings may be regular or extraordinary. The next meetings of the Board shall be convened by the Chairman of the Board at least once every three months. Extraordinary (emergency) meetings of the Board are convened by the Chairman with the request of the members of the Board, and new members are appointed within 30 (thirty) calendar days if the number of members is reduced to half. Decisions of the Board shall be taken by a simple majority of votes, with each member having one vote. Members are not allowed to abstain from voting. In the case of a tie, the vote of the Chairman of the Board shall be decisive.

(MANAGING ORGANIZATION) The organization, management and development of the activities of ANAS HTP is carried out by the Managing Organization. The Managing Organization of ANAS HTP is the state commercial legal entity “High Technology Park” (HTP) which organizes, manages and develops its activities. To ensure more efficient management of ANAS HTP, the operator can be hired on a competitive basis by decision of the competent authority. The Managing Organization and operator are responsible for the operation of ANAS HTP. The main activity of the operator is to perform the functions according to the agreement with ANAS HTP.

In addition, the Managing Organization assists its residents in complying with the legislation, coordinates their interaction with state and local self-government bodies, develops public-private partnership in the relevant field, maintains a register and certifies them in accordance with the Cabinet of Ministers, organizes the interaction of science and industry in the field of high technology and innovations, maintains and develops the scientific and technical potential of HTP, supports the role of innovative technologies in

the economy, as well as increases the state's information resources in this area.

The Managing Organization organizes the analysis of new scientific achievements on high technology and innovations at higher education institutions, as well as other research institutions and the application of these results in ANAS HTP.

- **(Organizational Structure, Departmentalization, Units) As of 2020, major tasks of each unit. job descriptions, job specifications of each staff, etc.**

The Director is responsible person for the results of all activities of the ANAS HTP. On behalf of ANAS HTP, he represents the organization in relation to third parties, including in court, prepares documents for approval of the General Assembly, and signs official documents.

The Deputy-Director for Investment, commercial and production oversees the attraction of investments to ANAS HTP, the implementation of commercial activities of the park, and the organization of effective work with the production process carried out by residents.

The Deputy-Director for General and Strategic Development is responsible for the organization and maintenance of HTP's infrastructure, as well as the implementation of strategic goals. The Deputy-Director for General and Strategic Development oversees the implementation of international projects, including the establishment of cooperation with foreign partners and the implementation of international events.

The Legal Support Department organizes and implements the preparation of proposals to improve the legal framework of HTP, reviews and comments on draft normative legal acts and other normative documents, and controls the systematization of HTP's activities.

The Special Department provides the necessary official correspondence with law enforcement agencies and ensures the registration and constant control of confidential documents.

The Department of Financial and Economic Analysis is a structural unit that provides financial and economic coordination, accounting and reporting, tariff and investment

policy, planning and conducting procurement, as well as control over salaries.

The Maintenance Department controls the procurement, supply and use of goods (works, services) required for HTP, as well as ensuring labor protection. It also takes measures to ensure their legal use and documentation of property. This department provides orders for HTP, organizes the delivery of goods, devices, equipment and other inventory in the prescribed manner.

The Department of Patents, Certification, Marketing and Business Incubation systematizes patents obtained on the basis of inventions of ANAS scientists, creates a patent bank, provides written confirmation of compliance of products produced by ANAS HTP residents with official requirements, supports new business of entrepreneurs wishing to cooperate with ANAS HTP. The department also provides identification of consumer needs for ANAS HTP exchange activities, preparation of products in accordance with these needs, marketing, warehousing, transportation, evaluation of products, organization of pre-sales and after-sales service, as well as advertising activities

The Strategic Scientific Analytical Testing Center of ANAS operates as a unique laboratory in the territory of ANAS HTP, which has no analogues in the Caspian littoral states. The innovative laboratory, created with a multifaceted scientific approach, is internationally accredited and provided with advanced equipment. According to the existing resource balance in the laboratory, especially gas, oil and oil products are accurately tested.

In addition to the structure of the HTP, the Internal Auditor operates under the supervision of ANAS for the organization of audit work and carries out continuous control over the effectiveness of the internal control and risk management systems in cooperation with the leadership of the HTP. Standards and rules of the internal auditor's activity are based on the Law of the Azerbaijan Republic "On Internal Audit", normative documents of ANAS, relevant normative acts of the Central Bank and international internal audit standards.

- **How the structure of ANAS HTP needs to be changed in terms of governance, organization chart, tasks, job descriptions, etc.in 10 years (Year 2030)?**

The formation of the structure of ANAS HTP can be considered optimal. Because the activities of the organization are largely reflected in the structure. But some shortcomings occur along the work process. So, if the activity on international relations is in the format of a separate department, the strategy of international relations of ANAS HTP will be effective. A team consisted of 4-5 persons can ensure the implementation of international projects in groups, as well as the attraction of foreign residents, the exchange of experience and other activities with high quality.

It would be better to combine the legal and human resources issues under the name of the Legal Support and HR Department. Because personnel issues are mostly regulated by legal acts, and it is expedient to unify these two areas.

The Department of Patents, Certification, Marketing and Business Incubation combines four areas. Since each of these areas is a separate broad activity, it is necessary to systematize this department as two separate departments.

- **The volume of total budget of ANAS HTP as of 2020 (U.S. dollars?)**

The total budget of ANAS HTP is about 500,000 AZN (Approximately USD 0.3 million).

- **The sources of budget (from where?, portion?)**

The organization of ANAS HTP's activity is financed from the revenues of the Managing Organization and other non-prohibited sources. The Managing Organization creates conditions for funds, government agencies, foreign investors and international financial institutions to attract extra-budgetary funds. According to the legislation of the Republic of Azerbaijan, on the basis of an agreement signed with the residents of ANAS HTP, a certain amount is paid monthly to ANAS HTP from the total income from innovative projects implemented by them.

- **The usages of budget (to which activities? Portion?)**

The budget formed from these sources is used to maintain the infrastructure of ANAS HTP,

staff salaries, household expenses and other expenses.

- **The sufficiency of the current budget? Enough or not?**

It can be considered that the current budget, formed from the expenses of the above-mentioned sources, is partially sufficient for the maintenance of the infrastructure of ANAS HTP, staff salaries, household expenses and other expenses. However, it should be noted that one of the main tasks is to provide additional income by providing ANAS with extra-budgetary funds. From this point of view, it is impossible to say that the budget is sufficient in general, as the predicted results have not been achieved yet.

- **How can you mobilize the funds (or budget) for running ANAS HTP? How can you sustain ANAS HTP in terms of budget? Which funding sources are available in 10 years (Year 2030)?**

I think that, in order to optimize the budget of ANAS HTP during the next 10 years, the number of residents should be increased, potential donors should be attracted, and the activities of ANAS HTP's own production facilities should be strengthened. It should be noted that high-tech products create high prices. This, in turn, means earnings, employment, high incomes for scientists, increasing the living standards of scientists and the population, and thus enriching the budget. In addition, special attention should be paid to international grants and projects, through which the ANAS HTP budget should be enriched with additional revenues. In a globalizing world, there are sample opportunities to join and benefit from projects. The above-mentioned issues can be direct proposals that ensure the sustainability of the budget.

- **Who or which institutions gain power to distribute budget to ANAS HTP?**

The process of budget distribution in ANAS HTP is an internal matter of the organization and is carried out by the Department of Finance and Economic Analysis. The distribution of the budget is carried out in two simultaneous and closely interrelated stages: (1) Creation of budget revenues, (2) Use of budget funds. It is envisaged that all expenditures provided for in the use of budget revenues will be covered from the budget.

- **Total number of staffs (including director) as of 2020, Number of staff within each unit (or department), Academic backgrounds of each staff (majors, degrees, from which**

Currently, 40 employees work at ANAS HTP. Approximately, 5 employees operate at each department. The director is academician Ali Abbasov, doctor of technical sciences and professor. He is a well-known scientist in the field of computer science. He is the author of more than 200 scientific works. Besides, Mammad Bagirzadeh, Deputy Director for General and Strategic Development, is a PhD in Law and Arzu Alizadeh, director of the Strategic Scientific Analytical Testing Center, is a PhD in chemistry. Other employees do not have academic degrees. The main part of the workforce is technical staff.

- **Is the workforce enough to carry out the current works? How many staffs need to be more supplemented in 10 years? How do you fill the shortage of workforce?**

Looking at the current structure of ANAS HTP, we can observe that the number of existing employees is sufficient. The number of employees can be increased if additional departments are created. It is estimated that the number of employees could increase from 40 to 60 during next 10 years.

- **Who is an important person in allocating workforce for ANAS HTP? What efforts need to be made to be newly assigned?**

The director appoints, reassigns, dismisses, and orders incentives and penalties for the ANAS HTP employees. The Director, his Deputies, the Chief Accountant and the Internal Auditor of HTP are appointed and dismissed by ANAS. Heads of departments of ANAS HTP, as well as other employees are appointed and dismissed by the Director. If there is a need for vacancy for any position in HTP, a competition is announced through the website of ANAS and is carried out transparently. The appropriate candidates are appointed for different positions.

- **Cooperation/network activities between ANAS HTP and 4 scientific divisions of ANAS (e.g., Physical Math & Tech Science, Chemical Science, Earth Science, Biological & Medical Science, etc.)**

The science and technology focus of ANAS HTP is directed to biotechnology, nanotechnology, petrochemicals, robotics, agriculture, ICT-Software development, metallurgy, mechanics and health sciences. This focus should be modified, either reduced or expanded depending on certain factors. These factors are demand, availability of resources, profitability and capacity. Based on the above-mentioned factors, ANAS HTP closely cooperates with the Scientific Departments of ANAS. In order to regularly commercialize scientific ideas based on the proposals of the institutes, some projects are submitted and their application opportunities are evaluated. Information on patents obtained by scientists representing various scientific departments are submitted to ANAS HTP and are systematized on this basis of these documents. ANAS HTP interacts with four main departments of ANAS (Department of Physics, Mathematics and Technical Sciences, Department of Chemical Sciences, Department of Earth Sciences and Department of Biological and Medical Sciences).

- **Cooperation/network activities between ANAS HTP and the other Azerbaijani innovation actors (e.g., academies, industries, research institutes)**

ANAS HTP closely cooperates with universities, research centers and companies operating in Azerbaijan. For joint implementation of innovative projects, granting the status of "resident of ANAS," implementation of aggressive marketing policy, organization of procurement of scientific, competitive and innovative products, industrial mineral and lubricants produced by HTP residents in state enterprises, application of new technologies in agriculture and for implementation of other purposes, the cooperation network was established with SOCAR Baku Oil Refinery Plant, Institute of Petrochemical Processes of ANAS, "Smart Home" LLC, Western Caspian University, "Azerishig" OJSC, AzScienceNet science computer network of ANAS Institute of Information Technologies, Ministry of Ecology and Natural Resources, Ministry of Agriculture, Agroleasing Open Joint-Stock Company, Baku Stock Exchange, KUTVAN LLC and Baku Engineering University. Additionally, in order to support the development and procurement of innovative

products in ANAS HTP, the Agreements have been signed with Azerbaijan Technical University, Economic Reforms Research Institute of the Ministry of Economy of the Republic of Azerbaijan, Azerbaijan State University of Oil and Industry, and EPC Group LLC.

- **Cooperation/network activities between ANAS HTP and the international innovation actors (e.g., academies, industries, research institutes in various countries including South Korea)**

Currently, ANAS HTP cooperates with South Korea Science and Technology Policy Institute (STEPI), International Association of Science Parks and Areas of Innovation (IASP) Center for Analysis of Economic Reforms and Communication, South Korea Innopolis Foundation, European Union TAIEX Program, Islamic Corporation for the Development of the Private Sector, Moscow Center for Structural Transformation of Industry, Saint Petersburg Technopark, EPC Group of Companies, China-Russian Innovation Technopark, Anzali-Astara Technopark of Iran, Azerbaijan Office of the South Korea International Cooperation Agency (KOICA), SGS (Switzerland), BP, ACTIVA of Spain, General Robotics of Israel⁶⁸, Egyptian Company CHIVY, British Company Micro Focus & HPE Software, Czech Institute of Experimental and Applied Physics and other organizations⁶⁹. The scale of these relationships is expanding every year. The Memorandums of Understanding were also signed with China's Ningbo State Technology and Industry Company Hua Pei and with the Islamic Development Bank's ICD.⁷⁰

- **The cooperation activities between ANAS HTP and 4 scientific divisions of ANAS are enough? If not, how can you facilitate the network activities? In which cooperation areas?**

We consider that, despite a certain level of cooperation between the scientific departments of ANAS and HTP, it could not be appreciated sufficiently. Cooperation between the scientific departments and ANAS HTP, especially in the field of physics and chemistry, should be systematic. This network of cooperation can be established as the

⁶⁸ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/9223>

⁶⁹ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/7623>

⁷⁰ Azerbaijan National Academy of Sciences, <http://science.gov.az/news/open/11458>

organization of joint laboratories, experiments, and other forms, and there is a great demand for this activity.

- **The cooperation activities between ANAS HTP and Residents with ANAS HTP are enough? If not how can you facilitate the network activities? In which cooperation areas?**

Taking into consideration the existing infrastructure, it can be mentioned that, ANAS HTP can increase the quantity of residents. The enterprise is ready to accept any entrepreneurs and investors interested in the production of export-oriented, import-substituting and competitive products. Most importantly, production must be based on scientific results and innovations in this area. In addition, cooperation with residents should be aimed at creating favorable economic, legal and organizational conditions for innovation, developing and effectively deploying scientific and technological potential, and increasing the role of science and technology in economic development.

- **The cooperation activities between ANAS HTP and the other Azerbaijani innovation actors are enough? If not how can you enhance the network activities? In which cooperation areas?**

The cooperation between ANAS HTP and other innovation actors can be considered comprehensive. During last 4 years, many meetings were held, a number of agreements were signed, and, as the result of this activity, international cooperation gradually increased. However, this interaction needs to be renewed and expanded. According to the studies, in order to use the scientific, entrepreneurial and innovative potential in Azerbaijan, to strengthen the transfer and commercialization of innovation, research and technology, the cooperation of all actors in the ecosystem must be promoted to a higher level. This could be done through trainings, exchange of experiences, webinars, round tables, discussion of the package of proposals, and other formats.

- **The cooperation/network activities between ANAS HTP and the international innovation actors (countries including South Korea) are enough? If not, how can you strengthen network activities? In which areas?**

Though HTP's international cooperation is currently at a certain level, it cannot be

considered as sufficient. However, we are matining a very close ties with STEPI and this cooperation has continued since 2017. Regarding the development of cooperation with other countries, we can note that if the active work is carried out for the organization of international cooperation in high technology, the expansion and development of HTP would proceed in accordance with the priority directions of modern world technologies. The use of world experience, competition of science-intensive high-tech products in the international arena, the study of experiences and challenges arising in the knowledge-based, innovative development of developed countries, and the usage of them in the implementation of the development strategy of HTP, as well as cooperation with foreign partners, foreign technoparks and their associations would make it possible to to create a successful international cooperation strategy

1.6.2 Technology Transfer & Commercialization

- **TT & TC related unit/department/team within ANAS HTP?**

Yes, there is a team in ANAS HTP for TT & TC operations under the partnership with coordinators of the Presidium of ANAS and the Institutes of ANAS. In the previous years, ANAS had the Patent Department. After restructuring, this department has been canceled and its functions have been assigned to ANAS HTP. During the first year after the restructuring, ANAS HTP centralized the patents information of all ANAS Institutes and coordinated the functions of the previous department of ANAS HTP; Patent, Certification, Marketing and Business Incubation. Today, as result of the restructuring, a new department dedicated to TT & TC is planned to be set up with 3-5 people.

- **Volume of budget per year, number of concerned staffs/personnel related TT& TC?**

TT & TC activities are being operated by three staffs, but there is no dedicated budget for operations. The planned budget is only salary for these staffs. The average salary per person is 500 AZN per month and the annual budget is over 20,000 AZN.

- **Major activities relating TT & TC within ANAS HTP?**

ANAS HTP has centralized data on existing IP files of ANAS and, during the business meetings, it's ideal resources for discussing IP licensing and tech based venture business building. ANAS has established partnership initiatives with ministries (Agriculture, Defense, Economy) to pursue collaboration for innovative products and solutions. In this perspective, the everyday collaboration among the innovation ecosystem stakeholders is improving and it has also enhanced the demand for TT & TC activities as well. Also in the upcoming years, after dozens of economic reforms, there is an expectation of foreign investment into the country and with this capital, industries will modernize and the demand for new technologies and know-how will increase a lot. ANAS HTP TT & TC is expected to play such an important role in this operations. Today, ANAS HTP TT & TC is kind of training center for this field. Its business results are not satisfactory due to the current situation in the economy. Key activities include capacity building programs, intensive workshops, boot camps, and corporate networking events.

- **Level of capacities relating TT & TC of ANAS HTP in comparison with the other TT & TC agencies in Azerbaijan?**

ANAS has more than 40 institutions and organizations related with science. Also there are more than 360 labs under the management of ANAS. So, from the capacity points of view it is the largest organization in Azerbaijan to produce and deliver technology and conduct TT & TC. There is also Innovation Agency in Azerbaijan under the Ministry of Transportation, Communication and High Technologies, which also has a function for TT & TC. But ANAS HTP is a bit far from industry. It has strong background in fundamental sciences, but securing practicality is one of the main challenges of the whole science commercialization operations of the country. That's why we see patents and inventions but few of them can benefit the industry. In vice versa, industry also is not very active to do sponsored researches with ANAS to develop new technologies. Generally, we can say from the operational point of view, ANAS HTP is being operated, but it needs to be more productive to build innovation driven industries and economy.

- **How many TT & TC deals/matching activities (including Projects or Programs) have been carried out between two institutions over the last 10 years? Which items (which research results)?**

One of the residents of ANAS HTP is ANAS Industrial Plant (TSZ) and it produces petrochemical products based on the research conducted by the ANAS Institute of Petrochemical Processes. We can consider this as a successful resident, a major outcome of all TT & TC activities. Overall TT & TC based discussions in ANAS have been based on its existing 457 patents in recent years. In agriculture, the ANAS Institute of Soil Science and Agro Chemistry is the main technology transfer body, while ANAS STP is the key technology transfer body for the defense field and EPC for the mining field. Generally there are nearly 5-10 successful programs that effected TT & TC activities with major outcomes like resident companies.

- **What is the amount of TT & TC deals (in the U.S. dollars) that have been traded (per activity or in total over the last 10 years)?**

Main activities are related with the ANAS Industrial Plant (TSZ). Its turnover is more than USD 500, 000.

- **What is the number of patents developed in the last three years by the Scientific Divisions by the major field?**

Based on the most recent data, we have 457 (total) patent developed till 2017 in our patent database.

- **What is the number of TT and TC cases , whose technologies were developed within the Scientific Divisions itself and transferred to the industrial sector over the last three years?**

The main activities are related with the ANAS Industrial Plant (TSZ). In agriculture, the ANAS Institute of Soil Science and Agro Chemistry is the main technology transfer body, while ANAS STP is the key technology transfer body for the defense field and EPC for the mining field. Generally there are nearly 5-10 successful programs that effected TT & TC activities with major outcomes like resident companies.

- **How many TT & TC deals/matching activities (including Projects or Programs) have been carried out between two institutions over the last 10 years? Which items (which technologies)?**

The main activities are related with the ANAS Industrial Plant (TSZ) and the current status is very good. It's true that the brainstorming sessions are participated by experts on both sides and companies and the Institute of ANAS. But, the product line has been continuously defined within the existing features. Currently, there are no new upgrades of their products. It can be summarized that there are strong partnership links and new transfers will be done based on new demands in the market.

- **How many TT & TC deals/matching activities have been carried out between two institutions over the last 10 years? Which items (which technologies)?**

Especially in the field of agriculture, the ANAS Institute of Soil Science and Agro Chemistry works with the Ministry's Agriculture Research and Development Centers. They do joint researches and try to improve the ecological conditions of farming fields for better productivity in producing agricultural products. ANAS STP, DN Technologies works with Ministry of Defense in quad chapters field. Also, they develop simulation programs for military trainings.

- **What is the amount of TT & TC deals (in U.S. dollars) that have been traded (per activity or in total over the last 10 years)?**

It is estimated at around USD 3 million.

- **How many TT & TC deals/matching activities have been carried out between two institutions over the last 10 years? Which items (which technologies)?**

Millers Oils UK in the field of petrochemicals and ANAS organizations hold TT & TC related meetings and brainstorming sessions with their teams. Also, ANAS has relations with the regional organization of the Academy of Sciences and key operations are done based on mutual research projects in petrochemicals and agriculture.

The ANAS Institute of Information Technologies conducts researches for wireless control systems and its key implementation targets are oil production and agriculture. They have

partnership with the international corporations like BP and the local stakeholders like SOCAR.

- **What is the amount of TT & TC deals (in U.S. dollars) that have been traded (per activity or in total over the last 10 years)?**

It is nearly USD 5 –10 million.

- **Are there any research outcomes or performances in four scientific divisions within ANAS that can be transferred or commercialized? If yes, what should be done to promote technology transfer or commercialization by ANAS HTP in ten years? In which items or research results?**

ANAS has competitive potential in petrochemicals and agriculture and of course the implementation of new technologies to those fields are very crucial for ANAS to be successful. ANAS has staffs to implement this. But it needs more of younger, energetic, and creative researchers. The current situation in Azerbaijan is not good in terms of researchers' average age. Oftentimes, researchers are quite old and even their number is limited. If we want to make science wave, we need more skillful talents. Also the funding sources of research are limited and the private sector is not active in conducting sponsored researches. And lastly, for the global links, partnerships should be further promoted to enhance the transfer of knowledge, technology, and resources among the global players.

- **In case of resident companies within ANAS HTP (e.g., Millers Oil, DN Technologies, etc.) what should be done to strengthen technology transfer or commercialization by ANAS HTP in ten years? With which companies? In which technologies?**

Residents are active in the B2G business model and their main stakeholders are government agencies. For better innovations, they also should enter into the private sector in the country and as well as in the regional tenders. With limited demand from the government, it's not easy to make companies more innovative and it is necessary to create and promote the demand for TT & TC. In summary, it is necessary to promote competition, innovation, and demand for TT & TC.

- **Regarding with various Azerbaijani innovation actors (e.g., academies, industries, research institutes), what should be done to strengthen technology transfer or commercialization by ANAS HTP in ten years?**

First of all, the country should define its priority industries and should support the R&D activities. Now, we see this in mainly agriculture, but then it should continue with financial sector, healthcare, defense and etc. After clarification of the sectors, all innovation ecosystem actors should work under the scope of one main governmental strategy. At the current time, we see many duplications, overload in the ecosystem members and it's true, it's getting better. But it takes time as well to be more systematic in innovation related operations of country. Nowadays, we see nearly in all governmental agencies, they have team or body which is related with innovations and it's good news. Also, the Innovation Agency of Azerbaijan has a function like "innovation audit" and we do believe to its real innovation demand making potential in the future after it will be obligatory for organizations to do innovation audit like financial one.

- **How about with international innovation actors? What should be done to facilitate technology transfer or commercialization by ANAS HTP in ten years? With which organizations? In which technologies?**

Energy, agriculture, tourism, ICT are the main fields of economy that are priority areas and demand generators for new technologies by local and innovation actors. In this perspective, Asian TT & TC Centers are planned target of partnership in upcoming years due to business and cultural similarities. Also, global corporations are the main partnership bodies to enable cluster systems in the Azerbaijan Innovation Ecosystem.

- **What are the challenges that need to be solved for ANAS HTP to strengthen capacities regarding TT & TC activities?**

Limited sponsored research programs, limited talents, industry's low level of interest to make transfer available, and increasing competition in the global market.

- **Is there a legal basis for the construction of an industrial or scientific complex? If yes, are there incentives, such as tax cuts, for companies in industrial complexes?**

Yes, there are the Free Economic Zones and the Industrial Development Centers. For technology field, there are TPs. But from the clustering point of view, there is long way to make it more sustainable and more global. Also, becoming global corporations requires know-how, knowledge transfer, skillful talent development and long term ecosystem maturation.

- **Do you have plans to increase the number of incumbent companies or to create new industrial complexes?**

Yes, we mentioned earlier that Azerbaijan is in the economic transformation stage from a resource-driven economy to a knowledge-driven economy. To become a knowledge economy, we need to fulfill lots of requirements including some structural transformation. Azerbaijan has done some programs and projects in this field on a daily basis. Each year, the reform and transformation are being accelerated and the diversification of economy is increasing with the high performance of ICT industry. In this perspective, there is need for increasing the number of incumbent companies and creating new industrial complexes. Because the largest company of Azerbaijan is SOCAR and its annual revenue is -120 billion AZN. There is no such company in Azerbaijan, whose revenue is in a similar scale of the billions. That's why there is a long way to replace this industry with other more value added alternatives.

1.6.3 Public Procurement

- **Other than HTP, are there any other organizations in Azerbaijan that conduct public procurement like South Korea's Public Procurement Service? I would like to know if Azerbaijan has adopted a central or distributed procurement system.**

The System is not a centralized one but it covers more than 800 procuring entities. A procuring entity is a state enterprise and organization (administration), having 30 or more percent of state share in its charter fund, for purchasing goods (works and services) and in

charge of: Determining needs; Determining estimated price; Establishing its Tender Committee; Preparing tender documents; Publishing tender announcements and notices; Accepting tender proposals; and Carrying out evaluation and awarding the winner

- **The report of Year 1 project indicates that HTP has an authority to publicly purchase G2B items. Could you explain about the authority specifically?-Also, I would like to know if HTP has a function and authority to issue public tender and select companies for public procurement like South Korea's Public Procurement Service or the Defense Acquisition Program Administration. Or does HTP purchase products by itself?**

ANAS HTP also has to buy goods and services via public procurement. The rules related with procurement are same with the general approach. Public procurement exceeding 50,000 AZN have to proceed with the tender method while those in the range of 5 000 AZN – 50,000 AZN can proceed via the method of request for quotations. ANAS HTP does its public procurement by itself to buy goods and services.

- **What kind of products do you want to purchase? You can give us examples. Are you interested in purchasing products of residents of HTP? What benefits does HTP expect to get from the public procurement?**

Mainly, office materials, work items and technical products and services for the management of the facility (22 ha area). Yes, HTP is interested in purchasing products in oil and chemicals for its transport vehicles from AMEA TSZ and Millers Oil.

- **What are the main products produced by HTP residents? Please explain by category such as defense products, chemical materials, commercial items like chairs and desks. Also please let us know your residents specifically**
 - Defense products (ETP)
 - Chemical materials (Millers Oil, AMEA TSZ)
 - Electronics & Hardware (ETP)
 - Software (DN Technologies, 4Maps)

- **I assume there must be some demand agencies who would be interested if HTP is in charge of issuing public tender and contract like South Korea's Public Procurement Service. Could you explain about demand agencies you have and how you figure out demand?**

When there are needs for new goods and services, all state companies announce their needs on their website. Also, there is a centralized platform for consolidating all tender related information in one place, the online public procurement platform. So, each organization coordinates their tenders by themselves.

- **When selecting a resident company, do you consider its demand or suggestions regarding public procurement? I believe many of HTP residents are high-tech companies. In that case, I believe there would not be much demand for these companies' products in the public procurement market. If so, it might be necessary to provide innovative procurement in the form of R&D-linked way. It means purchase is available when a demand agency develops products. This question intends to inform that HTP residents need to develop technologies and products that public demand agencies require for the innovative procurement.**

Yes, it's one of the main points that we try to take into account. At this time, many of the existing ANAS HTP resident companies are not high tech companies and they mainly work in B2G. With the formalization of "extraterritorial acceptance" to technopark, ANAS HTP expects many high tech companies would be its residents in the future. But, also, if the company can indicate its potential success in the private market, it's okay. There will be good opportunities for them firstly in the private market. Later, when the demand for high tech products increases within the government, they will be very strong players in the market to supply this demand.

- **Are you funding R&D subsidy for your residents? If so, could you share with us the amount of funding per project and the number of projects.**

Currently, no. But we plan to do this via the Innovation Fund in the upcoming years.

- **Are there any regulations on public procurement at the national level? Also, are there any public procurement projects that HTP has implemented?**

ANAS HTP has been established on the basis of the ANAS Industrial Plant and it has enough resources to operate the current area with limited residents. That's why it has no massive procurement projects, whose value is more than 50 000 AZN. But our residents are participating in the tenders for SOCAR, AZAL, and Azerbaijan Railways companies.

1.6.4 Start-up & Incubation

- **The evolution process of Incubation Center and Start-up in Azerbaijan**

Azerbaijan is a country with 10 million population and over USD 40 billion GDP. It is one of the lead developing nations in the Caucasus region. The country's economy is boosted by natural resources and a significant part of its GDP comes from outputs of resource based economy. The transformation from resource-based economy to knowledge-based one was initiated in the early 2010s. In 2012, the first High-Tech Park of Azerbaijan was established and special support for startups started. Today, the country's startup ecosystem is developing rapidly and we can see the current status of ecosystem via the list given below (information provided by the Ministry of Transport, Communication, High-Technologies, MTHC): There are nearly 20 business incubation centers in Azerbaijan and 5 of them are located in the local provinces of the country. Half of these incubators are supported by the government. The rest of them are being operated mainly as part of the corporate social responsibility projects of big corporations. In recent years, hundreds of startups have received support from these incubators and they have produced some successful outputs. Generally, these incubators made fruitful outcomes in terms of increased awareness of tech entrepreneurship. In addition, many qualified experts have been developed through these entrepreneurial support centers and their programs.

- **Supporting Service of Incubation Center and Start-up in AZ**

In Azerbaijan, incubation centers are active in training and development programs. They transform university students to entrepreneurs with their intensive boot camps,

workshops and face-to-face networking events. The financial support is provided through grant programs like the contest, startup competitions and Youth Foundation, and Innovation Agency. Also, there are some labs, from whom entrepreneurs receive technical support to build their products and solutions. Some of these labs are listed below (Info - MTHC): In Azerbaijan, a majority of business incubation centers has amazing workspaces and co-working centers. Kitchen, phone center, entertainment center, meeting rooms, high – speed Internet are general services provided to their resident companies. For incubators, government has no support package yet, but discussions are underway regarding how to increase the interest of other civil and private organizations in the incubation centers through government support for the incubation ecosystem with their services and solutions. So, government's financial or other technical support packages are expected to be prepared in the upcoming years. Also, business incubation centers have a wide range of mentor pools from technology to business experts and from academicians to government officers. Currently, there is no centralized online tools for mentor matchmaking. By the end of this year, however, the government plans to build a centralized information database of the incubation ecosystem in Azerbaijan.

- **Certificate of Incubation Center**

In the current time, there is specific resident certificate issued to residents of incubation centers of ANAS HTP, which is a general phenomenon for other incubation centers in Azerbaijan. But government agencies are considering to provide one-stop service for startups at incubation centers by issuing a startup passport and startup certificate for them. At the same time, by the end of this year, a centralized database for innovation projects will be established in Azerbaijan. Once registered with this database, startups can enter their company profile including projects into the database and they can get their startup passports. Using this startup passport, companies can receive government support.

The competitiveness program for the residents of incubation center will be based on practicality approach. General curriculum can be formalized into three modules: (1)Technical skills development, (2)Business skills development, (3)Communication skills development. The planned tools for workshops are Business Model Canvas (BMC), Value

Proposition Canvas (VPC), and Lean Canvas Incubation Program (LC). The period of the program is for 3 months. One training session based on each module of the curriculum will be organized per week.

With its extensive mentor pool from Institutes of ANAS (2000+ young researcher) and business sectors of economy of country, a good guidance can be provided for startups. In every two weeks, a demo presentation will be made on the current status of development. At the end of each month, there will be a rehearsal of the demo day. At the end of the incubation program, there will be a demo day with pitch deck on the startup ecosystem in Azerbaijan, where startups are introduced. In addition, networking will be provided for the stakeholders of the ecosystem for venture matching. Each batch of incubation program will be open for 8 startup projects. Key requirements for applying for this program is that the applicant should be a group of at least 2 people with ideas in MVP (Minimal Viable Product) stage. In a year, there will be 4 batches with a total of 32 startup projects. An annual target of ANAS HTP BIC is to convert at least 8 startups to technopark residents.

- **Start-up & Incubation related unit/department/team within ANAS HTP?**

There is a team dedicated to startup & incubation operations.

- **Volume of budget per year, the number of concerned staffs/personnel related Start-up & Incubation?**

We have 3 persons with an annual budget of 20,000 AZN

- **Level of capacities relating Start-up & Incubation of ANAS HTP in comparison with the other Start-up & Incubation agencies/organizations in Azerbaijan?**

In terms of physical resources, it's one of the top incubation centers. But from the capital accessibility point of view, it's not well positioned. The given pictures below is the space of the Business Incubation Center. It is located in the academic city of Baku, close to 3 main universities of Azerbaijan: Baku State University, Azerbaijan Technical University, and Azerbaijan University of Architecture and Construction

1.6.5 Test & Certification

- **Test & Certification related unit/department/center within ANAS HTP?**

The ANAS Strategic Scientific Analytic Testing Center is the key body of ANAS HTP for Testing & Certification activities.

- **Volume of budget per year, number of concerned staffs/personnel related to Test & Certification?**

The annual budget is around 30,000 - 50,00 AZN, which is basically the salary of staffs. The average salary is in the range of 500 AZN- 1,000 AZN.

- **Level of capacities related to Test & Certification of ANAS HTP in comparison with other Test & Certification organizations in Azerbaijan?**

ANAS HTP is related in this field via the ANAS Strategic Scientific Analytic Testing Center. The operation of this center is coordinated by government bodies.

- **Does ANAS have legal basis for conformity assessment in itself?**

The ANAS Strategic Scientific Analytic Testing Center has this rights based on AZS ISO/IEC 17025:2009, for 21/01/2019-21/01/2022 period. Accreditation No: AZ 01.0382.01.19

- **What is the ANAS conformity assessment system? In other words, who are the TB (Test Body) or CB (Certification Body) operating entities, TB and CB accreditation bodies, or TB and CB regulatory authorities?**

The ANAS Strategic Scientific Analytic Testing Center is a operating body of the US (ASTM), Britain (IP), Germany (DIN), Russia (QOST), ISO, EN, and AZS and it provides solutions to public and private entities.

- **Are there technical regulations (or Standards) that should be considered in the conformity assessment process?**

US (ASTM), Britain (IP), Germany (DIN), Russia (QOST), ISO, EN, AZS

- **Are there separate technical regulations or standards development authorities?**

No.

- **Is there a separate voluntary certification scheme at the private level?**

Yes. The ANAS Strategic Scientific Analytic Testing Center is a key partner of public and private organizations and it has collaboration opportunities with other stakeholders. But its key testing fields are petrochemicals and toxicology.

- **How many test & accreditation services have been provided for customer/users over the last 10 years? Which items (which research areas)?**

It has been only a year since the ANAS Strategic Scientific Analytic Testing Center started its operations, so it has done few successful projects. But it has plan to expand its customer base through public-private partnership.

- **How much amount of money relating test & accreditation services (in U.S. dollars) has been created (per activity or in total over the last 10 years)?**

The ANAS Strategic Scientific Analytic Testing Center generates profits only sufficient to cover its operations. Its annual labor cost is around 30,000 AZN - 50,000 AZN.

- **Do you think there are need/demands for Certification services of ANAS HTP from the various customers (from universities or companies, etc.) in Azerbaijan?**

The ANAS Strategic Scientific Analytic Testing Center can be a key partner of the local ecosystem. We see the demand for this organization.

- **What are the challenges that need to be solved for ANAS HTP to facilitate its Test & Certification activities in ten years?**

Its technical resources should be further developed.

- **For example, is there a separate national level conformity assessment system such as the US FCC, European CE, CCC in China, and KC in South Korea? If yes, what is that?**

Though ANAS SSATC has the right to conduct conformity assessment, it is not the key

organization in charge of that. A conformity assessment determines whether goods and methods of their manufacturing meet relevant technical standards and fulfill relevant requirements set out by technical laws. The assessment is procured by a manufacturer prior to releasing its goods into the market. The assessment shall be carried out by the relevant accredited entities through: (i) certification and (ii) declaration of conformity. The high-risk products require mandatory certification and obtaining of conformity certificate while the goods categorized as medium- and low-risk require declaration of conformity approved by the conformity assessment bodies. Rules of issuance of certificates and declarations of conformity will be determined by the Cabinet of Ministers and information about the issued conformity documents will be entered into a special register accessible online. Foreign conformity documents will be recognized subject to their validity and correspondence to the terms of international agreements that Azerbaijan is a party to. Producers located abroad will have a right to pass the conformity assessment of their goods through their representatives in Azerbaijan.

Products, subject to a technical regulation control, unassessed or having failed to pass the conformity assessment, shall be restricted from being manufactured, imported, sold, or used in Azerbaijan. The Law refers to the criminal, administrative and civil liability of the persons who fail to comply with it.

- **What level of testing and certification body would you like to build? As an example, should the testing body establish an ISO / IEC 17025 quality management system and obtain accreditation from accreditation bodies? And also, for a certification body, should a quality management system based on ISO / IEC 17065 be established and accredited by the accreditation body?**

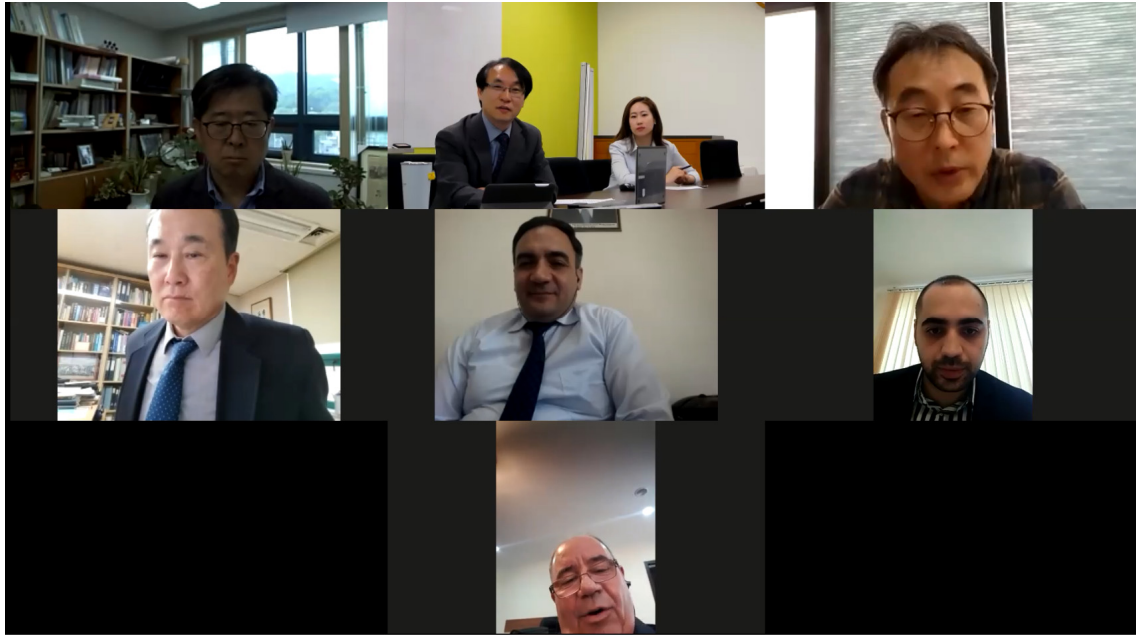
It depends on country. In Azerbaijan, it has been centralized.

- **As you know, the conformity assessment system differs from country to country, and also differs according to special laws at the national level. The question is, should the testing and certification bodies be designated and operated based on the law instead of accreditation by accreditation bodies?**

It depends on situations, of course. Operations based on the law can be acceptable.

1.7 Photos

Group Photo



Interview

Internal System

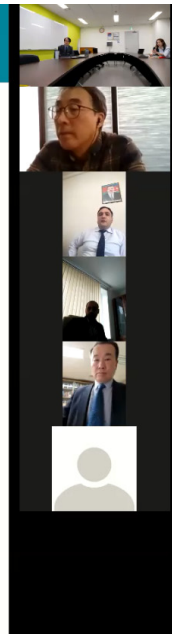
[Q1] Vision, Mission, Principles of ANAS-HTP

- Current vision, mission, mandates, strategies granted by ANAS?
- New director's vision, future plan?
- Recent environmental changes surrounding ANAS-HTP or ANAS? (SWOT)
- Given the changing environments, what should be a vision, mission, or principles in 10 years (Year 2030)?

Vision

Mission

Principles



2. STI Capacity Development Workshop

2.1 Background

As the first stage of the project, the South Korean researcher team specified the demand for the 2020 project through a kick-off workshop and field interview with the relevant experts. They also conducted fact finding on the status of the internal system, TT & TC, Start-up & Incubation, and Test & Certification of the ANAS HTP.

In order to enhance the STI capacity of ANAS HTP at an institutional level, ANAS HTP and STEPI analyzed the need for capacity development in the field of technology areas mentioned above. Azerbaijan's research team particularly made an emphasis to get South Korean experiences and skills for strategy set-up.

Reflecting demands and requests from ANAS HTP, STEPI prepared lectures on South Korea's TPs containing basic principles and strategies for the development of TPs. Furthermore, STEPI organized an in-depth interview session which took place between experts from STEPI team and professionals from ANAS HTP. It intended to diagnose the status of ANAS HTP and come up with strategies for the development of ANAS HTP.

2.2 Overview of the Capacity Development Workshop

- **Title:** 2020 K-Innovation ODA Program (Azerbaijan) Capacity Development Workshop Program
- **Dates:** September 3 (Thu.) – September 4 (Fri.), 2020
- **Venue:** Online Conference (Zoom)
- **Purpose:** In this invitational training, the capabilities of the exclusive organization are to be improved by providing the practical methodology for internal system, TT & TC, Test & Certification, and public procurement through a series of lectures as well as Q&A. Lectures on the experiences of South Korean TPs are also provided.

2.3 Attendees

STEPI Research Team (total of 6)

Name	Position/Organization	Role/Area of Expertise
Wangdon Kim	Chief Director, Division of Global Innovation Strategy, STEPI	Project management and planning Consultation and strategy establishment for the internal system of ANAS HTP
Jong Hwa Choi	Head, Office of Future Industry Research, STEPI	Consultation and strategy establishment for public procurement
Yejin Yoon	Researcher, Division of Global Innovation Strategy, STEPI	Research and project coordinating
Myungdo Oh	Professor, Department of Mechanical and Information Engineering, University of Seoul	Reviewing and advising overall project strategies
Jongin Choi	Professor, Department of Business Accounting, Hanbat National University	Consultation and strategy establishment for Start-up & Incubation
Sangho Choi	Head, Policy Planning Department, Daejeon Technopark	Consultation and strategy establishment for Technology Transfer & Commercialization, Test Certification

Lecturers (total of 4)

Name	Position/ Organization	Syllabus
Yang-jin Kim	Entrepreneurship Consultant, Gyeongbuk Technopark	Definition of South Korean Technopark South Korean Technopark as a legal entity Evolutionary Roadmap of South Korean Technopark The Case of GyeongBuk Technopark TP models in the future
Kyoung Joo Nam	Team Leader, Gyeonggi Technopark	Start-up money for starting South Korean TP Space funding by local government Leasing space for tenants Attraction of SME supportive programs Applying on ANAS HTP
Jong-Wha Lee	Associate Researcher, Daejeon Technopark	Business Incubation (BI) Business Incubator's Performance Business Incubation Performance Regional Innovation Platform The Triple Helix Model
Seong Byung-Ho	Director General, Daegu Technopark	Concept of Digital Transformation Digital Transformation: National level efforts Digital Transformation Cases Digital Transformation in the Daegu Technopark

Azerbaijan Research Team (total of 4)

Name	Position/Organization	Role/Area of Expertise
Ali Abbasov	Director, ANAS HTP	Project management
Bunyamin Seyidov,	Chief, Department of Relations with Eastern Countries, ANAS HTP	Consulting on Internal System of ANAS HTP
Ilaha Rzayeva	Head of Investment, Marketing and Commerce Department	Consulting on Test & Certification, Public Procurement of ANAS HTP
Vusal Suleymanli	Innovation Projects Coordinator, ANAS HTP	Consulting on Technology Transfer & Technology Commercialization

2.4 Schedule

September 3, 2020 (Lecture)

Time	Program	Speaker	Position
09:00-09:05 (AZ) 14:00-14:05 (KR)	Kick-off	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
09:05-09:10(AZ) 14:05-14:10(KR)	Opening Remarks	Myungdo Oh	Professor, University of Seoul
09:10-09:15(AZ) 14:10-14:15(KR)		Ilaha Rzayeva	Head of Investment, Marketing and Commerce Department, ANAS HTP
09:15-10:15(AZ) 14:15-15:15(KR)	Business model for Technopark	Yang-jin Kim	Entrepreneurship Consultant, Gyeongbuk Technopark
10:15-11:15(AZ) 15:15-16:15(KR)	Fundraising models for Technopark	Kyoung Joo Nam	Gyeonggi Technopark
11:15-12:15(AZ) 16:15-17:15(KR)	Resident hunting in Technopark	Jong-Wha Lee	Associate Researcher, Daejeon Technopark
12:15-13:15(AZ) 17:15-18:15(KR)	Digitalization of Technopark	Seong Byung-Ho	Director General, Daegu Technopark
13:15-13:20(AZ) 18:15-18:20(KR)	Wrap-up	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI

September 4, 2020 (Interview & Discussion)

Time	Program	Speaker	Position
09:30-09:35(AZ) 14:30-14:35(KR)	Kick-off	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
09:35-10:15(AZ) 14:35-15:15(KR)	Internal System	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
		Bunyamin Seyidov	Chief, Department of Relations with Eastern Countries, ANAS HTP
10:15-10:55(AZ) 15:15-15:55(KR)	Technology Transfer & Commercialization	Sangho Choi	Head, Policy Planning Department, Daejeon Technopark
		Vusal Suleymanli	Chief Innovation Officer, ANAS HTP
10:55-11:35(AZ) 15:55-16:35(KR)	Test & Accreditation	Sangho Choi	Head, Policy Planning Department, Daejeon Technopark
		Ilaha Rzayeva	Head of Investment, Marketing and Commerce Department at ANAS HTP
11:35-12:15(AZ) 16:35-17:15(KR)	Public Procurement	Jong Hwa Choi	Head, Office of Future Industry Research, STEPI
		Ilaha Rzayeva	Head of Investment, Marketing and Commerce Department at ANAS HTP
12:15-12:55(AZ) 17:15-17:55(KR)	Start-up & Incubation	Jongin Choi	Professor, Department of Business Accounting, Hanbat National University
		Vusal Suleymanli	Chief Innovation Officer, ANAS HTP
12:55-13:00(AZ) 17:55-18:00(KR)	Wrap-up	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI

2.5 Minutes

2.5.1 Business model for Technopark (Lecture)

- Definition of South Korean TP
 - Technopark (TP) is defined as a South Korean version of science and technology park (STP).
- The term of 'science and technology park' encompasses any kinds of high-tech clusters including technopark, research park, technology park, technopolis, science city, cyber park, hi-tech park, and research and development park.
- South Korean TP as a Legal Entity
 - TPs were established by the special law ("Act on Special Cases Concerning Support for Technoparks") in 1998.
 - Main tasks: Facilitator for an institutional systems of UIG, technology business incubator, building and providing infrastructure, training, and marketing
- Evolutionary Roadmap of South Korean TP
 - TPs in South Korea have gone through four stages of development: 1) Infrastructure builder and business Incubator, 2) U-I-G facilitator, 3) Vision provider, and 4) Regional integrated business platform.
 - Infrastructure builder and business incubator: A technologically intensive industry park that puts every essential function in a single spot for effective high tech industry development
 - U-I-G facilitator: University-industry-government cooperation facilitator
 - Regional development agency: Technoparks were asked to play the role of regional development agencies to implement the balanced development policy.
 - Regional integrated business platform: A platform based on enabling value creating interactions between startups and SMEs, between SMEs and large firms and between service providers and users
- The Case of GyeongBuk TP
 - Macro effects: GB TP's outcomes: Gumi, Chilgok , Gyeongsan , Yeongcheon , and Pohang are connected along the Gyeongbu Expressway, forming one important axis of the Yeongnam Industrial Belt.

- Overall Performances
 - The first unicorn TP in South Korea, start-up support & establishment after start-up, promising technologies & commercialization
- Main Functions
 - Engine of triple helix, agency for central-local government relationship: Role of innovation catalyst
 - Innovation networking hub: MOUS, partnership with global & local universities, research institutes, venture capitals, and firms.
 - Business support: Fund, design, commercialization, marketing, idea, etc.
- GB TP as a “Platform”
 - The corporate-support platform of GBTP provides one-stop service including initiating business idea, pioneering the market, and even exporting products.
 - Office-sharing, support prototyping, logistics warehouse DB, financing solution
- Strategies for the Platform Effectiveness
 - Pulling, facilitating, matching, data-based, user-oriented, smart specializing
 - Evidence-based policy making, facilitating & coordinating role
 - Policy feedbacks -> information gathering -> reporting -> analysis
 - Holistic Collaboration Model
 - Risk management, testbed

2.5.2 Fundraising Models for Technopark (Lecture)

- Fundraising and securing more budgets are always the most critical matter for the public sector to survive.
- 10 possible funding sources for the public sector
 - Heartfelt connector, Local Nationalizer, Market Maker, Resource Recycler, Beneficiary Broker, Policy Innovator, Public Provider, Big Bettor, Member Motivator, and Beneficiary Builder
- Who do TPs work for? Who are the clients?

- A TP refers to an industrial and technological complex where personal, material and technological resources are gathered into a single place.
- Technopark as a mediator: Funder(client) - Gov., Beneficiaries – SMEs, N/W for operation- Univ. & Research sector
- Since South Korean TPs started with the government's designation, most of them had seed money from the government.
- Seed money for starting South Korean TPs came from the central and regional governments.
- Seed money in Gyeonggi TP (1999 -2006) was spent for construction, equipment purchase and the basic operation for SME support.
- Facilities including resident space and shops for lease were established for stable revenue as well as residents' welfare.
- Research equipment was purchased. But in revenue wise, the equipment is no of little help. It was almost like a pro bono service for the public.
 - Bought minimum(22) equipment (USD 5million)
 - Hard to generate income from the public use of equipment
 - Operated equipment usage business for 10 years and terminated it in 2015.
- Each TP has worked hard to increase its funds. Main funding sources are governments. The governance structure has been determined accordingly.
- The contribution to infrastructure is an important determinant affecting the number of board members to be nominated.
- Space funding is important for stable running of a TP. By the government rules space is offered by local governments while equipment infrastructure is funded by the central government.
- Land has been funded by local government (city) either through free lease or donation to TP
- Portion of leasing profit in 2020
 - Contribution: USD 124,014K

- Budget 2020: USD166,838K
- How much profit earned from space leasing to residents?
 - Income from leasing spent for: maintenance (70%), GTP personnel: Facility Team (4), Not sufficient to cover the labor cost of the entire employees of GTP
 - As facilities are getting outdated, more expenses are required for repair.
 - Currently earnings cannot cover 100% of the repair cost.
- Attraction of SME supportive programs: government funds
 - 4 major fields, 14 divisions, and 50 programs (2020)
- Charging indirect expenses to local governments for their program outsourcing
- Annual operation-supportive contribution by regional government
- Fund & Laws
 - National laws, local ordinances of province and city on TP, internal rules, the National R&D laws and guidelines
- Structure strong enough (personnel) to get stable fund?
- Legal foundation is ready for taking others' outsourcing programs?
- Lease is a stable source of earning, but ANAS HTP needs to look for more leasing possibilities as part of its long term plan.

2.5.3 Resident Hunting in Technopark (Lecture)

- Motivation: The Input-Process-Output Model adapted to BI
- Motivation: Domains and Indicators for Assessing BI Performance
- Motivation: How does it help? [For your review before presentation]
 - Shown above in relation to 'motivation' is about the discussion on general dimension in the business incubation program.
 - The factors in the systematic model and the related domains are expected to be combined effectively to achieve the goals for the policy to strengthen the capacities of SMEs.

- What we are Interested in are the stories on how K-TPs have been doing for incubating.
- Discussed below are the ones that can be shared as part of South Korean experiences
- K-TPs and their Policy for Resident Enterprises
 - All functions and roles of K-TPs are based on the “Act on Special Cases concerning the Support for the Industrial Technology Complex,” enacted in 1998.
 - Based on the Act, all TP in South Korea establish the Rules and Regulations for their own operation and management.
 - The policies for residents’ move-in & out procedure is based on each TP’s Rules and Regulations.
 - It could be best stated that the policy for resident support and business incubation operation is based on the regional innovation platform model in general.
- Regional Innovation Platform Model of K-TP
- The Model constructed has its basis on the Triple Helix Model
 - Refer to Heriana (2015) who shows the following model
 - How TP is acting in relation to the suggestions and especially how K-TP is doing
 - Especially on the resident hunting (BI) and it is not simple but very challenging
- TP’s Major Functions based on National Industrial Policy
 - Evolution of K-TP’s Functions!
- Four Stages of Business Incubation by TP
 - (stages) Pre-Incubation, Main Incubation, Post Incubation, Mass Production
 - Pre-Incubation: Joint support by universities, research institutes, and TP
 - Main Incubation: Joint support by universities and TP
 - Post Incubation/ Mass Incubation: Support in association with TP and Industrial Complex
 - (In all stages) Government provides programs in administration and financial support areas; All sectors interact closely

- TP's Support System for Business Incubation's Mass Production
- DJTP's Major Support Policies for Residents
 - Vision: Fostering regional star enterprises through strengthening post-BI functions
 - Purpose: Internal innovation for post-BI, specialized support by moving in stages, by company types, attracting global export-oriented company growth
- TP's Internal System on Resident Support Projects
 - Improving Institutional System
 - Enhancing Welfare
 - Post-BI Specialization Support
 - Connecting to TP Support Projects
 - System Establishment

2.5.4 Digitalization of Technopark

- Digital transformation is the process of using digital technologies to create new business process, culture, and customer experiences to meet changing business and market requirements.
- The Fourth Industrial Revolution Period: The importance of digital transformation is emphasized as the era of super-connected, super-intelligent, super-converged, and super-border through ICT technology.
- Preparing for the New Normal after COVID-19 pandemic
 - Spreading "Untacted Culture" due to COVID-19 pandemic
 - Accelerating digital transformation through social distancing
- National level efforts
 - Germany: (strategy) Industry 4.0, (Prioritized fields) cyber physical system, smart factory, (Aims) Advance the manufacturing industry by supplying smart factories through the convergence of manufacturing and ICT to become a global role model
 - USA: (strategy) Advanced manufacturing, (Prioritized fields) manufacturing, reshoring (Aims) The private sector leads the core technologies of the industrial revolution such as IoT, AI, etc.

- China: (strategy) Made in China 2025, Internet +, (Prioritized fields) Integration of information and industrialization (Aims) Convergence of manufacturing industry and IT technology, promoting green manufacturing become a manufacturing powerhouse
- South Korea: (strategy) New Robot Strategy, (Prioritized fields) Robot-based new industrial revolution, (Aims) Solving social structural problems based on original technologies of the Fourth Industries such as robots and AI
- 2019 Digital Government Innovation Promotional Strategy
 - Goals: Promotion of digital transformation to improve public services
 - 6 strategies
- Digital Government Innovation Promotional Plan for the POST-COVID 19 Era
 - Goals: Leap to the world's leading country through digitalization
 - 4 strategies
- Cloud Industry Development Strategy
 - Goals: in the era of untact becoming commonplace, support for citizens and businesses to actively carry out activities based on cloud-contact
 - 6 strategies
- Digital New Deal Plan
 - Goal: Positioning digital transformation as a key element of national and industrial innovation
 - 4 strategies
- Digital Transformation Cases
 - Master Card
 - Starbucks Coffee
 - Microsoft
 - Nike
 - Smart Factory
 - Smart Store
- Digital Transformation in Daegu Technopark

- KMS systems
- ERP system
- R&D Supporting
- Company Supporting

2.5.5 Internal System (Interview)

- **Is ANAS a member of the Cabinet? Or is it one of the agency/organization supporting the president?**

ANAS is a higher scientific organization implementing the state policy of the Republic of Azerbaijan in the field of science. It is not a member of the Cabinet, but its authorities are equal with that of the Ministries.

- **Does ANAS participate in the Azerbaijan government's policymaking (e.g., enacting related laws or developing plans, etc.)? Does ANAS have the mandate/ mission to develop and implement STI related policies (such as R&D, entrepreneurship, innovation system, technology transfer and commercialization, etc.)?**

Of course, ANAS participates in Azerbaijan government's policymaking and ANAS has a mandate to develop and implement STI related policies. Regarding to this mission, ANAS carries out the following functions to strengthen the links between science and industry; i) create an innovative economy in the country, ii) create the necessary mechanisms for conducting applied scientific research in accordance with market needs, iii) take measures to stimulate the production of competitive products by increasing innovation activity, iv) establish the necessary mechanisms for the implementation of innovation activities, v) ensure the effective usage and development of innovation potential, vi) take measures for the transfer and mastering of advanced technologies, along with the creation of a favorable environment for the development of innovative entrepreneurship and related activities, and vii) make proposals on the organization of technoparks, innovation zones, incubation services, and technology transfer centers for the development and application of science-based products and technologies ;

- **Are there other STI related ministries or committees or agencies in Azerbaijan besides ANAS? What is the concerned ministry for policy development and implementation related to High Technology Park (such as industrial park), technology transfer and commercialization policies?**

The Ministry of Transport, Communications and High Technologies of the Republic of Azerbaijan and the State Agency for Public Service and Social Innovations under the President of the Republic of Azerbaijan are related organizations in Azerbaijan. The Innovation Agency was established under the Ministry of Transport, Communications and High Technologies of the Republic of Azerbaijan on the basis of the State Fund for Development of Information Technologies and the High-Tech Park Limited Liability Company in accordance with Decree № 325 of President of the Republic of Azerbaijan Ilham Aliyev dated November 6, 2018. The government-funded High Tech Research Center under the Ministry of Communications and Information Technologies (MCIT) of the Republic of Azerbaijan was established by resolution №228 of the Cabinet of Ministers dated September 23, 2008. The aim of the center is to import cutting-edge technologies to Azerbaijan and develop high technology industries.

- **Is the work scope of ANAS HTP limited to functions that manage and support residents under the ANAS HTP? Or furthermore, as kind of policy think-tank or agency for ANAS, does it also support a variety of other ANAS projects and national level policy developments or implementation relating STI issues?**

ANAS HTP is an area with the necessary infrastructure and material and technical base for conducting research and development work for the development and improvement of high technologies and the application (commercialization) of their results in industry, services and other fields. The purpose of ANAS HTP was to increase state support for sustainable economic development and competitiveness in the country, expansion of innovation and high technology based on modern scientific and technological achievements, scientific research and the creation of modern complexes for the development of new technologies. Since its establishment, ANAS HTP has been supporting the formation of innovative entrepreneurship by developing innovative technological innovations in our country, increasing innovative initiatives in the society and, most

importantly, implementing the commercialization of science. ANAS HTP, which is directly involved in the commercialization of science in Azerbaijan, has produced science-intensive, competitive and innovative products in Azerbaijan, and cooperates with the private sector and business networks to create innovative businesses. At the same time, ANAS HTP cooperates with young startups and universities with innovative business ideas.

At present, ANAS HTP continues its activities in the direction of development and improvement of new innovative technologies based on modern scientific and technological achievements.

- **We know that ANAS HTP was established by executive order of the current President of Azerbaijan in 2016. If the current regime could change in the future, could ANAS HTP survive or disappear? I mean its stability in terms of legal and institutional framework. If you think it could be vulnerable, what are the preemptive measures to increase its sustainability?**

I think the stability of ANAS HTP's activities in terms of legal and institutional framework will remain unchanged. If HTP can develop the commercialization of science, startups, business incubation and other activities, these would be preemptive measures.

- **In the last interim report, you identified the various sources of budget (e.g., rental proceed, loans, grants and donations, investment income, residents' fee, etc). Which sources are the most contributors for the budget?**

Currently, the biggest contributions to the budget are the revenues paid by residents.

- **One of the sources of funding for ANAS HTP could be international ODA funds-grant or loan. Which countries or institutions could be your promising donors? Which ODA projects have been carried out with international organizations, especially South Korean government?**

ANAS implements many projects with the European Union, NATO, the United Nations and other organizations and countries. The scientific links between the Republic of Azerbaijan and East Asian countries are developing day by day. For the past two years, 3 STEPI delegations and 2 KOICA delegations have visited to ANAS. Currently only one ODA project

is being implemented with STEPI.

- **If ANAS HTP could generate profit/revenue by running various programs (e.g., rental fee, investment incomes, or residents' fee, etc.), can the self-generated revenue be carried over to the next year and reinvest in ANAS HTP's activities? Or is that revenue returned back to the government's finances? Furthermore, can self-generated revenues be provided as incentives to motivate ANAS HTP employees?**

Of course, it is possible to generate profits and revenues by running various programs in the following years. These amounts will not be returned back to the government's finances, because ANAS HTP does not have such kind of obligation. The self-generated revenues will be offered as incentives to motivate ANAS HTP employees.

- **What is the Innovation Fund? What is this fund used for? What are the sources for Innovation Fund? How much possibility do you have for creating Innovation Fund? What are the preconditions for creating Innovation Fund in the near future?**

For the formation and optimization of the national innovation system, the optimization of public funding in the field of scientific innovation is required to i) support the development of fundamental research, ii) strengthen innovation cooperation between local businesses and science community, iii) attract large amounts of state and foreign capital in the field of science and innovation, and iv) promote technological specialization. In order to establish an internal innovation network, we need to attract small and medium-sized businesses to the field of innovation, and put priority on high technology. For these purposes, it was considered expedient to establish the Innovation Fund under the "High Technology Park of ANAS". This Fund was established two months ago. The Innovation Fund can be funded through the profits of ANAS HTP and some aids from other institutes.

- **And what is ANAS HTP's Investment Fund? What is this fund used for? What are the sources for Investment Fund? How much possibility do you have for establishing Investment Fund? What are the preconditions for establishment Fund in the near future?**

The Investment Fund is not a priority at the moment. It may be possible to establish it in several years.

2.5.6 Technology Transfer & Commercialization (Interview)

- I understand that the ANAS HTP plans to create a new department of about 5 people, which is larger than the current team related to TT & TC. What are you planning for your core business with the new department's TT & TC?

Management of all patent and licensing operations of ANAS. Implementation of the commercialization science via a well-developed investor & tech broker network

- Currently, I know that the operating budget of TT & TC is about 20,000 AZN per year, which is the level of labor costs. Can you get an extra budget? If possible, to what extent can you increase your budget?

Our team is working on it. Hopefully, we will double this budget next year.

- The key to technology transfer and technology commercialization is the number of patents and the value of technology held by ANAS HTP. In addition, we know that the number of patents currently held by ANAS HTP is about 457. Is it possible to secure additional technology patents through various institutes in the future?

Yes, ANAS Institutes get new patents every year and the number will increase. Also, with tech transfer, ANAS itself can play a tech broker role as well.

- What policies and projects is the Azerbaijani government promoting for ANAS HTP (or Institutes related to ANAS HTP)?

ANAS has projects with the Ministry of Agriculture, the Ministry of Economy, and State Oil Company (SOCAR). Main tech transfer partners from the government side are mentioned above.

- The ANAS Industrial Plant (TSZ) that produces petrochemical products, the resident of ANS HTP, as well as ANAS Institutes in the fields of agriculture and defense, are known to be the key players in technology transfer and technology commercialization. Is there a government-sponsored project (Public Sector) for TT & TC currently being carried out within the ANAS HTP organization itself?

No, at this time, there is no government-sponsored project.

- **In my opinion, in relation to the characteristics and role of ANAS HTP's organization, in order to activate TT & TC, I feel that ANAS HTP should grow into a technology evaluation or technology transfer organization. What is ANAS HTP thinking about this?**

For reference, it would be nice to refer to my PPT presentation in a recent workshop for information on technology evaluation and technology transaction. This is very important topic for us. Of course, the professional education and training program in this field are crucial to improve these operations.

- **We know that ANAS HTP is working with various universities, institutes and companies in industrial complexes. In terms of technology commercialization, it seems very important to have competent scientists and engineers.
Can you build a human network of university professors, researchers within the Institutes, and competent technicians in companies?**

Yes, we have already good academy and government networks. Now, it's important to engage private sector in these operations. For your reference, we will actively use our human network for workforce training or technical consulting in the future.

- **What is essential for ANAS HTP's TT & TC team from an organizational point of view through the STEPI project in South Korea? For example, workforce, budget, organizational capability, expert network, system or platform, etc.**

Expert network, system, platform, organizational capability.

- **What is required for the activation of technology transfer and technology commercialization in Azerbaijan? For example, 1) law and policy base, 2) government R&D investment, 3) government technology transfer and technology commercialization project, 4) public and private technology finance base, 5) technology evaluation/technology transaction system base, etc.**

All of the points mentioned above are crucial for local TT & TC system. Because, we need to build modern system to engage all ecosystem stakeholders in the open innovation ecosystem and enhance commercialization opportunities.

2.5.7 Start-up & Incubation (Interview)

- **Who is the target customer of Incubation and Startup? ANAS HTP.**

Students, colleagues of ANAS Institutes and other universities, and all players of the Azerbaijan Startup Ecosystem

- **Who is in charge of this job? His (her) Career?**

Vusal Suleymanli (www.vusalsuleyman.com)

- **How can make absorptive capacity for Incubation management?**

Development of a sustainable link with idea owners and investors. Building a vibrant “exit” ecosystem for startups and entrepreneurs.

- **What is uniqueness?**

The uniqueness of ANAS HTP Business Incubation Center is that it has a strong link with academia and science community.

- **What is history of Incubation policy in AZ and ANAS HTP**

The history of BIS dates back to 2012. Currently, there are more than 20 business incubation centers in Azerbaijan. ANAS HTP BIS was established in 2017. Additional information was provided in the previous report.

- **How many incubation companies in ANAS HTP?**

At this time, there is no active startup company at the incubation center due to the current pandemic situation.

- **Specific Company profile**

There are no specific company profiles, but generally, ICT-based companies are our priority.

- **Performance : Sales, employment, products, history and so on,**

Previous companies are mainly in ICT sector, and the average number of staffs is about 10.

They produce hardware and software solutions. Many of them are newly established companies. Also, their average sales are in the range of 10 000 AZN – 50 000 AZN per month.

- **What is investment ecosystem in ANAS HTP and related fund?**

Currently, the main player of the investment ecosystem is the Innovation Agency of Azerbaijan. At the present moment, ANAS HTP has no special fund for investment.

- **Incubation managers' career?**

Generally, their background comes from engineering or business sciences and they continue their career as an innovation manager, business development manager, marketing manager, or growth manager.

- **Status, competitive wage?**

Average salary is around 800 AZN.

- **What is performance index of Incubation Center in AZ and ANAS HTP?**

The planned performance indicators are presented below.

2.5.8 Test & Certification (Interview)

- **What is the priority of ANA-HTP in Test & Certification?**

- Securing and training professional workforce
- Efficient operation of the quality system
- Increasing test revenues through network creation to expand test demands
- Expansion of test areas such as ICT, chemicals, and medicine other than gas and LPG

- **Among the above fields, which area would you like SETPI to support most?**

- Building a well-designed professional Test & Certification service

- The ANAS Strategic Scientific Analytic Testing Center is a newly established center and we believe that South Korea's experience in this field would be of great help.

2.5.9 Public Procurement (Interview)

- **Is it correct that the State Procurement Agency (SPA) has disappeared and has a decentralized procurement system? If so, which organization is in charge of current procurement system design and policy currently?**

SPA was not completely eliminated, but the role was modified and transferred. Many roles have been transferred to the 'Antitrust Policy and Consumer Rights Protection Department' under the Ministry of Economy,

- **Is ANAS HTP a public procurement agency?**

HTP is a government organization and it is also correct that it has a purchase function

- **Does HTP actually act as a buyer of residents? In what context do you intend to deal with public procurement within this project?**

We would like to learn South Korea's policy and system design experience in terms of helping residents. It is not intended to establish an institution or policy.

- **Do residents pay a fee to HTP if they make a profit in the process of public purchase?**

ANAS HTP receives monthly management expenses from resident companies, but this is a fixed amount (rental fee, electricity tax, etc.). ANAS HTP does not receive fees for public procurement directly. It means we are not directly receiving revenue from public procurement. In the case of other high-tech parks, even if it is not a public purchase, it is known that a certain percentage of the total revenue is received from the residents (e.g., about 1%). We are also contemplating whether it is possible to receive a certain portion as a fee.

2.6 Photos

Group Photo (Day1)	Group Photo (Day2)
	
Wangdong Kim	Yang-jin Kim
	
Kyoung Joo Nam	Jong-Wha Lee
	

Seong Byung-Ho



Ilaha Rzayeva



Bunyamin Seyidov



Vusal Suleymanli

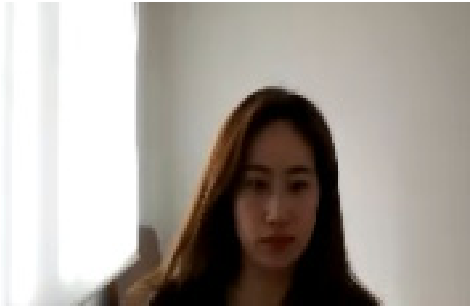


Jong Hwa Choi



Myungdo Oh



Jongin Choi	Sangho Choi
	
Yejin Yoon	Presentation Screen
	Expansion of Various Test Fields ❖ It is judged that ANAS Strategic Scientific Analytic Testing Center is supporting test services by building test facilities in the GAS and LPG fields. ❖ In addition, although the budget, etc. is insufficient, it is understood that it is planning to expand the test field in the near future. ❖ Assuming that other government departments have established and operated a separate test and certification infrastructure (for example, ICT-related departments provide separate ICT test and certification services), it may be difficult to expand tests in science and technology fields. ❖ Therefore, it is judged that the expansion of the test field will inevitably be limited to the fields of chemistry, bio, and medical. ❖ For reference, it is necessary to carefully expand the test field according to the test demand of the industry or the lack of test and certification infrastructure of other departments. 

3. The Dissemination Workshop

3.1 Overview of the Dissemination Workshop

- **Title:** 2020 K-Innovation ODA Program (Azerbaijan) Dissemination Workshop
- **Dates:** December 28 (Mon), 2020
- **Venue:** Online Conference (Zoom)
- **Purpose:** STEPI team held a Dissemination Workshop in December online to spread the results of policy consultation conducted in this project. The workshop consisted of presentations on project results by topics. STEPI and ANAS HTP also discussed future cooperation plans.

3.2 Attendees

STEPI Research Team (total of 6)

Name	Position/Organization	Role/Area of Expertise
Wangdon Kim	Chief Director, Division of Global Innovation Strategy, STEPI	Project management and planning Consultation and strategy establishment for the internal system of ANAS HTP
Jong Hwa Choi	Head, Office of Future Industry Research, STEPI	Consultation and strategy establishment for public procurement
Yeojin Yoon	Researcher, Division of Global Innovation Strategy, STEPI	Research and project coordinating
Myungdo Oh	Professor, Department of Mechanical and Information Engineering, University of Seoul	Reviewing and advising overall project strategies

Jongin Choi	Professor, Department of Business Accounting, Hanbat National University	Consultation and strategy establishment for Start-up & Incubation
Sangho Choi	Head, Policy Planning Department, Daejeon Technopark	Consultation and strategy establishment for Technology Transfer & Commercialization, Test Certification

Azerbaijan Research Team (total of 4)

Name	Position/Organization	Role/Area of Expertise
Ali Abbasov	Director, ANAS HTP	Project management
Bunyamin Seyidov,	Chief, Department of Relations with Eastern Countries, ANAS HTP	Consulting on Internal System of ANAS HTP
Ilaha Rzayeva	Independent Researcher	Consulting on Test & Certification, Public Procurement of ANAS HTP
Vusal Suleymanli	Independent Researcher	Consulting on Technology Transfer & Technology Commercialization

3.3 Schedule

September 3, 2020 (Lecture)

Time	Program	Speaker	Position
09:00-09:05 (AZ) 14:00-14:05 (KR)	Kick-off	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
09:05-09:10 (AZ) 14:05-14:10 (KR)	Opening Remarks	Myungdo Oh	Professor, University of Seoul
09:10-09:15 (AZ) 14:10-14:15 (KR)		Vasif Bilal oglu Abbasov	Director, ANAS HTP
09:15-09:45 (AZ) 14:15-14:45 (KR)	Public Procurement (Providing Interpretation)	Jong Hwa Choi	Head, Office of Future Industry Research, STEPI
09:45-10:15 (AZ) 14:45-15:15 (KR)	Internal System	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
10:15-10:35 (AZ) 15:15-15:35 (KR)	Start-up & Incubation	Jongin Choi	Professor, Department of Business Accounting, Hanbat National University
10:35-10:55 (AZ) 15:35-15:55 (KR)	Technology Transfer & Commercialization	Sangho Choi	Head, Policy Planning Department, Daejeon Technopark
10:55-11:15 (AZ) 15:55-16:15 (KR)	Test & Certification	Sangho Choi	Head, Policy Planning Department, Daejeon Technopark
11:15-11:30 (AZ) 16:15-16:30 (KR)	Q&A	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI
11:30-11:40(AZ) 16:30-16:40(KR)	Wrap-up	Wangdong Kim	Chief Director, Division of Global Innovation Strategy, STEPI

3.4 Minutes

3.4.1 Kick-off and Opening Remarks

- Kick-off : Wangdong Kim (Chief Director, Division of Global Innovation Strategy, STEPI)
- Opening Remarks: (1) Myungdo Oh (Professor, University of Seoul)
(2) Vasif Abbasov (Director, ANAS HTP)

3.4.2 Public Procurement

- Overview of the public procurement
 - classification of South Korea's public procurement system (finance /contract, policy support): overall description, related laws and agencies
- Key features of public procurement in South Korea from the perspective of policy support
 - Type and key features of the priority purchase system in South Korea (Policy support by company feature, product feature)
 - Priority purchasing system and procurement contract
- Size and status of public procurement in South Korea
 - Trend of public procurement market size in South Korea
- SME support system
 - Competition between SMEs
 - Other SME Support systems
- Venture and startup support system
 - SME establishment support act
- Priority purchase system for Technology Development Products
 - Background of the introduction of the priority purchase system technology development products
 - Priority Purchase of Technology Development Products and Product Certification and Designation
 - Linkage between R&D and Public Procurement

- Innovation-Oriented Public Procurement Policy
 - Background of Innovation-Oriented Public Procurement Policy
 - Establishment of the Fast Track and Test Purchase

3.4.3 Internal System

- Vision, Mission, Goals, and Strategies of ANAS HTP
 - Current Status of ANAS HTP
 - Vision Map of ANAS HTP (by 2035)
 - Roadmap of ANAS HTP (foundation, transformation, expansion)
- Structure of ANAS HTP
 - Organizational Characteristics of ANAS HTP by Phase: Foundation/ Transformation/ Expansion Stages (goal, activities/tasks, organizational structure&divisions, personnel)
 - Structural changes of ANAS HTP
 - Interrelationship Between Current and New Structure
- Budget of ANAS HTP
 - Current Status of Budget
 - Suggestions for Budget
 - Strategic Roadmap for Budget
- Workforce of ANAS HTP
 - Current Status of Workforce
 - Suggestions for Workforce
 - Strategic Roadmap for Workforce
- Network of ANAS HTP
 - Current Status of Networking
 - Suggestions for Networking
 - Strategic Roadmap for Networking

3.4.4 Start-up & Incubation

- Business Incubation
 - Start-up Ecosystem
 - South Korean Business Incubation Centers : Numbers and Performance
 - Evolution of KISED
 - Five Key Tasks of KISED
 - TIPS Incubation Process
 - KSF of Business Incubation
 - Diverse Incubation in DAEJEON , SOUTH KOREA
 - Global Entrepreneurship Index (GEI) Rank
- Strategic Plan for ANAS-HTP
 - INC-based Strategic Design
 - Training Program : TEC Algorithm
 - Building the Capacity of Leaders and Employees
 - Promoting Start-up Incubation Among ANAS-affiliated Researchers
 - Building a Foundation by Securing External Funds and Existing Profits
 - Close Cooperation with Domestic Start-up and Incubation Organizations
 - Strengthening Cooperation with Foreign Start-up and Incubation Organizations
 - Increasing the Awareness Level of Research through Enhanced Publicity
 - Roadmap

3.4.5 Technology Transfer & Commercialization

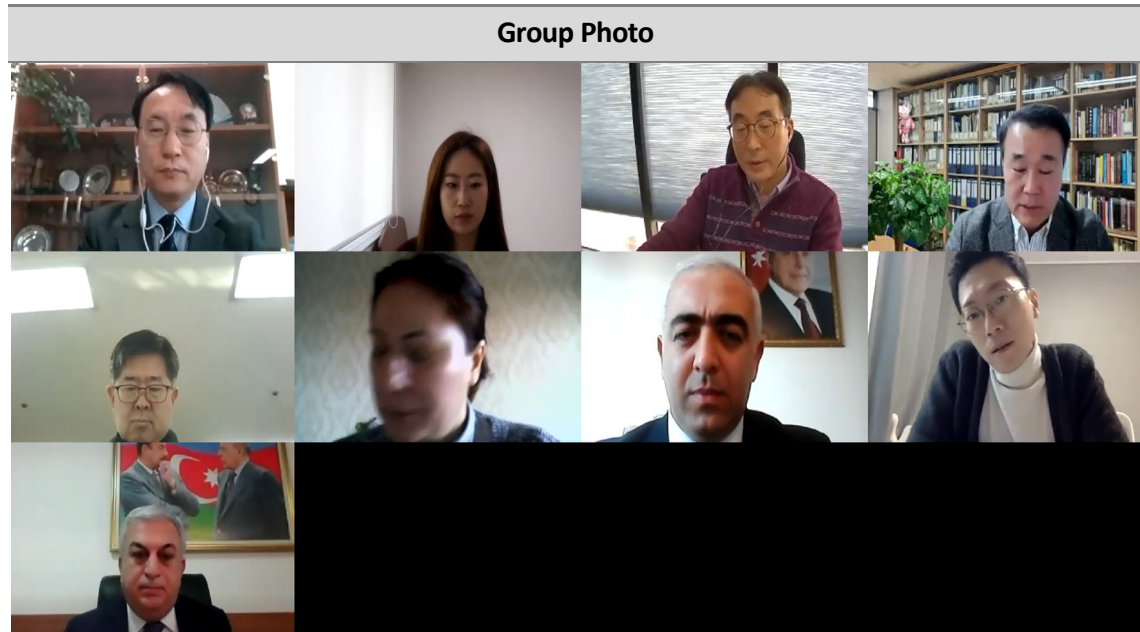
- Necessities
 - General overview: Importance of technology transfer and commercialization
 - Status and Necessities: current status, necessities
- Suggestions
 - Infrastructure Construction for TT&TC: Improvement of laws and regulations related to TT&TC

- Securing Technology Patents and Building Patent DB: Activation of technology patent development, Patent DB construction
- Establishment of Technology Value Evaluation System: planning, Construction of a technology evaluation platform
- Technology Transaction System Construction: planning, Construction of technology transaction system
- Roadmap
 - building infrastructure
 - acquiring technology patent and building a patent DB
 - building a technology valuation system

3.4.6 Test & Certification

- Necessities
 - General overview
 - Status and Necessities: current status, necessities
- Suggestions
 - Infrastructure Construction for Test & Certification: developing a plan for organizational structure and manpower operation
 - Quality System Operation in the Field of Testing and Certification: efficient quality system operation in the test field, New establishment of quality management system in certification field
 - Diversification of test areas and advancement of test facilities: Expansion of test areas such as ICT, Advanced testing facilities in petrochemical and LPG fields
 - Maximizing Test Fee Revenue
- Roadmap
 - building infrastructure
 - operating a quality managementsystem
 - diversifying testing areas and upgrading testing facilities
 - maximizing profits from testing service fee

3.5 Photos



A solid teal vertical bar on the left side of the page.

CHAPTER 5

Strategic Plan for ANAS HTP

Policy Consultation on the Promotion of Technology
Transfer and Commercialization in Azerbaijan



Chapter 5. Strategic Plan for ANAS HTP

1. Internal System⁷¹

1.1 Vision & Strategies

1.1.1 Necessity

In general, an organization evolves through four stages of life cycle as shown in [Figure 5-1]. The first is the foundation stage, during which leaders with robust leadership establish a new organization by mobilizing human and financial resources. In case of an organization in the public sector, it can be established only when leaders with strong influence succeed in convincing the government, mobilizing funds, and institutionalizing the organization.

Following the first comes the second stage of “take-off.” In this stage, an organization reaches a cross point of either making a giant leap forward or disappearing depending on its capacity of securing financial sustainability. If the organization can secure sustainable funds, it can gain a strong momentum for further growth. If not, it will simply disappear. Therefore, organizations need to do their best to secure financial sustainability. For example, when an organization has limitations in its workforce and capabilities, it needs to make focused investment in its competitive areas, secure funds with investment profits, and reinvest them to expand profits. This is quite similar to the take-off procedure of airplanes, which consumes quite a lot of fuel to lift the airplane off the runway.

However, for further growth going beyond the take-off stage, an organization needs systems and engines than can support its growth. This is also similar to the airplane’s

⁷¹ This chapter was written by Dr. Wangdong Kim of the STEPI

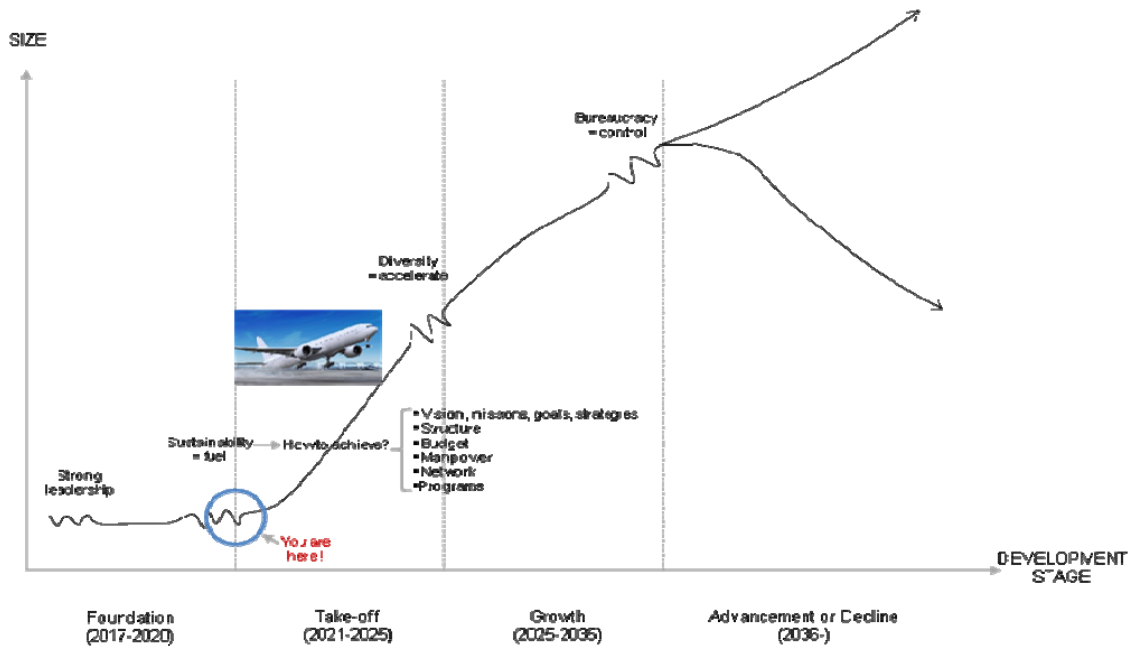
acceleration process, making the plane fly at higher altitude. During this growth stage, organizations can maximize their profits by diversifying operational programs and finally achieve further growth.

As an organization grows, however, its workforce, organizational complexity, and environmental uncertainty grow as well, which naturally leads to an organizational bureaucracy. The organization can either advance or decline depending on how effectively it can control this process. Therefore, a well-designed management system is introduced to upgrade the organization, and then decentralization or empowerment follows as part of this process.

Like this, an organization goes through various stages of development from birth to decline. If so, where is ANAS HTP currently standing at in this development lifecycle? Probably, it is now about to enter the take-off stage just passing through the foundation stage. Since its foundation in November 2016, ANAS HTP has established its institutional and organizational foundation and is now running some early-stage projects. However, ANAS HTP is not ready to take off yet as its financial sustainability remains at an insufficient level.

Then, what needs to be done for ANAS HTP to secure its financial sustainability and finally launch a take-off? Basically, ANAS HTP needs to pursue changes that can enhance the sustainability of its internal system. This means redefining its vision, missions, goals, and strategies and restructuring its systems, so ANAS HTP can secure financial sustainability. In addition, to meet the budget requirements, ANAS HTP needs to secure various sources of resources and strengthen the motivation and capacity of its workforce. Securing various income sources and visibility requires strengthening of its local and global networks. To meet these needs for change, this chapter designs and presents strategic plans for each element of ANAS HTP's internal system.

[Figure 5-1] Current Status of ANAS HTP by Stage



1.1.2 Suggestions

Then, what vision should ANAS HTP establish to make another leap? A vision means a future image or a dream an organization aspires to fulfill. A vision motivates members of an organization to dream an ideal state and work hard to achieve such state. ANAS HTP is an organization in charge of supporting and managing a high-tech park. For ANAS HTP, the research team proposes a vision of “becoming a top high-tech park managing company in Caspian Countries, not just in Azerbaijan, by 2035.” This means becoming a top-notch high-tech park across Iran, Russia, Kazakhstan, and Turkmenistan, which is surely a formidable challenge.

Then, what missions should ANAS HTP pursue in realizing this vision? In fact, ANAS has entrusted ANAS HTP to achieve four targets (Wikipedia, 2020). The main purpose of ANAS HTP is further increasing the state support for sustainable economic development and competitiveness in the country, expanding innovation and high technology areas based on modern scientific and technological achievements, carrying out scientific research, and creating modern complexes for the elaboration of new technologies (Wikipedia, 2020).

These missions need to be more specific and more appropriate than now to support ANAS HTP's vision of become a top high-tech park in Caspian countries in 15 years. In this context, the research team proposes the following three missions.

The first mission is to support businesses of ANAS HTP residents and other innovation actors in Azerbaijan. The primary mission of ANAS HTP is to support residents located in the technology park. To have a world-class competitiveness as a supporting organization, ANAS HTP needs to expand its mission to cover various innovation actors in Azerbaijan. The second mission of ANAS HTP is to support policy development for ANAS or the Azerbaijan government. ANAS HTP is under the supervision of ANAS (the Azerbaijan National Academy of Science) in charge of developing and implementing the country's STI policy. However, ANAS is currently lacking a think tank organization who can support its STI policy development. Therefore, supporting the policy development of the Azerbaijan government can be added to ANAS HTP's missions since ANAS HTP is an organization under the supervision of ANAS. The last mission is to facilitate networks between ANAS HTP and innovation actors in and outside of Azerbaijan. To effectively support residents located in the technology park, ANAS HTP needs to pursue a mission of playing the role of a hub organization who facilitates networks with various innovation actors in and outside of Azerbaijan.

Specific goals and efforts are needed to realize the above-mentioned vision and missions. In general, the more challenging and specific the goals are, the more motivating they become. Though they might look rather challenging, quantitative goals are usually more effective than abstract goals in realizing the vision and missions. If so, what scales can ANAS HTP set for its goals? The research team proposes four quantitative goals in scale.

The first goal is to increase ANAS HTP's annual budget from the current level of USD 0.3 million in 2020 to USD 1.5 million by 2035. For ANAS HTP, the most important priority is to expand its budget by creating profits as a company. A five-fold increase of its annual budget in 15 years may sound a bit challenging for ANAS HTP. However, a successful outcome would not be possible without an ambitious plan and aggressive efforts to implement the plan. The second goal is to increase the number of residents from 11 in

2020 to 30 by the year 2035. When ANAS HTP attracts more tenant companies as a high-tech park, it can create higher synergies among its tenants, which will then bring higher profits to ANAS HTP. An increase of about two-fold in the number of resident companies seems a feasible target to achieve. The third goal is to increase the number of technology transfer and commercialization from 11 in 2020 to 200 by 2035. One of the most important missions of ANAS HTP is to facilitate technology transfer and commercialization from its resident companies to external stakeholders or vice-versa. Therefore, an increase in the number of technology transfer and commercialization is an extremely important goal in scale. Now, the last goal of ANAS HTP is to expand the number of its incubation companies from 10 in 2020 to 100 by 2035. As a high-tech park, an increase of its incubation companies is another important quantitative goal for ANAS HTP.

Then, what strategies need to be pursued for ANAS HTP to realize afore-mentioned vision, missions, and goals? If it had sufficient budget, ANAS HTP would not have problem in trying various projects across different fields. Unfortunately, however, both its budget and the capacity of its workforce are limited. So, ANAS HTP needs to identify the most important services that fit its missions and focus its resources on these priority services where it has strengths. Based on this principle of “selection and focus,” the research team proposes the following four strategies to focus on.

The first strategy is scaling-up of technology transfer and commercialization service. ANAS HTP’s important mission is to support businesses of its resident companies and other local companies in Azerbaijan. To fulfill this mission, ANAS HTP needs to promote technology transfer and commercialization service, which is the foremost important requirement for ANAS HTP’s business supporting function. For this purpose, patents created by the various institutes under the four scientific divisions of ANAS can be transferred to resident companies of ANAS HTP or other local companies in Azerbaijan, who will then commercialize these patents.

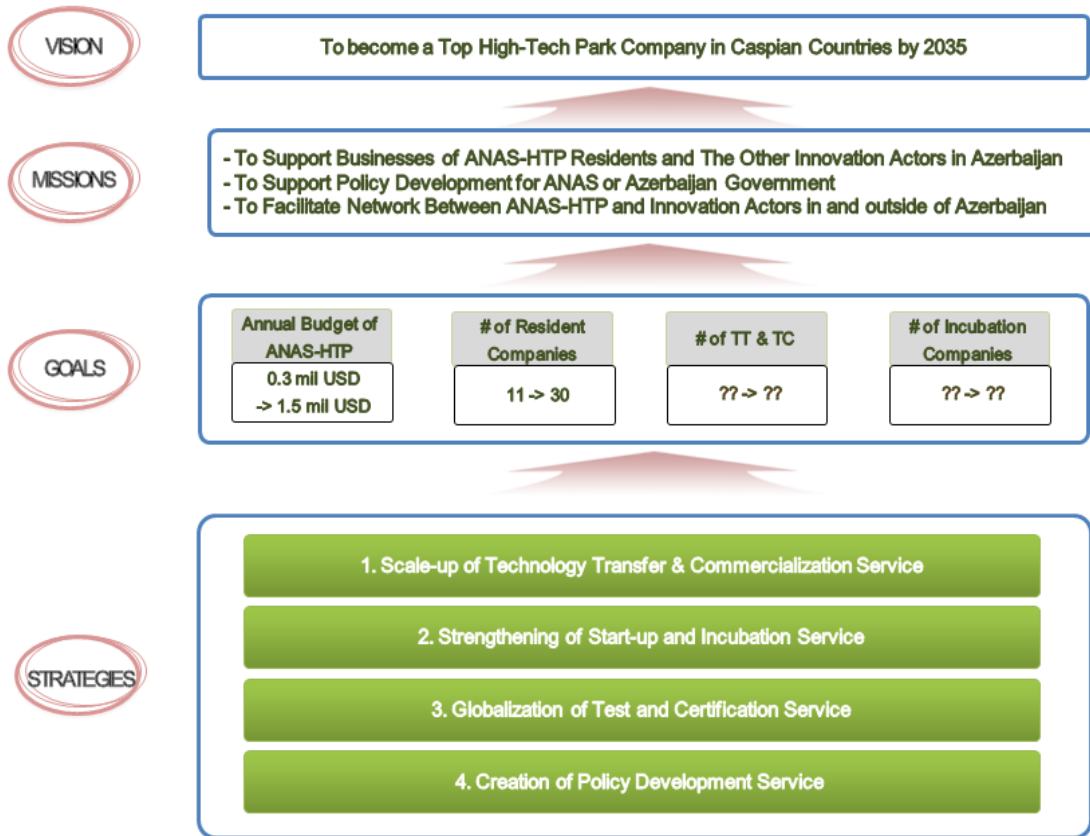
The second strategy is strengthening of start-up and incubation service. As a supporting organization of a high-tech park, ANAS HTP can provide incubation services for student researchers affiliated with the institutes under the four scientific divisions of ANAS. It can

also provide incubation services for students from colleges located near the technology park, which will help promote entrepreneurship and technology innovation in Azerbaijan.

The third strategy is the globalization of certification service. In fact, ANAS HTP possesses the top-notch facilities in Azerbaijan. ANAS HTP can create profits by providing certification service for its resident companies and other innovation actors in the country by using these top-class facilities.

The fourth strategy is the creation of policy development service. Currently, Azerbaijan is lacking a think tank organization who can support ANAS' STI policy planning and management. As an organization under the supervision of ANAS, ANAS HTP can add its new function of supporting the policy development of the Azerbaijan government. In fact, South Korean Technoparks are also supporting the policy development of local and provincial governments.

[Figure 5-2] Vision Map of ANAS HTP (by 2035)



By 2035, ANAS HTP will be able to achieve its vision, missions, and goals by focusing on these four strategies. Then, what process needs to be pursued to implement these four strategies? The research team presents a roadmap in three stages.

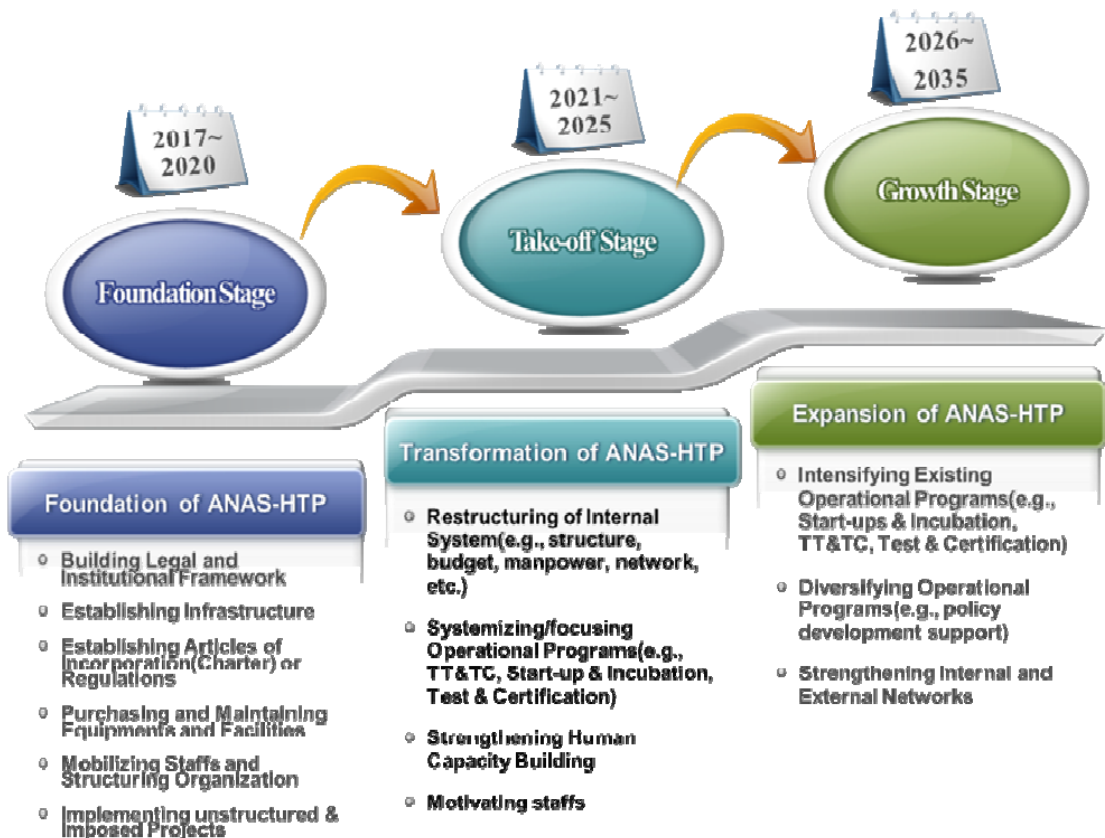
The first stage is the foundation stage from 2017 to 2020. During this period, the infrastructure (e.g., buildings) and legal and institutional framework of ANAS HTP was built after its foundation in November 2016. ANAS HTP also acquired its equipment and facilities on its property, recruited its personnel, and completed its organizational structuring. As of 2020, ANAS HTP is implementing some unstructured and imposed projects.

The second stage is the take-off stage from 2021 to 2025. In this stage, ANAS HTP will try a

transformation to make another leap forward. It will restructure its internal system (e.g., structure, budget, workforce, etc.) to enhance financial sustainability and intensify three operational programs including (1) TT & TC, (2) Start-up & Incubation, and (3) Test & Certification. Staffs need to be fully motivated to run these operational programs to their maximum impact.

The third stage is the growth stage from 2026 to 2035. During this stage, ANAS HTP will further expand its operational programs that it pushed forward in the take-off stage by adding the “policy development support function.” During this growth stage, internal and external networks will be strengthened as well.

[Figure 5-3] Roadmap of ANAS HTP by Phase



1.2 Structure

1.2.1 Necessity

The research team has presented ANAS HTP's vision, missions, goals, and strategies that need to be pursued by 2035 along with a three-staged roadmap with specific tasks to be implemented over the next 15 years. For a successful implementation of these tasks, ANAS HTP needs to design its organizational structure and define job descriptions for its internal organizations. [Figure 5-3] shows ANAS HTP's goals, activities and tasks, and organizational structure and divisions, and required personnel by each stage of its growth.

[Figure 5-4] Organizational Characteristics of ANAS HTP by Phase

	Foundation Stage (2017-2020)	Take-off Stage (2021-2025)	Growth Stage (2026-2035)
Goal	• Foundation of ANAS-HTP • Survival(maintain the status quo)	• Transformation of ANAS-HTP • Sustainability(fuel to propel)	• Expansion of ANAS-HTP • Effectiveness(accelerate to expand)
Activities/ Tasks	• Building Legal and Institutional Framework • Establishing Infrastructure • Establishing Articles of Incorporation(Charter) or Regulations • Purchasing and Maintaining Equipments and Facilities • Mobilizing Staffs and Structuring Organization • Implementing unstructured & Imposed Projects	• Restructuring of Internal System(e.g., structure, budget, manpower, network, etc.) • Systemizing/focusing Operational Programs(e.g., TT&TC, Start-up & Incubation, Test & Certification) • Strengthening Human Capacity Building • Motivating staffs	• Intensifying Existing Operational Programs(e.g., TT&TC, Start-up & Incubation, Test & Certification) • Diversifying Operational Programs(e.g., policy development support) • Strengthening Internal and External Networks
Organizational Structure & Divisions	2 Divisions 7 Departments Legal Support, Maintenance, Special, Strategic Scientific Analytical Testing Center, Patent Certification Marketing & BI, Finance & Economic Analysis Department	3 Divisions 8 Departments 1 Center Strategic Development, Government Support, External Cooperation & PR, TT&C, Start-up & Incubation, Test & Certification, General Affairs & HR, Financial Accounting, Resident Support Department	2 Divisions 7 Departments 3 Center Departments (e.g., Strategic Development, Government Support, External Cooperation & PR, General Affairs & HR, Financial Accounting, Resident Support) Centers (e.g., TT&TC, Start-up & Incubation, Test & Certification)
Personnel	46 personnel	60 personnel	80 personnel

As [Figure 5-4] above shows, the primary goal of the foundation stage from 2017 to 2020 was to establish ANAS HTP and secure its survival. For the foundation of ANAS HTP, legal and institutional framework was laid down and its charter was established at the

government level. Other key tasks were carried out including the construction of buildings for the administrative unit of ANAS HTP and purchasing and maintaining of its equipment and facilities. To efficiently carry out these activities, ANAS HTP recruited necessary personnel and completed organizational structuring. As of 2020, ANAS HTP has a total of 46 personnel in 2 divisions and 7 departments.

ANAS HTP's organizational structure during this stage, however, show the following characteristics which are considered as limitations. First, as it was still in an early stage of its growth, ANAS HTP was focused on legal, maintenance, finance, and economic issues, which were the basic requirements for its foundation and maintenance. As result, most of ANAS HTP's organizations during this stage were those who were spending its budget. Second, ANAS HTP did not have any profit-making services (e.g., technology transfer and commercialization, start-up & incubation, and test & certification) that could help secure its own budget. Third, ANAS HTP's think tank function of planning and creating the organizational vision and its profit-making services are still quite weak though a strategy planning function has been recently added in the form of a department. Fourth, ANAS HTP is relatively weak in its functions of building cooperation networks with innovation actors in and outside of Azerbaijan and promoting its own organizations. Fifth, though ANAS HTP directly reports to ANAS, ANAS HTP does not have an internal organization to support the policy development function of ANAS. These organizational characteristics work as barriers limiting ANAS HTP's income generation capacity. These barriers should be overcome for ANAS HTP to make another giant leap forward.

1.2.2 Suggestions

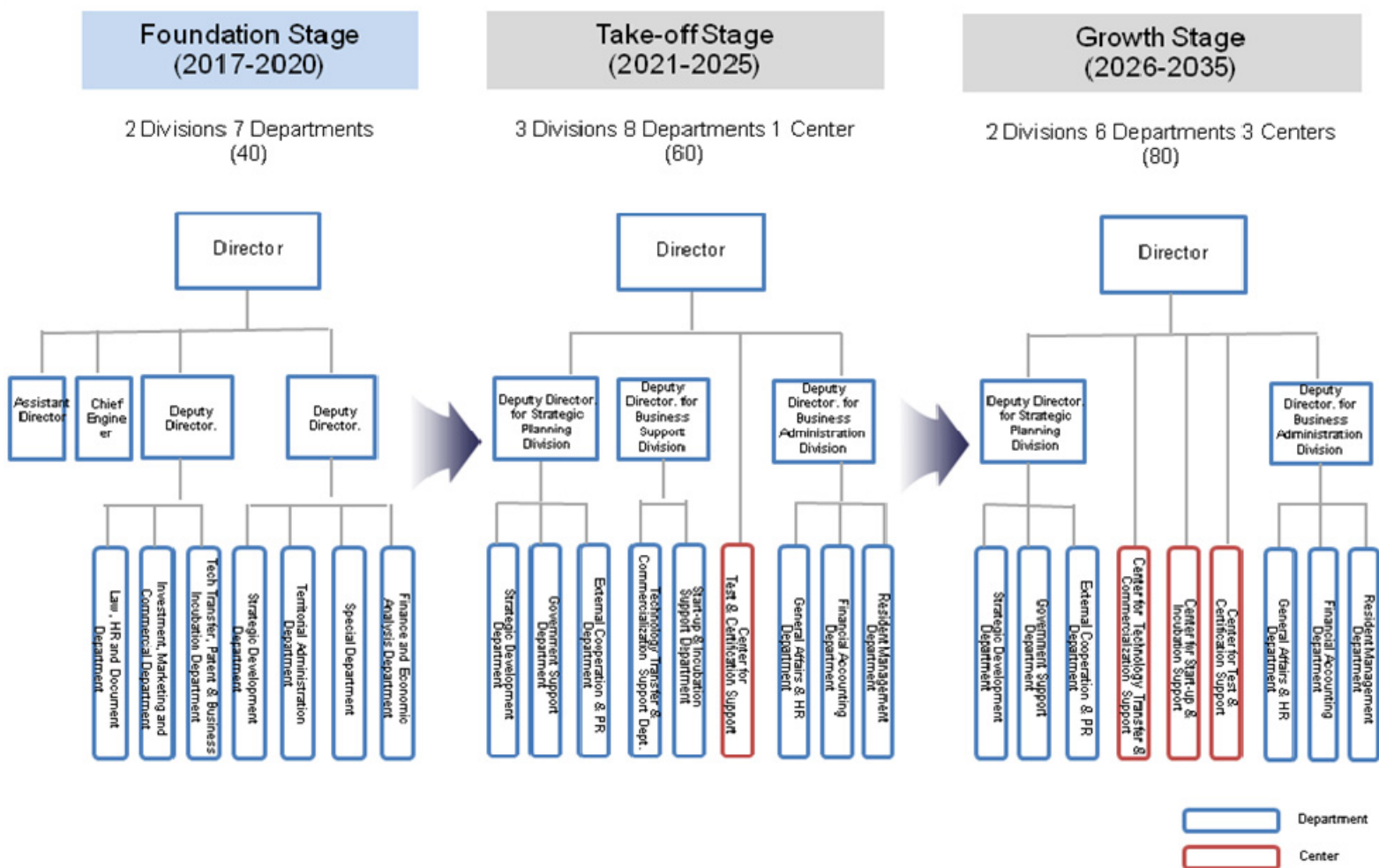
The research team proposes ANAS HTP to implement a structural transformation from an organization with 2 divisions and 7 departments into one with 3 divisions, 9 departments and 1 center during the take-off stage from 2021 to 2025. This structural transformation will help ANAS HTP overcome the above-mentioned limitations of the foundation stage and make another leap forward. As [Figure 5-5] suggests the primary goal of the second stage is to fuel the growth engine of ANAS HTP. In other words, during this second stage, ANAS HTP will explore possible solutions that can enhance its organizational sustainability.

For this purpose, ANAS HTP needs to restructure its internal system and pursue a strategy of focusing its limited funds and human capacities on three key operational programs (e.g., TT & TC, Start-up & Incubation, and Test & Certification). For effective implementation of all these activities, ANAS HTP needs to build various systems for building and strengthening human capacity and motivating its staffs. It is estimated that the maintenance of all these systems would require a total of 60 personnel.

To further expand the operational programs implemented in the second stage, the organization can be restructured and expanded into 3 divisions, 8 departments, and 1 center in the third stage. A key objective of the third stage is to accelerate the expansion of the operational programs implemented in the second stage. For this purpose, the TT & TC and Start-up & Incubation functions that were operated at the level of departments will be promoted to the level of centers. These centers will be run as financially independent units, so that they can seek for their sustainability through profit maximization. A new supportive function of assisting the S&T policy development function of ANAS can be added, which will further expand ANAS-HTP's organizational structure. In addition, ANAS-HTP's internal and external networks can be strengthened to promote its financial sustainability. The implementation of the expanded operational programs in the third stage would require about 80 personnel.

The following [Figure 5-5] shows the structural changes of ANAS HTP from the foundation stage to the growth stage.

[Figure 5-5] Structural Changes of ANAS HTP by Phase



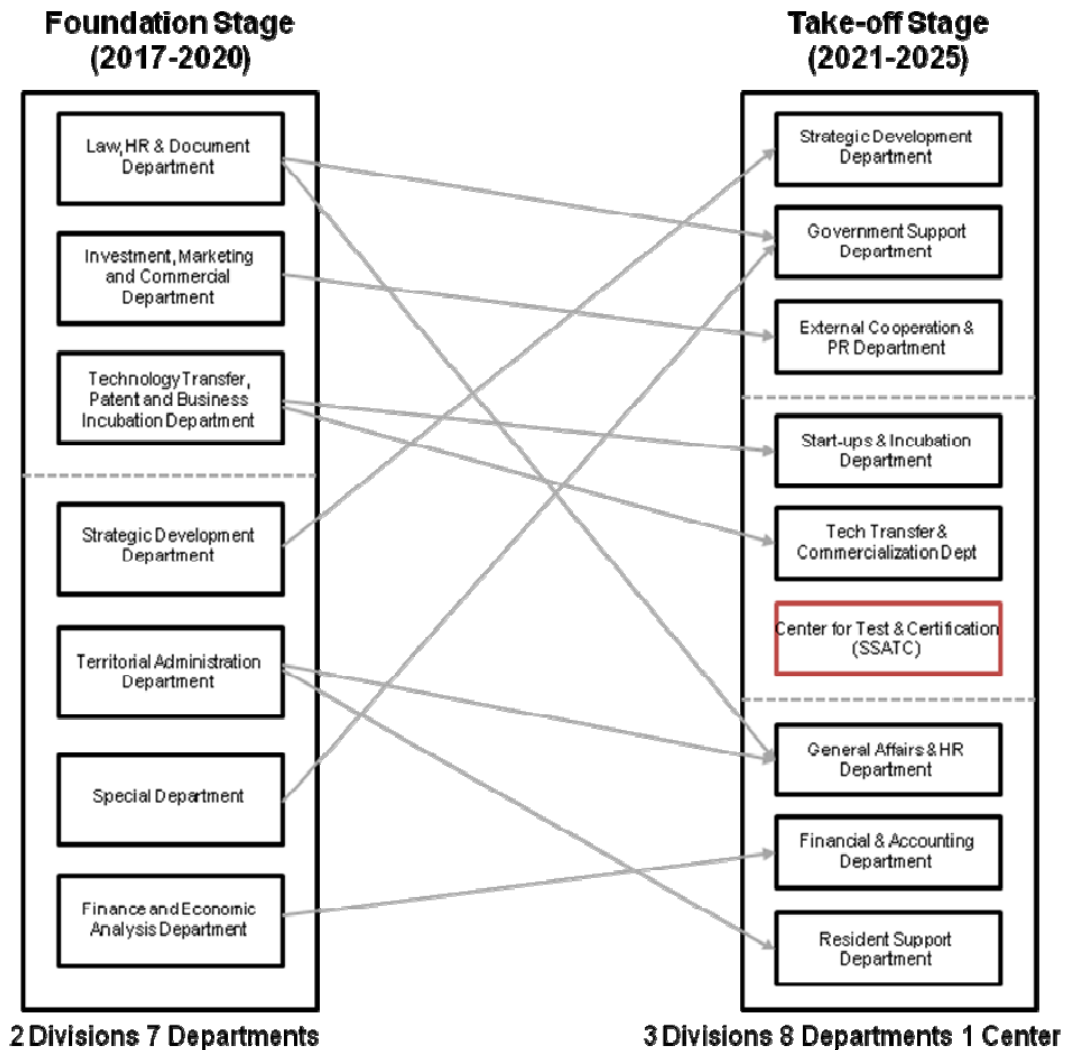
The key characteristics of the organizational structure in the take-off stage are as follows. First, ANAS HTP's strategic function is reinforced. To carry out the transformation and make another leap forward, ANAS HTP first and foremost needs a brain function that can design its future and develop and implement its strategies. The Strategic Planning Division is one that can play this role. This division has three departments such as i) the Strategic Development Department who develops the organization's future development direction and strategies, ii) the Government Support Department who responds to the government or ANAS projects, and iii) the External Cooperation & PR Department who will promote ANAS HTP's cooperation and networking with other innovation actors in and outside of Azerbaijan and handle the organization's PR activities.

Second, business support functions are systemized and integrated into one division, so that they can create synergies. A strategy called "selection and focus" has been presented to help ANAS HTP make a huge stride and this strategy calls for ANAS HTP to focus on three key services such as TT & TC, Start-up & Incubation, and Test & Certification. For an effective implementation of this strategy, the formation of three responsible departments is suggested, each of them covering respective service. These business support functions are placed under the supervision of Deputy Director, so that these functions are linked and coordinated together in an efficient and effective manner.

Third, the efficiency of the administrative support function is improved by placing the administrative function of the ANAS HTP management unit and that for supporting resident companies under the same division. These two functions are inter-related. One deputy director coordinating these two functions would double the efficiency of the administrative support function.

[Figure 5-6] shows the inter-relationship between the seven departments of the foundation stage and the eight departments and one center of the take-off stage.

[Figure 5-6] Inter-relationship between Current and New Structure



Compared to the second stage, some noticeable changes of the third stage are the promotion of TT & TC and Start-up & Incubation functions from the department level to the level of centers and their enhanced autonomy in the budget and workforce operation. These changes ensure profit maximization, which will then enhance ANAS-HTP's budgetary sustainability. The organizational designing and functions of these two centers will be reviewed in sections 2 and 3 of Chapter 5.

1.3 Budget

1.3.1 Necessity

An organization's budget is energy that fuels the organization's annual activities and programs. Therefore, it would not be an exaggeration to say that an organization's sustainability and success is often determined by its capacity of securing sufficient budget and executing it properly. Then, what is the status of ANAS HTP's budget and how financially sound is it? The [Table 5-1] below shows ANAS HTP's sources and uses of budget for 2020. ANAS HTP's total annual budget for 2020 is estimated at USD 205,411 and it shows the following characteristics.

First, in terms of the budget sources, amounts paid by ANAS HTP residents and other non-prohibited sources take the lion's share, accounting for 87% of the total annual budget, while proceeds from the rent of park areas (9.8%) and loans, grants, and donations (2.6%) take up the remaining share. Besides these, however, ANAS HTP has no other budget sources like revenues from the management organization's entrepreneurial activities in innovative areas or incomes from investment made by the managing organization in the infrastructure facilities of ANAS HTP, which ANAS HTP needs desperately for making profits on its own. This implies that most of the budget comes from passive budget sources like rent fees or registration fees while no profit is being created in HTP's core business areas.

Second, in terms of budget uses, 26.8% of the total budget is being spent as territorial administration cost for managing the residents. The rest of the budget is also used for functions that are needed in the early foundation stage including legal and human resources work and for work with documents (11%), investment, marketing, and commercial (11%), and finance and economic analysis (8.9%). In the meantime, ANAS HTP's primary business function, technology transfer, patent and business incubation, accounts for only 11%.

ANAS HTP's budget structure can be summarized as follows. Most of its budget is being used to cover the labor cost for carrying out those basic functions necessary for the

foundation stage while a small fraction of its budget is used for its core businesses. This imbalanced budget structure implies limitations for ANAS HTP when it tries to diversify its businesses. Therefore, ANAS HTP needs to explore ways to diversify its budget sources that can financially support its future growth.

[Figure 5-7] The Sources and Uses of Budget (Year 2020)

The sources of budget			The uses of budget		
Source	Amount(USD) /year 2020	%	Department	Amount(USD) /year 2020	%
Amounts paid by ANAS-HTP residents and other non-prohibited sources	180,000	87.6	Territorial Administration	55,058	26.8
Proceeds from the rent of park areas	20,000	9.8	Leadership	35,294	17.2
Loans, grants and donations	5,411	2.6	Law, Human Resources and for work with Documents	22,588	11.0
Revenues from the management organization's entrepreneurial activities in innovative areas	0		Investment, Marketing and Commercial	22,588	11.0
Incomes from investment made by the managing organization in the infrastructure facilities of ANAS-HTP	0		Technology Transfer, Patent and Business Incubation	22,588	11.0
			Finance and Economic Analysis	18,352	8.9
			Strategic Development	18,352	8.9
			Special	10,588	5.2
	205,411	100		205,411	100

1.3.2 Suggestions

Then, how can ANAS HTP diversify its budget sources?

First, ANAS HTP needs to secure funds necessary for building its business infrastructure and strengthening its capacities by focusing on free and paid aid like ODA grants, loans, and donations. Currently, ANAS HTP is sourcing 2.6% of its total budget from overseas aid funds. For short-term, ANAS HTP needs to increase this ratio so that it can use the overseas aid funds as seed money for building infrastructure and strengthening capacities required for conducting its businesses. For example, ANAS HTP can use STEPI's K-Innovation Program or KDI's Knowledge Sharing Program (KSP) to receive policy consultations. Then, based on these policy consultations, ANAS HTP can use KOICA's mid-

and long-term free aid programs to secure large-scale funding. For paid aid or loans, ANAS HTP can use the EDCF Program of South Korea Eximbank. STEPI can probably play the role of a bridge between ANAS HTP and South Korean aid programs when ANAS HTP is looking for free or paid aids. In addition to the South Korean aid programs, ANAS HTP can also seek for aid from JICA of Japan and other multilateral funding institutions like the World Bank and the UNDP Program. These activities can be taken care of by ANAS HTP's External Cooperation & PR Department.

Second, registration fees or rental fees paid by tenant companies can be increased to cover the direct costs including labor cost to be incurred in conducting businesses. As of 2020, income from these registration and rental fees account for 97.4% of ANAS HTP's total income. So, these income sources can still be used as effective budget sources. To increase income from these sources, ANAS HTP needs to provide attractive services and differentiated tax benefits for its tenant companies and carry out PR activities to advertise itself to potential residents. In addition, ANAS HTP can provide space for business incubation for college students or future entrepreneurs in Azerbaijan as part of its business space rental program, which can also create income. The External Cooperation & PR Department of ANAS HTP will supervise these activities.

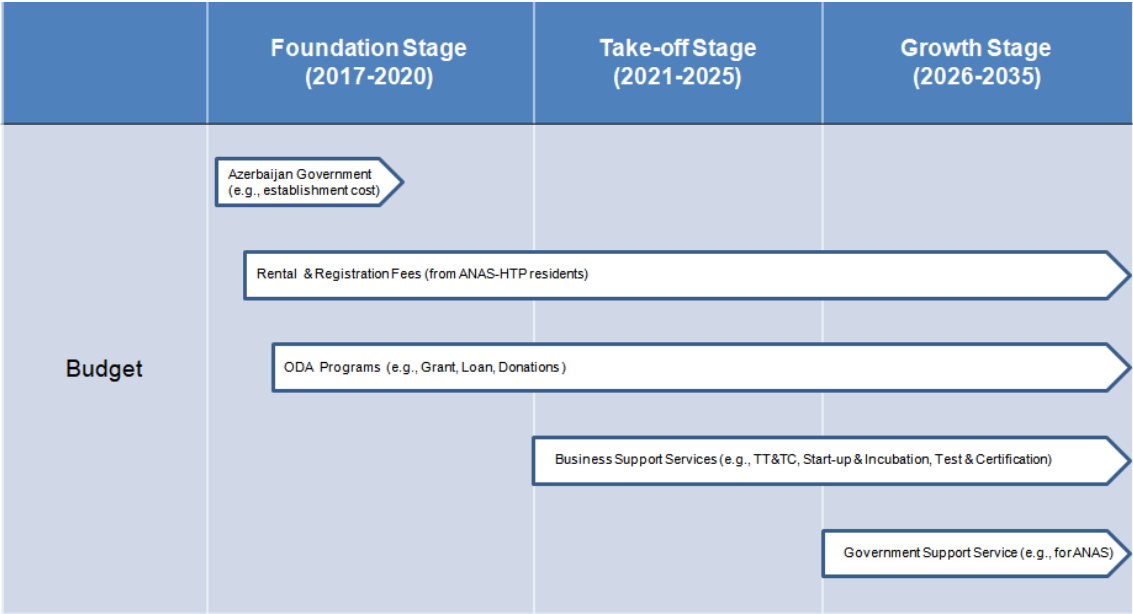
Third, once the infrastructure and capacities as well as budget required for conducting businesses are secured using external ODA aid funds or fees from its tenant companies, ANAS HTP can then expand its internal income by carrying out three key business support services (e.g., TT & TC, Start-up & Incubation, and Test & Certification). For ANAS HTP, who is a park management organization, internally created income from these business support service programs needs to be the most important income source. Unfortunately, however, the income from this internal source is nonexistent (0%) as of 2020. Activities related to these business support services will be carried out respectively by three departments such as TT & TC Department, Start-up & Incubation Department, and Test & Certification Department.

Fourth, ANAS HTP will be able to create income from its government support service, for example, from supporting the policy development of the Azerbaijan government once

ANAS HTP accumulates sufficient capacities in the afore-mentioned three business support services, which are the focus areas of ANAS HTP during the early stages. ANAS HTP is an institute under the supervision of ANAS who is responsible for planning and implementing Azerbaijan's S&T policy. Currently, the country is lacking a think tank organization capable of supporting ANAS and other related government ministries in their development and implementation of STI and industrial policies. So, ANAS HTP will be able to function as a policy development and management support organization for ANAS and other government ministries. For example, ANAS HTP can support policy development and management of ANAS by mobilizing the Innovation Fund of ANAS. In addition, ANAS HTP will be able to create extra income by providing various support services for other government ministries and agencies like TT & TC, start-up & incubation, test & certification, and patents. These activities will be carried out by the Government Support Department.

These four key functions can be mapped out by each stage of ANAS HTP's growth as shown in [Figure 5-8]. In the foundation stage, ANAS HTP received funding support from the Azerbaijan government through a block funding to cover its establishment costs. Since then, ANAS HTP has been using the rental and registration fees from tenants and ODA funds from external sources to cover its budget. During the take-off stage, ANAS HTP will have to accelerate its income generation by providing business support services, while reinforcing the two important income sources like rental and registration fees and ODA funds. From the growth stage, ANAS HTP will have to further diversify its budget sources by providing government support service.

[Figure 5-8] Roadmap for Budget



1.4 Workforce

1.4.1 Necessity

An organization consists of people and it is also run by these people. Therefore, a success of an organization largely depends on how capable, passionate, and dedicated its members are. Then, how competitive is ANAS HTP’s workforce? [Table 5-2] shows a workforce composition of ANAS HTP. As of 2020, ANAS HTP has a total of 46 staffs and their key characteristics can be summarized as follows.

[Figure 5-9] Number of Staff by Department (as of October 2020)

Number of Staff by Department		
Department	No. of Staff (as of Oct. 2020)	%
Territorial Administration	17	37.0
Leadership	5	10.9
Law, Human Resources and for work with Documents	5	10.9
Investment, Marketing and Commercial	5	10.9
Technology Transfer, Patent and Business Incubation	5	10.9
Finance and Economic Analysis	3	6.5
Strategic Development	4	8.7
Special	2	4.3
	46	100

First, ANAS HTP does not have a sufficient number of staffs in its three core business support services (e.g., TT & TC, Start-up & Incubation, and Test & Certification) and most of its staffs are currently concentrated on the management and maintenance of tenant companies. As [Figure 5-9] shows, the Territorial Administration Department responsible for managing the management organization and the residents accounts for 37% of the total workforce. In addition, staffs are mostly involved in the establishment and maintenance related activities, for example, human resources and for work with documents (10.9%), investment, marketing and commercial (10.9%), finance and economic analysis (6.5%), strategic development (8.7%), and special (4.3%). On the contrary, staffs of the Technology Transfer, Patent and Business Incubation taking care of the core business support services account for only 10.9%. This implies a heavy concentration of ANAS HTP workforce on management and maintenance activities rather than its core services.

Second, a majority of ANAS HTP people are administrative and technical staffs rather than experts of respective fields. As of May 2020, ANAS HTP has only four doctorate degree holders while the rest of its people are non-degree holders. In fact, most of ANAS HTP staffs are technical staffs in charge of managing the technology park. This workforce

composition may become a major barrier for ANAS HTP to emerge as a top-notch high-tech park among Caspian countries in the future. This implies that the future success of ANAS HTP relies on how it can secure capable personnel and strengthen and upgrade their capacities.

Third, the turnover rate of ANAS HTP staffs remains high. In particular, the ANAS HTP Director who has an important responsibility of representing and leading the organization has been replaced frequently, undermining the consistency of the organization's strategic direction. During the implementation of this project over two years since 2019, there have been four times of director replacements. What makes the situation worse is the frequent turnover of staffs in core service functions, which also limits ANAS HTP's business consistency and sustainability. The frequent replacement of the ANAS HTP Director stems from a structural issue that the appointment of the ANAS HTP Director is linked to that of the ANAS President. The frequent turnover of the ANAS HTP staffs also implies their lack of loyalty or motivation to work for the organization. These issues need to be addressed to ensure a sustainable growth of ANAS HTP.

1.4.2 Suggestions

Then, what are the possible solutions for these workforce issues of ANAS HTP?

First, ANAS HTP needs to adopt a strategy of developing talents inside rather than attracting them from outside for enhanced expertise in the three core service functions like TT & TC, Start-up & Incubation, and Test & Certification. With sufficient fund, ANAS HTP would be able to attract experts from outside. However, ANAS HTP's budget is not sufficient and the country does not have that many experts within its territory. Given these limitations, a strategy of strengthening the capacity of existing workforce would be more effective than recruiting experts from outside sources. One of the effective approaches to this strategy is the use of ODA projects. For example, ANAS HTP will be able to strengthen the human capacity of its people by sending them to participate in STEP's K-Innovation Program, KDI's KSP Program, and KOICA training programs. ANAS HTP can also operate partnership programs with other similar organizations in Azerbaijan (e.g., other high-tech

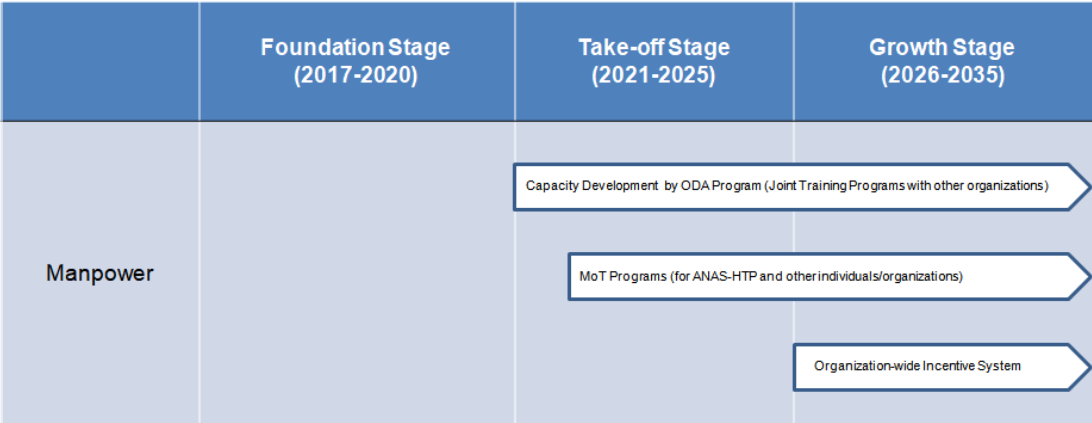
park management institutions, TLOs on college campuses, incubator organizations, and Technoparks). Through this partnership, when these organizations launch capacity-building programs as part of their ODA projects, ANAS HTP will be able to use these programs to expand training opportunities for its staffs.

Second, once it accumulates expertise in the three core service functions using ODA programs, ANAS HTP will be able to launch a program called the “Management of Technology (MoT) Program,” which will help strengthen human capacity of not only its internal staffs but also students and future entrepreneurs throughout the country. Currently, a handful of universities in Azerbaijan are running programs similar to the MoT Program as special courses, but most of these programs are not sustainable. The Bachelor of Science in Information Technology (BSIT) Program run by ADA University is the only representative course being operated in Azerbaijan. The course contents of the MoT Program may include R&D management, project management, technology innovation, technology transfer & commercialization, intellectual property management, start-up & incubation, and entrepreneurship. If ANAS HTP can accumulate its capacity in these areas by implementing related projects, it can then start to attract experts from outside or develop internal experts to run the MoT Program. This program will serve as not only an internal capacity-building program, but it would also constitute a profit-marking business model for ANAS HTP. This program will be operated and managed by the General Affairs & Human Resource Development Department.

Third, ANAS HTP needs to create an incentive system that can enhance the organizational loyalty and motivation of its capable staffs. One of the biggest issues of ANAS HTP is the high turnover rate of its employees. The frequent replacement of the ANAS HTP Director, whose appointment is linked to that of the ANAS President, is not an easy issue to solve. Nevertheless, a new incentive system that can boost people’s motivation can be a solution for addressing the high turnover rate. This incentive system ensures a fair distribution of rewards for outstanding performance by capable individuals or teams. So, a compensation system can be institutionalized to make sure that profits a team or an individual has created through outstanding performance are fairly distributed among those contributors in the form of year-end bonuses.

[Figure 5-10] shows activities in these three solution categories mapped out by development stage of ANAS HTP. During the take-off stage from 2021 to 2015, ANAS HTP will work on the capacity development of its people by using ODA programs and run the MoT Program. Through this program, ANAS HTP will be able to contribute to the capacity-building of the ANAS HTP staffs and external customers as well. During the growth stage from 2026 to 2035, ANAS HTP will be able to boost the motivation of its members by establishing an organization-wide incentive system.

[Figure 5-10] Roadmap for Workforce



1.5 Network

1.5.1 Necessity

An organization’s network capacity refers to its capabilities of securing and disseminating various kinds of resources. So, the organization’s extensive and robust networks can positively affect its growth. This network capacity is especially important during the take-off stage, when an organization needs to enhance the financial sustainability that can support its leap forward. If that is the case, what is the current status of ANAS HTP in terms of network situation? This section reviews the characteristics of ANAS HTP’s networks that it forms with key stakeholders.

First, a review of networking activities between ANAS HTP and the four scientific divisions of ANAS (e.g., Div. of Physical Math & Tech Science, Div. of Chemical Science, Div. of Earth

Science, and Div. of Biological & Medical Science) reveals that their network activities are still confined to building a database of information on patents generated by institutes under these four scientific divisions of ANAS. This implies that patents generated by these institutes under ANAS have not been transferred to other institutes nor commercialized.

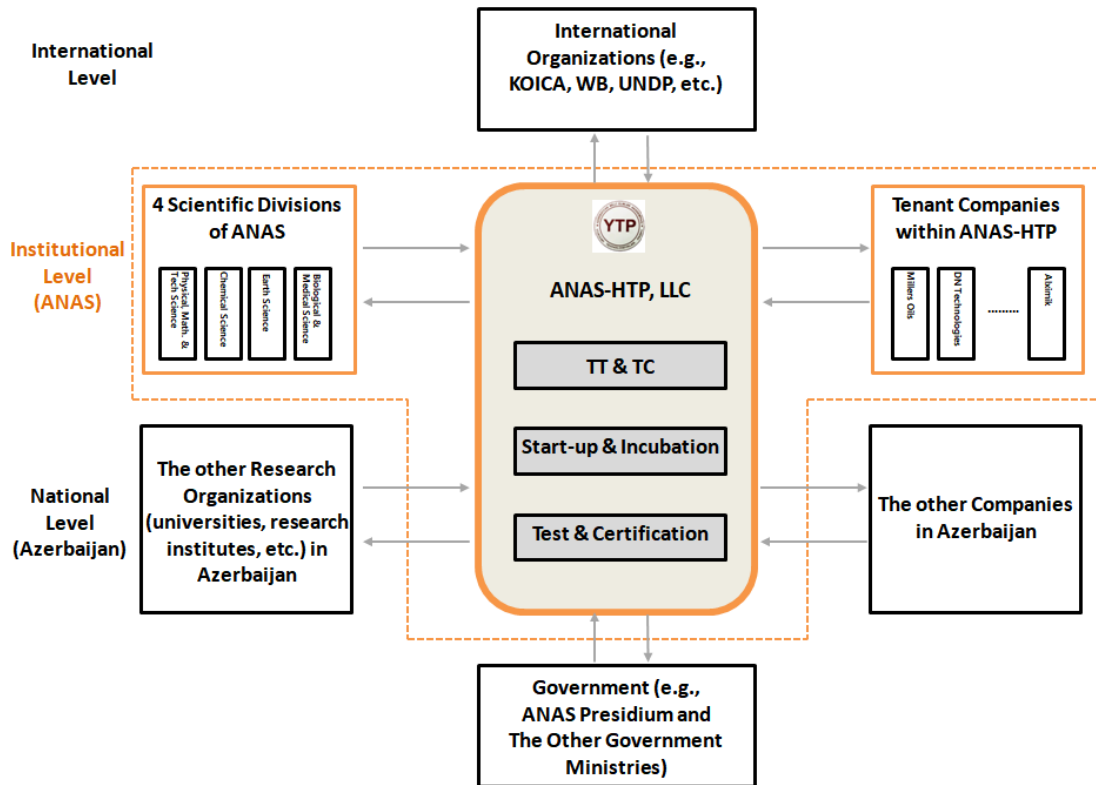
Second, an analysis of the networking activities between ANAS HTP and ANAS Presidium shows that ANAS HTP is not providing any necessary support for ANAS Presidium while the latter performs its functions of S&T policy planning and managements for the country. Moreover, the level of ANAS HTP's participation in the Innovation Fund operated by ANAS remains low.

Third, as part of networking activities with its tenant companies (i.e. residents), ANAS HTP provides some supportive services like marketing service for tenant companies' products. However, networking activities between other innovation actors and ANAS HTP tenant companies for technology transfer and commercialization are almost nonexistent.

Fourth, to build networks with other research institutes in Azerbaijan (e.g., innovator actors in industry, academia and research institutes), ANAS HTP is pursuing cooperation with SOCAR Baku Oil Refinery Plant, Institute of Petrochemical Processes of ANAS, Smart Home LLC, Western Caspian University, Azerishig OJSC, Baku Engineering University, and the Ministry of Agriculture, but active business relations have not been formed yet.

Lastly, ANAS HTP's networking with foreign and international organizations shows that ANAS HTP has been cooperating with various stakeholders including the Science and Technology Policy Institute (STEPI), South Korea Innopolis Foundation, EU TAIEX Program, the Moscow Center for Structural Transformation of Industry, Saint Petersburg Technopark, China-Russian Innovation Technopark, Anzali-Astara Technopark of Iran, Azerbaijan Office of KOICA, SGS (Switzerland), and ACTIVA of Spain. However, ANAS HTP's networking with these organizations is not sufficient yet to support ANAS HTP's capacity and infrastructure building in three core services functions.

[Figure 5-11] Stakeholders of ANAS HTP



1.5.2 Suggestions

Then, what can be done to promote ANAS HTP's networking activities during the imminent take-off stage?

First, ANAS HTP needs to promote its cooperative relations with international organizations, especially with ODA related organizations (e.g., KOICA, JICA, WB, and UNDP). For example, ANAS HTP is recommended to regularly visit the KOICA headquarter in South Korea, KOICA Azerbaijan Office, and the South Korean Embassy in Azerbaijan and ask for assistance. In addition, ANAS HTP needs to maintain its cooperation with KOICA programs in South Korea, STEPI's K-Innovation Program and KDI's KSP Program through steady and regular contacts with them. These programs are designed for ODA policy consulting and training, so ANAS HTP's close cooperation with these programs will help ANAS HTP

strengthen its human capacity in the three core service functions. In addition, KOICA programs can be of great help in terms of providing ANAS HTP with necessary physical infrastructure (e.g., machinery, equipment, and buildings). Particularly, STEPI will be able to play a mediating role between ANAS HTP and South Korean ODA organizations when ANAS HTP is pursuing cooperation with these South Korean counterparts.

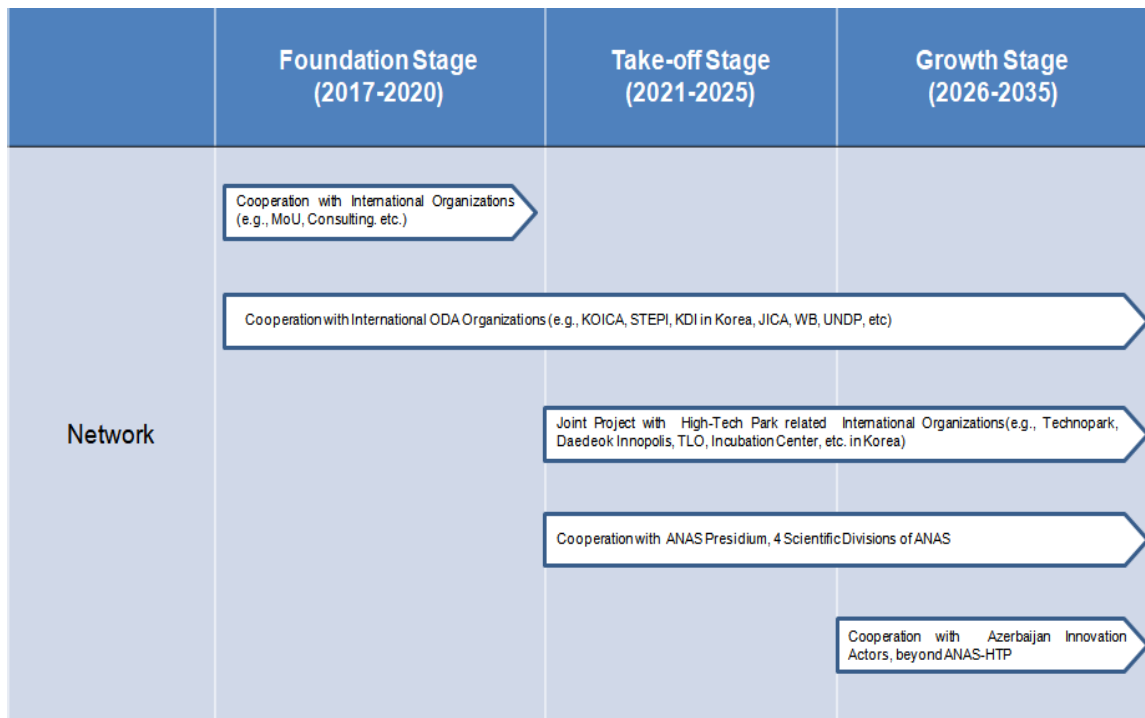
Second, ANAS HTP needs to build networks with world leading high-tech parks and their related organizations. For example, ANAS HTP can sign the Memorandum of Understanding (MoU) and consistently pursue cooperation projects with South Korean counterparts like Technoparks, Daedeok Innopolis, Science Parks, the Asian Science Park Association (ASPA), Incubation Centers, TLOs, and Industrial Parks. Through these cooperation projects, ANAS HTP will be able to learn important know-how and operational skills from the South Korean partner organizations in three core service areas. Funding for these cooperation projects can be secured through KOICA programs. Here, STEPI will be playing the role of a mediator again.

Third, ANAS HTP needs to discover new cooperation activities with ANAS Presidium and other institutes under the four scientific divisions of ANAS. For example, new projects need to be developed to promote the use of the Innovation Fund currently run by ANAS Presidium. To meet the needs of ANAS Presidium, S&T policy development and economic feasibility analysis can be done in the form of contract research while the Innovation Fund can financially support them. To maintain networking activities with the four scientific divisions of ANAS, ANAS HTP can establish a patent management system and effectively manage patent information. At the same time, ANAS HTP needs to provide necessary services to promote technology transfer and commercialization in its interactions with other institutes.

Fourth, it is necessary to promote cooperation networks with Azerbaijan innovation actors in industry, academia, and research institutes beyond ANAS HTP. For example, it is important to sign MoUs and pursue joint projects with high-tech parks and prestigious colleges (e.g., ADA University), TLOs and incubators in Azerbaijan.

A roadmap of these activities by stage is presented in [Figure 5-12]. The foundation stage from 2017 to 2020 was focused on cooperation activities with internal organizations including MoU signing and consulting. From the take-off stage from 2021 to 2025, active cooperation and networking need to be pursued with international ODA agencies and international high-tech park related organizations. In addition, new cooperation projects need to be developed and supported along with ANAS Presidium and institutes under the four scientific divisions of ANAS and profits need to be created from these newly developed joint projects. During the growth stage from 2026 to 2035, cooperation programs need to be jointly implemented with the Azerbaijan innovation actors going beyond ANAS HTP.

[Figure 5-12] Roadmap for Network



2. Technology Transfer & Commercialization⁷²

2.1 Necessity

The research team has proposed “scaling-up of technology transfer & commercialization service” as an important strategy that will make ANAS HTP emerge as a top HTP in the Caspian countries by 2035. Then, what makes technology transfer and commercialization service an important strategy?

The arrival of the Fourth Industrial Revolution has intensified technology competition among countries as well as companies, which is also accelerating technology development at an enormous speed. Now, with the increasing demands for efforts in responding to the rapidly changing technology environment, the size of the global R&D investment has steadily expanded along with the increased number of patent application and registration. Contrary to the increased R&D investment and improved R&D outcomes like technology patents, the life cycle of products and markets has been gradually shortened, which often interferes successful technology commercialization and even shortens the life cycle of companies. Efficient acquisition and use of technologies are considered as prerequisites for an active response to the changing market. To preemptively lead and dominate the newly emerging markets driven by technology convergence, global R&D companies are now moving swiftly by adopting the strategy of ‘buy R&D’. The emergence of new disruptive technologies that can dramatically change the existing industrial landscape and the increased importance of interdisciplinary convergence of different technologies are making ‘open innovation’ and rapid ‘technology commercialization’ core elements of competitiveness.

Then, what is the current status of ANAS HTP and its capacity in terms of technology transfer and commercialization? During the foundation stage, ANAS HTP has operated a

⁷² This chapter was written by Dr. Sangho Choi of the Daejeon Technopark

team in charge of technology transfer and commercialization consisted of a minimum number of personnel. ANAS HTP's annual budget for technology transfer and commercialization activities is estimated at around 20,000 AZN, which is only enough to cover the labor cost. This budget is mainly funded through the rental fees from resident companies and organizations as well as other relevant projects. Currently, ANAS HTP is managing a total of 457 patents developed by ANAS HTP. The patent development activities of ANAS HTP residents are expected to be intensified. Therefore, there arises a need for ANAS HTP to modernize its patent management by building a database of patents developed by ANAS HTP and set up and operate a system for technology valuation and transactions, which will ensure a sustainable operation of ANAS HTP.

2.2 Suggestions

Then, what specific efforts are needed to scale up ANAS HTP's technology transfer and commercialization service? The research team suggests the following action plans in four different areas; i) building infrastructure for technology transfer and commercialization, ii) acquisition of technology patents and building a patent database, iii) building a technology valuation system, and iv) building a technology transaction system.

2.2.1 Building infrastructure for technology transfer and commercialization

2.2.1.1 Developing a plan for organizational structure and operation

To be able to develop into a sustainable organization (i.e. an independent center) from the current TFT within a 10-year time period, ANAS HTP's 'Technology Transfer & Commercialization Team' needs to map out a plan for its organizational structure and operation that can help acquire sufficient budget for its stable operation and development and create business opportunities including various projects.

The research team suggests that the development of this plan be completed by the end of 2022 and the current team be expanded and restructured into 1 division (with 7 staffs)

composed of three teams by 2025. These teams will be responsible for database management, technology valuation, and operation of technology transactions, respectively.

It is further suggested that the division be evolved into an independent and financially sustainable center (with 14 staffs) with three divisions by 2035 by creating various income generating opportunities.

In mapping out this organizational restructuring plan, roles and responsibilities of different organizational units and their staffs need to be clearly defined by development stage. At the same time, an optimal number of staffs required for a successful implementation of planned projects needs to be accurately estimated. In fact, the estimation of an optimal workforce is closely related with the staffs' technical and professional maturity and skill level.

When an organization is restructured into one in a bigger scale, it always necessitates extra budget including increased labor costs. This implies that such extra budget requirements need to be funded through various sources. So, ANAS HTP needs to launch new initiatives in the forms of R&D projects and technology transfer and commercialization projects through collaboration with various partners like government, public research institutes, private sector companies, and overseas research institutes. It is recommended that ANAS HTP use these project budgets to attract experts in various fields, expand its organization, and build necessary infrastructure.

2.2.1.2 Promoting the legal status

Laying the legal foundation for the establishment of ANAS HTP itself may be important. However, what is more important for transforming it into a sustainable organization specialized in technology transfer and commercialization with growth potential is to enact and revise relevant laws on promoting technology transfer and commercialization activities in science and technology field. If ANAS HTP's legal status is solidified by being officially designated as an organization specialized in technology valuation and transactions in science and technology filed in the future, it will be able to generate

income from these technology valuation and transaction services, which will play a critical role in securing its financial sustainability on a long-term basis.

When it comes to the refinement of laws related to technology transfer and commercialization, it is recommended that a comprehensive review of the relevant Azerbaijani laws be conducted by ANAS HTP's Technology Transfer and Commercialization Team and ANAS HTP's staffs in charge of legal affairs and draft laws be prepared by 2022 by these legal staffs, who will work together to achieve the goal of enacting and revising related laws by 2023. For this purpose, it is important to maintain active communication with the government ministries in charge, convince them of the need for the enactment and revision of relevant laws, and ask them to provide their input into the legal enactment and revision process. It is also recommended to shorten the time to be taken for the legal enactment and revision process by building and operating collaborative networks with other stakeholders including the government ministries and the parliament.

In relation to the technology transfer and commercialization services to be provided by ANAS HTP, it is also recommended to lay the legal ground for i) developing a master plan for promoting technology transfer and transactions, ii) providing government budget support for promoting technology transfer and commercialization, iii) supporting the development of systems for technology valuation and transactions, iv) operating a designated organization specialized in technology valuation and transactions, and v) providing budget support for the development of related workforce, based on the "Science and Technology Promotion Act".

For your reference, South Korea has laid down the following legal grounds for technology transfer and commercialization by respective industry based on various related laws. These legal grounds have been laid down for the purpose of i) developing policies, ii) promoting technology development, iii) providing budgetary support for technology transfer and commercialization, iv) building and operating a system for technology transfer and commercialization, v) designating and developing organizations involved in technology valuation and transactions, vi) supporting organizations involved in technology valuation and transactions, vii) developing related workforce, and viii) promoting

international cooperation of the involved organizations.

Additionally, it is recommended that the following supportive needs be reviewed and reflected in the legal enactment and revision process for promoting government R&D; i) budget support for R&D in science and technology ii) support for obtaining technology patents, iii) management and operational support for running a comprehensive patent database, and iv) budget support for disseminating R&D outcomes. A thorough review of these needs is required to lay the policy and legal grounds tailored to the demands of the respective stakeholders including technology development actors as well as technology commercialization actors who are also the recipients of technology transfer.

2.2.1.3 Acquisition of operational workforce and strengthening their capacity

Under the phased growth plan of ANAS HTP's technology transfer and commercialization unit, it is critical to secure high caliber workforce and strengthen their capacity. Such workforce includes experts to be engaged in various initiatives such as promoting R&D projects, managing a patent database, and management and operation of systems for technology valuation, transfer, and transactions.

For this purpose, it is suggested that ANAS HTP's technology transfer and commercialization unit come up with a plan to secure operational workforce and strengthen their capacity by 2022. Then, it is recommended to hire 7 experts by 2025 for the efficient operation of three teams specialized in the operation of a patent database, technology valuation, and technology transactions, respectively. In order for this unit to evolve into an independent center to lead the nation's technology transfer and commercialization activities, it is recommended to develop a workforce plan to secure 14 experts by 2035.

Securing sufficient budget to cover the labor cost of these operational staffs is also an important task that can be achieved through i) implementing various R&D and technology commercialization projects and ii) generating income from fees for technology transfer and transactions services.

The research team also suggests the introduction of an incentive system by 2023, which will help attract experts and retain existing workforce. It is recommended that an incentive system be introduced to reward R&D outcomes like technology patents and technology transfer on top of the basic salaries now being paid to those engaged in various projects.

In addition, various channels need to be secured to attract experienced talents, retrain incumbent personnel through HR training programs, and outsource talents using local and global recruiting firms.

Based on the prediction that technology transfer and commercialization service would be proliferated in various sectors in Azerbaijan after 2035, a desirable long-term direction for ANAS HTP is to transform itself into an organization specialized in cultivating and certifying experts in technology valuation and transactions as the demand for these experts is expected to grow significantly not only from ANAS HTP itself but from other stakeholders like companies using ANAS HTP's technology transfer and commercialization services.

2.2.2 Acquisition of technology patents and building a patent database

2.2.2.1 Promoting the creation of technology patents

Prior to building a patent database, it is necessary to build and secure infrastructure for developing promising technologies that can be subjected to technology transfer and transactions in various technological fields and industries. Promoting technology transfer and transactions is not possible unless technology patents to be managed by ANAS HTP are first created by government research institutes, universities, and companies. The potential value of these technologies to be patented can only be enhanced when they are high quality technologies with high marketability and widespread impact.

To promote the development of such technology patents, infrastructure needs to be built so that various government-led R&D projects can be launched and high value added technologies can be created and patented through R&D activities of government agencies and universities. At the same time, ANAS HTP needs to build and secure its capacity to

manage and operate technology patents, whose management will be commissioned by research institutes and universities, through joint research or other forms of collaboration with these counterparts. ANAS HTP is also required to possess its capacity to create new projects through collaboration with private sector companies or overseas research institutes.

For the purpose of creating demands for R&D, ANAS HTP needs to build and expand collaborative networks with research institutes, universities, and companies by 2022. By taking advantage of these networks, it is recommended that ANAS HTP perform the following activities annually; i) conducting a survey on R&D needs, ii) developing project proposals and sending them to the government or private sector companies by forming project teams, iii) promoting successful biddings of R&D projects, iv) developing plans to disseminate R&D outcomes such as technology patents, and v) promoting active technology transfer and commercialization of R&D outcomes.

A two-step approach can be taken. As a short-term approach, ANAS HTP can place its focus on projects with government-funded research institutes, universities, and large companies till 2023. Afterwards, ANAS HTP will be able to expand its services by launching various R&D projects with startups and SMEs, who will play a pivotal role in the country's technology development. Of course, different approaches or strategies need to be considered depending on the ratio of Azerbaijani companies by their size.

For your reference, a proper budget size of R&D or technology commercialization projects is often estimated at around two to three times of the labor cost of the lead organization. Promoting R&D and technology commercialization projects is an important prerequisite not only for securing high caliber talents but also for pursuing the organizational development of ANAS HTP and solidifying its status as an organization specialized in technology valuation and transactions.

2.2.2.2 Building a patent database

A system and a platform for operating a patent database may be one of the most needed infrastructures by ANAS HTP in its journey to become an organization specialized in

technology transfer and commercialization service.

By the year 2022, it is recommended that ANAS HTP i) build an infrastructure to secure excellent technologies (or patents) by doubling or tripling the current number of patents under its management by launching various R&D and technology commercialization projects, and ii) conduct a detailed designing for building a patent database system and its platform.

By the year 2023, the research team suggests that ANAS HTP build a patent database system and its platform by using the budgets of R&D and technology commercialization projects. The limited number of patents currently under the possession of ANAS HTP implies that a reasonable cost is required for building the patent database system and the platform. ANAS HTP's in-house experts can also be mobilized for building such system and platform.

The following key considerations need to be made in building a patent database.

First, for the existing patents under its management, ANAS HTP needs to analyze and identify the following; i) key technology specifications, ii) a claim chart, iii) target areas of technology commercialization, iv) the current status of patent families, v) the current status of local and global patent registrations, vi) the current status of similar patents registered within and outside of Azerbaijan, and vii) the marketability analysis of technologies.

Second, in addition to its official website, ANAS HTP needs to open a website specifically designed for registering technologies to be included in the patent database. This website also needs to offer a separate search window for searching registered technologies. Through close consultations with local and global patent organizations, this website to be operated by ANAS HTP needs to be designed to offer a comprehensive search function linked to other domestic and foreign patent databases. This will help the potential clients of ANAS HTP's technology transfer service search and verify technologies of their interest easily through the website.

Third, ANAS HTP needs to build a linkage with technology developers to better support the patent database search function and provide an accurate analysis of patent technologies. ANAS HTP also needs to build and operate cooperation networks with patent-specialized law firms to be better prepared for technology patent disputes that may arise after technology commercialization.

In this regard, the South Korean case can be referenced. The South Korean Intellectual Property Office (KIPO), a government agency in charge of patent related matters, plays a leading role in i) providing information on patent application and registration, ii) supporting patent application home and abroad, iii) building and operating patent databases, iv) supporting concerned parties in patent disputes, and v) conducting training and educational sessions on intellectual property (IP). Currently, KIPO's patent database offers various services including a real-time search of patent application and registration status and registered patent technologies. For technology search, a category-based search is also available by the Cooperative Patent Classification (CPC) code, the International Patent Classification (IPC) code, the new classification for the Fourth Industrial Revolution, the product classification code, and the design classification code. KIPO also offers a variety of information including the IPC-KSIC (South Korean Standard Industrial Classification) concordance table and the technology-item-patent concordance table.

2.2.3 Building a technology valuation system

2.2.3.1 Planning technology valuation

In order for ANAS HTP to emerge as an organization specialized in technology transfer and commercialization service, it is necessary to build and operate a technology valuation system.

For this purpose, ANAS HTP needs to develop a detailed plan for establishing a technology valuation system by 2023, which will help the organization kick off its technology valuation service in a full-fledged manner from 2024. As a part of this plan, ANAS HTP is recommended to newly launch a team and build a platform for its service.

Various funding sources need to be secured to help build its technology valuation system. For example, ANAS HTP will be able to mobilize i) the government budget, ii) technology commercialization project budget, and iii) donations or grants from research institutes, universities, and companies. Of course, the actual funding will depend on how much efforts ANAS HTP makes in securing its budget for technology valuation service.

The exact definition of technology valuation may vary across countries. However, according to a widely-accepted general definition, technology valuation is often defined as an act of “estimating the economic value of a technology in value amount, grade, or score” based on the relevant laws including the “Act on Promoting Technology Transfer and Commercialization.” In other words, technology valuation may include technology impact assessment, but it can also be limited to the “evaluation of technological competence or value,” which refers to the capability of turning technology into assets from the market perspective. Especially, technology valuation needs to be supported by accompanying measures such as developing relevant laws and policies, building a technology valuation system, designating and operating technology valuation centers, and operating training centers for bringing up technology valuation experts.

Technology valuation can be classified into several different categories according to its purpose or use; i) the valuation of foreign investors’ industrial property rights, ii) collateral valuation of technology, iii) the valuation to estimate standard prices for technology transfer and transactions services, iv) the valuation for the transfer or disposition of technology-related businesses, and v) the valuation of other technologies and technology-based businesses. As such, ANAS HTP needs to build a comprehensive system of conducting various types of technology valuation that can meet the different valuation needs or purposes.

The technology valuation procedure may vary depending on the valuation type, but the following general steps need to be observed; i) a client’s request for the technology valuation service, which is received by the valuation service provider, ii) a preliminary technology valuation, iii) a main technology valuation based on the results of the preliminary valuation, iv) a deliberation of valuation results, and v) a notification of

finalized valuation results.

2.2.3.2 Building a technology valuation system

The establishment of a technology valuation system requires the development of a technology valuation model as the first step. Then, the model and the system need to be patented and the model needs to be continuously upgraded. Afterwards, an expanded model needs to be developed to be applied to different industries and different types of intellectual property rights. For your reference, there exist several technology valuation methods including an income approach, a market approach, and a cost approach, of which the income approach that estimates the value of a technology by projecting its future income is the most widely accepted one.

As the next step, based on the developed technology valuation model, it is necessary to develop a valuation system tailored to the specific valuation needs by company size, by corporate growth stage, and by industry. For your reference, a valuation system is often composed of the following steps.

- 1) Screening: a preliminary screening to evaluate the commercialization potential
- 2) Assessment: an in-depth qualitative assessment of the commercialization feasibility by analyzing key features of technology segments, their legal rights, their marketability, and their commercialization potential
- 3) Valuation: a quantitative assessment of the business value of technologies as well as the value of technology assets based on the feasibility assessment results

The quantitative assessment often includes a set of analyses like the estimated economic life cycle of technologies, their sales projection, their financial analysis, the flow of economic profit, their discount ratio, and their business value, technology factor, and technology value.

Third, it is necessary to develop and operate a specific estimation method that can help judge the technology commercialization potential, using the developed technology

valuation model. This method will help analyze the growth potential as well as the default possibility of technologies and companies by company size, by corporate growth stage, and by industry.

As part the technology valuation, corporate management competence, technological superiority, marketability, business potential, and profitability are analyzed, and internal and external environmental factors are also taken into consideration. Technologies are then classified into different grades based on the technology valuation results. For example, grades from A to E can be assigned to different technologies.

Fourth, it is necessary to develop and operate standard fees for technology valuation services, which will ensure reasonable profits for technology valuation service providers. The technology valuation service fees need to incorporate both elements of a fixed-amount fee and a contract-based fee.

For your reference, the fixed-amount portion is generally composed of screening fee, assessment fee, and other commissions and this portion is determined by the specific valuation service to be used. It is often recommended that the contract-based fee be calculated by adding up all the costs including labor cost and direct and indirect expenditures.

Lastly, it is extremely important to secure valuation experts. It is recommended that ANAS HTP build networks with valuation experts and form and operate a pool of valuation experts, while developing and operating its valuation framework. In addition, training programs can be developed and operated to bring up valuation experts.

2.2.4 Building a technology transactions system

2.2.4.1 Planning a technology transactions system

In order for ANAS HTP to become a leading organization specialized in technology transfer and commercialization service, it needs to build a technology transactions system.

To be able to start its technology transactions service in a full-fledged manner in 2024,

ANA-HTP needs to develop a detailed plan for setting up the system, newly launch a responsible team, and build a service platform by the year 2023.

Various funding sources can be secured if ANAS HTP mobilizes i) the government budget, ii) technology commercialization project budget, and iii) donations or grants from research institutes, universities, and companies.

Key elements of a technology transactions system include i) technology buyers who want to acquire technologies based on the technology valuation results, ii) technology sellers wishing to sell their technologies, and iii) a platform for transactions. In short, a technology transactions system is a kind of a platform for fostering technology transfer. Through this platform, technologies are transferred from technology owners to their counterparts through various forms like disposition, license, technology coaching, joint research, joint investment, and mergers and acquisitions (M&A).

A technology transfer contract is a contract signed between a possessor of R&D outcomes with property value and its counterparts who want to license the use of such R&D outcomes with both parties agreeing upon the contents of license, technology fees, and the amount and methods of payments. The types of technology transfer are often classified into i) the transfer of intellectual property rights, ii) the transfer of know-how, and iii) the transfer of both. Depending on the ownership of the transferred technology, it is classified into technology disposition (the sale of technology) and technology licensing. The technology licensing is again classified into an exclusive license (monopolistic use) and a non-exclusive license (non-monopolistic use) depending on the scope of license.

In general, technologies are transferred through the following procedure; i) identifying and registering technologies to be transferred, ii) receiving applications for technology transfer, iii) providing counseling on technology licensing, iv) holding interviews with inventors, v) signing technology transfer contracts, and vi) performing post-management after technology transfer.

2.2.4.2 Building a technology transactions system

Under the technology transaction system, technology transfer contracts are signed either through the transaction mediation by technology transfer agents based on technology valuation reports or through market-based technology auctions.

The role of technology transfer agents is extremely important in technology transactions as they provide supportive services including professional counseling, consultations, and guidance on technology transactions and commercialization as well as mediation and conciliation services for technology transfer. Usually, technology transfer agents are nationally licensed professionals, and they are authorized to perform their services and maintain their license only after they are registered with the relevant government ministries.

ANAS HTP needs to build a platform for technology valuation and transactions by operating a separate website for these services. A database of patents held by ANAS HTP can be loaded onto this website, so that it can be operated as an integrated service platform. Especially, it is recommended that the website provide links to technology transfer agents and introduce successful cases of technology transactions.

ANAS HTP can also charge technology buyers and sellers respectively with transaction fees including service fees and other expenditures. Depending on the type of technology transactions, ANAS HTP can ask for additional transaction service fees.

2.3 Roadmap

The above-mentioned activities related to technology transfer and commercialization can be performed in a stepwise manner as presented in [Figure 5-13].

First, as a part of its efforts to build infrastructure, ANAS HTP is recommended to i) complete the development of a plan for organizational structure and operation by the end of 2022, launch and operate 1 division and 3 teams from 2023, and restructure these

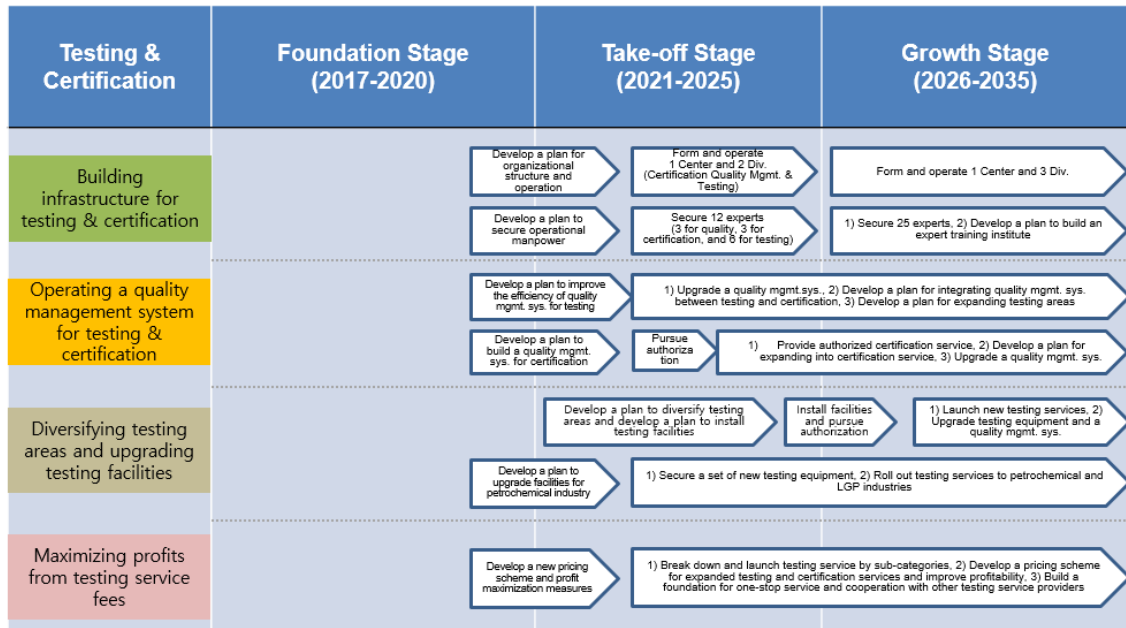
organizations into 1 center and 3 divisions and run these expanded units during the period from 2026 to 2035; ii) complete the enactment and revision of relevant laws by 2023, so that ANAS HTP can develop into an organization specialized in technology transfer and commercialization; and iii) complete the development of an operational workforce plan by 2022, hire 7 experts and start their operation from 2023, and increase the number of these experts to 14 in 2036.

Second, in relation to its effort to secure technology patents, ANAS HTP needs to develop a plan to foster technology patent creation and implement various projects by 2022. ANAS HTP also needs to continuously foster the establishment and operation of networks and implement R&D and technology commercialization projects from 2023. From 2026, it is recommended to expand its business scope to cover projects for startups and SMEs. As for its efforts to build a patent database, ANAS HTP needs to perform the following activities. It first needs to complete the detailed designing and establishment of its patent database by 2023. From 2024, the organization needs to continuously discover key patents and patent technologies to be included in the database, pursue an upgrading of the database, and continue to operate the database platform.

Third, when it comes to ANAS HTP's efforts to build a technology valuation system, it is suggested that ANAS HTP develop a detail plan for building the system by 2022 and complete the establishment of a technology valuation platform by 2023. From 2024, ANAS HTP is recommended to provide its technology valuation service in a full-fledged manner and pursue a system upgrade. It is also recommended to start developing a plan to operate a training center specialized in technology valuation in 2030.

Lastly, in relation to its efforts to build a technology transactions system, ANAS HTP is recommended to complete its system planning and detailed platform designing by 2022 and complete the establishment of the technology transactions platform by 2023. From 2024, it needs to provide technology transfer and transactions services in a full-fledged mode after hiring technology transfer agents. In 2030, ANAS HTP is recommended to start developing a plan to cultivate professional technology transfer agents.

[Figure 5-13] Roadmap for TT & TC



3. Start-up and Incubation⁷³

3.1 Necessity

This research team has proposed a support program for the Start-up and Incubation Service as an operation program that ANAS HTP must carry forward until 2035. Why is it so important to support the Start-up and Incubation Service?

Azerbaijan is pursuing a policy of transition to a knowledge-based economy to reduce its excessive dependence on the oil industry and facilitate social and economic development following the setback from the global financial crisis of 2008 to 2011. To this end, the country must achieve economic diversification and maintain high levels of employment and economic growth (Kuriakose, 2013). A high level of entrepreneurial activity is required to achieve these goals.

The reality of Azerbaijan, however, appears to be far from expectations. For example, Azerbaijan ranked 56th out of 137 countries worldwide on the Global Entrepreneurship Index (GEI) Rank of all countries 2019. According to the sub-index results, it ranked 87th on Entrepreneurship Attitudes, 50th on Entrepreneurship Abilities, and 48th on Entrepreneurship Aspiration. Put simply, Azerbaijan is not doing as great compared to other countries.

The reasons for these low levels of entrepreneurial activity and inclination are as follows: first, a lack of domestic competition; second, investment systems that do not adequately facilitate business development (e.g., high interest rates, risk-averse loan policies, lack of risk capital, and others); and third, a lack of industry-relevant skills among workers.

As of 2018, the gross domestic product (GDP) per capita of Azerbaijan was \$4,721 (World Bank, 2018). Nevertheless, for Azerbaijan to achieve a GDP per capita of \$10,000 in the

⁷³ This chapter was written by Dr. Jongin Choi of Hanbat University

future, it must support and cultivate entrepreneurship within the country.

[Figure 5-14] The Structure of the New Global Entrepreneurship Index (GEI)*

GLOBAL ENTREPRENEURSHIP INDEX	Sub-indexes	Pillars	Variables (Ind./Inst.)
	ATTITUDES SUB-INDEX	OPPORTUNITY PERCEPTION	OPPORTUNITY RECOGNITION
			FREEDOM (ECONOMIC FREEDOM *PROPERTY RIGHTS)
		STARTUP SKILLS	SKILL PERCEPTION
			EDUCATION (TERTIARY EDUCATION*QUALITY OF EDUCATION)
		RISK ACCEPTANCE	RISK PERCEPTION
			COUNTRY RISK
		NETWORKING	KNOW ENTREPRENEURS
			AGGLOMERATION (URBANIZATION*INFRASTRUCTURE)
		CULTURAL SUPPORT	CAREER STATUS
			CORRUPTION
	ABILITIES SUB-INDEX	OPPORTUNITY STARTUP	OPPORTUNITY MOTIVATION
			GOVERNANCE (TAXATION*GOOD GOVERNANCE)
		TECHNOLOGY ABSORPTION	TECHNOLOGY LEVEL
			TECHNOLOGY ABSORPTION
		HUMAN CAPITAL	EDUCATIONAL LEVEL
			LABOR MARKET (STAFF TRAINING*LABOUR FREEDOM)
	ASPIRATION SUB-INDEX	PRODUCT INNOVATION	COMPETITORS
			COMPETITIVENESS (MARKET DOMINANCE*REGULATION)
		PROCESS INNOVATION	NEW PRODUCT
			TECH TRANSFER
		HIGH GROWTH	NEW TECHNOLOGY
			SCIENCE (GERD*((AVERAGEQUALITY OF SCIENTIFIC INSTITUTIONS +AVAILABILITY OF SCIENTISTS AND ENGINEERS))
		INTERNATIONALIZATION	GAZELLE
			FINANCE AND STRATEGY (VENTURE CAPITAL*BUSINESS SOPHISTICATION)
		RISK CAPITAL	EXPORT
			ECONOMIC COMPLEXITY
			INFORMAL INVESTMENT
			DEPTH OF CAPITAL MARKET

*Individual variables are colored with white background while institutional ones with light blue background.

How, then, are the ANAS HTP's status and capacity related to the Start-up and Incubation Service?

ANAS HTP is considered a friendly setting that can promote start-up and incubation, particularly because research centers in many different fields are located around it.

However, the budget and workforce required to take charge of the start-up and incubation service are severely lacking. As of 2020, the budget, including technology transfer and patents other than business incubation, was \$22,588, which accounts for only 11% of the total budget of the ANAS HTP in 2020 (about \$300,000). Moreover, only five employees are available to provide technology transfer and patent services, accounting for 10.9% of the total number of workers (46). Knowledge accumulation within the

organization is also hindered by frequent turnover of the relevant personnel, while securing a stable and competent workforce is difficult due to the low salaries on offer.

3.2 Suggestions

What efforts must ANAS HTP make to scale up its technology transfer and commercialization services? Below, we propose specific implementation plans by organizing them into various categories, namely, "technology transfer and commercialization infrastructure establishment," "technology patent acquisition and patent DB establishment," "technology value evaluation system establishment," and "technology transaction system establishment."

3.2.1 Composing the Start-Up/Incubation Team and Program Development

3.2.1.1 Composing the Start-Up Support and Incubation Specialization Team

As of 2020, ANAS HTP has a Department of Technology Transfer, Patent, and Business Incubation that is in charge of the start-up and incubation functions as well as technology transfer. However, to strengthen the start-up and incubation functions at the Take-off Stage, it is necessary to have a separate department that can pursue systemization. For example, we suggest that ANAS HTP combine the Start-ups and Incubation Department with a new start-up discovery team and an incubation specialization team. The new start-up discovery team will actively look for start-ups founded by researchers from ANAS-affiliated institutes or by graduate students (post-doctoral). The incubation team can generate revenue by providing services for various functions needed to launch a start-up business (e.g., entrepreneurship, business modeling, funding, marketing, intellectual property rights, etc.).

Suppose the incubation function is stabilized to a certain extent in the Take-off Stage (2021–2025). In that case, the incubation system may be expanded by subdividing it into Pre-Incubation, Incubation, and Post-Incubation. If functions like these become active,

ANAS HTP may become an independent specialized organization by developing and reorganizing the current department-based system into a center-based system (e.g., KISED in South Korea) in the Growth Stage (2026–2035).

Case: South Korea Institute of Startup and Entrepreneurship Development (KISED)

South Korea is known for having one of the world's best education systems and a very conducive business atmosphere (the World Bank ranks the country eighth globally, in terms of ease of doing business). With these advantages, South Korea should be teeming with aspiring entrepreneurs. However, cultural barriers such as an aversion to risk and failure have kept many would-be entrepreneurs from reaching their full potential. However, over the past decade, the country has bolstered its support for start-up enterprises and implemented several programs that foster an entrepreneurial spirit, thus changing the attitude toward entrepreneurship. In 2008, South Korea began implementing a series of policy packages known as the "Start-up South Korea Initiatives." These policies foster entrepreneurship with support structured around three themes: developing start-up resources, enhancing start-up capacity, and leveraging successful incubation. At the forefront of this effort is the South Korea Institute of Startup and Entrepreneurship Development, a public institute under the country's Small and Medium Business Administration. The institute is now the primary implementation agency for the Ministry of SMEs and Startups' start-up initiatives.

The first theme of support involves developing the requisite skills and resources to create a backdrop for successful start-up creation. The institute implemented five programs addressing various developmental aspects: (i) a program to support the commercialization of creative ideas through content production, registration, and intellectual property protection; (ii) a program to cultivate entrepreneurs by providing them with financing for business operations and access to resources at universities and research centers; (iii) a program to foster the spread of knowledge and know-how by linking successful ventures with fledgling start-ups, and encouraging the former to act as mentors and technology consultants; (iv) a program to encourage technology-oriented start-ups by providing access to patents held by universities and research organizations; and (v) a program to organize a variety of events to generate an entrepreneurial spirit among students.

The second type of support measure is enhancing the capacity of potential entrepreneurs through various education programs, clubs, and competitions. The Youth BizCool program grooms entrepreneurs at a young age by sponsoring clubs, activities, and education

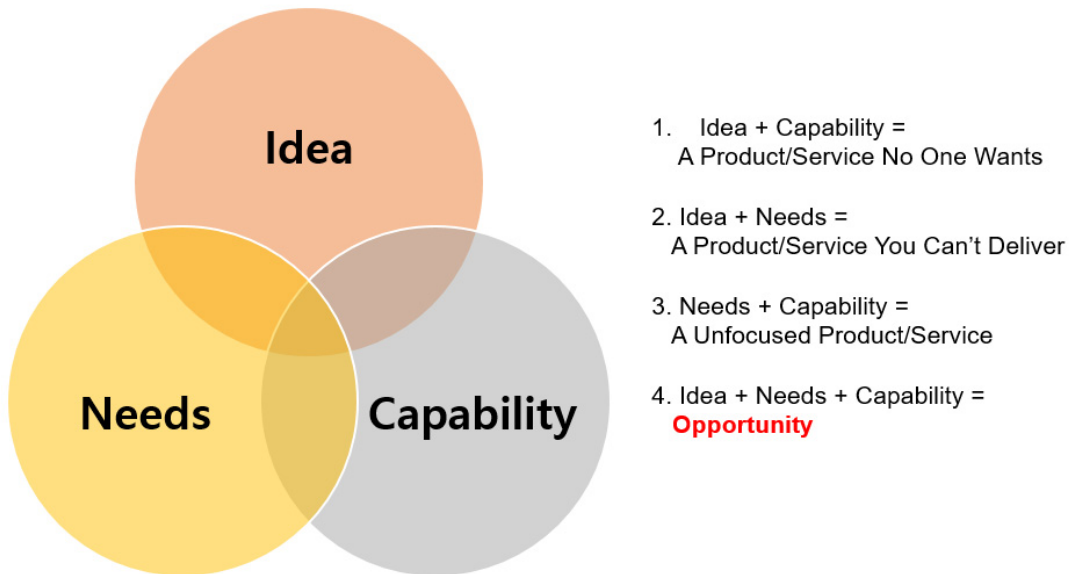
materials targeted at students from primary school through high school. University start-up support programs continue building skills, supporting activities such as clubs and overseas start-up training. Potential entrepreneurs who are not students can also participate in an education program for the general public, which offers mentoring and consulting on weekends. Other efforts also help build capacity in this phase, such as the South Korea National Startup Competition League, which holds annual competitions. Successful teams in this competition earn prize money and cost reimbursements for their pilot products.

The third theme of support enhances businesses' potential to succeed in the country's business incubators network. To improve the effectiveness of these incubators, the institute implemented a program to train and certify specialized business incubator managers. It offers the enterprises at these incubators support for marketing at home and abroad and provides commercialization support in the form of design and development assistance to promising businesses. The institute markets businesses that successfully commercialize products in newspapers, subway train ads, and e-books. Along with the start-up support mechanisms discussed above, the institute supports South Korean entrepreneurs through global outreach and cooperation with international partners to facilitate information exchange. Since 2008 the number of new business ventures and incorporations in South Korea has steadily grown despite the global economic crisis.

3.2.1.2 Start-Up Incubation Program Development

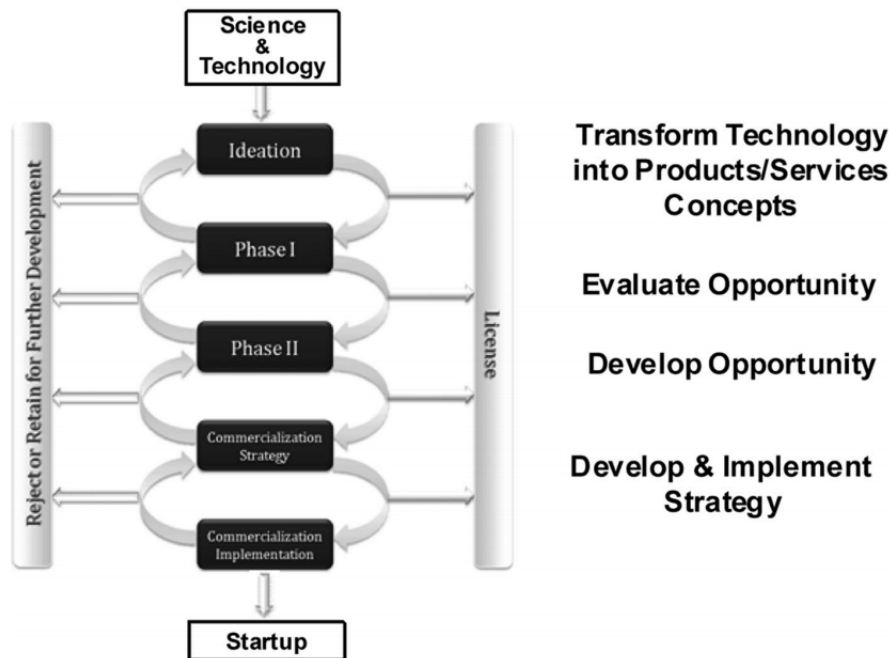
Once the start-up and incubation-related departments are organized, programs for providing start-up and incubation services must be developed. First, the concepts of the start-up and incubation program must be designed based on the three criteria of the INC Program: discovering unique ideas (I), identifying customer needs (N), and securing capabilities (C). As shown in the [Figure 5-15] below, it is difficult to develop as a start-up if any of the three elements—ideas (I), customer needs (N), and capabilities (C)—is missing. The business that satisfies all three criteria is most likely to meet with future success. Therefore, the start-up incubation program should cover content such as how to discover entrepreneurial ideas (I), how to understand customer needs (N), and how to develop capabilities (C).

[Figure 5-15] INC-based Program Design

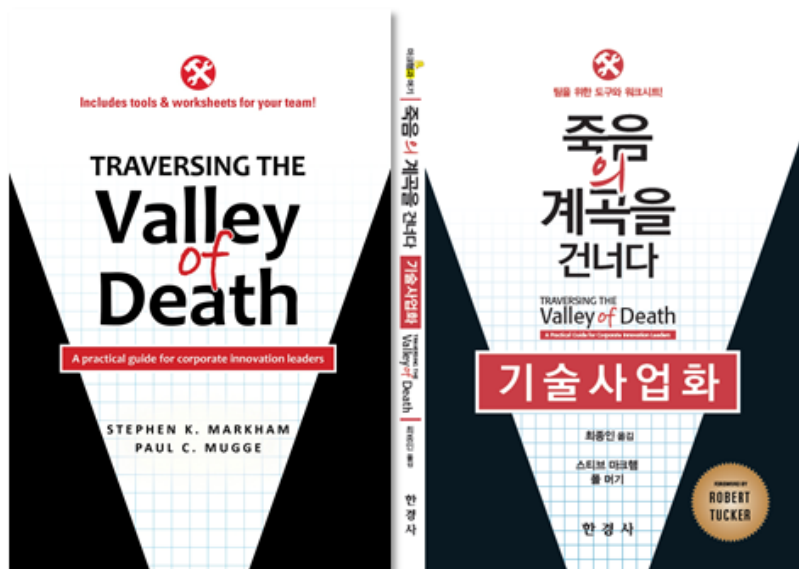


Second, we propose benchmarking the start-up education program to that of a foreign country and localizing such a program as the first step in introducing the start-up incubation program. For example, the Technology Entrepreneurship and Commercialization (TEC) Program is a technology commercialization and entrepreneurship program running in the Research Triangle Park (RTP) in the United States, which is an algorithm-based self-directed education system. ANAS HTP can translate the program into the Azerbaijani language in cooperation with the authors, Markham and Mugge, to use it as educational text. The program has been used as an educational textbook in South Korea since 2006, at research institutions located in the Daedeok R&D Special Zone. It is still being used as a textbook for start-up education in Techno Parks and Startup Graduate Schools. If ANAS HTP wants to cooperate with the American authors for translating the program, a South Korean expert may play a bridging role.

[Figure 5-16] TEC Algorithm



[Figure 5-17] Textbook and Translation



Third, as part of a specific start-up incubation program, the Post Doc Start-UP Program can be promoted among researchers in ANAS-affiliated institutes. There are many researchers and post-doc students currently working in these institutions. The start-up incubation program of ANAS HTP can be promoted by annually selecting those interested in start-ups and providing them with space and various services related to entrepreneurship. Providing such services can also be a source of income for ANAS HTP.

3.2.2 Building Employee Capacity and Focusing on Discovering ANAS Researchers

3.2.2.1 Building the Capacity of Leaders and Employees

In order to strengthen ANAS HTP's start-up and incubation program, it is first necessary to guarantee active support and continuous investment from the head of ANAS HTP. However, due to frequent changes in the management of ANAS HTP, sustainable leadership within the organization is currently limited. Additionally, the appointment of a bureaucratic leader may hinder the promotion of entrepreneurship. Therefore, to strengthen ANAS HTP's start-up and incubation functions, a leader with a good understanding of such processes must be in office for an extended period, presenting a cohesive and clear direction. Nevertheless, since the appointment of the head of ANAS HTP has structural limitations linked to the appointment of the ANAS Presidium, a system and culture in which the head of ANAS HTP is appointed notwithstanding the appointment of ANAS Presidium must be constructed, in order to advance the ANAS HTP.

Second, it is necessary to strengthen the capabilities of leaders and managers in charge of start-up and incubation. To nurture and successfully produce future entrepreneurs, the leaders and managers who train them must be highly capable themselves. However, the current crop of ANAS HTP leaders and managers have not been able to develop their capabilities to the required extent due to continuous replacements within the organization, which results in a limitation in providing quality services. Therefore, to strengthen the capabilities of leaders and managers, ANAS HTP must first actively utilize ODA capacity-

building and educational programs (e.g., KOICA Training Program, KDI KSP Program, STEPI K-Innovation Program, and others). Furthermore, it is vital to recruit and retain professionals with expertise and varied experience from major global universities or related institutions.

Third, it is necessary to strengthen the internal compensation (incentive) system to secure and retain competent human resources. In other words, efforts should be made to minimize the turnover of competent human resources by harmonizing external and internal rewards, which will help them create a vision for their careers.

Fourth, to secure and retain competent human resources, it is necessary to establish an internal compensation (incentive) system and provide training opportunities through international cooperation. Training opportunities can be created by seeking active cooperative relationships with start-up training institutions in the United States (e.g., Kauffman Foundation, Research Triangle Park, NC State, and others) or major South Korean institutions (e.g., Daedeok Innopolis, KISED, and others).

3.2.2.2 Promoting Start-up Incubation Among ANAS-affiliated Researchers

Researchers (graduate students) in ANAS-affiliated institutes can be considered potential customers of ANAS HTP's start-up and incubation services. ANAS is currently affiliated with four institutes related to science, technology, and engineering (e.g., Physics, Math & Tech Science, Chemical Science, Earth Science, Biological and Medical Science). Many young researchers such as graduate or post-doc students are currently carrying out various studies and research activities in these institutes. ANAS HTP can promote entrepreneurship by encouraging them to establish start-ups and providing entrepreneurship education, initial funding, and space.

Moreover, it is necessary to operate a program among researchers and faculty within ANAS-affiliated institutes, to actively seek people with excellent ideas and support them as the Idea Champion. This Idea Champion can establish a start-up through the intensive support offered by ANAS HTP, and grow into a promising entrepreneur in the future. This champion can also play a role in recognizing and emphasizing the importance of

entrepreneurship by becoming a successful reference model for other prospective entrepreneurs.

3.2.3 Securing Budget Continuously

3.2.3.1 Building a Foundation by Securing External Funds and Existing Profits

The provision of services related to start-up and incubation also entails personnel expenses, business expenses, and infrastructure building expenses. Hence, it is important to secure the budget through various sources of income. How can funding be secured?

First, to cover the costs of human-resource capacity building and infrastructure, it may be effective to secure the budget via ODA-related organizations abroad. For example, it is a good idea to actively utilize South Korea's ODA programs, Japan's JICA, World Bank, and other ODA programs run by the UNDP.

Additionally, increasing the registration fees or rental fees of tenant companies, as the primary source of income, can contribute to meeting the labor and direct costs necessary for future business execution. As of 2020, revenues from tenant companies cover 97.4% of ANAS HTP's total revenue. It would be necessary to increase this revenue by increasing the number of tenant companies in the future.

Moreover, it is essential to secure start-up- and incubation-related research funds through the Azerbaijani government and multinational corporations. To strengthen ANAS HTP's capacity for providing start-up and incubation services, relevant research and basic infrastructure are essential. For example, it is necessary to develop a related curriculum, prepare guidelines, and establish a system. ANAS HTP will build such an infrastructure by receiving research funding from the Azerbaijani government or multinational corporations.

3.2.3.2 Generating Own Revenues by Providing Start-up and Incubation Services

Once the budget required for capacity building and infrastructure construction is secured via overseas ODA-related organizations and existing tenant companies, ANAS can secure the direct budget required for its projects by providing core services.

For example, ANAS HTP can generate its profits by providing entrepreneurship and incubation-related services to researchers from ANAS-affiliated institutes or prospective entrepreneurs in Azerbaijan. In other words, by providing space and start-up-related services to prospective entrepreneurs (e.g., start-up funding, management consulting, technical consulting, marketing consulting, and others), it can earn not only rent income but also profits via secured future equity if a company becomes successful.

3.2.4 Expansion of Networks with Both Domestic and Foreign Organizations

3.2.4.1 Close Cooperation with Domestic Start-up and Incubation Organizations

For ANAS HTP to enhance its start-up and incubation capabilities and improve the quality of services, it must strengthen cooperation with other domestic start-up and incubation organizations. There are cases in which existing human resources move to other organizations; this can be a way to utilize these human resources in a community sense. For example, by appointing INNOLAND staff in Baku as its advisory members, ANAS can utilize their know-how.

Additionally, suppose ANAS HTP wishes to use the ODA education and capacity-building programs to improve employees' capabilities. In that case, it can double the education and training opportunities by operating a cooperative program with other similar institutions in Azerbaijan (e.g., other high-tech park management institutions, university TLO, incubators, technoparks, and others). Therefore, when other organizations carry out capacity-building programs as ODA projects, they can participate in those programs together. Furthermore,

such cooperation will enable ANAS to provide higher quality services to tenant companies by sharing information and capabilities related to start-up and incubation.

3.2.4.2 Strengthening Cooperation with Foreign Start-up and Incubation Organizations

ANAS HTP will acquire budgets and essential skills by strengthening cooperation with start-up and incubation institutions abroad. For example, it is possible to acquire relevant knowledge by activating personnel exchange programs with various entrepreneurship organizations overseas (e.g., the Kauffman Foundation in the United States and the South Korea Entrepreneurship Foundation in South Korea and others). Additionally, ANAS HTP's entrepreneurs' capabilities can be bolstered by introducing the education and incubation programs (e.g., RTP and TEC programs in the United States) of relevant organizations abroad.

Moreover, establishing a worldwide network with global start-up incubation organizations will improve ANAS' capabilities and vitalize start-up incubation programs. For example, joint projects with start-up incubation centers within technoparks (e.g., incubation centers in Gyeonggi TP), universities (e.g., incubation centers at Hanbat University), and research institutions (e.g., KIST Incubation Center) and others, can promote personal exchanges between South Korea and Azerbaijan as well as the entrepreneurship climate.

3.2.4.3 Increasing the Awareness Level of Research through Enhanced Publicity

In order to invigorate entrepreneurship and start-up incubation not only in ANAS HTP but also in Azerbaijan as a whole, a shift in researchers' perceptions is needed. It is necessary to raise awareness that a shift in researchers' perceptions can lead to entrepreneurship opportunities beyond research and technology developments, and this shift needs to be culturally established. A prime example is the British Enterprise Culture. First, ANAS HTP must carry out an awareness-raising campaign among researchers (graduate students) in ANAS-affiliated Institutes. Next, it should cooperate with similar institutions in Azerbaijan (e.g., INNOLAND) to spread the movement.

3.3 Roadmap

The roadmap for each stage of growth for start-up and incubation-related activities is shown in the figure below.

First, in terms of putting the start-up/incubation teams together, the Start-up and Incubation Department must be separately established in the Take-off Stage, with the new start-up discovery team and incubation specialization team as the underlying layer. In the Growth Stage, ANAS must be expanded to an independent, center-based system. With regards to start-up incubation programs, INC-based programs must be designed during the Take-off Stage, and foreign start-up incubation programs must be introduced and translated into the Azerbaijan language. After this, the Post Doc Start-UP Program must be implemented among researchers within ANAS.

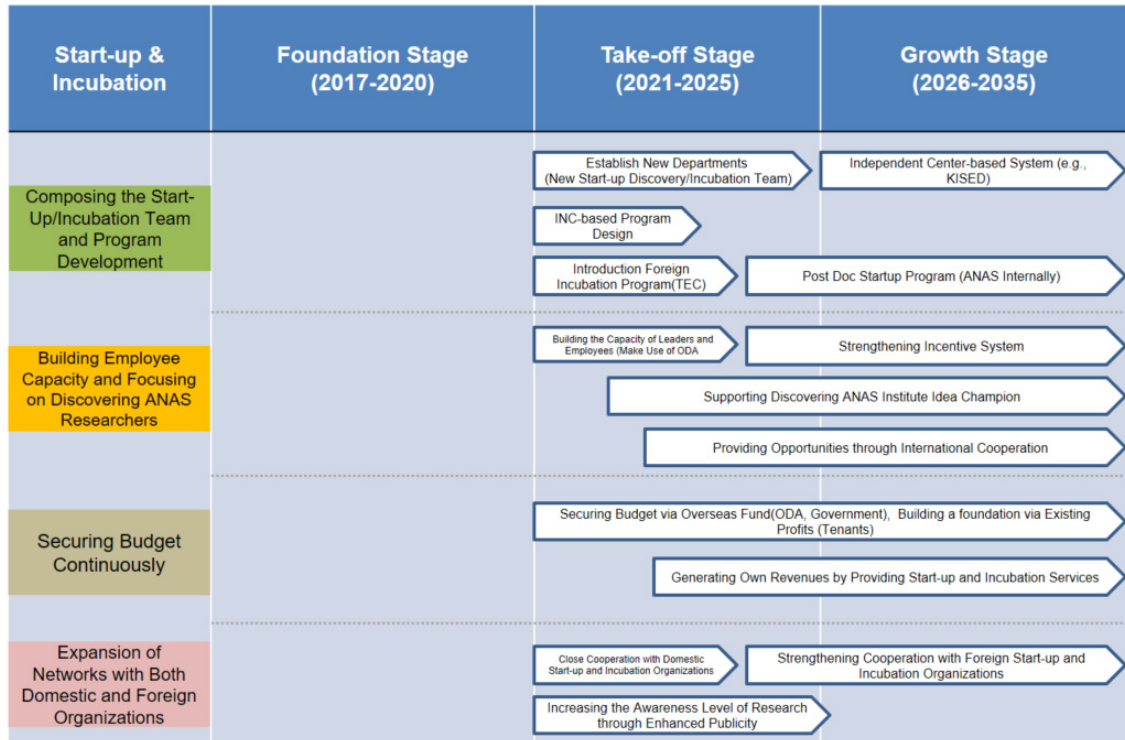
Second, with regard to strengthening staff members' capacity building, it is necessary to enhance the abilities of leaders and members using various ODA programs from the beginning of the Take-Off stage. Additionally, it is necessary to establish a reward (incentive) system to boost the morale of employees. Simultaneously, there must be a program to seek and intensively support Idea Champions in ANAS institutes from the middle of the Take-off Stage, and provide training opportunities for employees through international cooperation.

Third, to continuously secure the relevant budget, it is essential to seek funding from ODA programs or the Azerbaijani government from the beginning of the Take-Off Stage. It is also necessary to secure labor and infrastructure costs by maintaining the existing income sources generated by tenant companies. Once its start-up and incubation service functions achieve some level of maturity, the budget can be secured through the profits generated through these services.

Finally, in terms of establishing a network with related domestic institutions, it is critical to strengthen cooperation with domestic start-up and incubation organizations such as INNOLAND. This can gradually lead to cooperation with foreign organizations related to start-up and incubation. Concurrently, programs (e.g., Enterprise Culture) must be

conducted to increase the level of awareness of researchers regarding start-ups and incubation, through enhanced publicity efforts at the beginning of the Take-Off Stage.

[Figure 5-18] Roadmap for Startup & Incubation



4. Test & Certification⁷⁴

4.1 Necessity

The research team has proposed testing and certification services as a strategic area that ANAS HTP needs to focus on from the take-off stage. Then, why are the testing and certification services such an important business area for ANAS HTP under the current circumstances of Azerbaijan?

First of all, the testing and certification service industry is considered as one of the most important service industries as it helps guarantee product quality, verifies conformity to standards, and ensures product credibility using standardized specifications. Also, it is often referred to as the most representative case of infrastructure-intensive industries because it requires setting up a set of expensive facilities to make profits through testing and certification services.

In order for the testing center of ANAS HTP to emerge as the nation's top organization specialized in testing and certification services, ANAS HTP needs to clearly understand the industrial landscape of Azerbaijan. Since the basic unit of testing service is a specific product of a specific industry, it is necessary to accurately estimate the size of locally produced products as well as the size of products that are imported or exported through international trade and identify industries producing or circulating these products. Such understanding or estimation of products and industries is important because it helps ANAS HTP accurately forecast testing and certification demands. In fact, an accurate understanding and forecasting of testing and certification needs of specific industries is extremely important in making cost-effective investments in expensive facilities. It is also important because it will ensure ANAS HTP's testing center develops into a sustainable certification specialized organization by creating profits from its certification services using those expensive testing facilities.

⁷⁴ This chapter was written by Dr. Sangho Choi of the Daejeon Technopark

What is critical to the success of a testing and certification service institute is its capacity to secure the credibility of the testing and certification results. Securing credible testing and certification results requires a formidable number of experts, intensive practice of testing and certification services, and an efficient operation of a quality management system.

If so, what is the current status and capacity of ANAS HTP's testing and certification services? Still in the early stage of its foundation, ANAS HTP is currently operating a testing center with a minimum number of personnel. The annual budget of this center is around 30,000 - 50,000 AZN, which is barely sufficient to cover the labor cost of the center's 6 staffs. This budget is mainly funded through the rental fees from resident companies and organizations as well as other relevant projects.

The testing center of ANAS HTP has installed modernized testing facilities for petrochemical and LPG industries, whose service capacity is extensive enough to cover 27 testing specifications. The center has been recognized as an authorized testing institute for petrochemical and LPG industries by the nation's certification authority in accordance with ISO/IEC 17025. It now aspires to become an authorized certification institute as well by upgrading its quality management system in compliance with ISO/IEC 17065 and securing needed experts.

For petrochemical and LPG industries, the center strives to enhance the coverage and quality of its testing service by stepping up those testing facilities currently under operation. It also wishes to expand its testing service to other industries like ICT, healthcare and biotechnology, and agriculture.

4.2 Suggestions

Then, what initiatives need to be pursued by ANAS HTP for the globalization of its testing and certification services from the kick-off stage? To answer this question, the research team has proposed the following action plans; i) building testing and certification infrastructure, ii) developing an operational plan of the quality management system for

testing and certification services, iii) diversifying and upgrading testing facilities, and iv) maximizing profits from testing service fees.

4.2.1 Building testing and certification infrastructure

4.2.1.1 Developing a plan for organizational structure and operation

The testing center of ANAS HTP aims to develop into a sustainable and financially independent organization by securing a stable income from its high-quality testing services based on its facility investment. It also aspires to achieve a stable operation and organizational development. To achieve these goals, ANAS HTP needs to map out a plan for organizational structure and workforce operation of its testing center.

The research team recommends that the development of this plan be completed by the end of 2022 and the current testing center be expanded into a bigger organization (with 12 staffs) composed of two divisions by 2025. These two divisions will be responsible for i) certification and quality management, and ii) testing of petrochemical products, respectively.

It is suggested that the size of the center be further expanded (with 25 staffs), so that it establishes itself as an independent and financially sustainable center with three divisions by 2035 by securing various income sources. During the early stage of organizational structuring, it is recommended that the center be composed of two divisions, one for certification and quality management and the other for testing service specialized in petrochemical and LPG industries, and workforce be deployed between these two divisions according to their expertise.

When designing an organization in charge of building and operating a quality management system, it is recommended to structure the organization and deploy its workforce in consideration of the future system scalability. When ANAS HTP's testing service is rolled out to other industries like ICT in the future, a separate quality management unit may not be necessary, but a module-based expansion of a testing unit responsible for ICT industry

should become available. This, of course, should be supported by a timely supply of required workforce.

It is recommended that an organization dedicated to certification service be placed under the quality management division in the early stage, but later be established as an independent unit from the quality management division to better respond to the increased certification service needs. The quality management division should be allowed to operate a quality management system for testing and certification, respectively.

The most efficient approach to organizational structuring and workforce operation of the testing and certification center would be to follow the recommendations of ISO/IEC 17025 and ISO/IEC 17065. As the testing center is currently affiliated with ANAS HTP, its organization and business process have been designed in a manner that enhances the overall administrative efficiency of ANAS HTP. However, when the testing center is expanded into an organization performing both testing and certification services, its organizational structure and operation mode need to be aligned with the quality management system by conforming to the respective international standards for testing and certification.

4.2.1.2 Securing experts

The operation of a testing and certification institute requires securing workforce with expertise in testing, certification, and quality management. It is thus recommended that a plan be mapped out by the testing center of ANAS HTP by 2022 to secure such experts and strengthen their capacity. In accordance with this plan, a total of 12 experts, 3 for quality management, 3 for certification, and 6 for testing service, need to be secured by 2025. In preparation for the expansion of testing service into ICT industry, it is further recommended to increase the number of experts in quality management, certification, and testing to 25 by 2035, based on an accurate estimation of workforce requirements. The research team also suggests the introduction of an incentive system by 2023, which will help attract experts and retain existing workforce.

The labor cost of the operational workforce can be funded through i) the budgetary

support during the early stage of testing and certification services, and ii) income from service fees in the later stage of the expanded testing and certification.

In fact, securing the credibility of testing results is a critical factor for the success of the testing service. This means it is extremely important to draw accurate test results of the product performance by accurately applying testing methods conforming to the testing standards and consistently maintaining the accuracy of testing facilities. For this purpose, it is necessary to secure operational workforce as well as top-tier experts with expertise in testing activities in respective fields.

It is recommended to recruit testing staffs with some related educational background (e.g. completion of studies in the related majors), support their participation in external training programs, and have the technology officer develop and lead internal training programs for testing staffs. The training programs need to cover both theory and practice. In compliance with the methods presented in ISO/IEC 17025, the curriculum for training testing staffs needs to focus on factors directly affecting testing results. Some of these factors include i) management and maintenance of testing facilities, ii) verification of testing methods, iii) understanding of testing standards and methods, iv) drafting testing reports and management of raw data, v) understanding of measurement uncertainty, and vi) conducting comparative performance assessments of testing staffs.

Above all, securing qualified testing experts is extremely important, so collaboration with partners institutes or universities is recommended to ensure a stable supply of these experts. ANAS HTP's technology officers of the respective fields need to build their capacity so that they will be able to take part in the development and update of the national standards of Azerbaijan including the testing standards, and contribute to the global standardization efforts including the development of global standards. These technology officers are also required to be capable enough to run internal training programs that will continuously help enhance testing capacity of ANAS HTP's testing staffs. For further growth, the expanded testing and certification center will have to pursue cooperation with testing equipment developers and manufacturers and develop technology standards (or technology regulations) through local and global standardization

efforts. Qualified technology officers need to possess a high level of expertise that can help them develop not only the testing specifications but also the testing equipment.

When it comes to the issue of securing certification staffs, it is recommended that ANAS HTP select some of its experienced testing staffs or experts, who will serve as certification staffs and officers.

In the case of a quality management officer, ANAS HTP is recommended to secure an experienced expert with a good understanding of the quality management system. If this is not possible, it is recommended that ANAS HTP develop quality management capacity internally by building operational know-how and providing in-house training and educational programs under the lead of the top management. In this case, the job security of the quality management officer needs to be guaranteed.

When securing an optimal number of testing staffs, technology officers, and quality management officers, ANAS HTP needs to comply with ISO/IEC standards as well as the minimum qualification requirements recommended by the relevant Azerbaijani authority. From 2035, it is recommended that the testing center of ANAS HTP develop training programs for cultivating technology officers and quality management officers in science and technology field, and evolve into a certification organization.

4.2.2 Developing an operational plan for a quality management system in testing and certification

4.2.2.1 Developing a plan for an efficient operation of a quality management system for testing

In general, a testing institute is required to build and operate a quality management system conforming to the intentional standard of ISO/IEC 17025 and its operation needs to be authorized by the official certification entity. In fact, a quality management system is composed of various documents including manuals, procedural documents, guidelines, and other record-keeping documents. A key objective of operating a quality management

system is to help the testing institute provide accurate and credible testing service.

The testing center of ANAS HTP needs to develop a plan for an efficient operation of its authorized quality management system by 2022 and this plan needs to consider the possible expansion of the quality management system into certification service.

A quality management officer and a technology officer need to demonstrate a good understanding of their quality management system. A quality management officer needs to do his or her best to ensure an efficient management and operation of the quality management system and should be always prepared to present plans for improving the operational efficiency of the system. The quality management officer should always be ready to collect information on additional requirements or other technology regulations presented by the certification authority and upgrade the quality management system in accordance with these regulations.

In the meantime, a technology officer should be aware of important factors affecting the testing results and make sure these factors are properly recorded and monitored by testing staffs. For example, it is necessary for the technology officer to closely check and manage the following; i) managing and maintaining testing equipment, ii) verifying the efficacy of testing methods, iii) generating testing results, and iv) managing and maintaining raw data. At the same time, it is necessary to regularize and systemize these management activities, which will help reduce testing staffs' workload.

In relation to the maintenance of testing equipment, the following actions need to be taken. For example, the revision requests of testing equipment need to be systematized, so that they are delivered to the revision service providers every year and revision services are thus provided in a timely manner. When these revision requests are made, the system should be able to check the accuracy and measurement uncertainty of the equipment for which the revision requests are made, based on the testing standards. In addition, interim inspections need to be conducted regularly to ensure the traceability of the testing equipment. For this purpose, the methods and the criteria of interim inspections need to be clearly defined, so that testing staffs can check at any time whether the traceability of

the testing equipment is properly maintained.

All the documents that are required for the purpose of operating the quality management system need to be developed from the early stage in consideration of the possible system scalability. While changes to the manuals and procedural documents should be minimized, the addition of new guidelines may be necessary in case of the future expansion of the current testing service. Therefore, the system scalability needs to be ensured to support this possible expansion of the testing service. When designing the system scalability, it is also necessary to consider the need for managing other record-keeping documents including the maintenance records of additionally purchased equipment and the testing score sheets.

4.2.2.2 Developing a plan to newly establish a quality management system for certification

The testing center of ANAS HTP is known to have been certified to perform testing and certification services (or suppliers' declaration of conformity, SDoC) by the nation's official conformity assessment authority. The details of the conformity assessment may vary across countries, but the research team has found out the conformity assessment currently available in Azerbaijan covers both certification and SDoC. Therefore, it is critical that the testing center of ANAS HTP develops into an institute capable of providing both certification and SDoC services.

Currently, the testing center of ANAS HTP is providing some certification service, but the testing center is not an officially authorized and registered certification institute. Therefore, there exists a need to establish a quality management system for certification conforming to ISO/IEC 17065 and obtain an official authorization from the certification authority. For this purpose, it is recommended to develop a plan for establishing a quality management system for certification service by 2022, obtain an official authorization from the certification entity by 2023, and start providing certification service as an officially registered certification institute from 2024.

ISO/IEC 17065, an international standard of a quality management system for

certification, is quite similar to ISO/IEC 17025 in some details except for their differences in the technological fields they cover. Therefore, a good understanding of this international standard will make it easier to establish a quality management system for certification and obtain an authorization from the official certification entity. So, it is recommended that ANAS HTP establish its quality management system for certification service through outsourcing, if it can secure enough budget and hire a professional consulting firm operating in Azerbaijan.

One of the most important considerations in rolling out the certification service would be meeting the relevant legal requirements. Based on these legal requirements, it is necessary to check whether the types of inspection or certification are appropriate for expanding into the certification service. Given the varying conformity assessment systems across countries, some countries designate certification institutes based on their relevant laws instead of having the official certification entity to approve and authorize them. So, meeting the legal requirements becomes critical in the former case. In this regard, ISO/IEC 17065 can possibly be referenced as a proxy of these legal requirements.

Another important task is to secure certification experts and staffs. It is recommended to appoint the technology officer in charge of testing service as a new certification officer and replace the testing technology officer with a newly appointed vice technology officer for testing.

4.2.3 Diversifying testing service areas and upgrading testing facilities

4.2.3.1 Expanding the scope of testing service into ICT and other industries

The testing center of ANAS HTP is currently providing testing service for petrochemical and LPG industries, using its testing facilities tailored to these industries. It is recommended that ANAS-HTP explore the possibility of expanding its testing service to meet the increasing testing demands of other industries as a future strategy.

One of the most important considerations for ANAS HTP in planning the expansion of its

testing service is how to secure a stable funding for its testing organization. As aforementioned, it is important to first identify industries with the biggest testing demands. In parallel, it needs to conduct a competitive analysis of other testing and certification organizations operating in Azerbaijan. Especially, in its pursuit of becoming an integrated testing and certification service institute, ANAS HTP needs to analyze and understand the legal status of the expanded organization as well as other related laws and regulations in the concerned industries. In addition, it needs to secure testing staffs and experts capable of performing the expanded testing service for other industries.

For this purpose, the testing center of ANAS HTP needs to develop a plan to expand the scope of its testing service and secure stable budget sources including the existing income from service fees. From 2026, ANAS HTP needs to make major investments in testing facilities and start building a quality management system in newly added testing areas. In 2028, it can obtain an authorization from the certification authority and finally launch a package of new testing services. For the initial facility investment, ANAS HTP may receive budget support from other institutions, but it needs to work hard to maximize profits from service fees by fully utilizing its testing facilities.

If ANAS HTP decides to expand its testing service to specific industries based on the testing demand and competitive analyses, it needs to plan for securing testing equipment to meet such testing needs because testing and certification service industry is an infrastructure-based industry as aforementioned. One of the most important considerations in securing testing facilities is to acquire testing facilities suitable for the testing standards and specifications in the concerned testing areas.

For example, when ANAS HTP's testing service is rolled out to ICT industry, it needs to provide the following testing services like i) RF measurement, ii) EMC measurement, and iii) electrical safety testing, by conforming to the specifications of ICT products. It is critical to possess advanced testing equipment to be able to provide these testing services. Especially, the EMC measurement requires the installation of an anechoic chamber along with other relevant measurement equipment, costing a formidable amount of budget. Considering that EMC testing is required in all kinds of electrical products circulated in

Azerbaijan, it is recommended that ANAS HTP install necessary equipment for EMC testing, but this should be based on an accurate profit estimation of service fees relative to facility investment. In addition, electrical safety testing is also an important service area as it is applied to all kinds of ICT products as well as other electrical products. This implies the expansion potential of the electrical safety testing service. Therefore, it is necessary to conduct a thorough preliminary research when ANAS HTP considers expanding its testing service to include electrical safety testing.

For building the testing equipment, both new and used testing equipment can be acquired. Long-term maintenance service is often available for most of the testing equipment by their manufacturers. It is recommended that ANAS HTP refer to the relevant EN specifications as well as EU directives on testing specifications and equipment when it considers an expansion of its testing service to certain industries. In Azerbaijan, there may exist some testing specifications specifically defined and required by respective industries. Therefore, it is recommended that ANAS HTP acquire and install testing equipment catered to such specifications.

4.2.3.2 Upgrading testing facilities for petrochemical and LPG industries

Currently, the testing center of ANAS HTP is operating a set of modernized testing facilities to provide testing service for petrochemical and LPG industries. However, its testing facilities are in shortage to provide testing service for certain items. As mentioned earlier, it is recommended that ANAS HTP expand the testing facility of its testing center by considering the testing center's shortage of testing facility as well as the need for making stable profits from the testing service fees relative to the increasing testing demands.

Throughout the world, almost all the testing institutes aim to provide one-stop testing service, which requires a major investment in expensive testing facilities and equipment. When there are limited testing demands, such costly investment in testing facilities may negatively affect the testing institutes' profitability. Therefore, a cautious approach is needed in making investment decisions on testing facility. In consideration of this, it is recommended that ANAS HTP develop a plan to upgrade its testing facilities while securing stable profits from testing service fees utilizing the existing facilities by 2022. Then, it is

recommended to use these profits to acquire new testing facilities and start providing expanded testing service from 2023.

To secure extra testing facilities, it is recommended to first conduct a research to identify nearby institutions holding testing facilities in need because ANAS HTP can use these facilities held by other testing institutions through outsourcing contracts. This will enable ANAS HTP to provide one-stop testing service. However, in case where this outsourcing option is not available, the research team suggests ANAS HTP purchase used testing equipment or utilize equipment held by some private companies. It can even consider getting facility support through global ODA projects.

4.2.4 Maximizing profits from testing service fees

In order for the testing center of ANAS HTP to emerge as a testing and certification specialized organization, it needs to pursue financially sustainable growth by maximizing its profits from service fees. For this purpose, a testing demand analysis needs to be conducted with user industries every year. Even after the scope of testing service is expanded in the future, it is important to continue the same type of demand analysis.

The advertisement of testing service provided by the testing center of ANAS HTP should target industries who are the users of those testing services, so that ANAS HTP can make stable profits from its testing service fees. At the same time, collaboration or partnership with other testing and certification institutes need to be explored to provide one-stop testing and certification service. It is also recommended to build a foundation for mutual recognition of testing and certification services across borders in the future. For this purpose, it is recommended that ANAS HTP develop a plan for maximizing its profits from service fees by 2022 and, based on this plan, it can explore various ways to diversify its testing and certification services. At the same time, a proper pricing scheme for service fees needs to be developed to better support the expansion into certification service and other testing areas.

Especially, for the purpose of maximizing profits, testing service needs to be broken down

into several sub-categories such as i) testing service to verify R&D, ii) pre-testing service, iii) testing service for certification, and iv) review service. Service fees need to be estimated and charged for respective sub-categories. In expanding the scope of testing service, the same approach can be taken.

For your reference, the review service refers to the certification institute's review of testing results in cases where a conformity assessment is conducted based only on the testing results thanks to the mutual recognition of conformity assessment across borders.

The unit price of testing service fee for specific service needs to be calculated based on i) the price of equipment in use (considering depreciation cost of the equipment), ii) labor cost, and iii) other expenditures. In addition, the competitive dynamics with other testing organizations also needs to be considered in the pricing scheme of the testing service fees.

A wider user accessibility to ANAS HTP 's service is also critical for maximizing profits. For this purpose, PR and marketing activities need to be promoted as well. It is recommended to create opportunities that can stimulate industries' testing and certification service demands by building and operating an optimal network with them.

4.3 Roadmap

The afore-mentioned activities that have been proposed to support testing and certification services can be performed in a stepwise manner as presented in the [Figure 5-19] below.

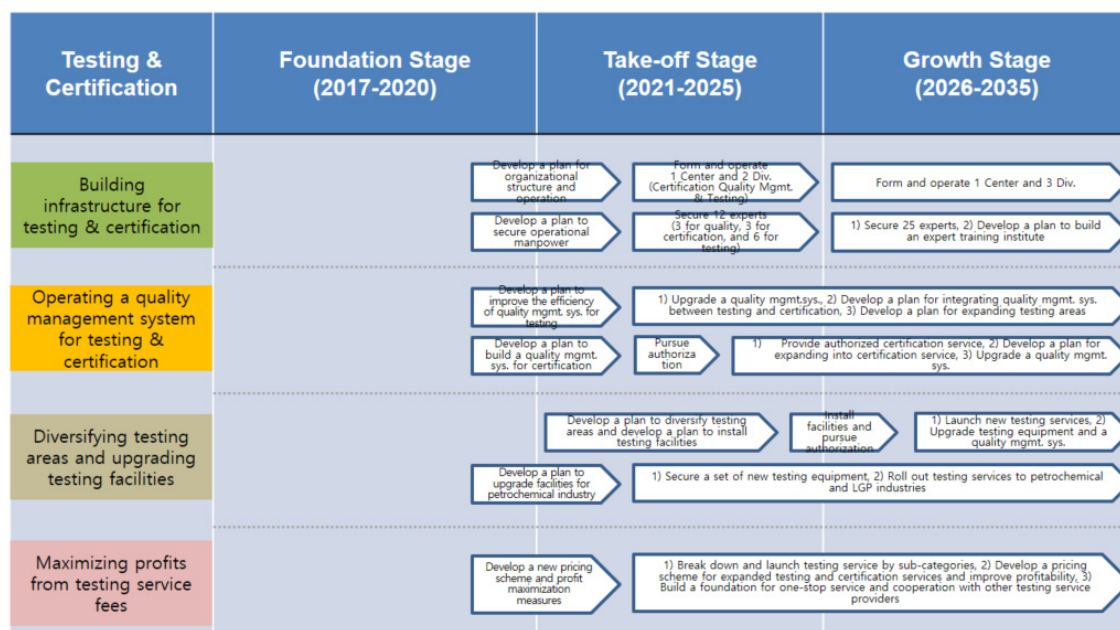
First, in relation to building infrastructure for certification, ANAS HTP is recommended to i) complete the development of a plan for organizational structure and operation by the end of 2022, launch and operate 1 center and 2 divisions from 2023, and restructure them into 1 center and 3 divisions and run this expanded organization during the period from 2026 to 2035, and ii) complete the development of a workforce operational plan by 2022, secure 12 experts and start their operation from 2023, and increase the number of these experts to 25 in 2026 and continue their operation till 2035.

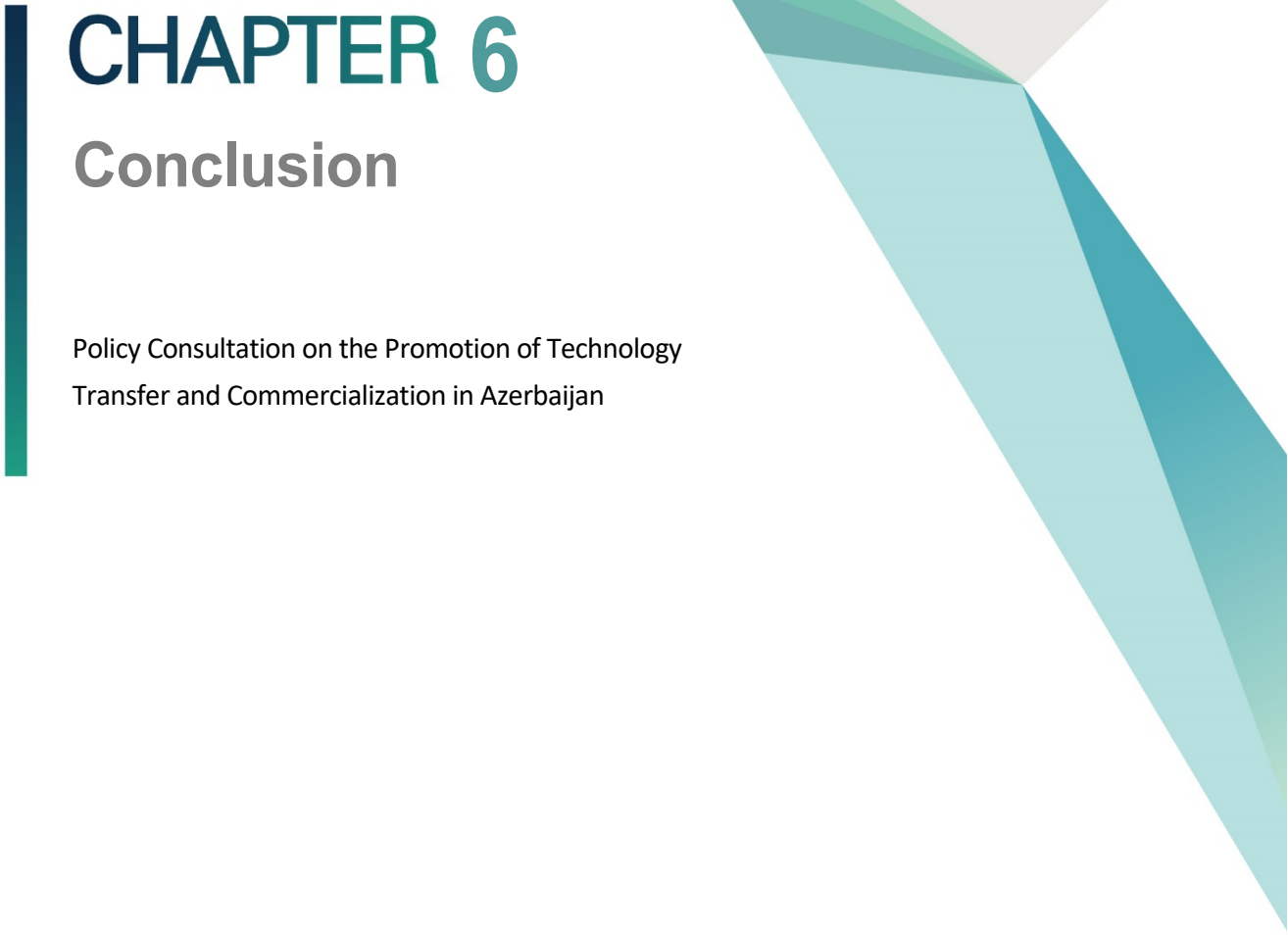
Second, in relation to its efforts to run a quality management system for testing, ANAS HTP needs to develop a plan to improve the efficiency of the quality management system for testing by 2022. From 2023, it can explore feasible approaches to an integrated operation of testing and certification services as result of upgraded quality management system and expanded testing and certification services. With regards to its certification related efforts, ANAS HTP needs to develop a plan to establish a quality management system for certification by 2022, obtain an authorization for its quality management system in certification from a certification authority by 2023, and launch its official certification service in a full-fledge manner from 2024.

Third, as for ANAS HTP's efforts to diversify testing areas, it is suggested that ANAS HTP develop a plan for expanding its testing service to ICT industry by 2025, start installing equipment and pursue the authorization of expanded testing service from 2026, and launch a new set of testing services from 2028. In relation to upgrading testing facilities, it is recommended to develop a plan for expanding existing testing facilities for petrochemical and LPG industries by 2022 and launch the expanded testing service using newly installed equipment from 2023.

Lastly, when it comes to measures for maximizing profits from testing service fees, ANAS HTP needs to come up with a pricing scheme of its service fees by breaking down its testing service into several sub-categories by 2022 and then improve its profitability based on the diversified testing service and expanded service into certification.

[Figure 5-19] Roadmap for Test & Certification



A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and green, also pointing downwards.

CHAPTER 6

Conclusion

Policy Consultation on the Promotion of Technology
Transfer and Commercialization in Azerbaijan



Chapter 6. Conclusion

This is the Year-2 part of the two-year long K-Innovation ODA Program (2019-2020) titled “Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan.”

In the first year (2019), the project was focused on i) analyzing Azerbaijan’s STI policy and the TT & TC ecosystem at the national level, affecting ANAS-HTP’s TT & TC activities, and ii) sharing South Korea’s TT & TC experiences to learn the theoretical approaches and methodologies that South Korea adopted in pursuing TT & TC. From the policy mix perspective of STI policy, TT & TC, and Start-up & Incubation, the project placed its priority on analyzing and strengthening the capabilities of government, academia, industry, and research institutes in Azerbaijan.

In the meantime, the second year (2020) was focused on developing strategic plan for the successful operation of ANAS-HTP at the institutional level, based on the research outcomes from the Year 1 activities. So, the project team analyzed the current state of ANAS-HTP to help build an internal system covering its vision, mission, principles, structure, budget, workforce, and network and present its future development direction. Then, the project team has identified the following key operational programs; i) TT & TC, ii) Start-up & Incubation, and iii) Test & Certification. Based on the analysis of the state of these operational programs, the project team has also proposed their implementation direction.

Then, what are the immediate actions that need to be taken by ANAS-HTP for its take-off? First of all, strong commitment to changes and active cooperation are needed on the side of ANAS-HTP’s Director and its top management team. The project team has now presented a strategic plan for ANAS-HTP as a blueprint and a future orientation to be pursued by ANAS-HTP, but it still remains as a plan. Even a perfect plan can become

useless if it stays on the shelf and is not put into practice. This clearly points to the importance of implementing a plan. In recognition of such importance of implementation, ANAS-HTP Director and its top management team are advised to take necessary actions that can facilitate the implementation of the proposed plan. Unfortunately, however, there have been too frequent changes in the ANAS-HTP leadership, which has hindered the consistent and sustainable implementation of any plans. ANAS-HTP's potential for sustainable growth is likely to be limited unless the service term of ANAS-HTP Director is guaranteed (e.g. for three years) and institutionalized.

Second, ANAS-HTP should exert its efforts to secure necessary budget for its take-off. A perfect plan would be of no use if it fails to secure budget to financially support the implementation. All the programs the project team has proposed in this report can be successfully implemented only when they are financially supported. Then, what would be the best approach to the successful implementation of the proposed programs? Maybe, the most feasible approach would be to implement a follow-up project. Now, strategic plans have been presented for each of the three key operational programs (e.g., TT & TC, Start-up & Incubation, Test & Certification) as outcomes of the project activities over the past two years. Now, for a follow-up project, it is necessary to carry out a set of follow-up activities that are required for the implementation of the three key operational programs, for example, methodology learning, field training, and cooperative projects. (see [Table 6-1]).

[Table 6-1] An Example of Follow-up Project Activities

Category	Description
Title of Project	2022 – 2024 K-Innovation Partnership Program: Operational Program of ANAS-HTP in Azerbaijan: Implementation and Capacity Building
Project Period/ Total Budget	- Project period: 2022 – 2024 (3 years) - Total budget: 1.5 million
Purpose of Project	This project aims to strengthen the operational capacities and competitiveness of the Azerbaijan National Academy of Sciences High Technologies Park (ANAS-HTP) through essential training programs for increasing operational expertise, on-the-job training (OJT), and collaborative projects.

Category	Description
Rationales of Project	• The “2019–2020 K-Innovation ODA Program” was operated by STEPI and ANAS-HTP. - Year 1 (2019): STEPI conducted a national-level analysis of Azerbaijan’s STI policy and TT/TC ecosystem, and shared South Korea’s experiences. - Year 2 (2020): Based on the research results from Year 1, the project focused on developing a strategic plan for the successful operation of ANAS-HTP at an institutional level. In other words, STEPI designed an internal system and three operational programs* for ANAS-HTP and presented a roadmap to achieve program objectives. * Technology transfers and commercialization, startups and incubation, and testing and certification services. • Follow-up tasks are critical for moving the project from the planning phase to the execution phase and for effectively implementing the strategic plan outlined above. - The abovementioned, two-year project (2019–2020) prioritizes the formulation of a strategic plan and roadmap for the future successful operation of STEPI’s partner, ANAS-HTP. - The plan proposes three operational programs (services) to be pursued: technology transfers and commercialization, startups and incubation, and testing and certification services. - If the three aforementioned operational programs remain a mere “plan” without any implementation, the ODA project will have been in vain.
Contents of Project	• Year 1: Capacity building for technology transfer and commercialization services - Knowledge transfers (capacity-building workshops), drafting of business guidelines - OJT, promotion of South Korea-Azerbaijan cooperation projects, etc. • Year 2: Capacity building for startups and incubation services - Knowledge transfers (capacity-building workshops), drafting of business guidelines - OJT, promotion of South Korea-Azerbaijan cooperation projects, etc. • Year 3: Capacity building for testing and certification services - Knowledge transfers (capacity-building workshops), drafting of business guidelines - OJT, promotion of South Korea-Azerbaijan cooperation projects, etc.

Third, ANAS-HTP should work hard to build infrastructure necessary for implementing the proposed programs by actively participating in various opportunities including ODA projects run by South Korean donor institutes (e.g. KOICA' policy consulting project and KDI's KSP Project), concessional loan programs operated by South Korean partners (e.g. EXIMBANK's EDCF project), and other ODA projects led by international organizations (e.g. JICA, WB, and UNDP projects). As a part of its efforts to build necessary capacities, ANAS-HTP should sign MOUs with South Korean counterparties engaged in TT & TC, Start-up & Incubation, and Test & Certification (e.g. Techno parks, TLOs, and Business Incubators), and actively pursue joint projects with them. In this endeavor, STEPI can play a bridging role between ANAS-HTP and its counterparties.

Over the past two years, STEPI and ANAS-HTP have successfully implemented this project through close cooperation and collaborative partnership. It is an ardent wish of the project team that the outcomes of this project serve as a momentum for ANAS-HTP to emerge as a top player among all the HTPs in the Caspian countries.

REFERENCES

- Choi, Jong-in & Markham, S. 2019. Creating a corporate entrepreneurial ecosystem: The case of entrepreneurship education in the RTP, USA. *Journal of Open Innovation: Technology, Market, and Complexity*, 5(3): 62-76.
- Choi, Jong-in & Park, C. The Key Success Factors of University Entrepreneurship Education: Implication from USA University Cases. *Asia Pac. J. Bus. Ventur. Entrep.* 2013, 8, 85–96.
- Internal document of GTP.
- Internal document of South Korea Technopark Association (KTPA).
- ITEP (2007), Planning the second stage Techno park construction project Planning the second.
- KISTEP (2019), Technology Commercialization Policy Trend, KISTEP Technoogy Trend Brief, 2019-8.
- KIAT (2009), South Korea Technopark 2020 long-term development strategy and implementation plan .
- KIAT (2019), Public technology transfer and commercialization status and tasks.
- KIIP (2019), Government R&D Patent Management Status and Implications.
- KTPA (2015), Establishment of South Korean Techno Park (TP) model and establishment of KVIP operation know-how transfer plan.
- KTPA (2016), Techno Park, The growth engine of South Korea.
- KTPA (2017), The role and function of TP for fostering local businesses.
- Markham, S.K.; Mugge, P.C. 2015, *Traversing the Valley of Death*; CIMS: Raleigh, NC, USA.
- Ministries (2017), The 6th technology transfer and commercialization promotion plan.
- Ministry of Science and ICT (2018), A study on ways to revitalize technology commercialization through cooperation between the public and the private sector.
- Ministry of SMEs and Startups (2018), Technopark (TP) function enhancement plan.

Ministry of Trade, Industry and Energy (2018), Public technology transfer commercialization status survey, Statistical Information Report.

Ministry of Transport, Communication, High-Technologies (MTCHT) in Azerbaijan.

National Assembly Budget Office (2019), Analysis of national research and development projects.

Seokchan Ko (2017), A Study on the Re-establishment of Functions of South Korean Techno Park.

Son, Yoon Jeong & Choi, Jong-in, 2020, University Entrepreneurial Ecosystem and Business Incubator's Performance: The Case Study of Hanbat National University, South Korean Journal of HRD. 23(2), 1-23.

STEPI (2019), Technology financing proposal for efficient technology commercialization, STEPI Insight.

Website

ADA University, <https://www.ada.edu.az>

ANAS HTP, <https://www.ameaytp.az/>

Azerbaijan National Academy of Sciences, <http://science.gov.az>

South Korean Standards and Certifications, <https://standard.go.kr>

Gyeongnam Technopark, <https://www.gntp.or.kr/>

President of the Republic of Azerbaijan, <https://president.az>